

Optimal Transport for Machine Learners

Gabriel Peyré
CNRS and ENS, PSL Université
gabriel.peyre@ens.fr

June 7, 2026

Abstract

Optimal transport is a foundational mathematical theory that connects optimization, partial differential equations, and probability. It offers a powerful framework for comparing probability distributions and has become an important tool in machine learning, especially for designing and evaluating generative models. This book covers the fundamental mathematical aspects of OT, including the Monge and Kantorovich formulations, Brenier’s theorem, dual and dynamic formulations, the Bures metric on Gaussian distributions, and gradient flows. It also introduces numerical methods such as linear programming, semi-discrete solvers, and entropic regularization. Applications in machine learning include training neural networks via gradient flows, token dynamics in transformers, GANs, diffusion and flow-matching models.

Guide to the Literature and Scope

Several books already cover optimal transport from complementary viewpoints. The two-volume monograph of Rachev and Rüschendorf [176, 177] gives a broad probabilistic treatment of mass transportation and its applications. Villani’s books [211, 212] are the standard references for the modern mathematical theory, from Kantorovich duality to curvature, concentration and geometric analysis. Santambrogio’s text [190] offers a concise applied-mathematics route through the same foundations, with a strong emphasis on PDEs and variational arguments. Ambrosio, Gigli and Savaré [6] develop the metric-space theory of gradient flows that underlies the dynamical part of the subject.

On the computational side, Peyré and Cuturi [170] provide the reference account of numerical OT, entropic regularization and applications in data sciences. Galichon’s book [87] explains the economic and matching-theoretic viewpoint, while the statistical theory of OT is developed in the recent monograph of Weed, Niles-Weed and Rigollet [215]. The long article of Liero, Mielke and Savaré [140] is the standard reference for unbalanced and conic extensions. Recent surveys complement these books by emphasizing scalable algorithms and machine-learning applications [125, 158], as well as the role of OT in imaging and graphics [35]. These references remain the natural places to find exhaustive proofs, historical details and specialized variants.

The aim here is different and more selective. The book keeps the core mathematics explicit, but organizes it around the questions that repeatedly arise in machine learning: how to compare singular empirical measures, how to compute differentiable transport losses, how regularization changes optimization and statistics, how dual potentials become discriminators, and how transport geometry produces flows of particles, neurons and tokens. The intended contribution is therefore not a replacement for the references above, but a compact bridge between rigorous OT and the geometric intuitions needed to use it in modern ML.

1 Optimal Matching between Point Clouds

This opening section isolates the simplest form of optimal transport: pairing two finite point clouds. The stakes are algorithmic and geometric at once: one sees the combinatorial nature of transport, the

special simplicity of the line, and the first hints that convex relaxation will be necessary in higher dimension. Classical assignment algorithms such as the Hungarian method and auction methods [130, 24] provide the computational backdrop, while the geometric examples prepare the Kantorovich relaxation.

1.1 Monge Problem for Discrete Points

This subsection formulates matching as Monge's deterministic transport problem on two equally weighted clouds. The one-dimensional case is a transparent reference case where the optimal map can be read off by sorting.

Matching Problem Given a cost matrix $(C_{i,j})_{i \in [n], j \in [m]}$ and assuming $n = m$, the optimal assignment problem aims to find a bijection σ within the set $\text{Perm}(n)$ of permutations of n elements that solves

$$\min_{\sigma \in \text{Perm}(n)} \frac{1}{n} \sum_{i=1}^n C_{i, \sigma(i)}. \quad (1)$$

One could naively evaluate the cost function above using all permutations in the set $\text{Perm}(n)$. However, this set has size $n!$, which becomes enormous even for small values of n . In general, the optimal σ is not unique.

1D Case In 1D, for convex cost, the matching defines a monotonic map.

Proposition 1 (Monotone matching on the line). *Assume that the points $(x_i)_i$ and $(y_j)_j$ are pairwise distinct. If the cost is of the form $C_{i,j} = h(x_i - y_j)$, where $h : \mathbb{R} \rightarrow \mathbb{R}^+$ is strictly convex (for example, $C_{i,j} = |x_i - y_j|^p$ for $p > 1$), then any optimal σ defines a strictly increasing map $x_i \mapsto y_{\sigma(i)}$ (and thus is unique), i.e.,*

$$\forall (i, i'), \quad (x_i - x_{i'})(y_{\sigma(i)} - y_{\sigma(i')}) > 0.$$

Proof. Indeed, if this property is violated, i.e., there exists (i, i') such that $(x_i - x_{i'})(y_{\sigma(i)} - y_{\sigma(i')}) < 0$, then one can define a permutation $\tilde{\sigma}$ by swapping the match, i.e., $\tilde{\sigma}(i) = \sigma(i')$ and $\tilde{\sigma}(i') = \sigma(i)$, yielding a strictly better cost, as proved in the following fact. Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be strictly convex and let $x < x'$ and $y < y'$. Then

$$h(x - y) + h(x' - y') < h(x - y') + h(x' - y).$$

We set the gap $d := y' - y > 0$ and define for every $s \in \mathbb{R}$

$$D(s) := \frac{h(s) - h(s - d)}{d} \quad \text{and} \quad \Delta := h(x - y') + h(x' - y) - h(x - y) - h(x' - y') = d(D(x' - y) - D(x - y)).$$

Because h is strictly convex, D is strictly increasing. Since $x - y < x' - y$, monotonicity yields $D(x - y) < D(x' - y)$, that is $\Delta > 0$. $\square$

This property extends by continuity to convex (not strictly convex) costs such as $|x - y|$, but in this case, the matching is not necessarily unique. Note that if h is concave instead of being convex, then the behavior is entirely different, and the optimal match tends to exchange the positions. In this case, there exists an $O(n^2)$ algorithm.

Therefore, the algorithm to compute an optimal transport is to sort the points, i.e., find some pair of permutations σ_X, σ_Y such that

$$x_{\sigma_X(1)} \leq x_{\sigma_X(2)} \leq \dots \quad \text{and} \quad y_{\sigma_Y(1)} \leq y_{\sigma_Y(2)} \leq \dots$$

and then an optimal match is mapping $x_{\sigma_X(k)} \mapsto y_{\sigma_Y(k)}$, i.e., an optimal transport is $\sigma = \sigma_Y \circ \sigma_X^{-1}$. The total computational cost is thus $O(n \log(n))$, using, for instance, the quicksort algorithm. Note that if $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is an increasing map, one can apply this technique to costs of the form $h(|\varphi(x) - \varphi(y)|)$ with a change of variable. A typical application is grayscale histogram equalization. The empirical cumulative distribution

of the luminance values of an image is transported to a prescribed target histogram, often a uniform or reference-image histogram. In one dimension, the monotone rearrangement above gives the exact transport map, so the operation is both computationally simple and geometrically faithful: it matches distributions of intensities rather than individual pixels.

Note that if h is strictly convex, then all optimal assignments are increasing, and if the points are all distinct, this increasing map is unique. If h is not strictly convex, for instance $c(x, y) = |x - y|$, non-increasing optimal assignments can also exist. This happens, for example, in the book-shifting problem with overlapping uniform distributions, where the mass in the intersection can stay fixed.

This efficient strategy to compute the OT in 1-D does not extend to higher dimensions. In 2-D, as already noted by Monge, one has the following property.

Proposition 2 (Non-crossing optimal matchings). *In dimension 2, for $c(x, y) = \|x - y\|$, if σ is an optimal assignment, then segments $[x_i, y_{\sigma(i)}]$ cannot cross.*

Proof. If two segments $[x_i, y_{\sigma(i)}]$ and $[x_j, y_{\sigma(j)}]$ cross at an interior point z , then the triangle inequality gives

$$\|x_i - y_{\sigma(j)}\| + \|x_j - y_{\sigma(i)}\| < \|x_i - y_{\sigma(i)}\| + \|x_j - y_{\sigma(j)}\|.$$

The assignment obtained by swapping $(\sigma(i), \sigma(j))$ therefore has a strictly smaller cost, which contradicts optimality. $\square$

This property alone is, however, not enough to lead to an efficient algorithm. Non-crossing is only a necessary local test, not a compact certificate of optimality. For instance, if n sources and n targets are placed alternately on the boundary of a convex polygon, the number of non-crossing perfect matchings is the Catalan number

$$C_n = \frac{1}{n+1} \binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}^{3/2}}.$$

Thus even after forbidding crossings, an exhaustive search remains exponential. The two-segment swap in the proof above is nevertheless useful: it explains why a crossing matching cannot be optimal, but it does not select among the exponentially many planar matchings that survive this local test.

1.2 Matching Algorithms

This subsection briefly locates matching within classical combinatorial optimization. Its main point is that efficient algorithms exist, but their cleanest analysis is obtained only after introducing the linear-programming viewpoint.

Efficient algorithms exist to solve the optimal matching problem. The most well-known are the Hungarian method and auction algorithms [130, 23, 24]. Auction algorithms use prices on the target points: each source bids for the target with largest reduced profit, the target price is increased, and the process terminates when the ε -complementary slackness conditions are satisfied. For integer costs, choosing $\varepsilon < 1/n$ gives an exact optimum after a finite number of bids [24]. Section 5.2 revisits this algorithm after Kantorovich duality and explains why it is a dual price method, parallel in spirit to Sinkhorn scaling.

Hungarian primal-dual method. The Hungarian method is best understood as a certificate-building algorithm for the assignment linear program. It maintains a partial matching M and dual prices (u_i, v_j) satisfying

$$u_i + v_j \leq C_{i,j} \quad \forall i, j.$$

The equality graph $E(u, v) = \{(i, j) : u_i + v_j = C_{i,j}\}$ contains the edges whose reduced cost is zero. The algorithm only augments M along alternating paths made of equality edges. Starting from an unmatched source, it grows an alternating tree with source set S and target set T . If the tree reaches an unmatched

target, the matching is augmented along the path. If no such edge exists, the dual variables are shifted by the smallest slack

$$\delta = \min_{i \in S, j \notin T} (C_{i,j} - u_i - v_j), \quad u_i \leftarrow u_i + \delta \ (i \in S), \quad v_j \leftarrow v_j - \delta \ (j \in T).$$

This update preserves all inequalities $u_i + v_j \leq C_{i,j}$, keeps the current alternating tree tight, and creates at least one new equality edge leaving S . Maintaining these slacks incrementally gives the standard $O(n^3)$ implementation for an $n \times n$ assignment problem.

Proposition 1 (Correctness and complexity of the Hungarian primal-dual method). *Assume the Hungarian method terminates with a perfect matching σ contained in the equality graph*

$$E(u, v) = \{(i, j) ; u_i + v_j = C_{i,j}\},$$

where (u, v) is dual feasible, i.e. $u_i + v_j \leq C_{i,j}$ for all (i, j) . Then σ is an optimal assignment. Moreover, the usual Hungarian updates terminate after finitely many augmentations. With maintained slacks, the method uses $O(n^3)$ arithmetic operations.

Proof. For any permutation τ , dual feasibility gives

$$\sum_i C_{i,\tau(i)} \geq \sum_i (u_i + v_{\tau(i)}) = \sum_i u_i + \sum_j v_j.$$

This is the weak duality lower bound. If σ is contained in the equality graph, then

$$\sum_i C_{i,\sigma(i)} = \sum_i u_i + \sum_j v_j,$$

so the primal cost of σ reaches the dual lower bound and is optimal.

It remains to justify finite termination and the complexity bound. At each successful augmentation, the matching cardinality increases by one, so there are at most n augmentations. During one augmentation phase, the algorithm grows an alternating tree in the equality graph. If no augmenting path is available, the dual update uses the smallest slack of an edge leaving the current tree. For edges inside the tree, adding δ to source labels and subtracting δ from target labels preserves tightness; for edges from S to T^c , the definition of δ preserves feasibility and makes at least one new edge tight; all other inequalities are unchanged or become looser. Thus the reachable sets strictly grow between two failed augmentation attempts, and they can grow at most n times within one phase. If the current slacks $\min_{i \in S} (C_{i,j} - u_i - v_j)$ are updated when a source enters S , each tree expansion costs $O(n)$. A phase therefore costs $O(n^2)$, and the n phases give $O(n^3)$ operations. Hence the method reaches a perfect optimal matching. $\square$

Their derivation and analysis are greatly simplified by introducing the Kantorovich relaxation and its associated dual problem. The transportation simplex and network-flow viewpoints go back to Dantzig's transportation linear program and modern minimum-cost-flow algorithms [69, 25, 162]. These finite matching and assignment ideas also underlie color transfer between images. Grayscale equalization is one-dimensional, but transferring a full color palette requires transporting empirical measures in a three-dimensional color space, for instance RGB or Lab. Early color-transfer methods use affine statistics or iterated one-dimensional projections [181, 174]; replacing these projections by a genuine three-dimensional OT coupling gives a more intrinsic way to match colors while preserving the geometry of the palette [175].

2 Monge Problem between Measures

The goal of this section is to pass from finite matching to transport between arbitrary probability laws. The central stakes are to define measures, push-forwards and Monge maps carefully enough that the discrete

picture survives, while exposing why deterministic maps can fail to exist. Monge’s original formulation [157] and modern treatments [211, 212, 190, 176] are the conceptual background for this transition.

The presentation of the previous section could only handle two sets with the same number of points. To relax this to a more general setting, one needs to consider probability distributions, so that the points are weighted by masses.

2.1 Measures

Measures are the language that lets point clouds, densities and singular objects be handled uniformly. We only recall the facts needed later: integration, total variation, densities and probabilistic laws.

Histograms We use the terms histogram and probability vector interchangeably for any element $\mathbf{a} \in \Sigma_n$ of the probability simplex

$$\Sigma_n := \left\{ \mathbf{a} \in \mathbb{R}_+^n ; \sum_{i=1}^n a_i = 1 \right\}.$$

Discrete measure, empirical measure A discrete measure with weights $\mathbf{a}$ and locations $x_1, \dots, x_n \in \mathcal{X}$ reads

$$\alpha = \sum_{i=1}^n a_i \delta_{x_i} \quad (2)$$

where δ_x is the Dirac mass at position x , intuitively a unit of mass infinitely concentrated at location x . Such a measure is a probability measure if, additionally, $\mathbf{a} \in \Sigma_n$, and more generally a positive measure if all weights a_i are nonnegative. An “empirical” probability distribution is uniform on a point cloud, i.e. $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$. In practice, in many applications, it is useful to be able to manipulate both the positions x_i (“Lagrangian” discretization) and the weights a_i (“Eulerian” discretization). Lagrangian modification is usually more powerful (because it leads to adaptive discretization) but it breaks the convexity of most problems.

General measures We consider Borel measures $\alpha \in \mathcal{M}(\mathcal{X})$ on a metric space $(\mathcal{X}, d)$, i.e. one can compute $\alpha(A)$ for any Borel set A (which can be obtained by applying countable union, countable intersection, and relative complement to open sets). Unless otherwise stated, the measures are finite. A Dirac measure δ_x is then defined as $\delta_x(A) = 1$ if $x \in A$ and 0 otherwise, and this extends by linearity for discrete measures of the form (2) as

$$\alpha(A) = \sum_{x_i \in A} a_i$$

We denote $\mathcal{M}_+(\mathcal{X})$ the subset of all positive measures on $\mathcal{X}$, i.e. $\alpha(A) \geq 0$ (and $\alpha(\mathcal{X}) < +\infty$ for the measure to be finite). The set of probability measures is denoted $\mathcal{M}_+^1(\mathcal{X})$, which means that any $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ is positive, and that $\alpha(\mathcal{X}) = 1$.

Radon measures Using Lebesgue integration, a Borel measure can be used to compute the integral of measurable functions (i.e. such that level sets $\{x ; f(x) < t\}$ are Borel sets), and we denote this pairing as

$$\langle f, \alpha \rangle := \int f(x) d\alpha(x).$$

Integration of such a measurable f against a discrete measure α computes a sum

$$\int_{\mathcal{X}} f(x) d\alpha(x) = \sum_{i=1}^n a_i f(x_i).$$

This applies in particular to continuous test functions, which are Borel measurable. Integration against a finite measure on a compact space thus defines a continuous linear form $f \mapsto \int f d\alpha$ on the Banach space of continuous functions $(\mathcal{C}(\mathcal{X}), \|\cdot\|_\infty)$, indeed $|\int f d\alpha| \leq \|f\|_\infty |\alpha|(\mathcal{X})$. On compact spaces, the converse is true: every continuous linear form on $\mathcal{C}(\mathcal{X})$ is represented by integration against a finite signed Radon measure. This is the Riesz–Markov–Kakutani representation theorem [184, 32]. It identifies $\mathcal{M}(\mathcal{X})$ with the Banach dual of $\mathcal{C}(\mathcal{X})$, and this duality pairing $\langle f, \alpha \rangle$ will be crucial for the convex duality arguments used later.

Relative densities. A measure α which is a weighting of another reference measure λ is said to have a density, denoted $d\alpha(x) = \rho_\alpha(x)d\lambda(x)$, or $\rho_\alpha = \frac{d\alpha}{d\lambda}$. On $\mathbb{R}^d$ the reference λ is often Lebesgue measure dx . This means that

$$\forall h \in \mathcal{C}(\mathcal{X}), \quad \int_{\mathcal{X}} h(x) d\alpha(x) = \int_{\mathcal{X}} h(x) \rho_\alpha(x) d\lambda(x).$$

Total variation norm. The norm inherited from the duality $\mathcal{M}(\mathcal{X}) = \mathcal{C}(\mathcal{X})^*$ is the total variation norm. We use the notation

Definition 1 (Total variation). *For a finite signed Radon measure α on a compact space $\mathcal{X}$,*

$$\|\alpha\|_{\text{TV}} := \sup_{f \in \mathcal{C}(\mathcal{X})} \{ \langle f, \alpha \rangle ; \|f\|_\infty \leq 1 \}.$$

Equivalently one can put an absolute value around $\langle f, \alpha \rangle$ in the supremum: if f is admissible, so is $-f$. This formula is useful because it computes the norm of a measure as the operator norm of the corresponding linear form on $\mathcal{C}(\mathcal{X})$.

The same norm also has a direct measure-theoretic expression. The absolute value of a signed measure is

$$|\alpha|(A) := \sup_{A = \cup_i B_i} \sum_i |\alpha(B_i)|,$$

where the supremum is over finite or countable measurable partitions of A . Thus, if $\alpha = \sum_i a_i \delta_{x_i}$ with distinct atoms, $|\alpha| = \sum_i |a_i| \delta_{x_i}$; if $d\alpha(x) = \rho(x)d\lambda(x)$, then $d|\alpha|(x) = |\rho(x)|d\lambda(x)$.

Proposition 2 (Dual and measure definitions of total variation). *For a finite signed Radon measure α on a compact space $\mathcal{X}$,*

$$\|\alpha\|_{\text{TV}} = |\alpha|(\mathcal{X}).$$

Consequently $(\mathcal{M}(\mathcal{X}), \|\cdot\|_{\text{TV}})$ is isometrically the Banach dual of $(\mathcal{C}(\mathcal{X}), \|\cdot\|_\infty)$.

Proof. The inequality $\|\alpha\|_{\text{TV}} \leq |\alpha|(\mathcal{X})$ follows from

$$\left| \int f d\alpha \right| \leq \int |f| d|\alpha| \leq \|f\|_\infty |\alpha|(\mathcal{X}).$$

For the reverse inequality, write the Jordan decomposition $\alpha = \alpha^+ - \alpha^-$, so that $|\alpha| = \alpha^+ + \alpha^-$. The measurable sign $s = \frac{d\alpha}{d|\alpha|}$ takes values in $\{-1, 1\}$ outside a $|\alpha|$ -null set and satisfies $d\alpha = s d|\alpha|$. By regularity of Radon measures on compact spaces, s can be approximated in $L^1(|\alpha|)$ by continuous functions f_k with $\|f_k\|_\infty \leq 1$. Hence $\int f_k d\alpha \rightarrow \int s d\alpha = |\alpha|(\mathcal{X})$, which proves the equality. The final statement is the Riesz–Markov–Kakutani representation theorem with this norm. $\square$

For two absolutely continuous measures $d\alpha = \rho_\alpha d\lambda$ and $d\beta = \rho_\beta d\lambda$, this gives the concrete formula

$$\|\alpha - \beta\|_{\text{TV}} = \int_{\mathcal{X}} |\rho_\alpha(x) - \rho_\beta(x)| d\lambda(x).$$

For two discrete measures written on the same union of supports, $\alpha = \sum_k a_k \delta_{z_k}$ and $\beta = \sum_k b_k \delta_{z_k}$, with missing coefficients set to zero,

$$\|\alpha - \beta\|_{\text{TV}} = \sum_k |a_k - b_k|.$$

Some probability texts call $\frac{1}{2}\|\alpha - \beta\|_{\text{TV}}$ the total variation distance; here $\|\cdot\|_{\text{TV}}$ denotes the Banach norm on signed measures.

Probabilistic interpretation. Radon probability measures represent the laws of random variables. A random variable with values in $\mathcal{X}$ is a measurable map $X : \Omega \rightarrow \mathcal{X}$ from an abstract probability space $(\Omega, \mathbb{P})$, and its law is the Radon probability measure $\alpha = X_{\#}\mathbb{P}$ satisfying $\mathbb{P}(X \in A) = \alpha(A)$. Integrals are then expectations:

$$\int_{\mathcal{X}} f(x) d\alpha(x) = \mathcal{E}[f(X)].$$

If $Y = T(X)$, then the law of Y is $T_{\#}\alpha$.

2.2 Push Forward

Push-forwards encode how maps move mass. This short subsection is the bridge between deterministic maps and linear operations on measures.

For some continuous map $T : \mathcal{X} \rightarrow \mathcal{Y}$, we define the pushforward operator $T_{\#} : \mathcal{M}(\mathcal{X}) \rightarrow \mathcal{M}(\mathcal{Y})$. For a Dirac mass, one has $T_{\#}\delta_x = \delta_{T(x)}$, and this formula is extended to arbitrary measures by linearity. In some sense, moving from T to $T_{\#}$ is a way to linearize any map at the price of moving from a (possibly) finite-dimensional space $\mathcal{X}$ to the infinite-dimensional space $\mathcal{M}(\mathcal{X})$, and this idea is central to many convex relaxation methods, most notably Lasserre's relaxation. For discrete measures (2), the pushforward operation consists simply in moving the positions of all the points in the support of the measure

$$T_{\#}\alpha := \sum_i a_i \delta_{T(x_i)}.$$

For more general measures, for instance for those with a density, the notion of push-forward plays a fundamental role in describing spatial modifications of probability measures. The formal definition reads as follows.

Definition 2 (Push-forward). *For $T : \mathcal{X} \rightarrow \mathcal{Y}$, the push forward measure $\beta = T_{\#}\alpha \in \mathcal{M}(\mathcal{Y})$ of some $\alpha \in \mathcal{M}(\mathcal{X})$ satisfies*

$$\forall h \in \mathcal{C}(\mathcal{Y}), \quad \int_{\mathcal{Y}} h(y) d\beta(y) = \int_{\mathcal{X}} h(T(x)) d\alpha(x). \quad (3)$$

Equivalently, for any measurable set $B \subset \mathcal{Y}$, one has

$$\beta(B) = \alpha(\{x \in \mathcal{X} ; T(x) \in B\}). \quad (4)$$

Note that $T_{\#}$ preserves positivity and total mass, so that if $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ then $T_{\#}\alpha \in \mathcal{M}_+^1(\mathcal{Y})$.

Proposition 3 (Push-forward formula for densities). *Let α and β have densities ρ_{α} and ρ_{β} on open subsets of $\mathbb{R}^d$. Assume that T is a C^1 diffeomorphism and that $\beta = T_{\#}\alpha$. Then, for all x ,*

$$\rho_{\alpha}(x) = |\det(T'(x))| \rho_{\beta}(T(x)). \quad (5)$$

Equivalently, for $y = T(x)$,

$$\rho_{\beta}(y) = \rho_{\alpha}(T^{-1}y) \frac{1}{|\det T'(T^{-1}y)|}.$$

Proof. Explicitly doing the change of variable $y = T(x)$, so that $dy = |\det(T'(x))|dx$ in formula (3) for measures with densities $(\rho_{\alpha}, \rho_{\beta})$ on $\mathbb{R}^d$ (assuming T is smooth and a bijection), one has for all $h \in \mathcal{C}(\mathcal{Y})$

$$\begin{aligned} \int_{\mathcal{Y}} h(y) \rho_{\beta}(y) dy &= \int_{\mathcal{Y}} h(y) d\beta(y) = \int_{\mathcal{X}} h(T(x)) d\alpha(x) = \int_{\mathcal{X}} h(T(x)) \rho_{\alpha}(x) dx \\ &= \int_{\mathcal{Y}} h(y) \rho_{\alpha}(T^{-1}y) \frac{dy}{|\det(T'(T^{-1}y))|}, \end{aligned}$$

which shows that

$$\rho_\beta(y) = \rho_\alpha(T^{-1}y) \frac{1}{|\det(T'(T^{-1}y))|}.$$

Since T is a diffeomorphism, one obtains equivalently

$$\rho_\alpha(x) = |\det(T'(x))| \rho_\beta(T(x))$$

where $T'(x) \in \mathbb{R}^{d \times d}$ is the Jacobian matrix of T (the matrix formed by taking the gradient of each coordinate of T). This implies, denoting $y = T(x)$

$$|\det(T'(x))| = \frac{\rho_\alpha(x)}{\rho_\beta(y)}.$$

□

Remark 1 (Probabilistic interpretation). The law (i.e. probability distribution) of a random variable X , equivalently, is the push-forward of $\mathbb{P}$ by X , $\alpha = X_\# \mathbb{P}$. Applying another push-forward $\beta = T_\# \alpha$ for $T : \mathcal{X} \rightarrow \mathcal{Y}$, following (3), is equivalent to defining another random variable $Y = T(X)$, namely $\omega \in \Omega \mapsto T(X(\omega)) \in \mathcal{Y}$. Thus β is the law of Y . Drawing a random sample y from Y is thus simply achieved by computing $y = T(x)$ where x is drawn from X .

2.3 Monge's Formulation

Monge's problem asks for a deterministic map transporting one law onto another while minimizing a prescribed cost. This is the original formulation introduced by Monge in his memoir on the “déblai et remblai” problem [157]. The formulation is natural but fragile: it is non-convex and may be infeasible, which is precisely why Kantorovich's relaxation will be needed.

Monge problem. Monge problem (1) is extended to the setting of two arbitrary probability measures (α, β) on two spaces $(\mathcal{X}, \mathcal{Y})$ as finding a map $T : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes

$$\inf_T \left\{ \int_{\mathcal{X}} c(x, T(x)) d\alpha(x) ; T_\# \alpha = \beta \right\}. \quad (6)$$

The constraint $T_\# \alpha = \beta$ means that T pushes forward the mass of α to β , and makes use of the push-forward operator (3).

Proposition 4 (Empirical Monge maps and matchings). *Assume that the source atoms $x_1, \dots, x_n$ are distinct and that*

$$\alpha = \frac{1}{n} \sum_{i=1}^n \delta_{x_i} \quad \text{and} \quad \beta = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}.$$

If $T_\# \alpha = \beta$, then for each distinct target value z in the support of β , exactly $n\beta(\{z\})$ source atoms are mapped to z . In particular, if the y_j are distinct, then there exists a permutation $\sigma \in \text{Perm}(n)$ such that $T(x_i) = y_{\sigma(i)}$ for all i . Conversely, every such assignment of source atoms to target atoms with the correct masses defines a feasible Monge map on the support of α , and in the distinct-target case

$$\int_{\mathcal{X}} c(x, T(x)) d\alpha(x) = \frac{1}{n} \sum_i c(x_i, y_{\sigma(i)}).$$

If source locations are repeated, they should first be merged into atoms with larger masses; such atoms cannot be split by a Monge map.

Proof. Since $T_{\#}\alpha = \beta$, all images $T(x_i)$ must belong to the support of β ; otherwise the push-forward would give positive mass to a point outside that support. For any target atom z ,

$$\beta(\{z\}) = \alpha(T^{-1}(\{z\})) = \frac{1}{n} \#\{i : T(x_i) = z\}.$$

This proves the counting statement. If all target atoms have mass $1/n$, each target receives exactly one source atom, which is a permutation. The converse and the cost identity follow by direct substitution. $\square$

Proposition 5 (Existence of transport maps from non-atomic sources). *Let α and β be Borel probability measures on Polish spaces, and assume that α is atomless, equivalently on Polish spaces $\alpha(\{x\}) = 0$ for all x . Then there exists a measurable map T such that $T_{\#}\alpha = \beta$.*

Proof. A standard measure-isomorphism theorem identifies the atomless probability space generated by α with Lebesgue measure on $[0, 1]$, modulo null sets [32]. It is therefore enough to construct a map from $[0, 1]$ to the target law β . Choose a Borel isomorphism i between the target support and a Borel subset of $[0, 1]$, set $\nu = i_{\#}\beta$, and take the one-dimensional quantile map Q_{ν} . Since Q_{ν} belongs to $i(\text{supp } \beta)$ almost surely, $i^{-1} \circ Q_{\nu}$ is well defined up to null sets and has law β . Composing with the source isomorphism gives the desired measurable map T . $\square$

Remark 2 (Existence is not attainment). In general, an optimal map T solving (6) might fail to exist. The constraint set can be empty, for instance if $\alpha = \delta_x$ and β is not a single Dirac. Proposition 5 shows that non-atomicity of the source removes this obstruction, but it does not by itself ensure that the Monge infimum is attained.

Example 1 (A splitting obstruction). A classical example, discussed for instance in Santambrogio [190], takes α uniform on a vertical segment and β equal to the average of the uniform measures on two parallel vertical segments placed symmetrically to the left and to the right. For the quadratic cost, the relaxed Kantorovich optimizer splits each source point into its two symmetric targets. A deterministic Monge map cannot split one point into two destinations, so minimizing sequences must oscillate between the two sides and the Monge problem has no optimizer.

Example 2 (Semi-discrete Monge maps). The Monge formulation is not symmetric in α and β . It makes sense, for instance, when α has a density with respect to Lebesgue measure and β is discrete. On $\mathcal{X} = \mathcal{Y} = \mathbb{R}^d$, let $\beta = \sum_j b_j \delta_{y_j}$ be supported on $\{y_1, \dots, y_m\}$. A map T such that $T_{\#}\alpha = \beta$ defines a segmentation of the space into m cells

$$C_j := T^{-1}(y_j) \quad \text{where} \quad \alpha(C_j) = b_j.$$

This is the semi-discrete setting, which can be understood as an optimal quantization problem. Section 6 explains how the cells become Laguerre cells for prescribed masses and ordinary Voronoi cells when the masses are free.

If one exchanges the roles of α and β so that α is discrete, then no valid T exists in general: it is not possible to push forward a discrete measure to a measure with density.

Monge distance. In the special case $c(x, y) = d^p(x, y)$ where d is a distance, we denote

$$\tilde{\mathcal{W}}_p^p(\alpha, \beta) := \inf_T \left\{ \mathcal{E}_{\alpha}(T) := \int_{\mathcal{X}} d(x, T(x))^p d\alpha(x) ; T_{\#}\alpha = \beta \right\}. \quad (7)$$

If the constraint set is empty, then we set $\tilde{\mathcal{W}}_p^p(\alpha, \beta) = +\infty$. The following proposition records the metric properties that remain true for the Monge formulation. The important caveat is that the quantity is directed and can be infinite.

Proposition 6 (Directed Monge distance). *Assume $\mathcal{X} = \mathcal{Y}$ is a compact metric space. The quantity $\tilde{\mathcal{W}}_p$ is nonnegative, vanishes only on the diagonal, and satisfies the triangle inequality. It is therefore a directed extended distance: it need not be symmetric and may take the value $+\infty$.*

Proof. Nonnegativity is immediate. If $\tilde{\mathcal{W}}_p(\alpha, \beta) = 0$, choose feasible maps T_k with $\int d(x, T_k(x))^p d\alpha(x) \rightarrow 0$. For every continuous function h , compactness gives uniform continuity, hence

$$\left| \int h d\beta - \int h d\alpha \right| = \left| \int h(T_k(x)) - h(x) d\alpha(x) \right| \rightarrow 0.$$

Thus $\alpha = \beta$. We prove the triangle inequality. If $\tilde{\mathcal{W}}_p(\alpha, \beta) = +\infty$, then either $\tilde{\mathcal{W}}_p(\alpha, \gamma) = +\infty$ or $\tilde{\mathcal{W}}_p(\gamma, \beta) = +\infty$, because otherwise there are maps (S, T) such that $S_\# \alpha = \gamma$ and $T_\# \gamma = \beta$ and then $(T \circ S)_\# \alpha = \beta$ so that $\tilde{\mathcal{W}}_p(\alpha, \beta) \leq \mathcal{E}_\alpha(T \circ S) < +\infty$. We can therefore restrict our attention to the case where $\tilde{\mathcal{W}}_p(\alpha, \gamma) < +\infty$ and $\tilde{\mathcal{W}}_p(\gamma, \beta) < +\infty$ because otherwise the inequality is trivial. For any $\varepsilon > 0$, consider ε -minimizers $S_\# \alpha = \gamma$ and $T_\# \gamma = \beta$ such that

$$\mathcal{E}_\alpha(S)^{\frac{1}{p}} \leq \tilde{\mathcal{W}}_p(\alpha, \gamma) + \varepsilon \quad \text{and} \quad \mathcal{E}_\gamma(T)^{\frac{1}{p}} \leq \tilde{\mathcal{W}}_p(\gamma, \beta) + \varepsilon.$$

Using $(T \circ S)_\# \alpha = \beta$, the feasibility of this composed map and the triangle inequality give

$$\begin{aligned} \tilde{\mathcal{W}}_p(\alpha, \beta) &\leq \left(\int d(x, T(S(x)))^p d\alpha(x) \right)^{1/p} \\ &\leq \left(\int (d(x, S(x)) + d(S(x), T(S(x))))^p d\alpha(x) \right)^{1/p}. \end{aligned}$$

Then, using Minkowski inequality for the L^p spaces with measure α

$$\|f + g\|_{L^p(\alpha)} \leq \|f\|_{L^p(\alpha)} + \|g\|_{L^p(\alpha)}$$

and with $f(x) \triangleq d(x, S(x))$ and $g(x) \triangleq d(S(x), T(S(x)))$ one has

$$\begin{aligned} \tilde{\mathcal{W}}_p(\alpha, \beta) &\leq \left(\int d(x, S(x))^p d\alpha(x) \right)^{1/p} + \left(\int d(S(x), T(S(x)))^p d\alpha(x) \right)^{1/p} \\ &= \left(\int d(x, S(x))^p d\alpha(x) \right)^{1/p} + \left(\int d(y, T(y))^p d\gamma(y) \right)^{1/p} \\ &\leq \tilde{\mathcal{W}}_p(\alpha, \gamma) + \tilde{\mathcal{W}}_p(\gamma, \beta) + 2\varepsilon. \end{aligned}$$

Letting $\varepsilon \rightarrow 0$ gives the result. $\square$

This quantity $\tilde{\mathcal{W}}_p$ is problematic because it can have the value $+\infty$. It is the purpose of the Kantorovich formulation exposed below to remedy this issue, to define a symmetric and well-behaved distance $\mathcal{W}_p$.

2.4 Existence and Uniqueness of the Monge Map

This subsection records the special regimes where Monge's deterministic formulation becomes well posed. Brenier's theorem is the central result: for the squared Euclidean cost, absolute continuity of the source restores existence, uniqueness and convex-potential structure.

Brenier's theorem. The following celebrated theorem of [39, 40] ensures that in $\mathbb{R}^d$ for $p = 2$, if the source measure has a density, then Monge's problem has a unique solution, and it gives a precise description of the structure of this solution.

Theorem 1 (Brenier). *In the case $\mathcal{X} = \mathcal{Y} = \mathbb{R}^d$ and $c(x, y) = \|x - y\|^2$, if α has a density with respect to the Lebesgue measure, then there exists a unique optimal Monge map T . This map is characterized by being the unique gradient of a convex function $T = \nabla \varphi$ such that $(\nabla \varphi)_\# \alpha = \beta$.*

Proof. The proof uses the Kantorovich relaxation and duality developed in Section 5.4 and Equation (51). Kantorovich duality provides optimal potentials (f, g) with equality $f(x) + g(y) = \|x - y\|^2$ on the support of any optimal plan. After subtracting the harmless quadratic terms, this equality is equivalent to the Fenchel equality $\varphi(x) + \varphi^*(y) = \langle x, y \rangle$ for a convex function φ . Hence the support of every optimal plan lies in the graph of the subdifferential $\partial\varphi$. Since α has a density and convex functions are differentiable Lebesgue-almost everywhere, $\partial\varphi(x)$ is a singleton for α -almost every x . The plan is therefore concentrated on the graph of $T = \nabla\varphi$ and defines a Monge map. Any two optimal plans are concentrated on the same single-valued graph α -almost everywhere, which gives uniqueness of the map. This is the standard route in Brenier's polar factorization theorem; extensions to general strictly convex costs are developed in [91, 90, 43, 80]. $\square$

Definition 3 (Measures not charging hypersurfaces). *A Borel measure α on $\mathbb{R}^d$ is said not to charge hypersurfaces if $\alpha(S) = 0$ for every countably $(d - 1)$ -rectifiable set S , i.e. every set covered up to $\mathcal{H}^{d-1}$ -null sets by countably many C^1 hypersurfaces.*

Remark 3 (A sharp source hypothesis). The absolute-continuity assumption in Brenier's theorem can be weakened: for the quadratic cost, it is enough that α does not charge hypersurfaces [91, 212, 190]. The reason is that the set where a convex potential has a genuinely multi-valued subdifferential is contained in a countable union of such lower-dimensional pieces. This condition is close to sharp. If the source gives positive mass to a segment or a hypersurface, the subdifferential may be multi-valued on a set of positive α -mass, and the optimal relaxed plan may need to split mass, as in Example 1.

Remark 4 (Beyond the quadratic Euclidean cost). Brenier's theorem is the cleanest statement because the squared Euclidean cost turns optimal maps into gradients of convex functions. For costs $c(x, y) = \|x - y\|^p$ with $p > 1$, the same strategy gives a unique optimal map under absolute continuity of the source, but the map is written in terms of a c -convex potential rather than directly as $\nabla\varphi$. On a Riemannian manifold, the local analogue for the cost $d_{\mathcal{M}}(x, y)^2/2$ uses the exponential map:

$$T(x) = \exp_x(-\nabla\varphi(x)),$$

where φ is c -convex. The main additional issues are the cut locus and regularity of geodesics; this is why the Euclidean statement is usually presented first. These extensions are part of McCann's displacement convexity theory and the general theory of optimal maps on manifolds [148, 212].

Brenier's theorem, which states that an optimal transport map must be the gradient of a convex function, should be understood in light of the fact that convex functions naturally generalize increasing functions to dimensions larger than one. For instance, the gradient of a convex function is a monotone gradient field in the sense

$$\forall (x, x') \in \mathbb{R}^d \times \mathbb{R}^d, \quad \langle \nabla\varphi(x) - \nabla\varphi(x'), x - x' \rangle \geq 0.$$

Remark 5 (Monotone fields need not be gradients). In dimensions larger than 1, not all monotone fields are gradients of convex functions. Consider in $\mathbb{R}^2$ the rotation matrix

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

The linear map $T(x) = R_\theta x$ is monotone as soon as $|\theta| \leq \pi/2$, because

$$\langle R_\theta x - R_\theta x', x - x' \rangle = \langle R_\theta(x - x'), x - x' \rangle = \cos(\theta) \|x - x'\|^2 \geq 0.$$

However, for $\theta \neq 0$, R_θ is not symmetric and therefore cannot be the gradient of a scalar potential. Indeed, if a linear field Ax equals $\nabla\varphi(x)$, then its Jacobian A must be symmetric; equivalently, for a quadratic potential $\varphi(x) = \langle Bx, x \rangle/2$, one has $\nabla\varphi(x) = ((B + B^\top)/2)x$. Thus monotonicity is weaker than Brenier optimality in dimension $d \geq 2$.

Note also that this theorem can be extended in many directions. McCann's displacement convexity theory clarifies the geometric meaning of interpolating along Brenier maps [148]; this is the language of

Wasserstein geodesics used later for geodesic convexity in Section 10.4 and for barycenters in Section 9.2. Caffarelli's regularity theory explains when the convex potential is smooth [42]. One can also consider costs of the form $c(x, y) = h(x - y)$ where h is a strictly convex smooth function, for instance, $c(x, y) = \|x - y\|^p$ with $1 < p < +\infty$.

Proposition 7 (Caffarelli regularity, informal form). *Let $\Omega, \Lambda \subset \mathbb{R}^d$ be bounded domains, with Λ convex, and let $\alpha = \rho_\alpha dx$ and $\beta = \rho_\beta dy$ have densities bounded above and below by positive constants. If $T = \nabla\varphi$ is the Brenier map from α to β , then φ is strictly convex and T is locally Hölder continuous. Under stronger smoothness and uniform convexity assumptions on the domains and densities, φ is smooth in the interior and T is a smooth diffeomorphism between the domains.*

Proof. This is a proof sketch of Caffarelli's regularity theory [42]. When densities exist, the Brenier potential solves the Monge–Ampère equation (8). Convexity of the target prevents sections of φ from folding or disconnecting, while upper and lower bounds on the densities give ellipticity estimates for the Monge–Ampère measure. Caffarelli's localization and section estimates then imply strict convexity and Hölder continuity of the gradient. If the boundary and densities are smoother and uniformly convex, the equation becomes uniformly elliptic after localization, and Schauder-type bootstrapping yields higher regularity. $\square$

This regularity discussion should also be read as a warning. Brenier's theorem provides existence and uniqueness under the mild condition that the source has a density, but regularity is a different matter: it requires density bounds, smoothness and convex geometry that are rarely satisfied by empirical, manifold-supported or neural generative distributions. In machine-learning applications, one should therefore expect the exact OT map to be only weakly defined, possibly unstable, and often better represented by a coupling, an entropic approximation or a learned parametric surrogate than by a smooth map. Indeed, φ is in general non-smooth, but it is convex and therefore locally Lipschitz on the interior of its domain, so $\nabla\varphi$ is well defined α -almost everywhere.

If α does not have a density, then a Monge map might fail to exist. The correct relaxed object is an optimal Kantorovich plan concentrated on the graph of the set-valued map $\partial\varphi$. At points where $\partial\varphi(x)$ contains several target locations, the plan can split the mass starting from x among those locations. Thus the subdifferential still describes the geometry of optimality, but the transport object is a coupling rather than a single-valued map.

Monge–Ampère equation. For measures with densities, using (5), one obtains that φ is the unique (up to the addition of a constant) convex function which solves the following Monge–Ampère-type equation

$$\det(\partial^2\varphi(x))\rho_\beta(\nabla\varphi(x)) = \rho_\alpha(x) \quad (8)$$

where $\partial^2\varphi(x) \in \mathbb{R}^{d \times d}$ is the Hessian of φ . The convexity constraint forces $\det(\partial^2\varphi(x)) \geq 0$ and is necessary for this equation to have a solution and be well-posed. Numerical schemes for this PDE connect OT with fully nonlinear elliptic solvers; see [21, 20] for representative discretizations. The Monge–Ampère operator $\det(\partial^2\varphi(x))$ can be understood as a nonlinear degenerate Laplacian. In the limit of small displacements, consider

$$\varphi(x) = \frac{\|x\|^2}{2} + \varepsilon\psi(x), \quad T(x) = \nabla\varphi(x) = x + \varepsilon\nabla\psi(x).$$

The density equation becomes

$$\rho_\alpha(x) = \det(I + \varepsilon\nabla^2\psi(x)) \rho_\beta(x + \varepsilon\nabla\psi(x)).$$

Using $\det(I + \varepsilon A) = 1 + \varepsilon \operatorname{tr}(A) + o(\varepsilon)$ and Taylor expanding the target density gives

$$\rho_\alpha = \rho_\beta + \varepsilon(\rho_\beta \Delta\psi + \langle \nabla\rho_\beta, \nabla\psi \rangle) + o(\varepsilon) = \rho_\beta + \varepsilon \operatorname{div}(\rho_\beta \nabla\psi) + o(\varepsilon).$$

Equivalently, if both densities are perturbations of a common smooth density ρ and $\rho_\beta - \rho_\alpha = \varepsilon\eta + o(\varepsilon)$, the first-order equation is the weighted Poisson equation

$$-\operatorname{div}(\rho\nabla\psi) = \eta.$$

For the uniform reference density $\rho = 1$, this reduces to $-\Delta\psi = \eta$, which is the precise sense in which Monge–Ampère linearizes to a Laplacian.

2.5 One-Dimensional Transport and Quantiles

In one dimension, optimal transport is completely explicit. The cumulative distribution function orders the mass, and the optimal map is obtained by matching quantile levels. This special case is both a computational tool and the model for several linearized constructions used later.

For a measure α on $\mathbb{R}$, we introduce the cumulative function

$$\forall x \in \mathbb{R}, \quad \mathcal{C}_\alpha(x) := \int_{-\infty}^x d\alpha, \quad (9)$$

which is a function $\mathcal{C}_\alpha : \mathbb{R} \rightarrow [0, 1]$. Its pseudo-inverse $\mathcal{C}_\alpha^{-1} : [0, 1] \rightarrow \mathbb{R} \cup \{-\infty\}$

$$\forall r \in [0, 1], \quad \mathcal{C}_\alpha^{-1}(r) = \min_x \{x \in \mathbb{R} \cup \{-\infty\} ; \mathcal{C}_\alpha(x) \geq r\}.$$

That function is also called the quantile function of α . The following proposition shows that these functions define push-forwards toward the uniform distribution $\mathcal{U}$ on $[0, 1]$.

Proposition 8 (Quantile push-forward). *One has $(\mathcal{C}_\alpha)_\#^{-1}\mathcal{U} = \alpha$, where $\mathcal{U}$ is the uniform distribution in $[0, 1]$. If α has a density, then $(\mathcal{C}_\alpha)_\#\alpha = \mathcal{U}$.*

Proof. For simplicity, we assume α has a strictly positive density, so that $\mathcal{C}_\alpha$ is a strictly increasing continuous function. Denoting $\gamma := (\mathcal{C}_\alpha)_\#^{-1}\mathcal{U}$ we aim at proving $\gamma = \alpha$, which is equivalent to $\mathcal{C}_\gamma = \mathcal{C}_\alpha$. One has

$$\mathcal{C}_\gamma(x) = \int_{-\infty}^x d\gamma = \int_{\mathbb{R}} 1_{]-\infty, x]} d((\mathcal{C}_\alpha^{-1})_\#\mathcal{U}) = \int_0^1 1_{]-\infty, x]}(\mathcal{C}_\alpha^{-1}(z)) dz = \int_0^1 1_{[0, \mathcal{C}_\alpha(x)]}(z) dz = \mathcal{C}_\alpha(x)$$

where we use the fact that

$$-\infty \leq \mathcal{C}_\alpha^{-1}(z) \leq x \iff 0 \leq z \leq \mathcal{C}_\alpha(x).$$

□

If α has a density, this shows that the map

$$T = \mathcal{C}_\beta^{-1} \circ \mathcal{C}_\alpha \quad (10)$$

satisfies $T_\#\alpha = \beta$.

For the cost $c(x, y) = |x - y|^2$, since this T is increasing (hence the gradient of a convex function since we are in 1-D), by Brenier's theorem, T is the solution to Monge problem (at least if we impose that α has a density, otherwise it might lead to a solution of the Kantorovich problem by properly defining the pseudo-inverse). This closed-form formula is also optimal for any cost of the form $h(|x - y|)$ for increasing h . For discrete measures, one cannot apply directly this reasoning (because α does not have a density), but if the measures are uniform on the same number of Dirac masses, then this approach is equivalent to the sorting formula.

Proposition 9 (One-dimensional Wasserstein formulas). *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R})$ have finite p -th moments. For every $p \geq 1$,*

$$\mathcal{W}_p(\alpha, \beta)^p = \int_0^1 |\mathcal{C}_\alpha^{-1}(r) - \mathcal{C}_\beta^{-1}(r)|^p dr = \|\mathcal{C}_\alpha^{-1} - \mathcal{C}_\beta^{-1}\|_{L^p([0, 1])}^p. \quad (11)$$

For $p = 1$, this is equivalently

$$\mathcal{W}_1(\alpha, \beta) = \|\mathcal{C}_\alpha - \mathcal{C}_\beta\|_{L^1(\mathbb{R})} = \int_{\mathbb{R}} |\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)| dx = \int_{\mathbb{R}} \left| \int_{-\infty}^x d(\alpha - \beta) \right| dx. \quad (12)$$

Proof. When α has a density, the map $T = \mathcal{C}_\beta^{-1} \circ \mathcal{C}_\alpha$ pushes α onto β and is nondecreasing. Hence it is optimal for every convex increasing cost $h(|x - y|)$, in particular for $|x - y|^p$. Since $r = \mathcal{C}_\alpha(x)$ is uniformly distributed on $[0, 1]$ under $x \sim \alpha$, the change of variables $x = \mathcal{C}_\alpha^{-1}(r)$ gives the first formula. General measures follow by approximating them narrowly with smooth measures while preserving the p -th moments, or equivalently by using the monotone rearrangement theorem for generalized quantiles.

For $p = 1$, use the layer-cake identity. If q_α and q_β are the two quantile functions, then

$$\int_0^1 |q_\alpha(r) - q_\beta(r)| dr = \int_{\mathbb{R}} \lambda(\{r : q_\alpha(r) \leq x < q_\beta(r)\} \cup \{r : q_\beta(r) \leq x < q_\alpha(r)\}) dx.$$

The measure of the displayed set is exactly $|\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)|$ for almost every x . This proves (12). $\square$

Proposition 10 (Knothe–Rosenblatt triangular rearrangement). *Let $\alpha, \beta \in \mathcal{P}(\mathbb{R}^d)$. Assume, for simplicity, that the first marginal of α and the one-dimensional conditional laws of α used below are atomless. There exists a triangular map*

$$T(x_1, \dots, x_d) = (T_1(x_1), T_2(x_1, x_2), \dots, T_d(x_1, \dots, x_d))$$

such that $T_\# \alpha = \beta$ and, for each k , the function $x_k \mapsto T_k(x_1, \dots, x_k)$ is nondecreasing for α -almost every value of $(x_1, \dots, x_{k-1})$.

Proof. The construction is recursive. For $k = 1$, let T_1 be the monotone rearrangement between the first marginal of α and the first marginal of β . Suppose that $T_1, \dots, T_{k-1}$ have already been built and write $x_{<k} = (x_1, \dots, x_{k-1})$ and $T_{<k} = (T_1, \dots, T_{k-1})$. By the induction hypothesis, $(T_{<k})_\# \alpha_{<k} = \beta_{<k}$, where $\alpha_{<k}$ and $\beta_{<k}$ are the first $(k-1)$ -coordinate marginals. Let $\alpha_{x_{<k}}^k$ and $\beta_{y_{<k}}^k$ be regular conditional laws of the k -th coordinate given the previous coordinates. Define $T_k(x_{<k}, \cdot)$ as the one-dimensional monotone rearrangement from $\alpha_{x_{<k}}^k$ to $\beta_{T_{<k}(x_{<k})}^k$.

The map $T_k(x_{<k}, \cdot)$ is nondecreasing by the one-dimensional rearrangement theorem. The chain rule for disintegrations then shows that, after step k , the first k coordinates of $T_\# \alpha$ have the same law as the first k coordinates of β . At $k = d$ this gives $T_\# \alpha = \beta$. Target atoms are already covered by generalized quantiles. If a source conditional law has atoms that must be split, the same recursive construction defines a triangular Markov kernel rather than a deterministic map. $\square$

This construction transports successively along coordinate axes and is therefore often called axis-wise transport. It depends on the chosen ordering of the coordinates, and is not in general optimal for the quadratic cost. It is nevertheless a useful limiting object: Brenier maps for increasingly anisotropic quadratic costs converge to triangular rearrangements under suitable assumptions [49]. The quantile formula above means that through the map $\alpha \mapsto \mathcal{C}_\alpha^{-1}$, the Wasserstein distance is isometric to a linear space equipped with the L^p norm. For $p = 2$, the Wasserstein distance for measures on the real line is thus a Hilbertian metric. This makes the geometry of 1-D optimal transport very simple, but also very different from its geometry in higher dimensions, which is not Hilbertian.

For $p = 1$, formula (12) shows that $\mathcal{W}_1$ is a norm on signed measures with zero total mass once they are identified with their cumulative primitives (see paragraph 6.4 for the generalization to arbitrary dimensions).

It is possible to define other norms which behave similarly, in the sense that they metrize convergence in law under suitable tightness assumptions. For instance, $\|\mathcal{C}_\alpha - \mathcal{C}_\beta\|_{L^p(\mathbb{R})}$ gives respectively the Wasserstein, Cramer–von Mises and Kolmogorov–Smirnov distances for $p = 1, 2, \infty$.

2.6 Vector Quantiles and Linearized Transport

The one-dimensional quantile function represents a probability measure by the monotone map sending a fixed reference law to it. In dimension $d > 1$, Brenier’s theorem gives the analogous construction after choosing a reference probability ρ_0 , typically the uniform law on a convex body or a standard Gaussian.

Vector quantiles. Assume that the reference law ρ_0 is absolutely continuous. For a target law μ with finite second moment, its vector quantile relative to ρ_0 is the Brenier map

$$T_\mu = \nabla \varphi_\mu, \quad (T_\mu)_\# \rho_0 = \mu,$$

or equivalently the solution of

$$\min_{T_\# \rho_0 = \mu} \int \|x - T(x)\|^2 d\rho_0(x).$$

This construction is canonical only after fixing ρ_0 : changing the reference law changes the coordinates used to represent μ . Vector quantile regression uses the same idea conditionally, replacing scalar conditional quantiles by conditional Brenier maps and thereby encoding multivariate ranks and depths [46].

Linearized Wasserstein coordinates. The same reference measure leads to the linearized OT embedding

$$\mu \mapsto T_\mu - \text{Id} \in L^2(\rho_0; \mathbb{R}^d).$$

Distances between embedded maps are not exactly equal to Wasserstein distances in general, but they give a useful tangent-space approximation. In one dimension this construction is exact because quantile functions give the isometry (11). Section 9.6 uses this reference-map viewpoint to build fast linearized barycenters: for varying weights, one averages maps in $L^2(\rho_0)$ and pushes ρ_0 forward by the averaged map.

2.7 Gaussian Measures and the Bures Metric

Gaussian measures form the most important finite-dimensional family preserved by quadratic optimal transport. The mean moves linearly, while the covariance evolves according to the Bures–Wasserstein geometry of positive semidefinite matrices.

One-dimensional Gaussians. We first consider the nondegenerate source case where $\alpha = \mathcal{N}(m_\alpha, s_\alpha^2)$ and $\beta = \mathcal{N}(m_\beta, s_\beta^2)$ are two Gaussians in $\mathbb{R}$ with $s_\alpha > 0$. Then one verifies that

$$T(x) = \frac{s_\beta}{s_\alpha}(x - m_\alpha) + m_\beta$$

satisfies $T_\# \alpha = \beta$. Furthermore, it is the derivative of the convex function

$$\varphi(x) = \frac{s_\beta}{2s_\alpha}(x - m_\alpha)^2 + m_\beta x,$$

so that according to Brenier’s theorem, for the cost $c(x - y) = (x - y)^2$, T is the unique optimal transport, and the associated Monge distance is, after some computation

$$\tilde{W}_2^2(\alpha, \beta) = \int_{\mathbb{R}} \left(\frac{s_\beta}{s_\alpha}(x - m_\alpha) + m_\beta - x \right)^2 d\alpha(x) = (m_\alpha - m_\beta)^2 + (s_\alpha - s_\beta)^2.$$

The distance formula extends by continuity to Dirac masses, i.e. if $s_\alpha = 0$ or $s_\beta = 0$, although the affine Monge map itself only pushes a Dirac source to another Dirac. The OT geometry of Gaussians is thus the Euclidean distance on the half plane $(m, s) \in \mathbb{R} \times \mathbb{R}_+$.

This should be contrasted with the geometry of KL, where singular Gaussians (for which $s = 0$) are infinitely distant.

Multivariate Gaussians. If $\alpha = \mathcal{N}(\mathbf{m}_\alpha, \Sigma_\alpha)$ and $\beta = \mathcal{N}(\mathbf{m}_\beta, \Sigma_\beta)$ are two Gaussians in $\mathbb{R}^d$, we now look for an affine map

$$T : x \mapsto \mathbf{m}_\beta + A(x - \mathbf{m}_\alpha). \quad (13)$$

This map is the gradient of the convex function $\varphi(x) = \langle \mathbf{m}_\beta, x \rangle + \langle A(x - \mathbf{m}_\alpha), x - \mathbf{m}_\alpha \rangle / 2$ if and only if A is symmetric positive semidefinite.

Proposition 11 (Affine push-forward of Gaussians). *One has $T_\# \alpha = \beta$ if and only if*

$$A \Sigma_\alpha A^\top = \Sigma_\beta. \quad (14)$$

Proof. An affine function maps a Gaussian to a Gaussian so that it remains to show that the mean and covariance of $T_\# \alpha$ are those of β . We consider X to be a random vector with law α , so that, denoting $Y := T(X)$ a random vector with law $T_\# \alpha$, its mean is

$$\mathbb{E}(Y) = \mathbb{E}(\mathbf{m}_\beta + A(X - \mathbf{m}_\alpha)) = \mathbf{m}_\beta + A\mathbb{E}(X - \mathbf{m}_\alpha) = \mathbf{m}_\beta$$

while its covariance is

$$\mathbb{E}((Y - \mathbf{m}_\beta)(Y - \mathbf{m}_\beta)^\top) = \mathbb{E}([A(X - \mathbf{m}_\alpha)][A(X - \mathbf{m}_\alpha)]^\top) = A\mathbb{E}((X - \mathbf{m}_\alpha)(X - \mathbf{m}_\alpha)^\top)A^\top = A\Sigma_\alpha A^\top.$$

□

The covariance equation (14) is quadratic in A . Under the symmetry constraint imposed by Brenier's theorem it becomes $A\Sigma_\alpha A = \Sigma_\beta$. The next proposition solves this equation and computes the resulting transport cost.

Proposition 12 (Gaussian $\mathcal{W}_2$ formula and Bures covariance term). *Assume that Σ_α and Σ_β are positive definite. The unique symmetric positive-definite solution of*

$$A\Sigma_\alpha A = \Sigma_\beta$$

is

$$A = \Sigma_\alpha^{-\frac{1}{2}} \left(\Sigma_\alpha^{\frac{1}{2}} \Sigma_\beta \Sigma_\alpha^{\frac{1}{2}} \right)^{\frac{1}{2}} \Sigma_\alpha^{-\frac{1}{2}}. \quad (15)$$

The affine map $T(x) = \mathbf{m}_\beta + A(x - \mathbf{m}_\alpha)$ is the optimal quadratic-cost transport from $\mathcal{N}(\mathbf{m}_\alpha, \Sigma_\alpha)$ to $\mathcal{N}(\mathbf{m}_\beta, \Sigma_\beta)$, and

$$\mathcal{W}_2^2(\alpha, \beta) = \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2, \quad (16)$$

where

$$\mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2 := \text{tr} \left(\Sigma_\alpha + \Sigma_\beta - 2(\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2})^{1/2} \right). \quad (17)$$

Proof. Multiplying $A\Sigma_\alpha A = \Sigma_\beta$ on the left and right by $\Sigma_\alpha^{1/2}$ gives

$$(\Sigma_\alpha^{1/2} A \Sigma_\alpha^{1/2})^2 = \Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2}.$$

Since A is symmetric positive, $\Sigma_\alpha^{1/2} A \Sigma_\alpha^{1/2}$ is symmetric positive and is therefore the unique positive square root of the right-hand side. This gives (15). Conversely, the matrix in (15) is symmetric positive and satisfies the covariance equation.

By Proposition 3 or by the covariance calculation above, this affine map pushes α to β . It is the gradient of the convex quadratic potential

$$\varphi(x) = \langle \mathbf{m}_\beta, x \rangle + \frac{1}{2} \langle A(x - \mathbf{m}_\alpha), x - \mathbf{m}_\alpha \rangle,$$

so Brenier's theorem implies optimality. If $X \sim \alpha$, then

$$\begin{aligned}\mathbb{E}\|X - T(X)\|^2 &= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \mathbb{E}\|(I - A)(X - \mathbf{m}_\alpha)\|^2 \\ &= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \text{tr}((I - A)\Sigma_\alpha(I - A)^\top) \\ &= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \text{tr}(\Sigma_\alpha) + \text{tr}(A\Sigma_\alpha A) - 2\text{tr}(A\Sigma_\alpha) \\ &= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \text{tr}(\Sigma_\alpha) + \text{tr}(\Sigma_\beta) - 2\text{tr}((\Sigma_\alpha^{1/2}\Sigma_\beta\Sigma_\alpha^{1/2})^{1/2}),\end{aligned}$$

which is the desired expression. The formula for singular covariance matrices follows by adding ηI and letting $\eta \downarrow 0$. $\square$

The covariance term $\mathcal{B}$ is the Bures–Wasserstein metric on positive semidefinite matrices [41, 27, 82]. It is useful because it separates the Euclidean motion of the mean from the intrinsic transport geometry of covariance ellipsoids.

Proposition 13 (Metric and convexity properties of the Bures term). *The function $\mathcal{B}$ is a distance on positive semidefinite covariance matrices. Moreover, $\mathcal{B}^2$ is jointly convex: for $t \in [0, 1]$,*

$$\mathcal{B}^2((1-t)\Sigma_0 + t\Sigma_1, (1-t)\Lambda_0 + t\Lambda_1) \leq (1-t)\mathcal{B}^2(\Sigma_0, \Lambda_0) + t\mathcal{B}^2(\Sigma_1, \Lambda_1).$$

Proof. The key identity is the Procrustes representation

$$\mathcal{B}^2(\Sigma, \Lambda) = \min_{Q^\top Q = I} \|\Sigma^{1/2} - \Lambda^{1/2}Q\|_F^2.$$

Indeed, expanding the square gives

$$\text{tr} \Sigma + \text{tr} \Lambda - 2 \max_{Q^\top Q = I} \text{tr}(\Sigma^{1/2}Q^\top \Lambda^{1/2}),$$

and the orthogonal Procrustes formula identifies the maximum with $\text{tr}((\Sigma^{1/2}\Lambda\Sigma^{1/2})^{1/2})$. Symmetry, positivity and separation follow immediately from this representation. The triangle inequality follows by choosing two almost optimal orthogonal matrices Q_1, Q_2 and writing

$$\|\Sigma^{1/2} - \Lambda^{1/2}Q_2Q_1\|_F \leq \|\Sigma^{1/2} - \Lambda^{1/2}Q_1\|_F + \|\Lambda^{1/2} - \Lambda^{1/2}Q_2\|_F.$$

Letting the two choices become optimal proves the metric property.

For convexity, use the equivalent factor formulation

$$\mathcal{B}^2(\Sigma, \Lambda) = \min_{UU^\top = \Sigma, VV^\top = \Lambda} \|U - V\|_F^2.$$

Choose nearly optimal factors (U_0, V_0) and (U_1, V_1) for the two pairs. Define block factors

$$U_t = [\sqrt{1-t}U_0, \sqrt{t}U_1], \quad V_t = [\sqrt{1-t}V_0, \sqrt{t}V_1].$$

Then $U_t U_t^\top = (1-t)\Sigma_0 + t\Sigma_1$ and $V_t V_t^\top = (1-t)\Lambda_0 + t\Lambda_1$, while

$$\|U_t - V_t\|_F^2 = (1-t)\|U_0 - V_0\|_F^2 + t\|U_1 - V_1\|_F^2.$$

Taking the infimum over the initial factors proves joint convexity. $\square$

Remark 6 (Diagonal covariances and Hellinger geometry). In the case where $\Sigma_\alpha = \text{diag}(r_i)_i$ and $\Sigma_\beta = \text{diag}(s_i)_i$ are diagonal, the Bures metric reduces to the Euclidean distance between square roots,

$$\mathcal{B}(\Sigma_\alpha, \Sigma_\beta) = \|\sqrt{r} - \sqrt{s}\|_2.$$

This is the finite-dimensional Hellinger geometry on the nonnegative covariance coordinates: variances are compared after the amplitude change of variables $r_i \mapsto \sqrt{r_i}$.

3 Kantorovich Relaxation

Kantorovich’s relaxation is the decisive move that turns transport into convex optimization. The section explains how deterministic maps are replaced by couplings, why this fixes infeasibility and symmetry, and how it produces the Wasserstein distances. Historically, this linear-programming viewpoint grew from Kantorovich’s economic planning work [122] and is now the standard foundation of OT [211, 212, 177].

3.1 Discrete Relaxation

The discrete relaxation is the cleanest place to see mass splitting. It replaces permutations by a transportation polytope and reveals the linear-programming structure that algorithms exploit.

Monge’s discrete matching problem is problematic because it cannot be applied when $n \neq m$. One needs to take into account masses (a_i, b_j) to handle this more general situation. Monge’s continuous formulation (6), based on push-forward maps, is also problematic because there may be no transport map T such that $T_\# \alpha = \beta$. This happens, for instance, when α is a single Dirac mass that should be mapped to several Dirac masses. Accordingly, Monge’s formulation is not symmetric with respect to the exchange of α and β : one can map two Dirac masses to a single one, but not the other way around. Moreover, these are non-convex optimization problems, which are difficult to solve numerically.

The key idea of [122] is to relax the deterministic nature of transportation. Instead of requiring each source point x_i to be sent to one and only one target location, Kantorovich allows the mass at x_i to be dispatched across several locations. This moves from deterministic transport maps to probabilistic, or fuzzy, transport plans and allows what is now called mass splitting. This flexibility is encoded, in place of a permutation σ or a map T , by a coupling matrix $P \in \mathbb{R}_+^{n \times m}$, where $P_{i,j}$ describes the amount of mass flowing from x_i to y_j in the formalism of discrete measures $\alpha = \sum_i a_i \delta_{x_i}$, $\beta = \sum_j b_j \delta_{y_j}$. Admissible couplings are only constrained to satisfy conservation of mass

$$U(a, b) := \{P \in \mathbb{R}_+^{n \times m} ; P \mathbf{1}_m = a \quad \text{and} \quad P^\top \mathbf{1}_n = b\}, \quad (18)$$

where we used the following matrix-vector notation

$$P \mathbf{1}_m = \left(\sum_j P_{i,j} \right)_i \in \mathbb{R}^n \quad \text{and} \quad P^\top \mathbf{1}_n = \left(\sum_i P_{i,j} \right)_j \in \mathbb{R}^m.$$

The set of matrices $U(a, b)$ is bounded, defined by $n+m$ equality constraints, and therefore a convex polytope (the convex hull of a finite set of matrices).

Additionally, whereas the Monge formulation is intrinsically asymmetric, Kantorovich’s relaxed formulation is always symmetric, in the sense that a coupling P is in $U(a, b)$ if and only if $P^\top$ is in $U(b, a)$.

Kantorovich, aiming for economic planning, made a strong simplifying assumption: the cost of transportation should be linear in the amount of transported mass. Under this assumption, denoting $C_{i,j}$ the cost of moving a unit amount of mass from x_i to y_j , Kantorovich’s optimal transport problem now reads

$$L_C(a, b) := \min_{P \in U(a, b)} \langle C, P \rangle := \sum_{i,j} C_{i,j} P_{i,j}. \quad (19)$$

This is a linear program, and as is usually the case with such programs, its solutions are not necessarily unique.

Proposition 14 (Sparse optimal plans). *Problem (19) admits an optimal coupling P^* with at most $n+m-1$ nonzero entries.*

Proof. The feasible set $U(a, b)$ is a non-empty compact polytope, hence the linear objective admits an extreme-point minimizer by Proposition 17. The affine marginal constraints have rank $n+m-1$: there are

$n + m$ row and column equations, and the only linear redundancy is that both sets imply the same total mass identity $\sum_{i,j} P_{i,j} = 1$.

At an extreme point of a polytope defined by equality constraints and non-negativity constraints, the columns corresponding to positive variables must be linearly independent in the constraint matrix. Otherwise, one could move a small amount of mass along a non-zero signed perturbation supported only on the positive entries while preserving all marginals, contradicting extremality. Since the rank is $n + m - 1$, the support of an extreme point has cardinality at most $n + m - 1$. $\square$

Proposition 15 (North-west corner feasible plan). *Let $\mathbf{a} \in \mathbb{R}_+^n$ and $\mathbf{b} \in \mathbb{R}_+^m$ have the same positive total mass. The following greedy sweep constructs a coupling $\mathbf{P} \in \mathcal{U}(\mathbf{a}, \mathbf{b})$ with at most $n + m - 1$ positive entries and an acyclic positive support. Starting from $(i, j) = (1, 1)$ with residual masses $r_i = a_i$ and $s_j = b_j$, skip zero residuals and set*

$$P_{i,j} = \min(r_i, s_j),$$

subtract this value from both residuals, and advance every index whose residual has become zero. Repeat until all residual masses are exhausted.

Proof. All assignments are non-negative. At each step, the mass placed in entry (i, j) is subtracted from exactly one current row residual and one current column residual, so no row or column can receive more mass than prescribed. Conversely, an index is advanced only when its residual has been fully filled. When the algorithm stops, the total assigned mass is $\sum_i a_i = \sum_j b_j$, hence all row and column sums are exactly $\mathbf{a}$ and $\mathbf{b}$.

Each positive assignment exhausts at least one current row or one current column. Before the final assignment, at most $n - 1$ row advances and $m - 1$ column advances can occur without terminating the construction. Hence the number of positive entries is at most $(n - 1) + (m - 1) + 1 = n + m - 1$.

Finally, view the positive support as a bipartite graph whose left vertices are rows and whose right vertices are columns. Once a row or column index is advanced, it never appears again. Thus each new positive edge either starts a new connected component, or attaches at least one new vertex to the component currently being swept. No edge is ever added between two old vertices of the same component, so no cycle can be created. $\square$

The north-west corner rule does not use the cost matrix, and is therefore not meant to solve (19). Its role is algorithmic: an acyclic support corresponds to linearly independent marginal constraints. When the support has fewer than $n + m - 1$ positive entries, simplex implementations complete it with zero-mass basic variables to obtain a degenerate basic feasible solution. This provides a cheap initialization for transportation simplex and network simplex methods.

Linear programming algorithms. The reference algorithms to solve (19) are network simplex methods, whose pivoting rules are rooted in minimum-cost-flow theory [69, 25, 162]. They improve an initial basic feasible coupling, for instance the one produced by Proposition 15, by local pivots that reduce the cost. Strongly polynomial minimum-cost-flow algorithms reach complexities of order $O(n^3 \log n)$ in balanced regimes, while network simplex implementations are often faster in practice. Alternatives include interior-point methods, which are usually less competitive for this particular type of linear program.

1-D cases. In 1-D, if $c(x, y) = |x - y|^p$ on $\mathcal{X} = \mathcal{Y} = \mathbb{R}$ with $p \geq 1$, then an optimal transport map is given by an increasing map. As explained in Section 1.1, the case $n = m$ and $a_i = b_j = \frac{1}{n}$ is solved in $O(n \log(n))$ operations. In the general case, an optimal coupling matrix $\mathbf{P}$ can be computed similarly in $O(n \log(n) + m \log(m))$ by sorting the points and then sweeping the mass in a single pass from left to right. At each step, the algorithm matches the remaining mass of the current source atom with the remaining mass of the current target atom, and then advances whichever atom has been exhausted.

Permutation Matrices as Couplings We restrict our attention to the special case $n = m$ and $a_i = b_i = 1$ (up to scaling by $1/n$, these are thus probability measures). In this case, one can solve Monge's optimal matching problem (1), and it is convenient to rewrite it using permutation matrices. For a permutation $\sigma \in \text{Perm}(n)$, we write P_σ for the corresponding permutation matrix,

$$\forall (i, j) \in \llbracket n \rrbracket^2, \quad (P_\sigma)_{i,j} = \begin{cases} 1 & \text{if } j = \sigma_i, \\ 0 & \text{otherwise.} \end{cases} \quad (20)$$

We denote the set of permutation matrices as

$$\text{Perm}_n := \{P_\sigma ; \sigma \in \text{Perm}(n)\},$$

which is a discrete, hence non-convex, set. One has

$$\langle C, P_\sigma \rangle = \sum_{i=1}^n C_{i, \sigma_i}$$

so that (1) is equivalent to the non-convex optimization problem

$$\min_{P \in \text{Perm}_n} \langle C, P \rangle.$$

In contrast, $U(a, b) = \mathcal{B}_n$ is equal to the convex set of bistochastic matrices

$$\mathcal{B}_n := \{P \in \mathbb{R}_+^{n \times n} ; P \mathbf{1}_n = P^\top \mathbf{1}_n = \mathbf{1}_n\}$$

so that the Kantorovich problem reads

$$\min_{P \in \mathcal{B}_n} \langle C, P \rangle.$$

The set of permutation matrices is strictly included in the set of bistochastic matrices, and more precisely

$$\text{Perm}_n = \mathcal{B}_n \cap \{0, 1\}^{n \times n}.$$

This gives the immediate comparison between the cost of the Monge and Kantorovich problems

$$\min_{P \in \mathcal{B}_n} \langle C, P \rangle \leq \min_{P \in \text{Perm}_n} \langle C, P \rangle.$$

We will now show that there is an equality between these two costs, so that both problems are in some sense equivalent.

For this, we will make a detour through a more general linear optimization problem of the form $\min_{P \in \mathcal{C}} \langle C, P \rangle$ for some compact convex set $\mathcal{C}$. We first introduce the notion of extremal points, which are intuitively the vertices of $\mathcal{C}$

$$\text{Extr}(\mathcal{C}) := \left\{ P ; \forall (Q, R) \in \mathcal{C}^2, P = \frac{Q + R}{2} \Rightarrow Q = R \right\}.$$

So to show that $P \notin \text{Extr}(\mathcal{C})$ it suffices to split P as $P = \frac{Q + R}{2}$ with $Q \neq R$ and $(Q, R) \in \mathcal{C}^2$. We will assume the following fundamental result.

Proposition 16 (Existence of extreme points). *If $\mathcal{C}$ is a non-empty compact convex subset of a finite-dimensional vector space, then $\text{Extr}(\mathcal{C}) \neq \emptyset$.*

Proof. Among all non-empty faces of $\mathcal{C}$, choose one of minimal affine dimension (this is possible in finite dimension). If this face contained two distinct points, it would have a non-trivial exposed subface obtained by maximizing a linear functional that is not constant on the face, contradicting minimality. Hence the minimal face is a singleton, and its unique point is an extreme point of $\mathcal{C}$. $\square$

The fact that $\mathcal{C}$ is compact is crucial, for instance, the set $\{(x, y) \in \mathbb{R}_+^2 ; xy \geq 1\}$ has no extremal point.

We can now use this result to show the following fundamental result, namely that there is always a solution to a linear program which is an extremal point. Note that of course, the set of solutions (which is non-empty because one minimizes a continuous function on a compact) might not be a singleton.

Proposition 17 (Linear programs have extreme minimizers). *If $\mathcal{C}$ is compact, then*

$$\text{Extr}(\mathcal{C}) \cap \left(\underset{P \in \mathcal{C}}{\text{argmin}} \langle C, P \rangle \right) \neq \emptyset.$$

Proof. Let $\mathcal{S} := \underset{P \in \mathcal{C}}{\text{argmin}} \langle C, P \rangle$. We first note that $\mathcal{S}$ is convex (as always for an argmin) and compact because $\mathcal{C}$ is compact and the objective function is continuous so that $\text{Extr}(\mathcal{S}) \neq \emptyset$. We will show that $\text{Extr}(\mathcal{S}) \subset \text{Extr}(\mathcal{C})$. Indeed, let $P \in \text{Extr}(\mathcal{S})$ and assume that $P = (Q + R)/2$ with $Q, R \in \mathcal{C}$. By linearity of the objective and optimality of P , both Q and R must also minimize $\langle C, \cdot \rangle$ over $\mathcal{C}$, hence $Q, R \in \mathcal{S}$. Since P is extremal in $\mathcal{S}$, this implies $Q = R = P$, so $P \in \text{Extr}(\mathcal{C})$. $\square$

The following theorem states that the extremal points of bistochastic matrices are the permutation matrices. It implies as a corollary that the Monge and Kantorovich costs are the same, and that they share a common solution.

Theorem 2 (Birkhoff and von Neumann). *One has $\text{Extr}(\mathcal{B}_n) = \text{Perm}_n$.*

Proof. We first show the simplest inclusion $\text{Perm}_n \subset \text{Extr}(\mathcal{B}_n)$. Indeed it follows from the fact that $\text{Extr}([0, 1]) = \{0, 1\}$. Take $P \in \text{Perm}_n$, if $P = (Q + R)/2$ with $Q_{i,j}, R_{i,j} \in [0, 1]$, since $P_{i,j} \in \{0, 1\}$ then necessarily $Q_{i,j} = R_{i,j} \in \{0, 1\}$.

We now show $\text{Extr}(\mathcal{B}_n) \subset \text{Perm}_n$ by proving the contrapositive. Pick $P \in \mathcal{B}_n \setminus \text{Perm}_n$; we need to split $P = (Q + R)/2$ where Q, R are distinct bistochastic matrices. P defines a bipartite graph linking two sets of n vertices. This graph is composed of isolated edges when $P_{i,j} = 1$ and connected edges corresponding to $0 < P_{i,j} < 1$. If i is such a connected vertex on the left (similarly for j on the right), because $\sum_j P_{i,j} = 1$, there are necessarily at least two edges (i, j_1) and (i, j_2) emanating from it (similarly on the right there are at least two converging edges (i_1, j) and (i_2, j)). This means that by following these connections, one necessarily can extract a cycle (if not, one could always extend it by the previous remarks) of the form

$$(i_1, j_1, i_2, j_2, \dots, i_p, j_p), \quad \text{i.e.} \quad i_{p+1} = i_1.$$

We assume this cycle is the shortest one among all this (finite) ensemble of cycles. Along this cycle, the left-right and right-left edges satisfy

$$0 < P_{i_s, j_s}, P_{i_{s+1}, j_s} < 1.$$

The $(i_s)_s$ and $(j_s)_s$ are also all distinct because the cycle is the shortest. Let us pick

$$\varepsilon := \min_{1 \leq s \leq p} \{P_{i_s, j_s}, P_{i_{s+1}, j_s}, 1 - P_{i_s, j_s}, 1 - P_{i_{s+1}, j_s}\}$$

so that $0 < \varepsilon < 1$. We split the graph into two sets of edges, left-right and right-left

$$\mathcal{A} := \{(i_s, j_s)\}_{s=1}^p \quad \text{and} \quad \mathcal{B} := \{(i_{s+1}, j_s)\}_{s=1}^p.$$

We define the two matrices as

$$Q_{i,j} := \begin{cases} P_{i,j} & \text{if } (i, j) \notin \mathcal{A} \cup \mathcal{B}, \\ P_{i,j} + \varepsilon/2 & \text{if } (i, j) \in \mathcal{A}, \\ P_{i,j} - \varepsilon/2 & \text{if } (i, j) \in \mathcal{B}, \end{cases} \quad \text{and} \quad R_{i,j} := \begin{cases} P_{i,j} & \text{if } (i, j) \notin \mathcal{A} \cup \mathcal{B}, \\ P_{i,j} - \varepsilon/2 & \text{if } (i, j) \in \mathcal{A}, \\ P_{i,j} + \varepsilon/2 & \text{if } (i, j) \in \mathcal{B}, \end{cases}.$$

Because of the choice of ε , one has $0 \leq Q_{i,j}, R_{i,j} \leq 1$. Because each left-right edge in $\mathcal{A}$ is associated with a right-left edge in $\mathcal{B}$, (and the other way) the sum constraint on the row (and on the column) is maintained, so that $Q, R \in \mathcal{B}_n$. Finally, note that $P = (Q + R)/2$. $\square$

By putting together Proposition 17 and Theorem 2, one obtains that for the discrete optimal problem with empirical measures, Monge and Kantorovich problems are equivalent.

Corollary 1 (Kantorovich for matching). *If $m = n$ and $a = b = \mathbb{1}_n$, then there exists an optimal solution for Problem (19) P_{σ^*} , which is a permutation matrix associated to an optimal permutation $\sigma^* \in \text{Perm}(n)$ for Problem (1).*

Proof. The feasible set of (19) is the compact convex polytope $\mathcal{B}_n$. Proposition 17 ensures that the linear objective has an optimal extreme point of $\mathcal{B}_n$. By Theorem 2, this extreme point is a permutation matrix P_{σ^*} . Since the cost of P_{σ^*} is exactly $\sum_i C_{i, \sigma^*(i)}$, the associated permutation solves the assignment problem. $\square$

Equivalently, one can always find a permutation matrix that minimizes Kantorovich's problem (19) between two uniform measures $a = b = \mathbb{1}_n/n$, which shows that the Kantorovich relaxation is *tight* for assignment problems.

Remark 7 (General case). For general input measures, one does not have equivalence between Monge and Kantorovich problems, since the Monge constraint can be empty. In the finite-dimensional case, however, the support of an optimal coupling P still enjoys strong regularity; in particular, one can choose an optimal basic feasible plan whose bipartite support is cycle-free. This implies that there are always optimal solutions with at most $n + m - 1$ nonzero entries.

3.2 Relaxation for Arbitrary Measures

This subsection lifts the finite-dimensional coupling matrix to a joint probability measure. The payoff is that existence, duality and metric properties can be stated for arbitrary laws, including singular and continuous distributions.

Continuous couplings. The definition of $\mathcal{L}_c$ in (19) is extended to arbitrary measures by replacing the coupling matrix by a joint probability measure on the product space. The first step is to formalize what it means for this joint law to have prescribed marginals.

Definition 4 (Marginals of a joint measure). *Let $\pi \in \mathcal{M}_+^1(\mathcal{X} \times \mathcal{Y})$ and let $P_1(x, y) = x$, $P_2(x, y) = y$ be the coordinate projections. The marginals of π are*

$$\pi_1 := P_{1\#}\pi \in \mathcal{M}_+^1(\mathcal{X}), \quad \pi_2 := P_{2\#}\pi \in \mathcal{M}_+^1(\mathcal{Y}).$$

Equivalently, for all bounded continuous test functions (f, g) ,

$$\int_{\mathcal{X} \times \mathcal{Y}} f(x) d\pi(x, y) = \int_{\mathcal{X}} f d\pi_1, \quad \int_{\mathcal{X} \times \mathcal{Y}} g(y) d\pi(x, y) = \int_{\mathcal{Y}} g d\pi_2.$$

A useful mnemonic for the marginal constraint $\pi_1 = \alpha$ and $\pi_2 = \beta$, mimicking the discrete case where one sums along rows and columns, is

$$\int_{\mathcal{Y}} d\pi(x, y) = d\alpha(x) \quad \text{and} \quad \int_{\mathcal{X}} d\pi(x, y) = d\beta(y),$$

and the mathematically rigorous way to write this, which corresponds to the change of variables formula, is

$$\forall (f, g) \in \mathcal{C}_b(\mathcal{X}) \times \mathcal{C}_b(\mathcal{Y}), \quad \int_{\mathcal{X} \times \mathcal{Y}} f(x) d\pi(x, y) = \int_{\mathcal{X}} f d\alpha \quad \text{and} \quad \int_{\mathcal{X} \times \mathcal{Y}} g(y) d\pi(x, y) = \int_{\mathcal{Y}} g d\beta.$$

Using (4), these marginal constraints are also equivalent to imposing that $\pi(A \times \mathcal{Y}) = \alpha(A)$ and $\pi(\mathcal{X} \times B) = \beta(B)$ for sets $A \subset \mathcal{X}$ and $B \subset \mathcal{Y}$.

Replacing continuous functions by indicator functions, one can also rephrase this conservation of mass constraint as

$$\forall A \subset \mathcal{X}, B \subset \mathcal{Y}, \quad \pi(A \times \mathcal{Y}) = \alpha(A) \quad \text{and} \quad \pi(\mathcal{X} \times B) = \beta(B).$$

Definition 5 (Couplings). Given $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ and $\beta \in \mathcal{M}_+^1(\mathcal{Y})$, the set of couplings between α and β is

$$\mathcal{U}(\alpha, \beta) := \left\{ \pi \in \mathcal{M}_+^1(\mathcal{X} \times \mathcal{Y}) ; \pi_1 = \alpha \quad \text{and} \quad \pi_2 = \beta \right\}. \quad (21)$$

It is the continuous analogue of the transportation polytope (18).

Remark 8 (Probabilistic interpretation of couplings). If we write $X \sim \alpha$ to mean that the law of a random vector X is α , then $\pi \in \mathcal{U}(\alpha, \beta)$ means that π is the law of a pair (X, Y) whose coordinates have laws α and β . The coupling encodes the dependence between X and Y . The tensor product $\alpha \otimes \beta$ corresponds to independence, whereas a graph coupling $(\text{Id}, T)_\# \alpha$ corresponds to the deterministic relation $Y = T(X)$.

In the discrete case, when $\alpha = \sum_i a_i \delta_{x_i}$ and $\beta = \sum_j b_j \delta_{y_j}$, the constraint $\pi_1 = \alpha$ and $\pi_2 = \beta$ necessarily imposes that π is discrete, supported on the set $\{(x_i, y_j)\}_{i,j}$, and thus has the form $\pi = \sum_{i,j} P_{i,j} \delta_{(x_i, y_j)}$. The discrete formulation is thus a special case (and not some sort of approximation) of the continuous formulation.

The set $\mathcal{U}(\alpha, \beta)$ is always non-empty because it contains at least the tensor product coupling $\alpha \otimes \beta$ defined by $d(\alpha \otimes \beta)(x, y) = d\alpha(x)d\beta(y)$, i.e.

$$\begin{aligned} \forall h \in \mathcal{C}_b(\mathcal{X} \times \mathcal{Y}), \quad \int_{\mathcal{X} \times \mathcal{Y}} h(x, y) d(\alpha \otimes \beta)(x, y) &= \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} h(x, y) d\beta(y) \right) d\alpha(x) \\ &= \int_{\mathcal{Y}} \left(\int_{\mathcal{X}} h(x, y) d\alpha(x) \right) d\beta(y). \end{aligned}$$

Indeed, $(\alpha \otimes \beta)_1 = \alpha$ since

$$\forall f \in \mathcal{C}_b(\mathcal{X}), \quad \int_{\mathcal{X}} f(x) d(\alpha \otimes \beta)_1(x) = \int_{\mathcal{X} \times \mathcal{Y}} f(x) d\alpha(x) d\beta(y) = \int_{\mathcal{X}} f(x) d\alpha(x) \int_{\mathcal{Y}} d\beta = \int_{\mathcal{X}} f(x) d\alpha(x)$$

because $\int_{\mathcal{Y}} d\beta = 1$.

A very different (concentrated) type of coupling is defined when there exists a map $T : \mathcal{X} \rightarrow \mathcal{Y}$ such that $T_\# \alpha = \beta$ (i.e. the constraint set of Monge's problem (6) is non-empty). In this case, $\pi = (\text{Id}, T)_\# \alpha \in \mathcal{U}(\alpha, \beta)$. This coupling is defined through the integrated definition of push-forward as

$$\forall h \in \mathcal{C}_b(\mathcal{X} \times \mathcal{Y}), \quad \int_{\mathcal{X} \times \mathcal{Y}} h(x, y) d\pi(x, y) = \int_{\mathcal{X}} h(x, T(x)) d\alpha.$$

In particular, applying this formula to $h(x, y) = f(x)$ or $h(x, y) = g(y)$ shows that $\pi_1 = \alpha$ and $\pi_2 = \beta$.

A last important class of examples are semi-discrete problems (for instance to perform quantization or to fit statistical models), where α has a density with respect to Lebesgue and β is discrete. In this case, couplings π are singular; for instance, if $\mathcal{X} = \mathcal{Y} = \mathbb{R}^d$, they are supported on a union of d -dimensional graphs inside $\mathbb{R}^d \times \mathbb{R}^d$.

Continuous Kantorovich problem. The Kantorovich problem (19) is then generalized as

$$\mathcal{L}_c(\alpha, \beta) := \min_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y). \quad (22)$$

This is an infinite-dimensional linear program over a space of measures.

Remark 9 (Probabilistic interpretation of Kantorovich's problem). Using Remark 8, the same problem can be written as

$$\mathcal{L}_c(\alpha, \beta) = \min_{X \sim \alpha, Y \sim \beta} \mathbb{E}(c(X, Y)). \quad (23)$$

The minimization is not over the marginal laws, which are fixed, but over all possible dependences between the two random variables. OT therefore chooses the cheapest joint law among all couplings.

Proposition 18 (Existence on compact spaces). *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$. Then the Kantorovich problem (22) admits at least one minimizer.*

Proof. The constraint set is non-empty because it contains the product coupling $\alpha \otimes \beta$. It is also closed for the weak topology of measures because the marginal constraints are preserved under weak convergence. Since $\mathcal{X} \times \mathcal{Y}$ is compact, the set of probability measures on it is compact for the weak topology, and therefore $\mathcal{U}(\alpha, \beta)$ is compact. Finally, the functional $\pi \mapsto \int c d\pi$ is weakly continuous because c is continuous and bounded. The minimum is thus attained. $\square$

On non-compact domains, one needs to impose coercivity and moment conditions. For the Wasserstein cost $c(x, y) = d(x, y)^p$ on a Polish metric space, the natural domain is

$$\mathcal{P}_p(\mathcal{X}) = \left\{ \mu \in \mathcal{M}_+^1(\mathcal{X}) ; \int d(x, x_0)^p d\mu(x) < +\infty \right\}$$

for one, and hence every, reference point x_0 . If $\alpha, \beta \in \mathcal{P}_p(\mathcal{X})$, then the product coupling has finite p -cost up to the triangle inequality, so the Kantorovich value is finite. Existence of minimizers holds under standard lower-semicontinuity assumptions on c , using tightness of finite-moment sublevel sets [212, 190].

Monge–Kantorovich equivalence. The proof of Brenier’s theorem 1 (detailed in Section 5.4, Remark 16) relies on Kantorovich relaxation and duality. It proves that this relaxation is tight, in the sense that it has the same cost as the Monge problem.

Corollary 2 (Monge–Kantorovich equivalence under Brenier). *We assume α has a density with respect to Lebesgue measure, and that $c(x, y) = \|x - y\|^2$. We denote by T the optimal transport solving Monge’s formulation. Then $\pi = (\text{Id}, T)_\# \alpha$ is the unique optimal coupling solving the Kantorovich formulation. Monge and Kantorovich costs are the same.*

Proof. The proof of Brenier’s theorem shows that the support of an optimal π is contained in the subdifferential $\partial\varphi$ of a convex function φ , which in general is a set-valued mapping. When α does not have a density, then φ may be non-smooth, and non-smooth points where $\alpha(\{x\}) > 0$ lead to mass splitting. For instance, moving δ_0 to $(\delta_{-1} + \delta_{+1})/2$ can be achieved using $\varphi(x) = |x|$.

If α has a density, then this φ is differentiable α -almost everywhere. Denoting by $T = \nabla\varphi$ the unique optimal transport (which is a valid definition almost everywhere, and can be assigned arbitrary values at points of non-differentiability), the coupling

$$\pi = (\text{Id}, T)_\# \alpha \quad \text{i.e.} \quad \forall h \in \mathcal{C}(\mathcal{X} \times \mathcal{Y}), \quad \int_{\mathcal{X} \times \mathcal{Y}} h d\pi = \int_{\mathcal{X}} h(x, T(x)) d\alpha(x)$$

is optimal. Since any optimal coupling is supported on this graph and T is unique α -almost everywhere, the optimal Kantorovich coupling is unique and the two costs coincide. $\square$

In terms of random vectors, denoting (X, Y) a random vector with law π , it means that any such optimal random vector satisfies $Y = T(X)$ where $X \sim \alpha$ (and of course $T(X) \sim \beta$ by the marginal constraint).

This key result is similar to Birkhoff-von Neumann Theorem 1 in the sense that it provides conditions ensuring the equivalence between Monge and Kantorovich problems (note however that Birkhoff-von Neumann does not imply uniqueness). The settings are radically different: one is fully discrete while the other requires the source measure to be “continuous”, i.e. to have a density.

3.3 c -cyclical monotonicity

Cyclical monotonicity is the local geometric fingerprint of optimality. It converts a global minimization problem into finite exchange inequalities and is the bridge from Kantorovich plans to convex potentials.

Optimal transport plans behave well when one looks at any finite sub-collection of points in their support: the restriction is still an optimal matching for those points alone. For finitely supported marginals this is immediate, and it leads to the notion of c -cyclical monotonicity.

Support and c -cyclical monotonicity. To formalise this, one needs a precise notion of *support*, i.e. the set that carries the mass of the coupling. Because we work with Radon measures, the support is automatically closed.

Definition 6 (Support). *For a Radon measure π on $\mathcal{X} \times \mathcal{Y}$*

$$\text{supp}(\pi) := \{(x, y) : \pi(U \times V) > 0 \text{ for every open } U \ni x, V \ni y\}.$$

Definition 7 (c -cyclical monotonicity). *A set $\Gamma \subset \mathcal{X} \times \mathcal{Y}$ is c -cyclically monotone if, for every $k \geq 2$, every finite family $\{(x_i, y_i)\}_{i=1}^k \subset \Gamma$, and every permutation $\sigma \in \mathfrak{S}_k$,*

$$\sum_{i=1}^k c(x_i, y_i) \leq \sum_{i=1}^k c(x_i, y_{\sigma(i)}).$$

Any permutation is a product of cycles, so it suffices to verify the inequality for cycles,

$$\sum_{i=1}^k c(x_i, y_i) \leq \sum_{i=1}^k c(x_i, y_{i+1}).$$

Optimal matching to optimal transport. Let the marginals be uniform on n points: $\alpha = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$, $\beta = \frac{1}{n} \sum_{i=1}^n \delta_{y_i}$. In this case, there exist solutions to the Kantorovich problem that are permutations. Such a permutation defines, by optimality, a coupling with support $\Gamma := (x_i, y_{\sigma(i)})_i$ that is c -cyclically monotone. The following theorem says this is true for any optimal coupling, not only in the case of a finite set of points.

Theorem 3 (Rockafellar). *For any optimal plan π solving Kantorovich's problem, $\text{supp}(\pi)$ is c -cyclically monotone.*

Proof. We prove the contrapositive. Suppose that $\text{supp}(\pi)$ is not c -cyclically monotone. Then there exist points $(x_i, y_i)_{i=1}^k$ in the support and a permutation σ such that

$$\sum_i c(x_i, y_i) > \sum_i c(x_i, y_{\sigma(i)}).$$

By continuity of c , this strict inequality persists on small neighborhoods $U_i \times V_i$ of the points (x_i, y_i) , chosen disjoint enough so that each has positive π -mass. Let $\eta > 0$ be smaller than all these masses. Removing mass η from each $U_i \times V_i$ and re-inserting the same mass from U_i to $V_{\sigma(i)}$ preserves both marginals, because the operation only permutes target neighborhoods among the same source neighborhoods. The cost strictly decreases by the displayed gap up to the continuity error. This contradicts optimality of π . $\square$

Monotonicity. Assume the plan is induced by a measurable map $T : \mathcal{X} \rightarrow \mathcal{Y}$, i.e. $\pi = (\text{Id}, T)_\# \alpha$. For any k points $x_1, \dots, x_k$ in the domain, cyclical monotonicity reads

$$\sum_{i=1}^k c(x_i, T(x_i)) \leq \sum_{i=1}^k c(x_i, T(x_{i+1})), \quad x_{k+1} = x_1.$$

For $c(x, y) = \frac{1}{2}|x - y|^2$, taking $k = 2$, shows that for any x, y ,

$$\langle T(x) - T(y), x - y \rangle \geq 0,$$

so T is a monotone vector field. Brenier's theorem adds that, when α is absolutely continuous, $T = \nabla \varphi$ for a convex potential φ . (The converse fails in dimension $d \geq 2$: a small rotation is monotone yet not a gradient.)

One dimension. In one space dimension, with cost $c(x, y) = |x - y|^p$ for any $p \geq 1$, the two-point inequality becomes

$$|x - T(x)|^p + |y - T(y)|^p \leq |x - T(y)|^p + |y - T(x)|^p,$$

which is equivalent to $T(x) \leq T(y)$ whenever $x < y$. Thus T must be non-decreasing, recovering the classical monotone rearrangement.

3.4 Metric Properties: Wasserstein Distances

The final part of the section proves that OT costs are genuine distances when the ground cost comes from a metric. It also compares Wasserstein convergence with total variation and explains why OT is weak enough to move Dirac masses continuously.

OT defines a distance. An important feature of OT is that it defines a distance between histograms and probability measures as soon as the cost matrix satisfies certain suitable properties. Indeed, OT can be understood as a canonical way to lift a ground distance between points to a distance between histograms or measures. The proof of this result relies on a “gluing lemma”, which we first prove in the discrete case.

Lemma 1 (Discrete gluing lemma). *Given $(a, b, c) \in \Sigma_n \times \Sigma_p \times \Sigma_m$, let $P \in U(a, b)$ and $Q \in U(b, c)$. Then there exists a 3-D tensor coupling $S \in \mathbb{R}_+^{n \times p \times m}$ such that the 2-D marginals satisfy*

$$\sum_k S_{i,j,k} = P_{i,j} \quad \text{and} \quad \sum_i S_{i,j,k} = Q_{j,k}.$$

Note that this implies that the three 1-D marginals of S are (a, b, c) .

Proof. One verifies that

$$S_{i,j,k} = \begin{cases} \frac{P_{i,j} Q_{j,k}}{b_j} & \text{if } b_j \neq 0 \\ 0 & \text{otherwise} \end{cases} \quad (24)$$

is acceptable. Indeed, if $b_j \neq 0$

$$\sum_k S_{i,j,k} = \sum_k \frac{P_{i,j} Q_{j,k}}{b_j} = \frac{P_{i,j}}{b_j} (Q \mathbf{1}_m)_j = \frac{P_{i,j}}{b_j} b_j.$$

If $b_j = 0$, then necessarily $P_{i,j} = 0$ and $\sum_k S_{i,j,k} = 0 = P_{i,j}$. $\square$

Proposition 19 (Discrete Wasserstein distance). *We suppose $n = m$, and that for some $p \geq 1$, $C = D^p = (D_{i,j}^p)_{i,j} \in \mathbb{R}^{n \times n}$ where $D \in \mathbb{R}_+^{n \times n}$ is a distance on $\llbracket n \rrbracket$, i.e.*

1. $D \in \mathbb{R}_+^{n \times n}$ is symmetric;
2. $D_{i,j} = 0$ if and only if $i = j$;
3. $\forall (i, j, k) \in \llbracket n \rrbracket^3, D_{i,k} \leq D_{i,j} + D_{j,k}$.

Then

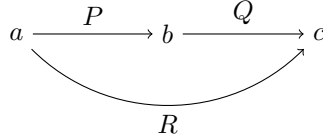
$$W_p(a, b) := L_{D^p}(a, b)^{1/p} \quad (25)$$

(note that W_p depends on D) defines the p -Wasserstein distance on Σ_n , i.e. W_p is symmetric, positive, $W_p(a, b) = 0$ if and only if $a = b$, and it satisfies the triangle inequality

$$\forall a, b, c \in \Sigma_n, \quad W_p(a, c) \leq W_p(a, b) + W_p(b, c).$$

Proof. For symmetry, since D^p is symmetric, we use the fact that if $P \in U(a, b)$ is optimal for $W_p(a, b)$, then $P^\top \in U(b, a)$ is optimal for $W_p(b, a)$. For definiteness, since $C = D^p$ has a null diagonal, $W_p(a, b) = 0$ is achieved by the diagonal coupling $P^* = \text{diag}(a) = \text{diag}(b)$ when $a = b$; by positivity of all off-diagonal elements of D^p , $W_p(a, b) > 0$ whenever $a \neq b$ because any admissible coupling then has a nonzero element outside the diagonal.

To prove the triangle inequality of Wasserstein distances for arbitrary measures, we consider $a, b, c \in \Sigma_n$, and let P and Q be two optimal solutions of the transport problems between a and b , and b and c respectively. We use the gluing Lemma 1 which defines $S \in \mathbb{R}_+^{n^3}$ with marginals $\sum_k S_{\cdot, \cdot, k} = P$ and $\sum_i S_{i, \cdot, \cdot} = Q$. We define $R = \sum_j S_{\cdot, j, \cdot}$, which is an element of $U(a, c)$.



Note that if one assumes $b > 0$ then $R = P \text{diag}(1/b)Q$.

The triangle inequality follows from

$$\begin{aligned}
W_p(a, c) &= \left(\min_{\tilde{R} \in U(a, c)} \langle \tilde{R}, D^p \rangle \right)^{1/p} \leq \langle R, D^p \rangle^{1/p} \\
&= \left(\sum_{i, k} D_{ik}^p \sum_j S_{i, j, k} \right)^{1/p} \leq \left(\sum_{i, j, k} (D_{ij} + D_{j, k})^p S_{i, j, k} \right)^{1/p} \\
&\leq \left(\sum_{i, j, k} D_{ij}^p S_{i, j, k} \right)^{1/p} + \left(\sum_{i, j, k} D_{j, k}^p S_{i, j, k} \right)^{1/p} \\
&= \left(\sum_{i, j} D_{ij}^p \sum_k S_{i, j, k} \right)^{1/p} + \left(\sum_{j, k} D_{j, k}^p \sum_i S_{i, j, k} \right)^{1/p} \\
&= \left(\sum_{i, j} D_{ij}^p P_{i, j} \right)^{1/p} + \left(\sum_{j, k} D_{j, k}^p Q_{j, k} \right)^{1/p} = W_p(a, b) + W_p(b, c).
\end{aligned}$$

The first inequality follows from the feasibility of R , the second is the usual triangle inequality for elements in D , and the third comes from Minkowski's inequality. $\square$

Proposition 19 generalizes from histogram to arbitrary measures that need not be discrete. For this, one needs the following general gluing lemma.

Lemma 2 (Gluing lemma). *Let $(\alpha, \beta, \gamma) \in \mathcal{M}_+^1(\mathcal{X}) \times \mathcal{M}_+^1(\mathcal{Y}) \times \mathcal{M}_+^1(\mathcal{Z})$ where $(\mathcal{X}, \mathcal{Y}, \mathcal{Z})$ are Polish spaces, i.e. separable topological spaces that can be metrized by complete distances. Given $\pi \in \mathcal{U}(\alpha, \beta)$ and $\xi \in \mathcal{U}(\beta, \gamma)$, then there exists a tensor coupling measure $\sigma \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ such that*

$$(P_{\mathcal{X}, \mathcal{Y}})_\# \sigma = \pi \quad \text{and} \quad (P_{\mathcal{Y}, \mathcal{Z}})_\# \sigma = \xi$$

where we denoted the projector $P_{\mathcal{X}, \mathcal{Y}}(x, y, z) = (x, y)$ and $P_{\mathcal{Y}, \mathcal{Z}}(x, y, z) = (y, z)$.

Proof. The proof of this fundamental result is involved since it requires using the disintegration of measure (which corresponds to conditional probabilities). The disintegration of measures is applicable because the spaces are Polish. We disintegrate π and ξ against β to obtain two families $(\pi_y)_{y \in \mathcal{Y}}$ and $(\xi_y)_{y \in \mathcal{Y}}$ of probability distributions on $\mathcal{X}$ and $\mathcal{Z}$. These families are defined by the fact that

$$\forall h \in \mathcal{C}(\mathcal{X} \times \mathcal{Y}), \quad \int_{\mathcal{Y}} \left(\int_{\mathcal{X}} h(x, y) d\pi_y(x) \right) d\beta(y) = \int h(x, y) d\pi(x, y).$$

and similarly for ξ . When $\beta = \sum_j b_j \delta_{y_j}$ and $\pi = \sum_{i,j} P_{i,j} \delta_{(x_i, y_j)}$, then this conditional distribution is defined on the support of β as $\pi_{y_j} = \sum_i \frac{P_{i,j}}{b_j} \delta_{x_i}$ (and similarly for ξ). The glued measure is then defined by the conditional-product formula

$$\forall g \in \mathcal{C}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}), \quad \int g(x, y, z) d\sigma(x, y, z) = \int g(x, y, z) d\pi_y(x) d\xi_y(z) d\beta(y).$$

For discrete measures, this matches the definition (24), since $\sigma = \sum_{i,j,k} S_{i,j,k} \delta_{x_i, y_j, z_k}$ where

$$S_{i,j,k} = \frac{P_{i,j}}{b_j} \frac{Q_{j,k}}{b_j} b_j.$$

□

Using this gluing lemma, we can now construct the Wasserstein distance in the general setting of arbitrary distributions on a Polish space.

Proposition 20 (Wasserstein distance on measures). *We assume $\mathcal{X} = \mathcal{Y}$, and that for some $p \geq 1$, $c(x, y) = d(x, y)^p$ where d is a distance on $\mathcal{X}$, i.e.*

- (i) $d(x, y) = d(y, x) \geq 0$;
- (ii) $d(x, y) = 0$ if and only if $x = y$;
- (iii) $\forall (x, y, z) \in \mathcal{X}^3, d(x, z) \leq d(x, y) + d(y, z)$.

Then

$$\mathcal{W}_p(\alpha, \beta) := \mathcal{L}_{d^p}(\alpha, \beta)^{1/p} \quad (26)$$

(note that $\mathcal{W}_p$ depends on d) defines the p -Wasserstein distance on $\mathcal{X}$, i.e. $\mathcal{W}_p$ is symmetric, positive, $\mathcal{W}_p(\alpha, \beta) = 0$ if and only if $\alpha = \beta$, and it satisfies the triangle inequality

$$\forall (\alpha, \beta, \gamma) \in \mathcal{M}_+^1(\mathcal{X})^3, \quad \mathcal{W}_p(\alpha, \gamma) \leq \mathcal{W}_p(\alpha, \beta) + \mathcal{W}_p(\beta, \gamma).$$

Proof. The symmetry follows from the fact that since d is symmetric, if $\pi(x, y)$ is optimal for $\mathcal{L}_{d^p}(\alpha, \beta)$, then $\pi(y, x) \in \mathcal{U}(\beta, \alpha)$ is optimal for $\mathcal{L}_{d^p}(\beta, \alpha)$. If $\mathcal{L}_{d^p}(\alpha, \beta) = 0$, then necessarily an optimal coupling π is supported on the diagonal $\Delta := \{(x, x)\}_x \subset \mathcal{X}^2$. We denote $\lambda(x)$ the corresponding measure on the diagonal, i.e. such that $\int h(x, y) d\pi(x, y) = \int h(x, x) d\lambda(x)$. Then since $\pi \in \mathcal{U}(\alpha, \beta)$ necessarily $\lambda = \alpha$ and $\lambda = \beta$ so that $\alpha = \beta$.

For the triangle inequality, we consider optimal couplings $\pi \in \mathcal{U}(\alpha, \beta)$ and $\xi \in \mathcal{U}(\beta, \gamma)$ and we glue them according to the Lemma 2. We define the composition of the two couplings (π, ξ) as $\rho := (P_{\mathcal{X}, \mathcal{Z}})_\# \sigma$. Note that if π and ξ are couplings induced by two Monge maps $T_{\mathcal{X}}(x)$ and $T_{\mathcal{Y}}(y)$, then ρ is itself induced by the Monge map $T_{\mathcal{Y}} \circ T_{\mathcal{X}}$, so that this notion of composition of coupling generalizes the composition of maps. The triangular inequality follows from

$$\begin{aligned} \mathcal{W}_p(\alpha, \gamma) &\leq \left(\int_{\mathcal{X} \times \mathcal{Z}} d(x, z)^p d\rho(x, z) \right)^{1/p} = \left(\int_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} d(x, z)^p d\sigma(x, y, z) \right)^{1/p} \\ &\leq \left(\int_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} (d(x, y) + d(y, z))^p d\sigma(x, y, z) \right)^{1/p} \\ &\leq \left(\int_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} d(x, y)^p d\sigma(x, y, z) \right)^{1/p} + \left(\int_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} d(y, z)^p d\sigma(x, y, z) \right)^{1/p} \\ &= \left(\int_{\mathcal{X} \times \mathcal{Y}} d(x, y)^p d\pi(x, y) \right)^{1/p} + \left(\int_{\mathcal{Y} \times \mathcal{Z}} d(y, z)^p d\xi(y, z) \right)^{1/p} = \mathcal{W}_p(\alpha, \beta) + \mathcal{W}_p(\beta, \gamma). \end{aligned}$$

□

This distance $\mathcal{W}_p$ defined through the Kantorovich problem (26) should be contrasted with the directed distance $\tilde{\mathcal{W}}$ obtained using Monge's problem (7). The Kantorovich feasible set is never empty, since it contains the product coupling, although the p -cost may still be infinite without moment assumptions on non-compact spaces. By contrast, Monge's constraint set $\{T; T_{\#}\alpha = \beta\}$ can be empty. When an optimal Monge map exists, Kantorovich gives the same value by choosing the graph coupling $(\text{Id}, T)_{\#}\alpha$; in this sense the Kantorovich problem is the convex relaxation of Monge's problem, with much better stability properties.

3.5 Metric Properties: Topology and Applications

This subsection shifts from metric axioms to topology and uses. Wasserstein distances metrize weak convergence under moment control, sit between weak and strong topologies, and provide quantitative estimates in probability and robust optimization.

Convergence in law topology. On a bounded metric space, all $\mathcal{W}_p$ distances define the same topology, although they are not equivalent as distances.

Proposition 21 (Equivalence of Wasserstein distances on compact spaces). *One has for $p \leq q$*

$$\mathcal{W}_p(\alpha, \beta) \leq \mathcal{W}_q(\alpha, \beta) \leq \text{diam}(\mathcal{X})^{\frac{q-p}{q}} \mathcal{W}_p(\alpha, \beta)^{\frac{p}{q}}$$

where $\text{diam}(\mathcal{X}) \triangleq \sup_{x,y} d(x, y)$.

Proof. The left inequality follows from Jensen inequality, $\varphi(\int c(x, y) d\pi(x, y)) \leq \int \varphi(c(x, y)) d\pi(x, y)$, applied to any probability distribution π and to the convex function $\varphi(r) = r^{q/p}$ with $c(x, y) = d(x, y)^p$, so that one gets

$$\left(\int d(x, y)^p d\pi(x, y) \right)^{\frac{q}{p}} \leq \int d(x, y)^q d\pi(x, y).$$

The right inequality follows from

$$d(x, y)^q \leq \text{diam}(\mathcal{X})^{q-p} d(x, y)^p.$$

□

The Wasserstein distance $\mathcal{W}_p$ is a weak distance: it compares singular distributions, such as discrete measures, and quantifies spatial shifts between supports. Its topology is studied in detail in [212, 190, 99], while empirical rates are quantified in [73, 84, 214, 33, 34].

Definition 8 (Weak* topology). $(\alpha_k)_k$ converges weakly* to α in $\mathcal{M}_+^1(\mathcal{X})$ (denoted $\alpha_k \rightharpoonup \alpha$) if and only if for any bounded continuous function $f \in \mathcal{C}_b(\mathcal{X})$, $\int_{\mathcal{X}} f d\alpha_k \rightarrow \int_{\mathcal{X}} f d\alpha$. On compact spaces, $\mathcal{C}_b(\mathcal{X}) = \mathcal{C}(\mathcal{X})$, which is why the boundedness is often left implicit there.

Remark 10 (A Riemann-sum weak limit). On $\mathcal{X} = \mathbb{R}$, the empirical measures on a regular grid satisfy

$$\frac{1}{n} \sum_{k=1}^n \delta_{k/n} \rightharpoonup \mathcal{U}_{[0,1]}.$$

Indeed, for every continuous bounded function f ,

$$\frac{1}{n} \sum_{k=1}^n f(k/n) \rightarrow \int_0^1 f(x) dx,$$

which is precisely the convergence of Riemann sums. This convergence is weak but not strong: for every n , the discrete measure and the uniform density are mutually singular, hence their total variation distance is equal to 2.

Remark 11 (Weak convergence for discrete measures). In the special case of a single Dirac, $\delta_{x^{(n)}} \rightarrow \delta_x$ is equivalent to $\int f d\delta_{x^{(n)}} = f(x^{(n)}) \rightarrow \int f d\delta_x = f(x)$ for any continuous f . This in turn is equivalent to $x^{(n)} \rightarrow x$. For a fixed number of atoms, if $\alpha_n = \sum_{i=1}^N a_i^{(n)} \delta_{x_i^{(n)}}$ and, after extracting a subsequence and relabeling, $a_i^{(n)} \rightarrow a_i$ and $x_i^{(n)} \rightarrow x_i$, then α_n converges weakly to $\sum_i a_i \delta_{x_i}$, with atoms at identical limits merged. Without a uniform bound on the number of atoms, weak limits of discrete measures can be non-discrete; empirical measures are the standard example.

In terms of random vectors, if $X_n \sim \alpha_n$ and $X \sim \alpha$ (not necessarily defined on the same probability space), weak convergence corresponds to convergence in law of X_n toward X .

Remark 12 (Central limit theorem). The central limit theorem states that if $(X_1, \dots, X_n)$ are i.i.d. random vectors with finite second moments, $\mathbb{E}(X_i) = 0$, and $\mathbb{E}(X_i X_i^\top) = \text{Id}$, then the rescaled average $Z_n \triangleq \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$ converges in law toward a Gaussian $\mathcal{N}(0, \text{Id})$. This means that the measure α_n representing the law of Z_n converges weakly toward the measure α of the centered normalized Gaussian.

Definition 9 (Strong topology). *The simplest distance on Radon measures is the total variation norm, which is the dual norm of the L^∞ norm on $\mathcal{C}(\mathcal{X})$ and whose topology is often called the “strong” topology*

$$\|\alpha - \beta\|_{\text{TV}} := \sup_{\|f\|_\infty \leq 1} \int f d(\alpha - \beta) = |\alpha - \beta|(\mathcal{X})$$

where $|\alpha - \beta|(\mathcal{X})$ is the mass of the absolute value of the difference measure. When $\alpha - \beta = \rho dx$ has a density, then $\|\alpha - \beta\|_{\text{TV}} = \int |\rho(x)| dx = \|\rho\|_{L^1(dx)}$ is the L^1 norm associated to dx . When $\alpha - \beta = \sum_i u_i \delta_{z_i}$ is discrete, then $\|\alpha - \beta\|_{\text{TV}} = \sum_i |u_i| = \|u\|_{\ell^1}$ is the discrete ℓ^1 norm.

The following proposition shows that the TV norm can be seen as a Wasserstein distance, but for a “degenerate” 0/1 metric.

Proposition 22 (Total variation as Wasserstein for the discrete metric). *Denoting d the 0/1 distance such that $d(x, x) = 0$ and $d(x, y) = 1$ if $x \neq y$, then*

$$\mathcal{W}_p(\alpha, \beta)^p = \frac{1}{2} \|\alpha - \beta\|_{\text{TV}}.$$

Proof. For the sake of simplicity, we do the proof for discrete measures with weights (a, b) and without loss of generality assume they have the same support $(x_i)_i$ and we denote $D \triangleq (d(x_i, x_j))_{i,j}$ which is 0 on the diagonal and one outside. Also since $d^p = d$ we consider $p = 1$. We denote $c_i = \min(a_i, b_i)$. By conservation of mass, for every $P \in U(a, b)$, $P_{i,i} \leq c_i$, thus

$$\langle P, D \rangle = \sum_{i \neq j} P_{i,j} = 1 - \sum_i P_{i,i} \geq 1 - \sum_i c_i.$$

We need to show that this bound is tight, namely to construct $\hat{P} \in U(a, b)$ such that $\text{diag}(\hat{P}) = c$. Let

$$\bar{a} \triangleq a - c = (a - b)_+ \geq 0 \quad \text{and} \quad \bar{b} \triangleq b - c = (b - a)_+ \geq 0$$

If $\bar{a} = \bar{b} = 0$, then $a = b$ and the diagonal coupling is optimal. Otherwise, one has

$$\frac{\bar{a} \otimes \bar{b}}{\langle \bar{a}, \mathbf{1} \rangle} \in U(\bar{a}, \bar{b})$$

and we remark that $\langle \bar{a}, \mathbf{1} \rangle = \langle \bar{b}, \mathbf{1} \rangle = 1 - \langle c, \mathbf{1} \rangle$. Thus denoting

$$\hat{P} \triangleq \text{diag}(c) + \frac{\bar{a} \otimes \bar{b}}{\langle \bar{a}, \mathbf{1} \rangle} \geq 0$$

satisfies

$$\hat{P}\mathbb{1} = c + \bar{a} = a \quad \text{and} \quad \hat{P}^\top \mathbb{1} = c + \bar{b} = b$$

so that $\hat{P} \in U(a, b)$ is a coupling so that $\text{diag}(\hat{P}) = \text{diag}(c)$ since $\text{diag}(\bar{a} \otimes \bar{b}) = 0$. We thus conclude that

$$W_1(a, b) = \langle D, \hat{P} \rangle = \sum_{i,j} \frac{\bar{a}_i \bar{b}_j}{\langle \bar{a}, \mathbb{1} \rangle} = \sum_i \bar{a}_i = \sum_i \bar{b}_i = \frac{1}{2} \sum_i (\bar{a}_i + \bar{b}_i) = \frac{1}{2} \|a - b\|_{\text{TV}}.$$

□

As explained in Remark 11, in the special case of Diracs, $\delta_{x_n} \rightharpoonup \delta_x$ is equivalent to $x_n \rightarrow x$. One can then contrast the strong topology with the Wasserstein distance if $x_n \neq x$,

$$\|\delta_{x_n} - \delta_x\|_{\text{TV}} = 2 \quad \text{and} \quad \mathcal{W}_p(\delta_{x_n}, \delta_x) = d(x_n, x).$$

This shows that for the strong topology, Diracs never converge, while they do converge for the Wasserstein distance. It is a powerful property of the Wasserstein distance: on compact spaces, it metrizes weak convergence.

Proposition 23 (Wasserstein metrizes weak convergence on compact spaces). *If $\mathcal{X}$ is compact, $\alpha_k \rightharpoonup \alpha$ if and only if $\mathcal{W}_p(\alpha_k, \alpha) \rightarrow 0$.*

Proof. For $p = 1$, this is exactly Corollary 3, which follows from the Kantorovich–Rubinstein dual representation and compactness of the unit Lipschitz ball modulo constants. For a compact space, Proposition 21 shows that all Wasserstein distances $\mathcal{W}_p$ induce the same convergent sequences. Hence weak convergence is equivalent to convergence in $\mathcal{W}_p$ for every $p \geq 1$. □

On non-compact spaces, one needs also to impose convergence of the p -th moments. More precisely, on a Polish metric space, $\mathcal{W}_p(\alpha_k, \alpha) \rightarrow 0$ if and only if $\alpha_k \rightharpoonup \alpha$ and, for some reference point x_0 ,

$$\int d(x, x_0)^p d\alpha_k(x) \longrightarrow \int d(x, x_0)^p d\alpha(x).$$

Note that there exist alternative distances which also metricize weak convergence. The simplest ones are Hilbertian kernel norms, which are detailed in Section 7.1.

On a discrete space, the strong and weak topologies coincide, and the following proposition relates the TV and Wasserstein distances.

Proposition 24 (Comparison with total variation on discrete spaces). *One has*

$$\frac{d_{\min}}{2} \|\alpha - \beta\|_{\text{TV}} \leq \mathcal{W}_1(\alpha, \beta) \leq \frac{d_{\max}}{2} \|\alpha - \beta\|_{\text{TV}} \quad \text{where} \quad \begin{cases} d_{\min} := \inf_{x \neq y} d(x, y) \\ d_{\max} := \sup_{x, y} d(x, y) \end{cases}$$

Proof. We denote $d_0(x, y)$ the distance such that $d_0(x, x) = 0$ and $d_0(x, y) = 1$ for $x \neq y$. One has

$$d_{\min} d_0(x, y) \leq d(x, y) \leq d_{\max} d_0(x, y)$$

so that integrating this against any $\pi \in \mathcal{U}(\alpha, \beta)$ and taking the minimum among those π gives the result using Proposition 22. □

This bound is sharp, as this can be observed by taking $\alpha = \delta_x$ and $\beta = \delta_y$, in which case the bound simply reads if $x \neq y$

$$d_{\min} \leq d(x, y) \leq d_{\max}.$$

This shows that the ratio between the two distances can blow up as $d_{\max}/d_{\min}$ increases. On non-discrete spaces, if $d_{\min} = 0$, then the two distances are not equivalent, in line with the fact that the strong and weak topologies do not coincide.

Distributional robustness and $\mathcal{W}_\infty$. Wasserstein distances are also used to define ambiguity sets around an empirical law. Given samples z_i and $\hat{\alpha}_n = \frac{1}{n} \sum_i \delta_{z_i}$, a distributionally robust optimization (DRO) problem replaces the empirical risk $\frac{1}{n} \sum_i \ell_\theta(z_i)$ by

$$\sup_{\beta: \mathcal{W}_p(\beta, \hat{\alpha}_n) \leq \rho} \int \ell_\theta(z) d\beta(z),$$

or, in Lagrangian penalized form, by $\sup_\beta \int \ell_\theta d\beta - \lambda \mathcal{W}_p(\beta, \hat{\alpha}_n)^p$. The constrained and penalized formulations are linked by the choice of multiplier λ , but are not the same problem for an arbitrary fixed λ . Both ask for performance against distributions that can be reached by transporting the empirical mass within a budget. The radius ρ is expressed in the geometry of the data space, so it can encode feature perturbations, domain shift or model misspecification [78, 30, 92].

The basic computational reason for the popularity of Wasserstein DRO is a dual reformulation. Under the usual upper-semicontinuity and growth assumptions on the loss, one has

$$\sup_{\beta: \mathcal{W}_p(\beta, \hat{\alpha}_n)^p \leq \rho^p} \int \ell_\theta d\beta = \inf_{\lambda \geq 0} \lambda \rho^p + \frac{1}{n} \sum_{i=1}^n \sup_z \{ \ell_\theta(z) - \lambda d(z, z_i)^p \}. \quad (27)$$

Thus the robust risk is an empirical risk in which each sample is replaced by its worst penalized perturbation. For $p = 1$ and an L_θ -Lipschitz loss, the Kantorovich–Rubinstein dual gives the transparent upper bound

$$\sup_{\beta: \mathcal{W}_1(\beta, \hat{\alpha}_n) \leq \rho} \int \ell_\theta d\beta \leq \frac{1}{n} \sum_i \ell_\theta(z_i) + \rho L_\theta,$$

which exhibits Wasserstein robustness as a Lipschitz regularizer. The bound is sharp for worst-case optimization over a full Lipschitz ball of losses, while a fixed loss may not realize all extremal Lipschitz directions.

In machine learning, the inner supremum in (27) has the form of an adversarial perturbation problem: the adversary moves each datum away from z_i but pays a transport penalty. This connects Wasserstein DRO to adversarial training and certified robustness [196, 195]. In sequential decision problems, the same idea places Wasserstein ambiguity sets around transition kernels or rewards, producing robust Bellman operators and robust reinforcement-learning models [217, 218].

The limiting distance

$$\mathcal{W}_\infty(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \operatorname{ess\,sup}_{(x, y) \sim \pi} d(x, y) \quad (28)$$

is the limit of $\mathcal{W}_p(\alpha, \beta)$ as $p \rightarrow \infty$ on bounded spaces, not the limit of the convex programs defining $\mathcal{W}_p^p$. It minimizes the worst displacement rather than an average displacement, and the resulting optimization is no longer a linear convex program because of the essential-supremum objective. This makes $\mathcal{W}_\infty$ less convenient algorithmically, but very natural in robust formulations where one wants to guarantee that every transported sample stays within a prescribed perturbation radius.

Theoretical application: quantitative central limit theorems. The weak topology only says whether laws converge; Wasserstein distances also quantify how fast they converge. The next result is a representative theoretical application: the central limit theorem becomes a rate estimate in $\mathcal{W}_1$. This is useful because $\mathcal{W}_1$ is exactly the supremum over 1-Lipschitz observables, so the bound controls the error of all stable numerical or statistical measurements of the normalized sum at once.

Proposition 25 (Berry–Esseen bound in $\mathcal{W}_1$). *Let $(X_i)_{i=1}^n$ be i.i.d. real random variables such that $\mathbb{E}X_i = 0$, $\mathbb{E}X_i^2 = 1$ and $\mathbb{E}|X_i|^3 < +\infty$. If α_n is the law of $n^{-1/2} \sum_i X_i$ and γ is the standard Gaussian law, then*

$$\mathcal{W}_1(\alpha_n, \gamma) \leq \frac{C \mathbb{E}|X_1|^3}{\sqrt{n}},$$

where C is a universal constant.

Proof. We give the standard Stein-method sketch. By Kantorovich–Rubinstein duality,

$$\mathcal{W}_1(\alpha_n, \gamma) = \sup_{\text{Lip}(h) \leq 1} |\mathbb{E}h(S_n) - \mathbb{E}h(G)|, \quad S_n = n^{-1/2} \sum_i X_i, \quad G \sim \gamma.$$

For each such h , solve Stein’s equation

$$f'_h(x) - xf_h(x) = h(x) - \mathbb{E}h(G).$$

The solution satisfies uniform derivative bounds depending only on the Lipschitz constant of h . Hence

$$\mathbb{E}h(S_n) - \mathbb{E}h(G) = \mathbb{E}[f'_h(S_n) - S_n f_h(S_n)].$$

Expanding $f'_h(S_n)$ and $f_h(S_n)$ by replacing the summands one at a time, the first- and second-order terms cancel because $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = 1$. The Taylor remainder is bounded by $C \sum_i \mathbb{E}|X_i/\sqrt{n}|^3$, which gives the displayed $n^{-1/2}$ rate. This is the Wasserstein form of the Berry–Esseen theorem; sharper constants and higher-order transport-distance refinements are studied in [31, 182]. $\square$

Applications and implications The practical value of Wasserstein distances is that they provide a geometric loss on probability measures. In barycenter problems, minimizing $\sum_s \lambda_s \mathcal{W}_2^2(\alpha, \beta_s)$ produces an average distribution that respects displacement rather than pointwise density interpolation, which is crucial for shape and image averaging [1, 66, 200]. In registration and morphing, OT penalizes how far mass travels, so it naturally encodes spatial deformations and correspondences [112, 172, 105]. In statistical fitting, one can estimate a parametric measure $\theta \mapsto \alpha_\theta$ from data β by minimizing $\theta \mapsto \mathcal{W}_p(\alpha_\theta, \beta)$; this remains meaningful even when the model and empirical measure are mutually singular, unlike likelihood or KL objectives [45, 86].

4 Sinkhorn

Entropic regularization makes optimal transport smooth, strictly convex and scalable. This section explains why adding entropy produces matrix scaling, how Sinkhorn iterations enforce the marginals, and why this simple algorithm became the computational workhorse of OT. The presentation connects the older matrix-scaling literature [197, 199, 198] with modern entropic OT [64, 170].

4.1 Entropic Regularization for Discrete Measures

Entropy turns a possibly non-unique linear program into a unique smooth problem. The price is bias, but the reward is differentiability and fast scaling algorithms.

Entropic Regularization for Discrete Measures. The idea of the entropic regularization of optimal transport is to use the Shannon-Boltzmann entropy

$$H(P) \triangleq - \sum_{i,j} P_{i,j} \log(P_{i,j}),$$

with the convention $0 \log(0) = 0$ as a regularizing function to obtain approximate solutions to the original transport problem (19)

$$L_C^\varepsilon(a, b) := \min_{P \in \mathcal{U}(a, b)} \langle P, C \rangle - \varepsilon H(P). \quad (29)$$

Proposition 26 (Existence and uniqueness of entropic OT). *Assume that a, b are probability histograms and that C is finite. For every $\varepsilon > 0$, problem (29) admits a unique minimizer. If all entries of a and b are positive, then this minimizer is positive on every entry.*

Proof. The transport polytope $U(a, b)$ is non-empty and compact, and the objective is continuous on it with the convention $0 \log 0 = 0$, so a minimizer exists. On the relative interior of the polytope,

$$-\partial^2 H(P) = \text{diag}(1/P_{i,j})$$

is positive definite on every non-zero feasible direction. Hence $-H$ is strictly convex on the polytope and $\langle P, C \rangle - \varepsilon H(P)$ is strictly convex, which implies uniqueness.

If $a_i, b_j > 0$ and a minimizer had $P_{i,j} = 0$, then for small $t > 0$ the perturbation $P_t = (1 - t)P + t a \otimes b$ remains feasible. The directional derivative of the entropic part at $t = 0$ is $-\infty$ because the derivative of $r \log r$ at 0 is $-\infty$ along a positive direction. Thus the objective decreases for small t , contradicting optimality. Therefore the minimizer is strictly positive. $\square$

Smoothing effect. By Proposition 26, problem (29) has a unique optimal solution. This smoothing, beyond providing uniqueness, actually leads to $L_C^\varepsilon(a, b)$ being a smooth function of a, b and C whenever these variables stay in the relative interior of their domains. In finite dimension, this follows from strict convexity and the envelope theorem applied to the dual problem. The effect of the entropy is to act as a barrier function for the positivity constraint. As we will show later, this forces the solution P to be strictly positive on the support of $a \otimes b$. We will also show that as $\varepsilon \rightarrow +\infty$, the solution satisfies $P \rightarrow a \otimes b$.

4.2 Sinkhorn's Algorithm

Sinkhorn's algorithm is alternating normalization of rows and columns. This subsection derives the scaling form of the optimizer and explains why each iteration only needs matrix-vector products.

The underlying matrix-scaling iteration has a long history, including iterative proportional fitting and the work of Sinkhorn and Knopp [197, 199, 198]. Its modern role in OT was transformed by Cuturi's entropic formulation [64]: the algorithm became a practical large-scale tool for machine learning, and also changed the way OT is viewed in ML, from a mostly geometric distance to a differentiable computational primitive.

The following proposition shows that the solution of (29) has a specific form, which can be parameterized using $n + m$ variables. That parameterization is therefore essentially dual, in the sense that a coupling P in $U(a, b)$ has nm variables but $n + m$ constraints.

Proposition 27 (Scaling form of entropic OT). *P is the unique solution to (29) if and only if there exists $(u, v) \in \mathbb{R}_+^n \times \mathbb{R}_+^m$ such that*

$$\forall (i, j) \in \llbracket n \rrbracket \times \llbracket m \rrbracket, \quad P_{i,j} = u_i K_{i,j} v_j \quad \text{where} \quad K_{i,j} := e^{-\frac{C_{i,j}}{\varepsilon}}, \quad (30)$$

and $P \in \mathcal{U}(a, b)$.

Proof. Without loss of generality, we assume $a_i, b_j > 0$ (otherwise, rows or columns with zero mass are fixed to zero and can be removed). By Proposition 26, the minimizer is strictly positive on the remaining support.

We can thus ignore the positivity constraint when introducing two dual variables $f \in \mathbb{R}^n, g \in \mathbb{R}^m$ for each marginal constraint so that the Lagrangian of (29) reads

$$\mathcal{E}(P, f, g) = \langle P, C \rangle + \varepsilon \sum_{i,j} P_{i,j} \log(P_{i,j}) + \langle f, a - P \mathbf{1}_m \rangle + \langle g, b - P^\top \mathbf{1}_n \rangle.$$

Considering first-order conditions (where we ignore the positivity constraint as explained above), we have

$$\frac{\partial \mathcal{E}(P, f, g)}{\partial P_{i,j}} = C_{i,j} + \varepsilon(\log(P_{i,j}) + 1) - f_i - g_j = 0.$$

which results, in an optimal P coupling of the regularized problem, in the expression $P_{i,j} = e^{\frac{f_i + g_j - C_{i,j}}{\varepsilon} - 1}$ which can be rewritten in the form provided in the proposition using non-negative vectors $u_i := e^{f_i/\varepsilon - 1}$ and $v_j := e^{g_j/\varepsilon}$. $\square$

The factorization of the optimal solution exhibited in Equation (30) can be conveniently rewritten in matrix form as $P = \text{diag}(u)K \text{diag}(v)$. u, v must therefore satisfy the following nonlinear equations which correspond to the mass conservation constraints inherent to $U(a, b)$,

$$\text{diag}(u)K \text{diag}(v)\mathbf{1}_m = a, \quad \text{and} \quad \text{diag}(v)K^\top \text{diag}(u)\mathbf{1}_n = b, \quad (31)$$

These two equations can be further simplified, since $\text{diag}(v)\mathbf{1}_m$ is v , and the multiplication of $\text{diag}(u)$ times Kv is

$$u \odot (Kv) = a \quad \text{and} \quad v \odot (K^\top u) = b \quad (32)$$

where $\odot$ corresponds to the entry-wise multiplication of vectors. This problem is known in the numerical analysis community as the matrix scaling problem (see [160] and references therein).

The problem of normalizing a positive matrix K has a long history, from iterative proportional fitting in statistics and economics [129, 219, 88] to modern matrix balancing algorithms [119, 63]. The problem of normalizing a positive matrix K by diagonal scaling is well known, in particular when $n = m$ and a and b are uniform. This corresponds to diagonal scaling toward bistochasticity, which is a very old problem. The previous result shows that there is a unique such scaled matrix P , thanks to the strong convexity of the regularized problem. The remaining question is how to compute this scaled matrix in practice. If some entries of K vanish (equivalently, if the cost matrix C can have infinite values), additional support conditions are needed; here we focus on the strictly positive case.

An intuitive way to try to solve these equations is to solve them iteratively, by modifying first u so that it satisfies the left-hand side of Equation (32) and then v to satisfy its right-hand side. These two updates define Sinkhorn's algorithm

$$u^{(\ell+1)} := \frac{a}{Kv^{(\ell)}} \quad \text{and} \quad v^{(\ell+1)} := \frac{b}{K^\top u^{(\ell+1)}}, \quad (33)$$

initialized with an arbitrary positive vector, for instance $v^{(0)} = \mathbf{1}_m$. The division operator used above between two vectors is to be understood entry-wise. Note that a different initialization will likely lead to a different solution for u, v , since u, v are only defined up to a multiplicative constant (if u, v satisfy (31) then so do $\lambda u, v/\lambda$ for any $\lambda > 0$). The next paragraphs explain why these iterations converge.

Complexity bounds for Sinkhorn and comparisons with accelerated first-order methods are discussed in [4, 75, 126]. A chief computational advantage of Sinkhorn's algorithm, besides its simplicity, is that the only expensive step is multiplication by the Gibbs kernel. Its complexity therefore scales like Cnm , where C is the number of Sinkhorn iterations. Section 8.3 gives a more precise convergence statement: for a fixed regularization strength ε , the entropic dual gap has an $O(1/k)$ bound with constants proportional to $1/\varepsilon$, while Hilbert-metric arguments give eventual linear convergence when the Gibbs kernel is uniformly positive. To approximate the unregularized OT value to accuracy δ , one must also balance the entropic bias, which is typically $O(\varepsilon)$ in finite dimension. Choosing ε proportional to δ and solving the entropic problem to accuracy $O(\delta)$ leads to the familiar iteration scaling of order $1/\delta^2$ for the unregularized value, up to logarithmic and cost-range factors [4, 75].

In many applications, however, one does not need a highly accurate optimizer for the OT subproblem. Downstream performance is often tied more to the geometric bias of the discrepancy than to exact optimality. In such settings, C is often moderate. This should be contrasted with interior point methods, which also operate by introducing a barrier function of the form $-\sum_i \log(P_{i,j})$. For a dense transportation linear program, these algorithms typically have worst-case complexity of order $O(n^6 \log(1/\delta))$ to reach accuracy δ , up to problem-dependent conditioning factors.

The second crucial aspect of Sinkhorn is that matrix-vector multiplication streams extremely well on GPU. Even better, if one is interested in computing many OT problems with a fixed cost matrix C , one can replace many matrix-vector multiplications with matrix-matrix multiplications, so that the computational gain can be substantial.

Remark 13 (Separable Gaussian kernels on grids). When the samples lie on a Cartesian grid and $c(x, y) = \|x - y\|^2$, the Gibbs kernel is Gaussian and factorizes along coordinates. If the grid has q points per axis in

dimension d , so that $N = q^d$ grid points are used, then

$$K(x, y) = \exp\left(-\frac{\|x - y\|^2}{\varepsilon}\right) = \prod_{\ell=1}^d \exp\left(-\frac{(x_\ell - y_\ell)^2}{\varepsilon}\right).$$

Multiplication by K can therefore be applied by successively multiplying along each coordinate direction, equivalently by applying one-dimensional Gaussian kernel operators along the axes. On a periodic or sufficiently padded uniform grid these are literal discrete convolutions. A direct dense one-dimensional multiplication costs $O(q^2)$ on each of the q^{d-1} coordinate lines, and this is repeated for d axes. Hence one Sinkhorn half-step costs

$$O(d q^{d+1}) = O(d N^{1+1/d})$$

instead of $O(N^2)$. With FFT-based or truncated Gaussian convolutions, the same separability can be pushed further, but the simple tensor-product estimate already explains why grid-based Sinkhorn can scale much better than a generic dense coupling.

4.3 Reformulation using relative entropy

The KL formulation identifies Sinkhorn as a projection method. It also prepares the continuous and unbalanced settings, where a reference measure is essential.

A convenient tool to reformulate and “normalize” this discrete entropy (which is crucial to formulate a continuous version of the problem below) is to consider the relative entropy, also called the Kullback–Leibler divergence, which is defined as

$$\text{KL}(\mathbf{P}|\mathbf{Q}) := \sum_{i,j} \mathbf{P}_{i,j} \log\left(\frac{\mathbf{P}_{i,j}}{\mathbf{Q}_{i,j}}\right) - \mathbf{P}_{i,j} + \mathbf{Q}_{i,j}. \quad (34)$$

with the convention $0 \log(0) = 0$ and $\text{KL}(\mathbf{P}|\mathbf{Q}) = +\infty$ if there exists some (i, j) such that $\mathbf{Q}_{i,j} = 0$ but $\mathbf{P}_{i,j} \neq 0$. For the specific case of comparing probability distributions, this further simplifies to

$$\text{KL}(\mathbf{P}|\mathbf{Q}) = \sum_{i,j} \mathbf{P}_{i,j} \log\left(\frac{\mathbf{P}_{i,j}}{\mathbf{Q}_{i,j}}\right).$$

The Shannon-Boltzmann neg-entropy is obtained when considering $\mathbf{Q} = \mathbf{1}_{n \times m}$, i.e.

$$\mathbf{H}(\mathbf{P}) = -\text{KL}(\mathbf{P}|\mathbf{1}_{n \times m}).$$

KL is a particular instance of both a φ -divergence (as defined in Section 7.3) and a Bregman divergence; up to standard affine rescalings, it is the canonical overlap between these two families. This special property is at the heart of the fact that this regularization leads to elegant algorithms and a tractable mathematical analysis.

Proposition 28 (Relative entropy is distance-like). *Let $\mathbf{P}, \mathbf{Q} \in \mathbb{R}_+^{n \times m}$ have the same total mass and assume $\mathbf{Q}_{i,j} > 0$ on the support of $\mathbf{P}$. Then $\text{KL}(\mathbf{P}|\mathbf{Q}) \geq 0$, with equality if and only if $\mathbf{P} = \mathbf{Q}$.*

Proof. Write $\varphi(s) = s \log s - s + 1$. Convexity gives $\varphi(s) \geq \varphi(1) + \varphi'(1)(s - 1) = 0$, and strict convexity gives equality only at $s = 1$. Hence

$$\text{KL}(\mathbf{P}|\mathbf{Q}) = \sum_{i,j} \mathbf{Q}_{i,j} \varphi(\mathbf{P}_{i,j}/\mathbf{Q}_{i,j}) \geq 0,$$

with equality only when $\mathbf{P}_{i,j}/\mathbf{Q}_{i,j} = 1$ for all entries with $\mathbf{Q}_{i,j} > 0$. The support convention rules out positive $\mathbf{P}$ where $\mathbf{Q} = 0$, so equality is equivalent to $\mathbf{P} = \mathbf{Q}$. $\square$

Equivalently, when P and Q have the same total mass, it reads

$$\text{KL}(P|Q) = \sum_{i,j} \varphi(P_{i,j}/Q_{i,j}) Q_{i,j}.$$

where $\varphi(s) = s \log(s)$. For any convex φ such that $\varphi(1) = 0$, one has indeed by Jensen

$$\sum_{i,j} \varphi(P_{i,j}/Q_{i,j}) Q_{i,j} \geq \varphi\left(\sum_{i,j} P_{i,j}/Q_{i,j} Q_{i,j}\right) = \varphi\left(\sum_{i,j} P_{i,j}\right) = \varphi(1) = 0.$$

For instance, one can use as reference measure the tensor product $a \otimes b = (a_i b_j)_{i,j}$ and consider

$$\min_{P \in U(a,b)} \langle P, C \rangle + \varepsilon \text{KL}(P|a \otimes b). \quad (35)$$

Note also that for later, it will be important to have this normalization when considering unbalanced OT.

But the choice of normalization (i.e. reference measure) has no importance for the selection of the optimal P since it only affects the objective by a constant, as shown in the following proposition. In particular, (35) and (29) have the same unique solution.

Proposition 29 (Reference measure shift for KL). *For $P \in U(a, b)$, one has*

$$\text{KL}(P|a \otimes b) = \text{KL}(P|a' \otimes b') - \text{KL}(a|a') - \text{KL}(b|b').$$

Consequently, changing the tensor-product reference measure only adds a constant on the transport polytope. In particular, (35) and (29) have the same unique minimizer.

Proof. This follows from

$$\sum_{i,j} P_{i,j} \log \frac{P_{i,j}}{a_i b_j} = \sum_{i,j} P_{i,j} \log \frac{P_{i,j}}{a'_i b'_j} + \sum_{i,j} P_{i,j} \left(\log \frac{a'_i}{a_i} + \log \frac{b'_j}{b_j} \right).$$

□

The choice of using the reference measure $a \otimes b$ is important to deal with situations where the support of a and b can change (so that some coordinate of a or b might vanish), and more importantly in the following section which deals with possibly continuous distributions.

One has the following convergence property.

Proposition 30 (Convergence with ε). *Assume, after removing zero-mass rows and columns, that a and b are positive and that C is finite. The unique solution P_ε of (29) converges to the optimal solution with maximal entropy within the set of all optimal solutions of the Kantorovich problem, namely*

$$P_\varepsilon \xrightarrow[\mathcal{P}]{\varepsilon \rightarrow 0} \underset{\mathcal{P}}{\text{argmin}} \{ -H(P) ; P \in U(a, b), \langle P, C \rangle = L_C(a, b) \} \quad (36)$$

so that in particular

$$L_C^\varepsilon(a, b) \xrightarrow{\varepsilon \rightarrow 0} L_C(a, b).$$

One has

$$P_\varepsilon \xrightarrow{\varepsilon \rightarrow \infty} a \otimes b. \quad (37)$$

Proof. Case $\varepsilon \rightarrow 0$. We consider a sequence $(\varepsilon_\ell)_\ell$ such that $\varepsilon_\ell \rightarrow 0$ and $\varepsilon_\ell > 0$. We denote P_ℓ the solution of (29) for $\varepsilon = \varepsilon_\ell$. Since $U(a, b)$ is bounded, we can extract a sequence (that we do not relabel for the sake of simplicity) such that $P_\ell \rightarrow P^*$. Since $U(a, b)$ is closed, $P^* \in U(a, b)$. We consider any P such that

$\langle C, P \rangle = L_C(a, b)$. Using the equivalent KL-normalized formulation (35), optimality of P and P_ℓ for their respective optimization problems (for $\varepsilon = 0$ and $\varepsilon = \varepsilon_\ell$) gives

$$0 \leq \langle C, P_\ell \rangle - \langle C, P \rangle \leq \varepsilon_\ell (\text{KL}(P|a \otimes b) - \text{KL}(P_\ell|a \otimes b)). \quad (38)$$

Since KL is continuous, taking the limit $\ell \rightarrow +\infty$ in this expression shows that $\langle C, P^* \rangle = \langle C, P \rangle$ so that P^* is a feasible point of (36). Furthermore, dividing by ε_ℓ in (38) and taking the limit shows that $\text{KL}(P^*|a \otimes b) \leq \text{KL}(P|a \otimes b)$, which shows that P^* is a solution of (36). Since the solution P_0^* to this program is unique by strict convexity of $\text{KL}(\cdot|a \otimes b)$ on the optimal face, one has $P^* = P_0^*$, and the whole sequence is converging.

Case $\varepsilon \rightarrow +\infty$. Evaluating at $a \otimes b$ (which belongs to the constraint set $U(a, b)$) the energy, one has

$$\langle C, P_\varepsilon \rangle + \varepsilon \text{KL}(P_\varepsilon|a \otimes b) \leq \langle C, a \otimes b \rangle + \varepsilon \times 0$$

and since $\langle C, P_\varepsilon \rangle \geq 0$, this leads to

$$\text{KL}(P_\varepsilon|a \otimes b) \leq \varepsilon^{-1} \langle C, a \otimes b \rangle \leq \frac{\|C\|_\infty}{\varepsilon}$$

so that $\text{KL}(P_\varepsilon|a \otimes b) \rightarrow 0$ and thus $P_\varepsilon \rightarrow a \otimes b$ since KL is a valid divergence. $\square$

4.4 General Formulation

The continuous entropic problem is the static Schrödinger problem. It gives a probabilistic interpretation of Sinkhorn as the most likely coupling under a noisy reference dynamics.

One can consider arbitrary measures by replacing the discrete entropy with the relative entropy with respect to the product measure $d\alpha \otimes d\beta(x, y) := d\alpha(x)d\beta(y)$, and propose a regularized counterpart to (22) using

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) := \min_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) + \varepsilon \text{KL}(\pi|\alpha \otimes \beta) \quad (39)$$

where the relative entropy is a generalization of the discrete Kullback-Leibler divergence (34)

$$\text{KL}(\pi|\xi) := \int_{\mathcal{X} \times \mathcal{Y}} \log \left(\frac{d\pi}{d\xi}(x, y) \right) d\pi(x, y) + \int_{\mathcal{X} \times \mathcal{Y}} (d\xi(x, y) - d\pi(x, y)), \quad (40)$$

and by convention $\text{KL}(\pi|\xi) = +\infty$ if π does not have a density $\frac{d\pi}{d\xi}$ with respect to ξ . It is important to realize that the reference measure $\alpha \otimes \beta$ chosen in (39) to define the entropic regularizing term $\text{KL}(\cdot|\alpha \otimes \beta)$ plays no specific role (because Proposition 29 still applies in this general setting), only its support matters. This problem is often referred to as the “static Schrödinger problem”, since π is intended to model the most likely coupling between gas particles observed only at two different times (the so-called lazy gas model). The parameter ε controls the temperature of the gas, and particles do not move in a deterministic straight line as in optimal transport for the Euclidean cost, but rather according to a stochastic Brownian bridge. This viewpoint goes back to Schrödinger’s reciprocal problem [194] and is surveyed in modern OT language in [136, 137]; stochastic-control formulations are developed in [55].

Remark 14 (Probabilistic interpretation). If $(X, Y) \sim \pi$ have marginals $X \sim \alpha$ and $Y \sim \beta$, then $\text{KL}(\pi|\alpha \otimes \beta) = \mathcal{I}(X, Y)$ is the mutual information of the couple, which is 0 if and only if X and Y are independent. The entropic problem (39) is thus equivalent to

$$\min_{(X, Y), X \sim \alpha, Y \sim \beta} \mathbb{E}(c(X, Y)) + \varepsilon \mathcal{I}(X, Y).$$

Using a large ε thus enforces the optimal coupling to describe independent variables, while, according to Brenier’s theorem, small ε rather imposes a deterministic dependency between the couple according to a Monge map.

4.5 Bregman Projection View of Sinkhorn

This subsection explains Sinkhorn as alternating Bregman projections. The main message is geometric: each row or column rescaling is the KL projection onto one affine marginal constraint, so convergence follows from the Pythagorean identity for Bregman divergences.

For the sake of simplicity, this section is written for discrete measures, but the analysis carries over to general measures. This analysis is revisited in Section 8.3 using convex duality.

Alternating KL projections. The projection viewpoint explains Sinkhorn as repeated enforcement of one marginal constraint at a time. It is not specific to the entropy, although the KL case is the one where the projections reduce to elementary row and column scalings.

Definition 10 (Bregman divergence). *Let Φ be a differentiable strictly convex function on a convex domain Ω . The Bregman divergence generated by Φ is*

$$D_\Phi(P|Q) := \Phi(P) - \Phi(Q) - \langle \nabla \Phi(Q), P - Q \rangle.$$

For the negative entropy $\Phi(P) = \sum_{i,j} P_{i,j} \log P_{i,j}$ on the positive orthant, one obtains $D_\Phi(P|Q) = \text{KL}(P|Q)$ up to the harmless convention at the boundary.

The main interest of Bregman divergences is that the geometry can encode constraints. A Legendre-type generator Φ blows up, or has an infinite derivative, at the boundary of its domain; for negative entropy this means that positivity is built into the divergence, so one projects onto affine marginal constraints without separately handling non-negativity.

The following proposition explains why adding a linear cost to a Bregman penalty merely shifts the reference point in dual coordinates. The usual Gibbs–KL reformulation is the entropy specialization.

Proposition 31 (Linear tilts of Bregman penalties). *Let Φ be differentiable and strictly convex, and let D_Φ be its Bregman divergence. Fix a reference point Q in the interior of the domain. Assume that there exists Q^C such that*

$$\nabla \Phi(Q^C) = \nabla \Phi(Q) - C/\varepsilon.$$

Then, for all P in the domain,

$$\langle P, C \rangle + \varepsilon D_\Phi(P|Q) = \varepsilon D_\Phi(P|Q^C) + \text{cst},$$

where the constant does not depend on P .

Proof. Subtract the two Bregman divergences:

$$D_\Phi(P|Q^C) - D_\Phi(P|Q) = \langle \nabla \Phi(Q) - \nabla \Phi(Q^C), P \rangle + \text{cst}.$$

Using $\nabla \Phi(Q) - \nabla \Phi(Q^C) = C/\varepsilon$ and multiplying by ε gives the claim. $\square$

For the negative entropy $\Phi(P) = \sum_{i,j} P_{i,j} \log P_{i,j}$, one has $D_\Phi = \text{KL}$. Taking $Q = a \otimes b$ gives the tilted reference

$$K_{a,b}^\varepsilon := (a \otimes b) \odot e^{-C/\varepsilon}.$$

Thus

$$\langle P, C \rangle + \varepsilon \text{KL}(P|a \otimes b) = \varepsilon \text{KL}(P|K_{a,b}^\varepsilon) + \text{cst}.$$

On the transport polytope, scaling $K_{a,b}^\varepsilon$ is equivalent to scaling the Gibbs kernel $K = e^{-C/\varepsilon}$ because the factors a_i and b_j can be absorbed into the Sinkhorn scalings.

This shows that the unique solution P_ε of (29) is a projection onto $U(a, b)$ of the tilted Gibbs reference

$$P_\varepsilon = \text{Proj}_{U(a,b)}^{\text{KL}}(K_{a,b}^\varepsilon) := \underset{P \in U(a,b)}{\text{argmin}} \text{KL}(P|K_{a,b}^\varepsilon). \quad (41)$$

The general convergence mechanism is the classical one of Bregman [38].

Proposition 32 (Cyclic Bregman projections on affine constraints). *Let Φ be a Legendre strictly convex generator on a finite-dimensional convex domain, and let $\mathcal{C}_1, \mathcal{C}_2$ be affine constraint sets whose intersection meets the domain. Define*

$$P_{k+1} = \text{Proj}_{\mathcal{C}_2}^{D_\Phi} \text{Proj}_{\mathcal{C}_1}^{D_\Phi}(P_k),$$

starting from an interior point P_0 , and assume the projections are well-defined. Then P_k converges to the Bregman projection of P_0 onto $\mathcal{C}_1 \cap \mathcal{C}_2$. In particular, the KL case converges for positive affine marginal constraints.

Proof. For an affine set $\mathcal{C}$, the Bregman projection $P^+ = \text{Proj}_{\mathcal{C}}^{D_\Phi}(P)$ satisfies the Pythagorean identity

$$D_\Phi(Q|P) = D_\Phi(Q|P^+) + D_\Phi(P^+|P) \quad \forall Q \in \mathcal{C}.$$

This follows by writing the first-order optimality condition: $\nabla\Phi(P^+) - \nabla\Phi(P)$ is orthogonal to the tangent space of $\mathcal{C}$, so the usual three-point identity for Bregman divergences has no cross term. Applying this identity successively with any $Q \in \mathcal{C}_1 \cap \mathcal{C}_2$ shows that $D_\Phi(Q|P_k)$ decreases and that the projection drops are summable. The affine residuals therefore vanish along the cycle, and every cluster point lies in both constraint sets. The Pythagorean identity then identifies any cluster point as the unique Bregman projection onto $\mathcal{C}_1 \cap \mathcal{C}_2$, by strict convexity. Thus the whole sequence converges. The KL statement is obtained by choosing the negative entropy generator. $\square$

Denoting

$$\mathcal{C}_a^1 := \{P ; P\mathbf{1}_m = \mathbf{a}\} \quad \text{and} \quad \mathcal{C}_b^2 := \{P ; P^\top \mathbf{1}_n = \mathbf{b}\}$$

the rows and columns constraints, one has $U(\mathbf{a}, \mathbf{b}) = \mathcal{C}_a^1 \cap \mathcal{C}_b^2$. One can use KL, or more generally Bregman, iterative projections [38, 185, 186]

$$P^{(\ell+1)} := \text{Proj}_{\mathcal{C}_a^1}^{\text{KL}}(P^{(\ell)}) \quad \text{and} \quad P^{(\ell+2)} := \text{Proj}_{\mathcal{C}_b^2}^{\text{KL}}(P^{(\ell+1)}). \quad (42)$$

Since the sets $\mathcal{C}_a^1$ and $\mathcal{C}_b^2$ are affine, Proposition 32 applies with $P_0 = K_{a,b}^\varepsilon$ and shows convergence to the solution of (41).

The two projectors are simple to compute since they correspond to scaling respectively the rows and the columns, as explained in this proposition.

Proposition 33 (KL projections are scalings). *One has*

$$\text{Proj}_{\mathcal{C}_a^1}^{\text{KL}}(P) = \text{diag}\left(\frac{\mathbf{a}}{P\mathbf{1}_m}\right)P \quad \text{and} \quad \text{Proj}_{\mathcal{C}_b^2}^{\text{KL}}(P) = P \text{diag}\left(\frac{\mathbf{b}}{P^\top \mathbf{1}_n}\right).$$

Proof. Consider the problem along each row or column vector to impose a fixed sum $s \in \mathbb{R}_+$

$$\min_p \{ \text{KL}(p|q) ; \langle p, \mathbf{1} \rangle = s \}.$$

The Lagrange multiplier equation for this problem reads

$$\log(p/q) + \lambda \mathbf{1} = 0 \implies p = uq \quad \text{where} \quad u = e^{-\lambda} > 0.$$

The constraint $\langle p, \mathbf{1} \rangle = s$ is equivalent to $\langle uq, \mathbf{1} \rangle = s$, i.e. $u = s / \sum_i q_i$, which gives the desired scaling formula $p = sq / \sum_i q_i$. $\square$

These iterations are equivalent to Sinkhorn iterations (33) since defining

$$P^{(2\ell)} := \text{diag}(u^{(\ell)})K \text{diag}(v^{(\ell)}),$$

one has

$$\begin{aligned} P^{(2\ell+1)} &:= \text{diag}(u^{(\ell+1)})K \text{diag}(v^{(\ell)}) \\ \text{and } P^{(2\ell+2)} &:= \text{diag}(u^{(\ell+1)})K \text{diag}(v^{(\ell+1)}) \end{aligned}$$

In practice, however, one should prefer using (33), which only requires manipulating scaling vectors and multiplying by a Gibbs kernel, and can often be accelerated when the kernel has separable, sparse, low-rank or geometric structure.

Such a convergence analysis using Bregman projection is of limited interest because it only works in finite dimension. For instance, the linear convergence speed one can obtain with these analyses (because the objective is strongly convex) degrades with the dimension and with ε . Section 8.3 gives a more robust dual analysis: the constants are expressed through the oscillation of the potentials and the cost range, and the resulting $O(1/k)$ estimate explains what can be guaranteed before any asymptotic linear regime becomes visible. It is also possible to decay ε during the iterates to improve the speed and rely on multiscale strategies in low dimensions.

4.6 Hilbert Metric View of Sinkhorn

Hilbert's projective metric gives a complementary convergence mechanism. Instead of following objective values, it measures distances between positive scaling vectors modulo global multiplication. Positive kernels are contractions in this geometry, yielding a global linear convergence statement.

As initially explained by [85], the global convergence analysis of Sinkhorn is greatly simplified using Hilbert projective metric on $\mathbb{R}_{+,*}^n$ (positive vectors), defined as

$$\forall (u, u') \in (\mathbb{R}_{+,*}^n)^2, \quad d_{\mathcal{H}}(u, u') := \|\log(u) - \log(u')\|_V \quad (43)$$

where the variation semi-norm is

$$\|z\|_V = \max(z) - \min(z).$$

Proposition 34 (Hilbert metric on the projective cone). *The function $d_{\mathcal{H}}$ defines a complete distance on the projective cone $\mathbb{R}_{+,*}^n / \sim$, where $u \sim u'$ means that $u = su'$ for some $s > 0$.*

Proof. The map $u \mapsto \log u$ identifies $\mathbb{R}_{+,*}^n / \sim$ with the quotient vector space $\mathbb{R}^n / \text{Span}(\mathbf{1}_n)$, because multiplying u by $s > 0$ adds the constant vector $\log(s)\mathbf{1}_n$. The variation semi-norm $\|z\|_V = \max_i z_i - \min_i z_i$ vanishes exactly on constant vectors, so it induces a norm on this quotient. Symmetry, the triangle inequality and separation therefore follow from the corresponding norm properties. Completeness follows because $\mathbb{R}^n / \text{Span}(\mathbf{1}_n)$ is finite-dimensional and all finite-dimensional normed spaces are complete. $\square$

It was introduced independently by [29] and [187] to provide quantitative proofs of the Perron-Frobenius theorem (convergence of iterations of positive matrices). Sinkhorn should be viewed as a nonlinear generalization of Perron-Frobenius.

Theorem 4 (Birkhoff contraction theorem). *Let $K \in \mathbb{R}_{+,*}^{n \times m}$, then for $(v, v') \in (\mathbb{R}_{+,*}^m)^2$*

$$d_{\mathcal{H}}(Kv, Kv') \leq \lambda(K) d_{\mathcal{H}}(v, v') \text{ where } \begin{cases} \lambda(K) := \frac{\sqrt{\eta(K)} - 1}{\sqrt{\eta(K)} + 1} < 1 \\ \eta(K) := \max_{i,j,k,\ell} \frac{K_{i,k} K_{j,\ell}}{K_{j,k} K_{i,\ell}}. \end{cases}$$

Proof. We recall the finite-dimensional Birkhoff–Hopf estimate. For a positive linear map A on a cone, define its projective diameter

$$\Delta(A) := \sup_{u,v>0} d_{\mathcal{H}}(Au, Av).$$

Then

$$d_{\mathcal{H}}(Au, Av) \leq \tanh(\Delta(A)/4) d_{\mathcal{H}}(u, v).$$

Indeed, after quotienting by positive scalings, write $r_k = u_k/v_k$ and normalize so that $e^{-h/2} \leq r_k \leq e^{h/2}$, where $h = d_{\mathcal{H}}(u, v)$. The ratio between two coordinates of Au/Av is a quotient of two weighted averages of the numbers r_k . A two-point extremal argument shows that the largest possible contraction is obtained

when the mass of the two weights is placed on the two endpoints $e^{-h/2}$ and $e^{h/2}$; the cross-ratio bound defining $\Delta(A)$ then gives

$$d_{\mathcal{H}}(Au, Av) \leq 2 \log \frac{e^{\Delta(A)/4} e^{h/2} + e^{-\Delta(A)/4} e^{-h/2}}{e^{\Delta(A)/4} e^{-h/2} + e^{-\Delta(A)/4} e^{h/2}} \leq \tanh(\Delta(A)/4)h.$$

For the matrix K , its projective diameter is

$$\Delta(K) = \log \eta(K), \quad \eta(K) = \max_{i,j,k,\ell} \frac{K_{i,k} K_{j,\ell}}{K_{j,k} K_{i,\ell}}.$$

Therefore $\tanh(\Delta(K)/4) = (\sqrt{\eta(K)} - 1)/(\sqrt{\eta(K)} + 1)$, which is the claimed contraction factor. $\square$

This result extends to arbitrary convex cones and affine mappings from the cone to its interior.

The following theorem proved by [85], makes use of this Theorem 4 to show the linear convergence of Sinkhorn's iterations.

Theorem 5 (Linear convergence of Sinkhorn). *One has $(u^{(\ell)}, v^{(\ell)}) \rightarrow (u^*, v^*)$ and*

$$d_{\mathcal{H}}(u^{(\ell)}, u^*) = O(\lambda(K)^{2\ell}), \quad d_{\mathcal{H}}(v^{(\ell)}, v^*) = O(\lambda(K)^{2\ell}). \quad (44)$$

One also has

$$d_{\mathcal{H}}(u^{(\ell)}, u^*) \leq \frac{d_{\mathcal{H}}(P^{(\ell)} \mathbf{1}_m, \mathbf{a})}{1 - \lambda(K)} \quad \text{and} \quad d_{\mathcal{H}}(v^{(\ell)}, v^*) \leq \frac{d_{\mathcal{H}}(P^{(\ell), \top} \mathbf{1}_n, \mathbf{b})}{1 - \lambda(K)}, \quad (45)$$

where we denoted $P^{(\ell)} := \text{diag}(u^{(\ell)})K \text{diag}(v^{(\ell)})$. Lastly, one has

$$\|\log(P^{(\ell)}) - \log(P^*)\|_{\infty} \leq d_{\mathcal{H}}(u^{(\ell)}, u^*) + d_{\mathcal{H}}(v^{(\ell)}, v^*) \quad (46)$$

where P^ is the unique solution of (29).*

Proof. Notice that for any $(v, v') \in (\mathbb{R}_{+,*}^m)^2$, one has

$$d_{\mathcal{H}}(v, v') = d_{\mathcal{H}}(v/v', \mathbf{1}_m) = d_{\mathcal{H}}(\mathbf{1}_m/v, \mathbf{1}_m/v'),$$

since indeed $d_{\mathcal{H}}(a/v, a/v') = d_{\mathcal{H}}(v, v')$. This shows that

$$d_{\mathcal{H}}(u^{(\ell+1)}, u^*) = d_{\mathcal{H}}\left(\frac{\mathbf{a}}{Kv^{(\ell)}}, \frac{\mathbf{a}}{Kv^*}\right) = d_{\mathcal{H}}(Kv^{(\ell)}, Kv^*) \leq \lambda(K) d_{\mathcal{H}}(v^{(\ell)}, v^*).$$

where we used Theorem 4. This shows (44). One also has, using the triangular inequality

$$\begin{aligned} d_{\mathcal{H}}(u^{(\ell)}, u^*) &\leq d_{\mathcal{H}}(u^{(\ell+1)}, u^{(\ell)}) + d_{\mathcal{H}}(u^{(\ell+1)}, u^*) \leq d_{\mathcal{H}}\left(\frac{\mathbf{a}}{Kv^{(\ell)}}, u^{(\ell)}\right) + \lambda(K) d_{\mathcal{H}}(u^{(\ell)}, u^*) \\ &= d_{\mathcal{H}}\left(\mathbf{a}, u^{(\ell)} \odot (Kv^{(\ell)})\right) + \lambda(K) d_{\mathcal{H}}(u^{(\ell)}, u^*), \end{aligned}$$

which gives the first part of (45) since $u^{(\ell)} \odot (Kv^{(\ell)}) = P^{(\ell)} \mathbf{1}_m$ (the second one being similar). The proof of (46) follows from [85, Lemma 3] $\square$

The bound (45) shows that some error measures on the marginal constraints violation, for instance, $\|P^{(\ell)} \mathbf{1}_m - \mathbf{a}\|_1$ and $\|P^{(\ell), \top} \mathbf{1}_n - \mathbf{b}\|_1$, are useful stopping criteria to monitor the convergence. This theorem shows that the Sinkhorn algorithm converges linearly, but the worst-case rate becomes exponentially bad as $\varepsilon \rightarrow 0$, since the global contraction factor is controlled by a term of the form $e^{-\|c\|_{\infty}/\varepsilon}$. In practice, one often observes a much better local linear regime after enough iterations. The same Hilbert-metric mechanism extends beyond finite matrices to positive integral operators under suitable compactness and positivity assumptions. An important limitation of this analysis is that it requires a uniformly bounded cost and a kernel bounded away from degeneracy; Gaussian distributions with quadratic cost therefore require a different approach.

5 Dual Problem

Duality turns the transport problem into a search for potentials rather than couplings. This section explains why potentials certify optimality, how c -transforms regularize them, and why the quadratic case reveals convex analysis behind Brenier maps. Linear-programming duality gives the discrete picture [26], while the continuous form is one of the central theorems of OT [211, 190].

5.1 Discrete dual

The discrete dual gives finite-dimensional certificates of optimality. Its complementary slackness conditions identify where an optimal coupling can put mass.

The Kantorovich problem (19) is a linear program so that one can equivalently compute its value by solving a dual linear program.

Proposition 35 (Discrete Kantorovich duality). *One has*

$$L_C(a, b) = \max_{(f, g) \in R(a, b)} \langle f, a \rangle + \langle g, b \rangle \quad (47)$$

where the set of admissible potentials is

$$R(a, b) := \{(f, g) \in \mathbb{R}^n \times \mathbb{R}^m ; \forall (i, j) \in \llbracket n \rrbracket \times \llbracket m \rrbracket, f \oplus g \leq C\} \quad (48)$$

Proof. For the sake of completeness, let us derive this dual problem using Lagrangian duality. The Lagrangian associated to (19) reads

$$\min_{P \geq 0} \max_{(f, g) \in \mathbb{R}^n \times \mathbb{R}^m} \langle C, P \rangle + \langle a - P \mathbf{1}_m, f \rangle + \langle b - P^\top \mathbf{1}_n, g \rangle. \quad (49)$$

For a linear program, if the primal constraint set is non-empty, one can always exchange the min and the max and get the same value. We thus consider

$$\max_{(f, g) \in \mathbb{R}^n \times \mathbb{R}^m} \langle a, f \rangle + \langle b, g \rangle + \min_{P \geq 0} \langle C - f \mathbf{1}_m^\top - \mathbf{1}_n g^\top, P \rangle.$$

We conclude by remarking that

$$\min_{P \geq 0} \langle Q, P \rangle = \begin{cases} 0 & \text{if } Q \geq 0 \\ -\infty & \text{otherwise} \end{cases}$$

so that the constraint reads $C - f \mathbf{1}_m^\top - \mathbf{1}_n g^\top = C - f \oplus g \geq 0$. $\square$

The primal-dual optimality relation for the Lagrangian (49) allows locating the support of the optimal transport plan

$$\text{Supp}(P) \subset \{(i, j) \in \llbracket n \rrbracket \times \llbracket m \rrbracket ; f_i + g_j = C_{i,j}\}. \quad (50)$$

The formulation (47) shows that $(a, b) \mapsto L_C(a, b)$ is a convex function (as a supremum of linear functions). From the primal problem (19), one also sees that $C \mapsto L_C(a, b)$ is concave.

5.2 Auction Algorithm and Dual Prices

The assignment algorithms mentioned in Section 1 become more transparent once one has dual variables. The auction algorithm is a dual price method: it updates target prices and maintains an approximate complementary-slackness certificate. The small tolerance ε removes ties, stabilizes the price updates and gives a quantitative optimality certificate [23, 24, 155].

Consider the square assignment problem with costs $C_{i,j}$ and rewrite it as the profit maximization problem with $a_{i,j} = -C_{i,j}$. The auction algorithm keeps prices p_j on the target points and a partial assignment. For an unassigned source i , define the best and second-best reduced profits

$$v_i = \max_j (a_{i,j} - p_j), \quad j_i \in \arg\max_j (a_{i,j} - p_j), \quad w_i = \max_{j \neq j_i} (a_{i,j} - p_j).$$

Source i bids for j_i and increases its price by the gap to the second-best target, plus a margin:

$$p_{j_i} \leftarrow p_{j_i} + v_i - w_i + \varepsilon.$$

The target j_i is then assigned to i , and its previous owner, if any, becomes unassigned. The iteration stops when all sources are assigned.

For fixed prices p , eliminating the bidder utilities u_i in the dual minimization gives the convex objective

$$D(p) = \sum_j p_j + \sum_i \max_j (a_{i,j} - p_j),$$

which comes from the dual constraints $u_i + p_j \geq a_{i,j}$. The auction update should thus be viewed as a price-adjustment method on this nonsmooth dual landscape, rather than as a generic gradient step: the proof of correctness is through approximate complementary slackness.

Definition 11 (ε -complementary slackness). *An assignment σ and prices p satisfy ε -complementary slackness if, for every source i ,*

$$a_{i,\sigma(i)} - p_{\sigma(i)} \geq \max_j (a_{i,j} - p_j) - \varepsilon.$$

Proposition 36 (Auction optimality certificate). *If a complete assignment σ satisfies ε -complementary slackness, then it is $n\varepsilon$ -optimal for the profit maximization problem, or equivalently $n\varepsilon$ -optimal for the original cost minimization problem. If all costs are integers and $\varepsilon < 1/n$, then σ is optimal.*

Proof. Let τ be any assignment. By ε -complementary slackness,

$$a_{i,\tau(i)} - p_{\tau(i)} \leq \max_j (a_{i,j} - p_j) \leq a_{i,\sigma(i)} - p_{\sigma(i)} + \varepsilon.$$

Summing over i cancels prices, because both σ and τ are permutations:

$$\sum_i a_{i,\tau(i)} \leq \sum_i a_{i,\sigma(i)} + n\varepsilon.$$

Thus no assignment has profit more than $n\varepsilon$ above that of σ . Since $a = -C$, the same statement says that the cost of σ is at most $n\varepsilon$ above the minimum cost. If the costs are integers, all assignment costs are integers; a gap strictly smaller than one therefore forces the gap to be zero. $\square$

During the bidding process the last bid made by a source makes its chosen target better, up to the margin ε , than all alternatives. Subsequent price increases can only make targets less attractive, and one checks by induction that currently assigned pairs satisfy ε -complementary slackness. The standard finite-termination proof normalizes prices by subtracting their minimum, observes that each bid increases one target price by at least ε , and bounds the normalized price spread in terms of the range of the profits; see [24, 155] for the full bound and implementation details.

Remark 15 (ε -scaling and relation with Sinkhorn). In practice one starts with a coarse ε and repeatedly decreases it, warm-starting the prices and assignment. This ε -scaling strategy is a homotopy method: large ε regularizes the combinatorial problem by enforcing a visible margin between the best and second-best reduced profits, while small ε recovers the exact dual certificate. If one wants a continuous-optimization analogy, the margin is closer to an exact-penalty or proximal continuation parameter than to a literal quadratic penalty.

Sinkhorn scaling plays a parallel role for entropic OT. There, the hard minimum in the dual c -transform is replaced by a soft minimum, or log-sum-exp, with temperature ε ; in the auction algorithm, the hard maximum is kept but the complementary slackness condition is relaxed by ε . Both methods therefore use an ε -controlled dual continuation, and both recover the unregularized transport certificate as $\varepsilon \rightarrow 0$ under the usual assumptions. The outputs are different: Sinkhorn produces dense entropic couplings, whereas auction keeps a sparse assignment throughout.

5.3 General formulation

The continuous dual is the analytic counterpart of the discrete linear program. It uses continuous potentials because measures are probed through integration.

To extend this primal-dual construction to arbitrary measures, it is important to realize that measures are naturally paired in duality with continuous functions, using the pairing $\langle f, \alpha \rangle := \int f d\alpha$.

Proposition 37 (Kantorovich duality). *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$. Then*

$$\mathcal{L}_c(\alpha, \beta) = \max_{(f, g) \in \mathcal{R}(c)} \int_{\mathcal{X}} f(x) d\alpha(x) + \int_{\mathcal{Y}} g(y) d\beta(y), \quad (51)$$

where the set of admissible dual potentials is

$$\mathcal{R}(c) := \{(f, g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y}) ; \forall (x, y), f(x) + g(y) \leq c(x, y)\}. \quad (52)$$

Here, (f, g) is a pair of continuous functions, often called “Kantorovich potentials”. The same formula extends under the usual lower-semicontinuity and integrability assumptions, replacing maxima by suprema when dual optimizers need not exist.

Proof. Weak duality is immediate: if $f(x) + g(y) \leq c(x, y)$ and $\pi \in \mathcal{U}(\alpha, \beta)$, then

$$\int f d\alpha + \int g d\beta = \int (f(x) + g(y)) d\pi(x, y) \leq \int c d\pi.$$

Taking the supremum over admissible potentials and the infimum over couplings gives “ $\leq$ ”.

For the reverse inequality, view the primal problem as a linear program over the locally convex space of Radon measures, paired with $\mathcal{C}(\mathcal{X} \times \mathcal{Y})$. The affine map $\pi \mapsto (\pi_1, \pi_2)$ is continuous for the weak topology, the feasible set is non-empty because it contains $\alpha \otimes \beta$, and the cost is continuous and bounded on compact sets. The separating-hyperplane theorem applied to the convex set of attainable triples

$$\left\{ (\pi_1, \pi_2, \int c d\pi + r) : \pi \geq 0, r \geq 0 \right\}$$

gives a continuous affine separator, hence functions (f, g) and a scalar multiplier which can be normalized so that $f \oplus g \leq c$. The separating inequality then states that the supremum over such potentials is at least the primal value. This proves equality. The same argument is the infinite-dimensional analogue of the finite linear-programming proof in Proposition 35. $\square$

The discrete case (47) corresponds to the dual vectors being samples of the continuous potentials, *i.e.* $(f_i, g_j) = (f(x_i), g(y_j))$. The primal-dual optimality conditions allow for tracking the support of the optimal plan, and (50) is generalized as

$$\text{Supp}(\pi) \subset \{(x, y) \in \mathcal{X} \times \mathcal{Y} ; f(x) + g(y) = c(x, y)\}. \quad (53)$$

Note that in contrast to the primal problem (22), showing the existence of solutions to (51) is non-trivial, because the constraint set $\mathcal{R}(c)$ is not compact and the objective is not coercive. Using the machinery of c -transforms detailed in Section 5.4, one can show that optimal (f, g) are necessarily Lipschitz regular, which enables the replacement of the constraint by a compact one.

5.4 c -transforms

The c -transform is the operation that improves potentials without changing feasibility. It is both a proof device for dual attainment and the route from duality to Brenier’s convex potentials.

Best-response potentials and the c -transform. Keeping a dual potential f fixed, one can maximize in closed form over the second potential in the dual problem (51), which leads to consider

$$\sup_{g \in \mathcal{C}(\mathcal{Y})} \left\{ \int g d\beta ; \forall (x, y), g(y) \leq c(x, y) - f(x) \right\}.$$

The constraint can be replaced by

$$\forall y \in \mathcal{Y}, \quad g(y) \leq f^c(y)$$

Definition 12 (c -transform). *For a function $f : \mathcal{X} \rightarrow \mathbb{R} \cup \{-\infty\}$, its c -transform is*

$$\forall y \in \mathcal{Y}, \quad f^c(y) := \inf_{x \in \mathcal{X}} c(x, y) - f(x). \quad (54)$$

For a function $g : \mathcal{Y} \rightarrow \mathbb{R} \cup \{-\infty\}$, the $\bar{c}$ -transform associated with $\bar{c}(y, x) = c(x, y)$ is

$$\forall x \in \mathcal{X}, \quad g^{\bar{c}}(x) := \inf_{y \in \mathcal{Y}} c(x, y) - g(y).$$

Since β is positive, the maximization of $\int g d\beta$ is thus achieved at those functions such that $g = f^c$ on the support of β , which means β -almost everywhere.

Proposition 38 (c -transforms solve the semi-relaxed problems). *For fixed f , the maximizers of the dual objective over all g such that $f \oplus g \leq c$ are exactly the functions satisfying $g = f^c$ β -almost everywhere. Equivalently, f^c gives the value of the one-marginal primal problem*

$$\inf_{\pi : \pi_2 = \beta} \int c(x, y) d\pi(x, y) - \int f(x) d\pi_1(x) = \int f^c(y) d\beta(y).$$

Symmetrically, for fixed g , the maximizers over f are the functions satisfying $f = g^{\bar{c}}$ α -almost everywhere.

Proof. The constraint $f(x) + g(y) \leq c(x, y)$ for all x is equivalent, for each y , to $g(y) \leq \inf_x c(x, y) - f(x) = f^c(y)$. Since β is nonnegative, the largest possible value of $\int g d\beta$ is obtained by saturating this pointwise upper bound on the support of β . The proof for $f = g^{\bar{c}}$ is identical after exchanging the two marginals.

For the primal formula, disintegrate any feasible π as $\pi(dx, dy) = \pi_y(dx)\beta(dy)$. Then

$$\int c d\pi - \int f d\pi_1 = \int \left(\int (c(x, y) - f(x)) d\pi_y(x) \right) d\beta(y) \geq \int f^c(y) d\beta(y).$$

If minimizers admit a measurable selection, equality is obtained by choosing π_y supported on minimizers of $x \mapsto c(x, y) - f(x)$. Otherwise one uses approximate measurable selections and lets the approximation error vanish. $\square$

The map $(f, g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y}) \mapsto (g^{\bar{c}}, f^c) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y})$ replaces dual potentials by “better” ones (improving the dual objective $\mathcal{E}$). Functions that can be written in the form f^c and $g^{\bar{c}}$ are called c -concave and $\bar{c}$ -concave functions.

Note that these partial minimizations define maximizers on the support of α and β , respectively, while the definitions (54) define functions on the whole spaces $\mathcal{X}$ and $\mathcal{Y}$. This is thus a canonical way to extend solutions of (51) to the whole space.

Furthermore, if c is Lipschitz, then f^c and g^c are also Lipschitz functions, as we now show. This property is crucial to prove the existence of solutions to the dual problem. Indeed, since one can impose this Lipschitz bound on the dual potentials, the constraint set is compact by the Ascoli theorem.

Proposition 39 (Lipschitz stability of c -transforms). *If c is L -Lipschitz with respect to the second variable, then f^c is L -Lipschitz.*

Proof. We apply to $F_x = c(x, \cdot) - f(x)$ the fact that if all the F_x are L -Lipschitz, then the Lipschitz constant of $F = \min_x F_x$ is at most L . Indeed, using the fact that $|\inf(A) - \inf(B)| \leq \sup |A - B|$ for two functions A and B , one obtains

$$|F(y) - F(y')| = \left| \inf_x (F_x(y)) - \inf_x (F_x(y')) \right| \leq \sup_x |F_x(y) - F_x(y')| \leq \sup_x Ld(y, y') = Ld(y, y').$$

$\square$

Euclidean case. The special case $c(x, y) = -\langle x, y \rangle$ in $\mathcal{X} = \mathcal{Y} = \mathbb{R}^d$ is of utmost importance because it allows one to study the $\mathcal{W}_2$ problem, since for any $\pi \in \mathcal{U}(\alpha, \beta)$

$$\int \|x - y\|^2 d\pi(x, y) = \text{cst} - 2 \int \langle x, y \rangle d\pi(x, y) \quad \text{where} \quad \text{cst} = \int \|x\|^2 d\alpha(x) + \int \|y\|^2 d\beta(y).$$

For this special choice of cost, one has $f^c = -(-f)^*$ where h^* is the Fenchel-Legendre transform

$$h^*(y) := \sup_x \langle x, y \rangle - h(x).$$

The function h^* is always convex, so f^c is always concave. For a general cost, functions of the form f^c are called c -concave.

Remark 16 (Proof of Brenier's theorem). In the case $c(x, y) = \|x - y\|^2$, subtracting the harmless quadratic terms reduces the geometry to the cost $c(x, y) = -\langle x, y \rangle$. The primal-dual relationship, together with the fact that one can replace (f, g) by $(f^{cc}, f^{ccc} = f^c)$, shows that

$$\text{supp}(\pi) \subset \{(x, y) ; \varphi(x) + \varphi^*(y) = \langle x, y \rangle\}$$

where $\varphi = -f^{cc}$ is convex and $-g = \varphi^*$. One always has $\varphi(x) + \varphi^*(y) \geq \langle x, y \rangle$ from the definition of the Legendre transform, and the set of y such that equality holds is precisely the subdifferential $\partial\varphi(x)$. In the special case where α has a density, a convex function is differentiable Lebesgue-almost everywhere, hence α -almost everywhere. Thus $\partial\varphi(x)$ is a singleton for α -almost every x , and it is legitimate to use $T = \nabla\varphi$ as an optimal transport map.

The failure of alternate optimization. A crucial property of the Legendre transform is that $f^{***} = f^*$, and that f^{**} is the convex envelope of f (the largest convex function below f). These properties carry over for the more general setting of c -transforms. The required convex-analytic background is standard in Rockafellar's theory [183].

Proposition 40 (Algebra of c -transforms). *The following identities, in which the inequality sign between vectors should be understood elementwise, hold, denoting $f^{c\bar{c}} := (f^c)^{\bar{c}}$:*

$$(i) f \leq f' \Rightarrow f^c \geq f'^c, \quad (ii) f^{c\bar{c}} \geq f, \quad (iii) g^{\bar{c}c} \geq g, \quad (iv) f^{c\bar{c}c} = f^c.$$

Proof. The first inequality (i) follows from the definition of c -transforms (because of the $-$ sign). To prove (ii), expanding the definition of $f^{c\bar{c}}$ we have

$$(f^{c\bar{c}})(x) = \min_y c(x, y) - f^c(y) = \min_y c(x, y) - \min_{x'} (c(x', y) - f(x')).$$

Now, since $-\min_{x'} c(x', y) - f(x') \geq -(c(x, y) - f(x))$, we recover

$$(f^{c\bar{c}})(x) \geq \min_y c(x, y) - c(x, y) + f(x) = f(x).$$

The relation $g^{\bar{c}c} \geq g$ is obtained in the same way. Now, to prove (iv), we first apply (ii) and then (i) with $f' = f^{c\bar{c}}$ to have $f^c \geq f^{c\bar{c}c}$. Then we apply (iii) to $g = f^c$ to obtain $f^c \leq f^{c\bar{c}c}$. $\square$

This invariance property shows that one can “improve” only once the dual potential this way. Indeed, starting from any pair (f, g) , one obtains the following iterates by alternating maximization

$$(f, g) \mapsto (f, f^c) \mapsto (f^{cc}, f^c) \mapsto (f^{cc}, f^{ccc}) = (f^{cc}, f^c) \dots \quad (55)$$

so that one reaches a stationary point. This failure is the classical behavior of alternating maximization on a non-smooth problem, where the non-smooth part of the functional (here the constraint) mixes the two variables. The workaround is to introduce smoothing, which is the classical method of augmented Lagrangian, and that we will develop here using entropic regularization, which corresponds to Sinkhorn's algorithm.

This can be visualized as two interleaved rows of dual updates: after one c -transform on each row, alternating maximization has already reached a fixed point.

6 Semi-discrete and $\mathcal{W}_1$

This section focuses on two computationally useful degeneracies of the dual problem. Semi-discrete OT turns a continuous-to-discrete map into finite-dimensional geometry, while $\mathcal{W}_1$ replaces convex potentials by Lipschitz functions and flow fields. The material connects computational geometry [13, 152, 153] with the Kantorovich–Rubinstein and Beckmann formulations [123, 15].

6.1 Semi-dual

The semi-dual eliminates one potential by an exact c -transform. It keeps concavity while removing explicit inequality constraints.

From the dual problem (51), that we write as

$$\sup_{f, g \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y})} \mathcal{E}(f, g)$$

where $\mathcal{E}$ takes into account the constraints one can “marginalize” it with respect to g by minimizing over it to obtain the following “semi-dual” problem

$$\sup_{f \in \mathcal{C}(\mathcal{X})} \tilde{\mathcal{E}}(f) := \mathcal{E}(f, f^c) = \sup_g \mathcal{E}(f, g) = \int_{\mathcal{X}} f d\alpha + \int_{\mathcal{Y}} f^c d\beta. \quad (56)$$

Note that marginalizing a concave problem retains concavity so that $\tilde{\mathcal{E}}$ is still concave. The major advantage of this new “semi-dual” problem is that it is unconstrained, which allows the use of simpler optimization algorithms, as we will now see.

6.2 Semi-discrete

The semi-discrete case is where dual potentials become weights of Laguerre cells. This gives both geometry and algorithms for quantization and density fitting.

A case of particular interest is when $\beta = \sum_j b_j \delta_{y_j}$ is discrete (of course the same construction applies if α is discrete by exchanging the role of α, β). One can adapt the definition of the $\bar{c}$ transform (54) to this setting by restricting the minimization to the support $(y_j)_j$ of β ,

$$\forall g \in \mathbb{R}^m, \forall x \in \mathcal{X}, \quad g^{\bar{c}}(x) := \min_{j \in [m]} c(x, y_j) - g_j. \quad (57)$$

This transform maps a vector g to a continuous function $g^{\bar{c}} \in \mathcal{C}(\mathcal{X})$. Note that this definition coincides with (54) when imposing that the space $\mathcal{X}$ is equal to the support of β .

Crucially, using the discrete $\bar{c}$ -transform, when β is a discrete measure, yields a finite-dimensional optimization,

$$\mathcal{L}_c(\alpha, \beta) = \max_{g \in \mathbb{R}^m} \mathcal{E}(g) := \int_{\mathcal{X}} g^{\bar{c}}(x) d\alpha(x) + \sum_j g_j b_j. \quad (58)$$

The Laguerre cells associated to the dual weights g

$$\mathbb{L}_j(g) := \{x \in \mathcal{X} ; \forall j' \neq j, c(x, y_j) - g_j \leq c(x, y_{j'}) - g_{j'}\}$$

induce a disjoint decomposition of $\mathcal{X} = \bigcup_j \mathbb{L}_j(g)$. When g is constant, the Laguerre cells decomposition corresponds to the Voronoi diagram partition of the space. For quadratic costs, varying the dual weights moves the walls between adjacent cells while keeping them parallel; this is the geometric mechanism by which the cell masses are adjusted.

This allows one to conveniently rewrite the minimized energy as

$$\mathcal{E}(g) = \sum_{j=1}^m \int_{\mathbb{L}_j(g)} (c(x, y_j) - g_j) d\alpha(x) + \langle g, b \rangle. \quad (59)$$

The following proposition provides a formula for the gradient of this concave function.

Proposition 41 (Gradient of the semi-discrete dual). *If α has a density with respect to Lebesgue measure and if c is smooth away from the diagonal, then $\mathcal{E}$ is differentiable and*

$$\forall j \in \llbracket m \rrbracket, \quad \nabla \mathcal{E}(g)_j = b_j - \int_{\mathbb{L}_j(g)} d\alpha.$$

Proof. One has

$$\mathcal{E}(g + \varepsilon \delta_j) - \mathcal{E}(g) - \varepsilon \left(b_j - \int_{\mathbb{L}_j(g)} d\alpha \right) = \sum_k \int_{\mathbb{L}_k(g + \varepsilon \delta_j)} c(x, y_k) d\alpha(x) - \int_{\mathbb{L}_k(g)} c(x, y_k) d\alpha(x).$$

Most of the terms on the right-hand side vanish (because most of the Laguerre cells associated with $g + \varepsilon \delta_j$ are equal to those of g) and the only terms remaining correspond to neighboring cells (j, k) such that $\mathbb{L}_j(g) \cap \mathbb{L}_k(g) \neq \emptyset$ (for the cost $\|x - y\|^2$ and $g = 0$ this forms the Delaunay triangulation). On these pairs, the right integral differs on a volume of the order of ε (since α has a density), and the function being integrated only varies on the order of ε (since the cost is smooth). So the right-hand side is of the order of ε^2 . $\square$

The first order optimality condition shows that to solve the dual semi-discrete problem, one needs to select the weights g to drive the Laguerre cell in a configuration such that $\int_{\mathbb{L}_j(g)} d\alpha = b_j$, i.e. each cell should capture the correct amount of mass. In this case, the optimal transport T such that $T_\# \alpha = \beta$ (which exists and is unique according to Brenier's theorem if α has a density) is piecewise constant and maps $x \in \mathbb{L}_j(g)$ to y_j .

In the special case $c(x, y) = \|x - y\|^2$, the decomposition in Laguerre cells is also known as a “power diagram”. In this case, the cells are polyhedral and can be computed efficiently using computational geometry algorithms; see [12, 13, 152, 153]. The most widely used algorithm relies on the fact that the power diagram of points in $\mathbb{R}^d$ is equal to the projection on $\mathbb{R}^d$ of the convex hull of the set of points $((y_j, \|y_j\|^2 - g_j))_{j=1}^m \subset \mathbb{R}^{d+1}$. There are numerous algorithms to compute convex hulls; for instance, that of [52] in two and three dimensions has complexity $O(m \log(Q))$, where Q is the number of vertices of the convex hull.

Stochastic optimization. The semidiscrete formulation (59) is also appealing because the energies to be minimized are written as an expectation with respect to the probability distribution α ,

$$\mathcal{E}(g) = \int_{\mathcal{X}} E(g, x) d\alpha(x) = \mathbb{E}_X(E(g, X)) \quad \text{where} \quad E(g, x) := g^{\bar{c}}(x) - \langle g, b \rangle, \quad (60)$$

and X denotes a random vector distributed on $\mathcal{X}$ according to α . Note that the gradient of each of the involved functional reads

$$\nabla_g E(x, g) = (\mathbb{1}_{\mathbb{L}_j(g)}(x) - b_j)_{j=1}^m \in \mathbb{R}^m$$

where $\mathbb{1}_{\mathbb{L}_j(g)}$ is the indicator function of the Laguerre cell. One can thus use stochastic optimization methods to perform the maximization, as proposed in [96]. This allows us to obtain provably convergent algorithms without the need to resort to an arbitrary discretization of α (either approximating α using sums of Diracs or using quadrature formula for the integrals). The measure α is used as a black box from which one can draw independent samples, which is a natural computational setup for many high-dimensional applications in statistics and machine learning.

Initializing $g^{(0)} = \mathbf{0}_m$, the stochastic gradient descent algorithm (SGD; used here as a maximization method) draws at step ℓ a point $x_\ell \in \mathcal{X}$ according to distribution α (independently from all past and future samples $(x_\ell)_\ell$) to form the update

$$g^{(\ell+1)} := g^{(\ell)} + \tau_\ell \nabla_g E(g^{(\ell)}, x_\ell). \quad (61)$$

The step size τ_ℓ should decay fast enough to zero to ensure that the “noise” created by using $\nabla_g E(x_\ell, g)$ as a proxy for the true gradient $\nabla \mathcal{E}(g)$ is canceled in the limit. A typical choice of schedule is

$$\tau_\ell := \frac{\tau_0}{1 + \ell/\ell_0}, \quad (62)$$

where ℓ_0 indicates roughly the number of iterations serving as a warmup phase. One can prove the convergence result

$$\mathcal{E}(g^*) - \mathbb{E}(\mathcal{E}(g^{(\ell)})) = O\left(\frac{1}{\sqrt{\ell}}\right),$$

where g^* is a solution of (60) and where $\mathbb{E}$ indicates an expectation with respect to the i.i.d. sampling of $(x_\ell)_\ell$ performed at each iteration.

6.3 Optimal Quantization

Optimal quantization asks for the best discrete approximation of a measure by m codepoints. It is the geometric core of vector quantization, compression and k -means clustering.

The optimal quantization problem of some measure α corresponds to the resolution of

$$\mathcal{Q}_m(\alpha) = \min_{Y=(y_j)_{j=1}^m, \mathbf{b} \in \Sigma_m} \mathcal{W}_p\left(\alpha, \sum_j \mathbf{b}_j \delta_{y_j}\right).$$

This problem is at the heart of the computation of efficient vector quantizers in information theory and compression and is also the basic problem to solve for clustering in unsupervised learning.

Proposition 42 (Quantization rate and curse of dimensionality). *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, and assume that $\alpha = \rho dx$ on Ω , with $0 < \rho_- \leq \rho \leq \rho_+ < +\infty$. Then, for fixed $p \geq 1$, there exist constants $0 < c \leq C < +\infty$ such that*

$$cm^{-1/d} \leq \mathcal{Q}_m(\alpha) \leq Cm^{-1/d}.$$

Proof. For the upper bound, partition Ω into m cells of diameter at most $Cm^{-1/d}$, up to boundary effects, and put one codepoint in each non-empty cell. Sending each point to the codepoint in its cell gives a transport cost bounded by $Cm^{-1/d}$.

For the lower bound, fix any set Y of m codepoints and write $d_Y(x) = \min_j \|x - y_j\|$. Since the density is bounded above, the mass of the t -neighborhood of Y is at most Cmt^d . Choosing $t_0 \simeq m^{-1/d}$ small enough gives $\alpha(\{d_Y > t\}) \geq c$ for $0 < t < t_0$. Hence

$$\int d_Y(x)^p d\alpha(x) = \int_0^\infty pt^{p-1} \alpha(\{d_Y > t\}) dt \geq ct_0^p \simeq cm^{-p/d}.$$

Taking the p -th root and minimizing over Y gives the result. $\square$

This deterministic rate mirrors the empirical OT sample-complexity rate of Proposition 58: both are governed by the spacing $m^{-1/d}$ of points in d dimensions. Quantization is best-case and deterministic, while empirical OT is random, but both display the same curse of dimensionality.

Quantization is a classical subject in approximation and compression; the OT viewpoint clarifies its connection with nearest-neighbor cells and barycentric updates [142]. This optimal quantization problem is convex with respect to $\mathbf{b}$, but is unfortunately non-convex with respect to $Y = (y_j)_j$. Its resolution is in general NP-hard. The only setting where this problem is substantially simpler is the one-dimensional case. Monotonicity fixes the ordering of the cells, so the problem reduces to optimizing interval endpoints and centroids; for the uniform law and the quadratic cost this gives equally spaced centroids.

Proposition 43 (Free masses give Voronoi cells). *For the cost used in $\mathcal{W}_p^p$, namely $c(x, y) = d(x, y)^p$, fix distinct codepoints $Y = (y_j)_{j=1}^m$; duplicate codepoints can be merged beforehand. Minimizing over the weights $\mathbf{b} \in \Sigma_m$ gives*

$$\min_{\mathbf{b} \in \Sigma_m} \mathcal{W}_p^p\left(\alpha, \sum_j \mathbf{b}_j \delta_{y_j}\right) = \int_{\mathcal{X}} \min_{1 \leq j \leq m} c(x, y_j) d\alpha(x).$$

An optimal coupling is induced by sending each x to a nearest codepoint. The corresponding cells are the Voronoi cells

$$\mathbb{V}_j(Y) = \{x; \forall j', c(x, y_j) \leq c(x, y_{j'})\},$$

up to arbitrary tie-breaking on common boundaries.

Proof. For any coupling between α and a measure supported on Y , the conditional destination of a point x belongs to Y , hence its conditional cost is at least $\min_j c(x, y_j)$. Integrating gives the lower bound. Conversely, choose a measurable nearest-codepoint map $T_Y(x) \in \operatorname{argmin}_j c(x, y_j)$, breaking ties measurably, and set $b_j = \alpha(T_Y^{-1}(y_j))$. Then $(T_Y)_\# \alpha = \sum_j b_j \delta_{y_j}$ and the induced transport reaches the displayed lower bound. $\square$

This also shows that the quantization energy can be rewritten in the intuitive nearest-centroid form

$$\mathcal{Q}_m(\alpha)^p = \min_Y \mathcal{F}(Y) := \int_{\mathcal{X}} \min_{1 \leq j \leq m} c(x, y_j) d\alpha(x).$$

At any local minimizer (at least if α has a density so that this function is differentiable) of this energy over Y , one sees that each y_j should be a centroid of its associated Voronoi region,

$$y_j \in \operatorname{argmin}_y \int_{V_j(Y)} c(x, y) d\alpha(x).$$

For instance, when $c(x, y) = \|x - y\|^2$, one sees that any local minimizer should satisfy the fixed point equation

$$y_j = \frac{\int_{V_j(Y)} x d\alpha(x)}{\int_{V_j(Y)} d\alpha}.$$

The celebrated k -means algorithm, also known as the Lloyd algorithm, iteratively applies this fixed point. It is not guaranteed to converge in complete generality, but with standard tie-breaking the objective decreases and the iterates typically reach a local minimum. A practical issue to obtain a good local minimizer is to seed a good initial configuration. The intuitive way to achieve this is to spread the points as much as possible, and a well-known algorithm to do so is the k -means++ method. For the squared Euclidean objective, its initialization has expected nearest-centroid energy within a logarithmic factor of the optimum [11].

6.4 Wasserstein-1 norm

The $\mathcal{W}_1$ distance has an especially transparent dual: 1-Lipschitz test functions. This makes it the meeting point between transport, PDE formulations and weak norms on signed measures.

c -transform for $\mathcal{W}_1$. Here we assume that d is a distance on $\mathcal{X} = \mathcal{Y}$, and we solve the OT problem with the ground cost $c(x, y) = d(x, y)$. The following proposition highlights key properties of the c -transform (54) in this setup. In the following, we denote the Lipschitz constant of a function $f \in \mathcal{C}(\mathcal{X})$ as

$$\operatorname{Lip}(f) := \sup \left\{ \frac{|f(x) - f(y)|}{d(x, y)} ; (x, y) \in \mathcal{X}^2, x \neq y \right\}. \quad (63)$$

Proposition 44 (c -transforms and 1-Lipschitz functions). *Suppose $\mathcal{X} = \mathcal{Y}$ and $c(x, y) = d(x, y)$. Then, there exists g such that $f = g^c$ if and only if $\operatorname{Lip}(f) \leq 1$. Furthermore, if $\operatorname{Lip}(f) \leq 1$, then $f^c = -f$.*

Proof. First, suppose $f = g^c$ for some g . Then, for $x, y \in \mathcal{X}$,

$$\begin{aligned} |f(x) - f(y)| &= \left| \inf_{z \in \mathcal{X}} [d(x, z) - g(z)] - \inf_{z \in \mathcal{X}} [d(y, z) - g(z)] \right| \\ &\leq \sup_{z \in \mathcal{X}} |d(x, z) - d(y, z)| \leq d(x, y). \end{aligned}$$

The first equality follows from the definition of g^c , the next inequality from the identity $|\inf A - \inf B| \leq \sup |A - B|$, and the last from the reversed triangle inequality. This shows that $\operatorname{Lip}(f) \leq 1$.

If f is 1-Lipschitz, for all $x, y \in \mathcal{X}$, $f(y) - d(x, y) \leq f(x) \leq f(y) + d(x, y)$, which shows that

$$\begin{aligned} f^c(y) &= \inf_{x \in \mathcal{X}} [d(x, y) - f(x)] \geq \inf_{x \in \mathcal{X}} [d(x, y) - f(y) - d(x, y)] = -f(y), \\ f^c(y) &= \inf_{x \in \mathcal{X}} [d(x, y) - f(x)] \leq \inf_{x \in \mathcal{X}} [d(x, y) - f(y) + d(x, y)] = -f(y), \end{aligned}$$

because $\inf_x d(x, y) = 0$ (for $x = y$) and thus $f^c = -f$.

Applying this property to $-f$ which is also 1-Lipschitz shows that $(-f)^c = f$ so that f is c -concave (i.e. it is the c -transform of a function). $\square$

Using the iterative c -transform scheme (55), one can replace the dual variable (f, g) by $(f^{cc}, f^c) = (-f^c, f^c)$, or equivalently by any pair $(f, -f)$ where f is 1-Lipschitz. This leads to the following alternative expression for the $\mathcal{W}_1$ distance

$$\mathcal{W}_1(\alpha, \beta) = \max_f \left\{ \int_{\mathcal{X}} f d(\alpha - \beta) ; \text{Lip}(f) \leq 1 \right\}. \quad (64)$$

This expression shows that $\mathcal{W}_1$ is actually a norm, i.e. $\mathcal{W}_1(\alpha, \beta) = \|\alpha - \beta\|_{\mathcal{W}_1}$, and that it is still valid for any measures (not necessarily positive) as long as $\int_{\mathcal{X}} \alpha = \int_{\mathcal{X}} \beta$. This norm is often called the Kantorovich-Rubinstein norm [123].

For discrete measures of the form (2), writing $\alpha - \beta = \sum_k m_k \delta_{z_k}$ with $z_k \in \mathcal{X}$ and $\sum_k m_k = 0$, the optimization (64) can be rewritten as

$$\mathcal{W}_1(\alpha, \beta) = \max_{(f_k)_k} \left\{ \sum_k f_k m_k ; \forall (k, \ell), |f_k - f_\ell| \leq d(z_k, z_\ell), \right\} \quad (65)$$

which is a finite-dimensional convex program with quadratic-cone constraints. It can be solved using interior point methods or, as we detail next for a similar problem, using proximal methods.

When using $d(x, y) = |x - y|$ with $\mathcal{X} = \mathbb{R}$, we can reduce the number of constraints by ordering the z_k 's via $z_1 \leq z_2 \leq \dots$. In this case, we only have to solve

$$\mathcal{W}_1(\alpha, \beta) = \max_{(f_k)_k} \left\{ \sum_k f_k m_k ; \forall k, |f_{k+1} - f_k| \leq z_{k+1} - z_k \right\},$$

which is a linear program. Furthermore, in this 1-D case, a closed-form expression for $\mathcal{W}_1$ using cumulative functions is given in (12).

$\mathcal{W}_1$ on Euclidean spaces In the special case of Euclidean spaces $\mathcal{X} = \mathcal{Y} = \mathbb{R}^d$, using $c(x, y) = \|x - y\|$, the global Lipschitz constraint appearing in (64) can be made local as a uniform bound on the gradient of f ,

$$\mathcal{W}_1(\alpha, \beta) = \sup_f \left\{ \int_{\mathbb{R}^d} f(d\alpha - d\beta) ; \|\nabla f\|_\infty \leq 1 \right\}. \quad (66)$$

Here the constraint $\|\nabla f\|_\infty \leq 1$ signifies that the norm of the gradient of f at any point x is upper bounded by 1, $\|\nabla f(x)\|_2 \leq 1$ for any x .

Considering the dual problem to (66), denoting $\xi := \alpha - \beta$, and using that

$$\iota_{\|\cdot\|_{\mathbb{R}^d} \leq 1}(u) = \max_v \langle u, v \rangle - \|v\|_{\mathbb{R}^d}$$

one has a maximization on flow vector fields $s : \mathbb{R}^d \rightarrow \mathbb{R}^d$

$$\begin{aligned} \mathcal{W}_1(\alpha, \beta) &= \sup_f \inf_{s(x) \in \mathbb{R}^d} \int_{\mathbb{R}^d} f d\xi - \int \langle \nabla f(x), s(x) \rangle dx + \int \|s(x)\|_{\mathbb{R}^d} dx \\ &= \inf_{s(x) \in \mathbb{R}^d} \int \|s(x)\|_{\mathbb{R}^d} dx + \sup_f \int f(x) (d\xi - \text{div}(s) dx) \end{aligned}$$

one obtains an optimization problem under a fixed divergence constraint

$$\mathcal{W}_1(\alpha, \beta) = \inf_s \left\{ \int_{\mathbb{R}^d} \|s(x)\|_{\mathbb{R}^d} dx ; \operatorname{div}(s) = \alpha - \beta \right\}, \quad (67)$$

which is often called the Beckmann formulation [15]. Here the vectorial function $s(x) \in \mathbb{R}^2$ can be interpreted as a flow field, describing locally the movement of mass. Outside the support of the two input measures, $\operatorname{div}(s) = 0$, which is the conservation of mass constraint. Once properly discretized using finite elements, Problems (66) and (67) become a nonsmooth convex optimization problem.

The previous formulations (66) and (67) of $\mathcal{W}_1$ can be generalized to the setting where $\mathcal{X}$ is a Riemannian manifold, i.e. $c(x, y) = d(x, y)$ where d is the associated geodesic distance (and then for smooth manifolds, the gradient and divergence should be understood as differential operators on manifolds).

Proposition 45 ($\mathcal{W}_1$ and Beckmann flow on a graph). *Let $G = (V, E)$ be a connected finite graph with positive edge lengths $(\ell_e)_{e \in E}$. The graph geodesic distance is*

$$d_G(i, j) = \min_{\gamma: i \rightsquigarrow j} \sum_{e \in \gamma} \ell_e.$$

For two probability vectors a, b on V , set $r = a - b$ and orient each edge $e = (i, j)$. If

$$(\nabla_G f)_e = f_j - f_i, \quad \operatorname{div}_G = -\nabla_G^*$$

are the finite-difference gradient and its negative adjoint, then a positive flow on the oriented edge $i \rightarrow j$ has positive divergence at i and negative divergence at j . With this convention,

$$\mathcal{W}_{1,G}(a, b) = \max_f \left\{ \sum_{i \in V} f_i r_i ; |f_i - f_j| \leq \ell_e \quad \forall e = (i, j) \right\} = \min_m \left\{ \sum_{e \in E} \ell_e |m_e| ; \operatorname{div}_G m = r \right\}.$$

The vector m_e is an oriented edge flow, and the constraint $\operatorname{div}_G m = r$ is conservation of mass at each vertex.

Proof. The edge constraint $|f_i - f_j| \leq \ell_e$ implies, by summing along paths, that $|f_i - f_j| \leq d_G(i, j)$ for all vertices. Conversely, any 1-Lipschitz function for d_G satisfies the edge constraints because each edge is a path of length ℓ_e . The first equality is therefore the Kantorovich–Rubinstein formula on the metric space (V, d_G) .

For the second equality, write the graph Beckmann problem and dualize its equality constraint with a potential f :

$$\inf_m \sum_e \ell_e |m_e| + \sup_f \sum_i f_i (r_i - (\operatorname{div}_G m)_i).$$

Using $\operatorname{div}_G = -\nabla_G^*$, the coupling term is $\sum_e m_e (\nabla_G f)_e$. The minimization over each scalar flow m_e is finite exactly when $|(\nabla_G f)_e| \leq \ell_e$, and is then equal to zero. The dual problem is therefore precisely the graph Lipschitz dual above. Strong duality holds because this is a finite-dimensional linear program with a non-empty feasible set: connectedness and $\sum_i r_i = 0$ allow the signed surplus to be routed along paths. This proves the graph Beckmann formula. $\square$

This graph formulation is the transshipment version of $\mathcal{W}_1$. It is the natural discrete analogue of (67): gradients are edge differences, divergences are incidence-matrix balances, and geodesic distance is the shortest-path length. It can be solved by min-cost flow methods on sparse graphs, and entropic or KL-projection variants lead to flow-Sinkhorn algorithms for graph $\mathcal{W}_1$ [15, 168].

7 Divergences and Dual Norms

This section compares OT with divergence-based and adversarial ways of measuring discrepancy. The main stake is topological: φ -divergences are cheap but strong, while dual norms and GAN objectives can be weak enough to compare singular measures. The discussion connects classical information divergences [62, 3] with modern integral probability metrics and generative modeling [203, 103, 10].

7.1 Dual norms (Integral Probability Metrics)

Dual norms generalize the $\mathcal{W}_1$ test-function principle. They are useful in statistics because they compare distributions by restricting the discriminator class.

Formulation (66) is a special case of a dual norm. A dual norm is a convenient way to design “weak” norms that can deal with arbitrary measures. For a symmetric convex set B of measurable functions, one defines

$$\|\alpha\|_B := \sup_f \left\{ \int_{\mathcal{X}} f(x) d\alpha(x) ; f \in B \right\}. \quad (68)$$

These dual norms are often called “integral probability metrics”; see [204, 203, 205].

Example 3 (Total variation). The total variation norm (Example 8) is a dual norm associated with the whole space of continuous functions

$$B = \{f \in \mathcal{C}(\mathcal{X}) ; \|f\|_{\infty} \leq 1\}.$$

Total variation is the canonical nontrivial example of a discrepancy that is both a φ -divergence and a dual norm; see [203].

Example 4 ($\mathcal{W}_1$ norm). $\mathcal{W}_1$ as defined in (66), is a special case of dual norm (68), using

$$B = \{f ; \text{Lip}(f) \leq 1\}$$

the set of 1-Lipschitz functions.

Example 5 (Flat norm and Dudley metric). If the set B is bounded and separates measures, then $\|\cdot\|_B$ is a norm on the whole space $\mathcal{M}(\mathcal{X})$ of finite measures. This is not the case of $\mathcal{W}_1$, which is only defined for α such that $\int_{\mathcal{X}} d\alpha = 0$ (otherwise $\|\alpha\|_B = +\infty$). This can be alleviated by imposing a bound on the value of the potential f , in order to define for instance the flat norm,

$$B = \{f ; \text{Lip}(f) \leq 1 \text{ and } \|f\|_{\infty} \leq 1\}. \quad (69)$$

It metrizes the weak convergence on the whole space $\mathcal{M}(\mathcal{X})$. Formula (65) is extended to compute the flat norm by adding the constraint $|f_k| \leq 1$. The flat norm is sometimes called the “Kantorovich–Rubinstein” norm [114] and has been used as a fidelity term for inverse problems in imaging [135]. The flat norm is similar to the Dudley metric, which uses

$$B = \{f ; \|\nabla f\|_{\infty} + \|f\|_{\infty} \leq 1\}.$$

The following proposition gives a useful compact-space criterion. The dual ball should be rich enough to separate continuous observables, but compact enough for weak convergence to imply uniform convergence over the discriminator class.

Proposition 46 (Metritzation by dual norms). *Assume $\mathcal{X}$ is compact, $B = -B$, and that all measures below are probability measures. (i) If $\mathcal{C}(\mathcal{X}) \subset \overline{\text{Span}(B)}$ (i.e. if the span of B is dense in continuous functions for the sup-norm $\|\cdot\|_{\infty}$), then $\|\alpha_k - \alpha\|_B \rightarrow 0$ implies $\alpha_k \rightarrow \alpha$. (ii) If $B \subset \mathcal{C}(\mathcal{X})$ is compact for $\|\cdot\|_{\infty}$, then $\alpha_k \rightarrow \alpha$ implies $\|\alpha_k - \alpha\|_B \rightarrow 0$.*

Proof. (i) If $\|\alpha_k - \alpha\|_B \rightarrow 0$, then by duality and the symmetry of B , for any $f \in B$,

$$|\langle f, \alpha_k - \alpha \rangle| \leq \|\alpha_k - \alpha\|_B,$$

so $\langle f, \alpha_k \rangle \rightarrow \langle f, \alpha \rangle$. By linearity, this convergence extends to $\text{Span}(B)$. Let $u \in \mathcal{C}(\mathcal{X})$ and choose $h \in \text{Span}(B)$ with $\|u - h\|_{\infty} \leq \eta$. Since α_k and α have mass one,

$$|\langle u, \alpha_k - \alpha \rangle| \leq |\langle h, \alpha_k - \alpha \rangle| + 2\eta.$$

Taking the limsup in k and then letting $\eta \rightarrow 0$ proves weak convergence. (ii) We assume $\alpha_k \rightharpoonup \alpha$ and we consider a sub-sequence α_{n_k} such that

$$\|\alpha_{n_k} - \alpha\|_B \longrightarrow \limsup_k \|\alpha_k - \alpha\|_B$$

Since B is compact, the maximum appearing in the definition of $\|\alpha_{n_k} - \alpha\|_B$ is reached, so there exists $f_{n_k} \in B$ such that $\langle \alpha_{n_k} - \alpha, f_{n_k} \rangle = \|\alpha_{n_k} - \alpha\|_B$. Once again, by compactness, we can extract from $(f_{n_k})_k$ a (not relabelled for simplicity) subsequence converging uniformly to some $f \in B$. One has $\|\alpha_{n_k} - \alpha\|_B = \langle \alpha_{n_k} - \alpha, f_{n_k} \rangle$, and this quantity converges to 0 because

$$\langle \alpha_{n_k} - \alpha, f_{n_k} \rangle = \langle \alpha_{n_k} - \alpha, f \rangle + \langle \alpha_{n_k}, f_{n_k} - f \rangle - \langle \alpha, f_{n_k} - f \rangle$$

and these three terms go to zero because $\alpha_{n_k} - \alpha \rightharpoonup 0$ for the first one and $\|f_{n_k} - f\|_\infty \rightarrow 0$ for the last two. $\square$

Corollary 3 (Wasserstein metrizes weak convergence). *On a compact space, the Wasserstein- p distance metrizes the weak convergence.*

Proof. For $\mathcal{W}_1$, take $B = \{f ; \text{Lip}(f) \leq 1\}$. The span of B contains all Lipschitz functions, and Lipschitz functions are dense in $\mathcal{C}(\mathcal{X})$ on a compact metric space. This gives the first implication in Proposition 46. For the converse implication, constants do not change the pairing with $\alpha_k - \alpha$ because both measures have mass one. We may therefore normalize every $f \in B$ by imposing $f(x_0) = 0$ for a fixed $x_0 \in \mathcal{X}$. The normalized class is uniformly bounded by $\text{diam}(\mathcal{X})$ and equicontinuous, hence compact in $\|\cdot\|_\infty$ by Ascoli–Arzelà. Proposition 46 then applies. Proposition 21 shows that $\mathcal{W}_p$ has the same topology as $\mathcal{W}_1$ on a compact space, so every $\mathcal{W}_p$ metrizes convergence in law. $\square$

7.2 Dual RKHS Norms and Maximum Mean Discrepancies

It is also possible to define “Euclidean” norms (built using quadratic functionals) on measures using the machinery of kernel methods and more specifically reproducing kernel Hilbert spaces (RKHS; see [193] for a survey of their applications in data sciences), of which we recall first some basic definitions.

Definition 13 (Positive and conditionally positive kernels). *A symmetric function k defined on $\mathcal{X} \times \mathcal{X}$ is said to be positive definite if for any $n \geq 0$, for any family $x_1, \dots, x_n \in \mathcal{X}$ the matrix $(k(x_i, x_j))_{i,j}$ is positive semidefinite, i.e. for all $r \in \mathbb{R}^n$*

$$\sum_{i,j=1}^n r_i r_j k(x_i, x_j) \geq 0, \quad (70)$$

The kernel is said to be conditionally positive definite if positivity only holds in (70) for zero-mean vectors r , i.e. such that $\langle r, \mathbb{1}_n \rangle = 0$.

The conditional version is the right notion for probability distances, because one applies the quadratic form to signed measures $\xi = \alpha - \beta$ of total mass zero. Adding terms of the form $a(x) + a(y)$ to the kernel does not change $\iint k(x, y) d\xi(x) d\xi(y)$ on such measures, and many natural distance kernels are only conditionally positive definite.

Example 6 (Riesz, energy and Matérn-type kernels). On $\mathbb{R}^d$, translation-invariant kernels are most transparent in Fourier variables. The Riesz family associated with $(-\Delta)^{-s}$ has multiplier $\|\omega\|^{-2s}$ and is positive on zero-mass measures for which the low-frequency singularity is integrable; this is the kernel counterpart of classical Riesz potentials [132, 188]. The energy distance corresponds to the conditionally positive kernel $k(x, y) = -\|x - y\|$, whose Fourier multiplier is proportional to $\|\omega\|^{-(d+1)}$; for $\xi = \alpha - \beta$,

$$-\iint \|x - y\| d\xi(x) d\xi(y)$$

is the squared energy distance up to a dimensional constant [208].

Shifted kernels replace $(-\Delta)^{-s}$ by $(-\Delta + \lambda I)^{-s}$ with $\lambda > 0$. Their Fourier multiplier $(\|\omega\|^2 + \lambda)^{-s}$ is bounded at the origin, hence the kernel is positive definite without imposing zero mass. These are the Matérn kernels; in closed form they are radial and involve a modified Bessel function [206, 216, 180]. The Laplacian kernel $e^{-\|x-y\|/\sigma}$ is a low-smoothness Matérn example, while the Gaussian kernel $e^{-\|x-y\|^2/(2\sigma^2)}$ is the infinite-smoothness limit after the usual rescaling of the Matérn smoothness parameter.

Definition 14 (Kernel norm and MMD). *If k is positive definite, or conditionally positive definite and restricted to signed measures of total mass zero, the associated kernel norm is*

$$\|\xi\|_k^2 := \int_{\mathcal{X} \times \mathcal{X}} k(x, y) d\xi(x) d\xi(y). \quad (71)$$

For two probability measures, $\|\alpha - \beta\|_k$ is the maximum mean discrepancy associated with k .

These norms are often referred to as “maximum mean discrepancy” (MMD) (see [106]) and have also been called “kernel norms” in shape analysis [100]. This expression (71) can be rephrased, introducing two independent random vectors (X, X') on $\mathcal{X}$ distributed with law α , as

$$\|\alpha\|_k^2 = \mathbb{E}_{X, X'}(k(X, X')).$$

Proposition 47 (Kernel norm as an RKHS dual norm). *Let $\mathcal{H}$ be the RKHS with reproducing kernel k , and assume the kernel mean embedding*

$$m_\xi := \int k(x, \cdot) d\xi(x)$$

is well-defined for the signed measure ξ . Then

$$\|\xi\|_k = \sup_{\|h\|_{\mathcal{H}} \leq 1} \int h(x) d\xi(x),$$

so $\|\cdot\|_k$ is the dual norm in the sense of (68) associated with the RKHS unit ball.

Proof. By the reproducing property,

$$\int h(x) d\xi(x) = \left\langle h, \int k(x, \cdot) d\xi(x) \right\rangle_{\mathcal{H}} = \langle h, m_\xi \rangle_{\mathcal{H}}.$$

Cauchy–Schwarz gives

$$\sup_{\|h\|_{\mathcal{H}} \leq 1} \int h d\xi = \|m_\xi\|_{\mathcal{H}}.$$

Finally,

$$\|m_\xi\|_{\mathcal{H}}^2 = \iint k(x, y) d\xi(x) d\xi(y),$$

which is exactly (71). □

We refer to [22, 118, 193] for more details on RKHS functional spaces.

Remark 17 (Universal kernels). According to Proposition 46, the MMD norm $\|\cdot\|_k$ metrizes weak convergence if the span of the dual ball B is dense in the space of continuous functions $\mathcal{C}(\mathcal{X})$. This means that finite sums of the form $\sum_{i=1}^n a_i k(x_i, \cdot)$ (for arbitrary choices of n and points $(x_i)_i$) are dense in $\mathcal{C}(\mathcal{X})$ for the uniform norm $\|\cdot\|_\infty$. For translation-invariant kernels over $\mathcal{X} = \mathbb{R}^d$, $k(x, y) = k_0(x - y)$, this is equivalent to having a non-vanishing Fourier transform, $\hat{k}_0(\omega) > 0$.

In the special case where α is a discrete measure, one thus has the simple expression

$$\|\alpha\|_k^2 = \sum_{i=1}^n \sum_{i'=1}^n a_i a_{i'} k_{i,i'} = \langle ka, a \rangle \quad \text{where} \quad k_{i,i'} := k(x_i, x_{i'}).$$

In particular, when $\alpha = \sum_{i=1}^n a_i \delta_{x_i}$ and $\beta = \sum_{i=1}^n b_i \delta_{x_i}$ are supported on the same set of points, $\|\alpha - \beta\|_k^2 = \langle k(a - b), a - b \rangle$, so that $\|\cdot\|_k$ is a Euclidean norm (proper if k is positive definite, degenerate otherwise if k is semidefinite) on the simplex Σ_n . To compute the discrepancy between two discrete measures, one can use

$$\|\alpha - \beta\|_k^2 = \sum_{i,i'} a_i a_{i'} k(x_i, x_{i'}) + \sum_{j,j'} b_j b_{j'} k(y_j, y_{j'}) - 2 \sum_{i,j} a_i b_j k(x_i, y_j). \quad (72)$$

7.3 φ -divergences

This subsection develops divergences based on pointwise density ratios. They are computationally simple and statistically classical, but they do not see small spatial displacements between singular measures.

We now consider a radically different class of methods to compare distributions, which are simpler to compute ($O(n)$ for discrete distributions) but never metrize weak-* convergence. Note that yet another way is possible, using Bregman divergence, which might metrize weak-* convergence when the associated entropy function is weak-* regular.

Definition 15 (Entropy function). *A function $\varphi : \mathbb{R} \rightarrow \mathbb{R} \cup \{\infty\}$ is an entropy function if it is lower semicontinuous, convex, $\text{dom } \varphi \subset [0, \infty[$, and satisfies the feasibility condition $\text{dom } \varphi \cap (0, +\infty) \neq \emptyset$. The speed of growth of φ at ∞ is described by*

$$\varphi'_\infty = \lim_{x \rightarrow +\infty} \varphi(x)/x \in \mathbb{R} \cup \{\infty\}.$$

If $\varphi'_\infty = \infty$, then φ grows faster than any linear function and φ is said to be *superlinear*. Any entropy function φ induces a φ -divergence (also known as Ciszár divergence [62, 3] or f -divergence) as follows.

Definition 16 (φ -Divergences). *Let φ be an entropy function. For $\alpha, \beta \in \mathcal{M}(\mathcal{X})$, let $\frac{d\alpha}{d\beta} \beta + \alpha^\perp$ be the Lebesgue decomposition¹ of α with respect to β . The divergence $\mathcal{D}_\varphi$ is defined by*

$$\mathcal{D}_\varphi(\alpha|\beta) := \int_{\mathcal{X}} \varphi\left(\frac{d\alpha}{d\beta}\right) d\beta + \varphi'_\infty \alpha^\perp(\mathcal{X}) \quad (73)$$

if α, β are nonnegative and ∞ otherwise.

The additional term $\varphi'_\infty \alpha^\perp(\mathcal{X})$ in (73) is the recession contribution of the perspective functional. It gives the weak-* lower-semicontinuous extension of the density-ratio integral when singular mass appears. This is essential for entropies with linear growth at infinity, such as the absolute value (77) defining the TV norm. If φ has superlinear growth, e.g. the usual entropy (76), then $\varphi'_\infty = +\infty$ so that $\mathcal{D}_\varphi(\alpha|\beta) = +\infty$ if α does not have a density with respect to β .

In the discrete setting, assuming

$$\alpha = \sum_i a_i \delta_{x_i} \quad \text{and} \quad \beta = \sum_i b_i \delta_{x_i} \quad (74)$$

are supported on the same set of n points $(x_i)_{i=1}^n \subset \mathcal{X}$, (73) defines a divergence on Σ_n

$$\mathcal{D}_\varphi(a|b) = \sum_{i \in \text{Supp}(b)} \varphi\left(\frac{a_i}{b_i}\right) b_i + \varphi'_\infty \sum_{i \notin \text{Supp}(b)} a_i, \quad (75)$$

where $\text{Supp}(b) := \{i \in \llbracket n \rrbracket ; b_i \neq 0\}$.

¹The Lebesgue decomposition theorem asserts that, given β , α admits a unique decomposition as the sum of two measures $\alpha^{\text{ac}} + \alpha^\perp$ such that α^{ac} is absolutely continuous with respect to β and $\alpha^\perp$ and β are singular.

Proposition 3 (Basic properties of φ -divergences). *If φ is an entropy function, then $\mathcal{D}_\varphi$ is jointly 1-homogeneous, convex and weak-* lower semicontinuous in (α, β) .*

Proof. One defines the associated perspective function

$$\forall (u, v) \in (\mathbb{R}_+)^2, \quad \psi(u, v) = \begin{cases} \varphi(u/v)v & \text{if } v \neq 0, \\ u\varphi'_\infty & \text{if } v = 0 \end{cases}$$

The joint 1-homogeneity follows from the definition of this perspective. We prove convexity in the discrete case, where

$$\mathcal{D}_\varphi(a|b) = \sum_i \psi(a_i, b_i),$$

and it is enough to show that ψ is convex on $(\mathbb{R}_+)^2$. We first prove this on $\mathbb{R}_+ \times \mathbb{R}_+^*$; the extension to $v = 0$ follows by lower semicontinuity of the recession value $u\varphi'_\infty$. Indeed, for any $\lambda \in [0, 1]$, $\tau = 1 - \lambda$

$$\varphi\left(\frac{\tau u_1 + \lambda u_2}{\tau v_1 + \lambda v_2}\right)(\tau v_1 + \lambda v_2) = \varphi\left(\frac{\tau v_1}{\tau v_1 + \lambda v_2} \frac{u_1}{v_1} + \frac{\lambda v_2}{\tau v_1 + \lambda v_2} \frac{u_2}{v_2}\right)(\tau v_1 + \lambda v_2) \leq \tau \varphi\left(\frac{u_1}{v_1}\right)v_1 + \lambda \varphi\left(\frac{u_2}{v_2}\right)v_2.$$

In the general measure case, weak-* lower semicontinuity is the standard lower-semicontinuity theorem for convex integral functionals with recession extension; in the discrete case it is immediate from the lower semicontinuity of ψ . $\square$

The following proposition records when $\mathcal{D}_\varphi$ is nonnegative.

Proposition 4 (Non-negativity of φ -divergences). *Assume that φ is normalized by $\varphi(1) = 0$. For probability distributions $(\alpha, \beta) \in \mathcal{M}_+^1(\mathcal{X})$, one has $\mathcal{D}_\varphi(\alpha|\beta) \geq 0$. If φ is strictly convex, then one has $\mathcal{D}_\varphi(\alpha|\beta) = 0$ if and only if $\alpha = \beta$. This property extends to arbitrary distributions $(\alpha, \beta) \in \mathcal{M}_+(\mathcal{X})$ if one furthermore imposes that $\varphi \geq 0$.*

Proof. Let $m = \alpha + \beta$ and write $a = \frac{d\alpha}{dm}$ and $b = \frac{d\beta}{dm}$. Using the perspective function ψ from the previous proof,

$$\mathcal{D}_\varphi(\alpha|\beta) = \int \psi(a, b) dm.$$

Since α and β are probabilities, $\int a dm = \int b dm = 1$. Jensen's inequality and $\psi(1, 1) = \varphi(1) = 0$ give

$$\mathcal{D}_\varphi(\alpha|\beta) \geq \psi\left(\int a dm, \int b dm\right) = 0.$$

If φ is strictly convex, equality in Jensen forces $a = b$ m -almost everywhere, hence $\alpha = \beta$. In the general non-probability case, if $\varphi \geq 0$ then the divergence is positive by construction. $\square$

Example 7 (Kullback–Leibler divergence). The Kullback–Leibler divergence $\text{KL} := \mathcal{D}_{\varphi_{\text{KL}}}$, also known as the relative entropy, was already introduced in (40) and (34). It is the divergence associated to the Shannon–Boltzmann entropy function φ_{KL} , given by

$$\varphi_{\text{KL}}(s) = \begin{cases} s \log(s) - s + 1 & \text{for } s > 0, \\ 1 & \text{for } s = 0, \\ +\infty & \text{otherwise.} \end{cases} \quad (76)$$

Example 8 (Total variation). The total variation distance $\text{TV} := \mathcal{D}_{\varphi_{\text{TV}}}$ is the divergence associated to

$$\varphi_{\text{TV}}(s) = \begin{cases} |s - 1| & \text{for } s \geq 0, \\ +\infty & \text{otherwise.} \end{cases} \quad (77)$$

It actually defines a norm on the full space of measures $\mathcal{M}(\mathcal{X})$ where

$$\text{TV}(\alpha|\beta) = \|\alpha - \beta\|_{\text{TV}}, \quad \text{where} \quad \|\alpha\|_{\text{TV}} = |\alpha|(\mathcal{X}) = \int_{\mathcal{X}} d|\alpha|(x). \quad (78)$$

If α has a density ρ_α on $\mathcal{X} = \mathbb{R}^d$, then the TV norm is the L^1 norm on functions, $\|\alpha\|_{\text{TV}} = \int_{\mathcal{X}} |\rho_\alpha(x)| dx = \|\rho_\alpha\|_{L^1}$. If α is discrete as in (74), then the TV norm is the ℓ^1 norm of vectors in $\mathbb{R}^n$, $\|\alpha\|_{\text{TV}} = \sum_i |a_i| = \|a\|_{\ell^1}$.

Remark 18 (Strong vs. weak topology). The total variation norm (78) defines the so-called “strong” topology on the space of measures. On a compact domain $\mathcal{X}$ of radius R , one has

$$\mathcal{W}_1(\alpha, \beta) \leq R \|\alpha - \beta\|_{\text{TV}}$$

so that this strong notion of convergence implies the weak convergence metrized by Wasserstein distances. The converse is, however, not true, since δ_x does not converge strongly to δ_y if $x \rightarrow y$ (note that $\|\delta_x - \delta_y\|_{\text{TV}} = 2$ if $x \neq y$). A chief advantage is that $\mathcal{M}_+^1(\mathcal{X})$ (once again on a compact ground space $\mathcal{X}$) is compact for the weak topology so that from any sequence of probability measures $(\alpha_k)_k$, one can always extract a converging subsequence, which makes it a suitable space for several optimization problems.

Main families of φ -divergences. Several classical divergences fit in the same template. The power-divergence family

$$\varphi_\gamma(s) = \frac{s^\gamma - \gamma s + \gamma - 1}{\gamma(\gamma - 1)} \quad (\gamma \neq 0, 1)$$

interpolates between the Pearson χ^2 divergence at $\gamma = 2$, the Hellinger-type behavior at $\gamma = 1/2$, and, by taking limits, the KL divergence as $\gamma \rightarrow 1$ and the reverse KL or Burg entropy $\varphi_0(s) = -\log s + s - 1$ as $\gamma \rightarrow 0$. The Hellinger divergence is often written separately with $\varphi_H(s) = (\sqrt{s} - 1)^2$, giving $\mathfrak{h}(\alpha, \beta) = \|\sqrt{\rho_\alpha} - \sqrt{\rho_\beta}\|_{L^2}$ when both measures have densities. The Jensen–Shannon divergence is the symmetrized and bounded KL-to-the-mixture divergence

$$\text{JS}(\alpha, \beta)^2 = \frac{1}{2} \text{KL}\left(\alpha \middle| \frac{\alpha + \beta}{2}\right) + \frac{1}{2} \text{KL}\left(\beta \middle| \frac{\alpha + \beta}{2}\right),$$

and is generated by a bounded entropy equivalent to $\varphi_{\text{JS}}(s) = s \log s - (s + 1) \log((s + 1)/2)$ up to an irrelevant affine term. Total variation corresponds to the non-smooth entropy $\varphi_{\text{TV}}(s) = |s - 1|$ and is exceptional because it is both a φ -divergence and an integral probability metric.

Remark 19 (φ -divergences versus Bregman divergences). Except for KL-type entropies, φ -divergences should not be confused with Bregman divergences. A φ -divergence compares measures pointwise through the density ratio $d\alpha/d\beta$ and is invariant under measurable changes of variables. A Bregman divergence is generated by a convex functional on a linear space and compares two points through first-order Taylor error. KL is special because the integral entropy $\alpha \mapsto \int \rho \log \rho$ produces a Bregman divergence whose density-ratio form is also a φ -divergence.

Proposition 5 (Dual expression). *A φ -divergence can be expressed using the Legendre transform*

$$\varphi^{*, \geq 0}(s) := \sup_{t \in \mathbb{R}^+} st - \varphi(t)$$

(notice that we restrict the function to the positive real) of φ as

$$\mathcal{D}_\varphi(\alpha|\beta) = \sup_{f: \mathcal{X} \rightarrow \mathbb{R}} \int_{\mathcal{X}} f(x) d\alpha(x) - \int_{\mathcal{X}} \varphi^{*, \geq 0}(f(x)) d\beta(x). \quad (79)$$

which equivalently reads that the Legendre transform of $\mathcal{D}_\varphi(\cdot|\beta)$ reads

$$\forall f \in \mathcal{C}(\mathcal{X}), \quad \mathcal{D}_\varphi^*(f|\beta) = \int_{\mathcal{X}} \varphi^{*, \geq 0}(f(x)) d\beta(x). \quad (80)$$

Proof. We first consider the superlinear case $\varphi'_\infty = +\infty$, so that $\mathcal{D}_\varphi(\alpha|\beta) = +\infty$ if α does not have a density $\rho \geq 0$ with respect to β , $d\alpha = \rho d\beta$. Thus the Legendre-Fenchel transform of $\mathcal{D}_\varphi(\cdot|\beta)$ reads

$$\begin{aligned}\mathcal{D}_\varphi^*(f|\beta) &= \sup_{\rho \geq 0} \int_{\mathcal{X}} f(x)\rho(x)d\beta(x) - \int_{\mathcal{X}} \varphi(\rho(x))d\beta(x) \\ &= \int_{\mathcal{X}} \sup_{\rho(x) \geq 0} (f(x)\rho(x) - \varphi(\rho(x))) d\beta(x) = \int_{\mathcal{X}} \varphi^{*, \geq 0}(f(x))d\beta(x).\end{aligned}$$

Fenchel–Moreau then gives the displayed dual expression. For a general entropy, the same argument is applied to the perspective with its recession term; the singular part is exactly encoded by the effective domain of $\varphi^{*, \geq 0}$. $\square$

7.4 GANs via Duality

GANs fit naturally into the dual viewpoint: the discriminator is a parameterized potential and the generator moves a reference measure. This subsection first explains the original divergence-based GAN objective, then contrasts it with integral probability metrics such as MMD and Wasserstein distances.

The goal is to fit a generative parametric model $\alpha_\theta = g_{\theta, \#}\zeta$ to empirical data $\beta = \frac{1}{m} \sum_j \delta_{y_j}$, where $\zeta \in \mathcal{M}_+^1(\mathcal{Z})$ is a fixed density over the latent space and $g_\theta : \mathcal{Z} \rightarrow \mathcal{X}$ is the generator, often a neural network.

Divergence-based adversarial losses. The original GAN objective [103] uses a discriminator $D_\xi : \mathcal{X} \rightarrow (0, 1)$ and solves

$$\min_{\theta} \max_{\xi} \int \log D_\xi(x) d\beta(x) + \int \log(1 - D_\xi(x)) d\alpha_\theta(x).$$

If the discriminator is optimized over all measurable functions, the optimal value is $-\log 4 + 2 \text{JS}(\beta, \alpha_\theta)^2$. Thus the idealized original GAN minimizes the Jensen–Shannon φ -divergence between the data law and the generated law.

More generally, any φ -divergence can be written in adversarial form through the dual formula (79):

$$\min_{\theta} \mathcal{D}_\varphi(\alpha_\theta|\beta) = \min_{\theta} \sup_f \int_{\mathcal{X}} f(x)d\alpha_\theta(x) - \mathcal{D}_\varphi^*(f|\beta) = \min_{\theta} \sup_f \int_{\mathcal{X}} f(g_\theta(z))d\zeta(z) - \frac{1}{m} \sum_j \varphi^*(f(y_j)).$$

Replacing the unrestricted potential f by a neural network f_ξ gives the usual parameterized discriminator. These losses estimate density ratios, which is statistically meaningful when the measures overlap but can saturate when the model and data are mutually singular; for example, the Jensen–Shannon divergence is maximal for disjoint supports.

Dual norms and integral probability metrics. Instead of a density-ratio divergence, one can minimize a dual norm (68), also called an integral probability metric,

$$\min_{\theta} \|\alpha_\theta - \beta\|_B = \min_{\theta} \sup_{f \in B} \int_{\mathcal{X}} f(x)d(\alpha_\theta - \beta)(x) = \min_{\theta} \sup_{f \in B} \int_{\mathcal{Z}} f(g_\theta(z))d\zeta - \frac{1}{m} \sum_j f(y_j).$$

MMD-GANs take B to be a unit ball in an RKHS [76]; Wasserstein GANs take B to be a Lipschitz ball, following Kantorovich–Rubinstein duality [10, 86]. The advantage of such choices is topological: for bounded continuous RKHS balls, or for bounded Lipschitz balls on compact spaces, the resulting objective is weak-* continuous. It can therefore compare singular empirical and generated measures through test functions, instead of requiring pointwise density ratios. The price is that the discriminator class must be controlled geometrically, either by a kernel norm, a Lipschitz constraint or a related regularization. For instance, Wasserstein GANs originally used weight clipping, constraining $\|\xi\|_\infty \leq 1$ as a proxy for enforcing $f_\xi \in B = \{f ; \text{Lip}(f) \leq 1\}$. This parameter set is both smaller than the true Lipschitz ball and non-convex, so clipping should be understood as a practical heuristic rather than a faithful implementation of the Kantorovich–Rubinstein dual constraint.

8 Advanced Topics on Entropic Regularization

This section collects the refinements that make entropic OT practical and statistically meaningful. It develops the dual smooth problem, revisits convergence, introduces the debiased Sinkhorn divergence and explains why regularization improves sample complexity. These topics connect algorithm design [68, 47] with modern statistical uses of regularized OT [95, 81].

8.1 Dual of Sinkhorn

The dual point of view replaces couplings by potentials and soft c -transforms. It is the right formulation for stabilized implementations and differentiation.

Discrete dual. The following proposition details the dual problem associated to (29).

Proposition 48 (Dual of entropic OT). *One has*

$$L_C^\varepsilon(a, b) = \max_{f \in \mathbb{R}^n, g \in \mathbb{R}^m} \langle f, a \rangle + \langle g, b \rangle - \varepsilon \sum_{i,j} \exp\left(\frac{f_i + g_j - C_{i,j}}{\varepsilon}\right) a_i b_j + \varepsilon. \quad (81)$$

The optimal (f, g) are linked to scalings (u, v) appearing in (30) through

$$u_i = a_i e^{f_i/\varepsilon} \quad \text{and} \quad v_j = b_j e^{g_j/\varepsilon}. \quad (82)$$

Proof. We introduce Lagrange multipliers and consider

$$\min_{P \geq 0} \max_{f, g} \langle C, P \rangle + \varepsilon \text{KL}(P | a \otimes b) + \langle a - P \mathbf{1}, f \rangle + \langle b - P^\top \mathbf{1}, g \rangle.$$

Finite-dimensional convex duality allows us to exchange the minimum over P with the maximum over (f, g) , giving

$$\max_{f, g} \langle f, a \rangle + \langle g, b \rangle + \varepsilon \min_{P \geq 0} \left(\text{KL}(P | a \otimes b) - \left\langle \frac{f \oplus g - C}{\varepsilon}, P \right\rangle \right) = \langle f, a \rangle + \langle g, b \rangle - \varepsilon \text{KL}^* \left(\frac{f \oplus g - C}{\varepsilon} | a \otimes b \right).$$

One concludes by using (80) for $\varphi(r) = r \log(r) - r + 1$

$$\text{KL}^*(H | a \otimes b) = \sum_{i,j} \varphi^*(H_{i,j}) a_i b_j.$$

Indeed, the scalar maximization

$$\varphi^*(s) = \sup_{r \geq 0} \{rs - r \log r + r - 1\}$$

has first-order condition $s - \log r = 0$, hence $r = e^s$ and $\varphi^*(s) = e^s - 1$. $\square$

Discrete soft c -transforms. Since the dual problem (81) is smooth and concave, one can perform alternating block maximization. For a fixed g , maximizing with respect to f leads to the following equation after zeroing the derivative with respect to f :

$$a_i - e^{\frac{f_i}{\varepsilon}} a_i \sum_j \exp\left(\frac{g_j - C_{i,j}}{\varepsilon}\right) b_j = 0$$

which leads to the explicit solution

$$f_i = -\varepsilon \log \sum_j \exp\left(\frac{g_j - C_{i,j}}{\varepsilon}\right) b_j.$$

We conveniently introduce the soft-min operator of some vector $h \in \mathbb{R}^m$

$$\min_b^\varepsilon(h) := -\varepsilon \log \sum_j e^{-h_j/\varepsilon} b_j$$

which is a smooth approximation of the minimum operator, and the optimal f for a fixed g is computed by a soft version of the c -transform

$$f_i = \min_b^\varepsilon(C_{i,\cdot} - g). \quad (83)$$

In a similar way, the optimal g for a fixed f is

$$g_j = \min_a^\varepsilon(C_{\cdot,j} - f). \quad (84)$$

Exponentiating these iterations recovers exactly the Sinkhorn algorithm. These iterations, however, become unstable for small ε . To apply the algorithm in this regime, one needs to stabilize it using the celebrated log-sum-exp trick. This follows from noticing that, similarly to the minimum operator, one has

$$\min_b^\varepsilon(h - \text{cst}) = \min_b^\varepsilon(h) - \text{cst}$$

and to replace the computation of $\min_b^\varepsilon(h)$ by its stabilized version (equal when using infinite precision computation) $\min_b^\varepsilon(h - \min(h)) + \min(h)$.

Continuous dual and soft-transforms. For generic (not necessarily discrete) input measures (α, β) , the dual problem (81) reads

$$\sup_{f,g \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y})} \int_{\mathcal{X}} f(x) d\alpha(x) + \int_{\mathcal{Y}} g(y) d\beta(y) - \varepsilon \int_{\mathcal{X} \times \mathcal{Y}} \left(e^{\frac{f \oplus g - c}{\varepsilon}} - 1 \right) d\alpha \otimes d\beta \quad (85)$$

This corresponds to a smoothing of the constraint $\mathcal{R}(c)$ appearing in the original problem (51), which is retrieved in the limit $\varepsilon \rightarrow 0$.

The corresponding soft c -transform, which minimizes this dual problem with respect to either f or g , reads

$$\begin{aligned} \forall y \in \mathcal{Y}, \quad f^{c,\varepsilon}(y) &:= -\varepsilon \log \left(\int_{\mathcal{X}} e^{\frac{-c(x,y) + f(x)}{\varepsilon}} d\alpha(x) \right), \\ \forall x \in \mathcal{X}, \quad g^{\bar{c},\varepsilon}(x) &:= -\varepsilon \log \left(\int_{\mathcal{Y}} e^{\frac{-c(x,y) + g(y)}{\varepsilon}} d\beta(y) \right). \end{aligned}$$

In the case of discrete measures, one retrieves the formulas (83) and (84).

The same soft-transform calculus also gives an entropic semi-discrete problem, in which stochastic optimization is applied to the remaining finite-dimensional potential.

The soft c -transform, besides being useful to define Sinkhorn's algorithm, is also important to show the existence of dual solutions.

Proposition 49 (Existence and uniqueness of entropic dual potentials). *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact and that c is continuous. The dual problem (85) has solutions, and the set of solutions is of the form $(f^* + \lambda, g^* - \lambda)$ for $\lambda \in \mathbb{R}$.*

Proof. Normalize potentials by imposing $\int f d\alpha = 0$. Replacing any pair (f, g) by the corresponding soft c -transforms does not decrease the dual objective, because each soft transform is the exact maximizer in one block variable. For transformed potentials, the oscillations are bounded by the oscillation of c :

$$\|f\|_V + \|g\|_V \leq 2(\sup c - \inf c).$$

Moreover the modulus of continuity of the soft transforms is controlled by the modulus of continuity of c ; for instance

$$|g^{\bar{c},\varepsilon}(x) - g^{\bar{c},\varepsilon}(x')| \leq \sup_y |c(x, y) - c(x', y)|.$$

After normalization, maximizing sequences are therefore uniformly bounded and equicontinuous. Arzela–Ascoli gives a uniformly converging subsequence, and continuity of the dual objective gives a maximizer. Uniqueness follows from strict convexity of $H \mapsto \int e^{H/\varepsilon} d\alpha \otimes \beta$ on the image of the map $(f, g) \mapsto H = f \oplus g - c$, modulo constants. If two optimal pairs exist, their midpoint is also optimal; strict convexity forces the two functions $f \oplus g$ to agree $\alpha \otimes \beta$ -almost everywhere. Since the potentials are continuous on compact supports, this implies $f - f'$ is constant and $g - g'$ is the opposite constant. $\square$

Remark 20 (Convexity properties). The $\log \int \exp$ operator behaves similarly to the $\max$ operator, and in particular it preserves convexity. Similarly to the unregularized case studied in Remark 16, in the particular case $c(x, y) = -\langle x, y \rangle$, the function $f^{c,\varepsilon}$ is always concave. Hence, when $c(x, y) = \|x - y\|^2/2$, the optimal potentials are of the form $f^*(x) = \|x\|^2 - \varphi^*(x)$ where φ^* is a convex function.

Remark 21 (Gaussian marginals). A useful property of these soft c -transforms in the case $c(x, y) = \|x - y\|^2$ and Gaussian marginals is that they preserve quadratic functions, since products and convolutions of Gaussian functions remain Gaussian. This implies that for Gaussian marginals, the optimal potentials (f^*, g^*) are quadratic, and that the optimal entropic-regularized coupling π^* is a Gaussian.

8.2 Gaussian Sinkhorn Problems

Gaussian marginals provide an explicit finite-dimensional model of Sinkhorn’s behavior. The soft c -transform preserves quadratic potentials, the optimal entropic coupling is Gaussian, and the value can be written using matrix square roots [120].

Proposition 50 (Quadratic closure of Sinkhorn iterates). *Let $\beta = \mathcal{N}(m_\beta, \Sigma_\beta)$ and take $c(x, y) = \|x - y\|^2$. If $g(y)$ is a quadratic polynomial such that the Gaussian integral below is finite, then the soft transform*

$$f(x) = -\varepsilon \log \int \exp\left(\frac{g(y) - \|x - y\|^2}{\varepsilon}\right) d\beta(y)$$

is a quadratic polynomial in x . In particular, starting Sinkhorn from $g_0 = 0$ gives

$$f_1(x) = \frac{\varepsilon}{2} \log \det\left(I + \frac{2\Sigma_\beta}{\varepsilon}\right) + \varepsilon \langle x - m_\beta, (\varepsilon I + 2\Sigma_\beta)^{-1}(x - m_\beta) \rangle.$$

Proof. The exponent is the sum of a quadratic polynomial in y and the logarithm of the Gaussian density of β . Completing the square in y evaluates the integral as a constant times the exponential of a quadratic polynomial in x . For $g_0 = 0$, the standard Gaussian identity

$$\mathcal{E} \exp\left(-\frac{\|x - Y\|^2}{\varepsilon}\right) = \det\left(I + \frac{2\Sigma_\beta}{\varepsilon}\right)^{-1/2} \exp(-\langle x - m_\beta, (\varepsilon I + 2\Sigma_\beta)^{-1}(x - m_\beta) \rangle)$$

with $Y \sim \beta$ gives the displayed expression. $\square$

Proposition 51 (Balanced entropic OT between Gaussians). *Let $\alpha = \mathcal{N}(m_\alpha, A)$ and $\beta = \mathcal{N}(m_\beta, B)$ with A, B positive definite, and let*

$$A^{1/2}B^{1/2} = U \operatorname{diag}(\sigma_i) V^\top$$

be a singular-value decomposition. For the balanced objective

$$\min_{\pi \in \mathcal{U}(\alpha, \beta)} \int \|x - y\|^2 d\pi(x, y) + \varepsilon \operatorname{KL}(\pi | \alpha \otimes \beta),$$

the optimizer is Gaussian with cross-covariance

$$K_\varepsilon = A^{1/2} U \operatorname{diag}(s_i) V^\top B^{1/2}, \quad s_i = \frac{\sqrt{\varepsilon^2 + 16\sigma_i^2} - \varepsilon}{4\sigma_i}.$$

The optimal value is

$$\|m_\alpha - m_\beta\|^2 + \operatorname{tr}(A) + \operatorname{tr}(B) + \sum_i \left(-2\sigma_i s_i - \frac{\varepsilon}{2} \log(1 - s_i^2) \right).$$

As $\varepsilon \downarrow 0$, $s_i \rightarrow 1$ and the covariance term converges to the Bures term in Proposition 12.

Proof. Let (X, Y) be any coupling with finite second moments and cross-covariance

$$K = \mathbb{E}[(X - m_\alpha)(Y - m_\beta)^\top].$$

Replacing (X, Y) by the Gaussian vector with the same mean and covariance leaves the quadratic cost unchanged. Since the marginals are fixed,

$$\operatorname{KL}(\pi|\alpha \otimes \beta) = -h(X, Y) + h(\alpha) + h(\beta),$$

where h denotes differential entropy whenever the entropy is finite. Among laws with fixed covariance, the Gaussian maximizes $h(X, Y)$; if the entropy is not finite, the relative entropy is already $+\infty$. Thus the Gaussian replacement cannot increase the objective, and it is enough to optimize over Gaussian couplings. Any such coupling has covariance

$$\begin{pmatrix} A & K \\ K^\top & B \end{pmatrix}.$$

The cost depends on K through $\operatorname{tr}(A) + \operatorname{tr}(B) - 2\operatorname{tr} K$, while

$$\operatorname{KL}(\pi|\alpha \otimes \beta) = -\frac{1}{2} \log \det(I - SS^\top), \quad K = A^{1/2} S B^{1/2}.$$

The block covariance constraint is equivalent to the singular values of S being at most one; finite entropy forces them to be strictly smaller than one. By von Neumann's trace inequality, the minimizer aligns S with the singular vectors of $A^{1/2} B^{1/2}$, so $S = U \operatorname{diag}(s_i) V^\top$. The problem separates into scalar minimizations

$$\min_{0 \leq s < 1} -2\sigma_i s - \frac{\varepsilon}{2} \log(1 - s^2).$$

The first-order condition is $2\sigma_i = \varepsilon s / (1 - s^2)$, whose positive solution is the displayed s_i . Substitution gives the value formula. $\square$

Remark 22 (Gaussian rates). Because the Sinkhorn iterates remain quadratic, Gaussian Sinkhorn reduces to a finite-dimensional nonlinear iteration on means and covariance parameters. In diagonal coordinates the covariance part becomes a collection of scalar contractions for the correlations s_i . The sharper continuous convergence estimates of Chizat, Delalande and Vaškevičius [58] show that the exponential rate is governed by the projective contraction of these Gaussian soft transforms: it improves when ε is large or the covariance scales overlap well, and deteriorates in the small-temperature limit where the entropic coupling approaches a deterministic Brenier map.

8.3 Convergence Analysis

This subsection revisits Sinkhorn convergence through three complementary lenses. Hilbert's metric gives a clean contraction, Fortet's order argument gives qualitative fixed-point convergence, and robust Bregman estimates give a non-asymptotic $O(1/k)$ dual-gap bound. The first two viewpoints explain asymptotic convergence mechanisms, while the last one is often the most useful for explicit complexity estimates.

Hilbert-metric linear convergence. The Hilbert metric convergence result is equivalent to convergence on the dual potentials (f, g) according to the variation norms. For bounded cost c (e.g. on compact spaces),

$$\|f_k - f^*\|_V = O(\lambda^k) \quad \text{and} \quad \|g_k - g^*\|_V = O(\lambda^k)$$

$$\left\| \log \frac{d\pi_k}{d\pi^*} \right\|_\infty = \|(f_k - f^*) \oplus (g_k - g^*)\|_\infty \leq \|f_k - f^*\|_V + \|g_k - g^*\|_V$$

where the contraction ratio is the Birkhoff factor of the Gibbs kernel $K_\varepsilon = e^{-c/\varepsilon}$. Namely, with $\eta = \eta(K_\varepsilon)$ as in Theorem 4 and $R := \sup c - \inf c$,

$$\lambda = \frac{\sqrt{\eta} - 1}{\sqrt{\eta} + 1} \leq \tanh(R/(2\varepsilon)) < 1.$$

One also has the following bounds

$$\|f_k - f^*\|_V \leq \frac{\left\| \log \frac{d\pi_{k,1}}{d\alpha} \right\|_\infty}{1 - \lambda}$$

which can be used to provide a posterior estimate of the rate of convergence and serve as a stopping criterion.

Order-monotone convergence. There is another, older way to understand convergence, going back to Fortet's proof of the Schrödinger system [83, 79, 138]. It does not primarily estimate a contraction factor. Instead, it uses the order structure of the soft transforms.

Proposition 52 (Monotone fixed-point route to Sinkhorn convergence). *Let c be bounded and continuous on compact spaces, and define the double Sinkhorn map, normalized by subtracting its α -mean,*

$$\mathcal{A}(f) := (f^{c,\varepsilon})^{\bar{c},\varepsilon} - \int (f^{c,\varepsilon})^{\bar{c},\varepsilon} d\alpha.$$

The map $\mathcal{A}$ is order preserving modulo additive constants. If an initialization is chosen so that $f_0 \leq \mathcal{A}(f_0)$ after normalization, then the iterates $f_{k+1} = \mathcal{A}(f_k)$ increase pointwise up to constants, remain uniformly bounded in oscillation, and converge to a fixed point. A convenient practical choice is $f_0 = 0$, followed if necessary by subtracting a sufficiently large constant so that the subsolution inequality holds. This fixed point is the entropic potential, hence the associated Sinkhorn scalings converge.

Proof. The soft c -transform is order reversing: if $g \leq g'$, then

$$-\varepsilon \log \int e^{(g-c)/\varepsilon} d\beta \geq -\varepsilon \log \int e^{(g'-c)/\varepsilon} d\beta.$$

The composition of two order-reversing transforms is therefore order preserving. The transform also commutes with additive constants in the projective sense, which is why one fixes a normalization. Starting from a subsolution gives $f_0 \leq f_1$, and order preservation gives $f_k \leq f_{k+1}$ for all k . Soft-transform oscillation bounds, controlled by $\sup c - \inf c$, prevent escape to infinity after normalization. Monotone pointwise convergence, compactness of equicontinuous soft transforms, and continuity of $\mathcal{A}$ then give a fixed point. Uniqueness of entropic potentials up to constants, Proposition 26 and the dual uniqueness statement above identify this fixed point with the Sinkhorn solution. $\square$

This proof is qualitative rather than quantitative, but it is conceptually useful: Sinkhorn is not only alternating projection or projective contraction; it is also a monotone fixed-point iteration on potentials once constants are quotiented out.

Robust sublinear Bregman rate. Sinkhorn is cyclic coordinate ascent on the smooth dual problem and, equivalently, alternating KL projection on the two marginal constraint sets. We denote the dual objective in (85) by

$$\mathcal{D}_\varepsilon(f, g) := \int f d\alpha + \int g d\beta - \varepsilon \int \left(e^{(f \oplus g - c)/\varepsilon} - 1 \right) d\alpha \otimes d\beta.$$

The following statement records the rate that is most useful for complexity estimates: before entering a possible linear regime, the dual objective gap decreases at order $1/k$. We state it for balanced entropic OT, which is the specialization needed here; related robust Bregman-projection rates are developed in [168, 4, 75], and statistical consequences of entropic smoothing are analyzed in [95, 28].

Proposition 53 (Pinsker inequality). *If $p, q \in \Sigma_n$, then*

$$\|p - q\|_1^2 \leq 2 \text{KL}(p|q).$$

Proof. It is enough to prove the scalar inequality

$$u \log u - u + 1 \geq \frac{(u-1)^2}{2(u+1)} \quad u \geq 0.$$

Indeed, after bringing the right-hand side to the left, the resulting function h satisfies $h(1) = h'(1) = 0$ and, for $u > 0$,

$$h''(u) = \frac{1}{u} - \frac{4}{(u+1)^3} = \frac{(u+1)^3 - 4u}{u(u+1)^3} \geq 0,$$

so h is convex and minimized at $u = 1$. Multiplying by q_i and summing with $u = p_i/q_i$ gives, first when all $q_i > 0$ and then by approximation in general,

$$\text{KL}(p|q) \geq \frac{1}{2} \sum_i \frac{(p_i - q_i)^2}{p_i + q_i}.$$

Cauchy–Schwarz yields

$$\|p - q\|_1^2 = \left(\sum_i \frac{|p_i - q_i|}{\sqrt{p_i + q_i}} \sqrt{p_i + q_i} \right)^2 \leq \left(\sum_i \frac{(p_i - q_i)^2}{p_i + q_i} \right) \left(\sum_i (p_i + q_i) \right),$$

and $\sum_i (p_i + q_i) = 2$. □

Proposition 54 (A compact $O(1/k)$ dual rate). *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact, that c is bounded, and write $R = \sup c - \inf c$. Let (f_k, g_k) be Sinkhorn dual iterates normalized by $\int f_k d\alpha = 0$, and let*

$$\Delta_k := \mathcal{D}_\varepsilon(f^*, g^*) - \mathcal{D}_\varepsilon(f_k, g_k)$$

be the dual suboptimality gap for the entropic dual objective (85). Then there exists a numerical constant C such that

$$\Delta_k \leq \frac{CR^2}{\varepsilon(k+1)}.$$

Proof. The proof uses three elementary ingredients. First, the soft c -transform bounds the oscillation of every normalized iterate and every normalized optimum:

$$\|f_k\|_V + \|g_k\|_V + \|f^*\|_V + \|g^*\|_V \leq CR.$$

Here $\|h\|_V := \sup h - \inf h$ is the variation seminorm, the same projective norm used in Hilbert’s metric in Section 4.6. It is natural because dual potentials can be shifted by constants without changing the coupling.

Second, each Sinkhorn half-step is an exact KL projection. The Pythagorean identity for KL projections gives the ascent identity

$$\mathcal{D}_\varepsilon(f_{k+1}, g_{k+1}) - \mathcal{D}_\varepsilon(f_k, g_k) = \varepsilon [\text{KL}(\pi^\star | \pi_k) - \text{KL}(\pi^\star | \pi_{k+1})],$$

where $\pi_k = e^{(f_k \oplus g_k - c)/\varepsilon} \alpha \otimes \beta$ and $\pi^\star$ is the optimal entropic coupling. The KL drop controls the marginal residuals through Pinsker's inequality, Proposition 53. Third, convexity of the exponential dual objective gives a one-step estimate of the form

$$\Delta_k^2 \leq \frac{CR^2}{\varepsilon} (\mathcal{D}_\varepsilon(f_{k+1}, g_{k+1}) - \mathcal{D}_\varepsilon(f_k, g_k)).$$

This is the usual Bregman-projection estimate: the dual gap is controlled by the product of a bounded dual radius and the marginal residual corrected by the next projection, and the residual squared is controlled by the KL drop. Summing the reciprocal inequality obtained from the last display yields

$$\frac{1}{\Delta_{k+1}} - \frac{1}{\Delta_k} \geq \frac{\varepsilon}{CR^2},$$

and therefore $\Delta_k \leq CR^2/(\varepsilon(k+1))$. $\square$

Corollary 4 (Approximating unregularized OT by regularized dual costs). *Consider discrete histograms $\mathbf{a} \in \Sigma_n$, $\mathbf{b} \in \Sigma_m$ and a finite cost matrix C . Let $\mathcal{D}_{\varepsilon,k}$ be the KL-normalized entropic dual value after k Sinkhorn cycles, and let Δ_k be its dual gap. Define the entropy-corrected lower bound*

$$L_{\varepsilon,k} := \mathcal{D}_{\varepsilon,k} - \varepsilon H(\mathbf{a}) - \varepsilon H(\mathbf{b}), \quad H(\mathbf{a}) = - \sum_i a_i \log a_i.$$

Then

$$0 \leq L_C(\mathbf{a}, \mathbf{b}) - L_{\varepsilon,k} \leq \varepsilon \log(nm) + \Delta_k.$$

Consequently, choosing $\varepsilon \leq \delta/(2 \log(nm))$ and running Sinkhorn until $\Delta_k \leq \delta/2$ gives a δ -accurate lower bound on the unregularized OT value. Under Proposition 54, it is sufficient to take

$$k+1 \geq \frac{2CR^2}{\varepsilon\delta},$$

hence $k = O(R^2 \log(nm)/\delta^2)$ with the above choice of ε , up to constants and logarithmic stabilization factors.

Proof. The KL-normalized objective differs from the entropy convention (29) by the constant $\varepsilon H(\mathbf{a}) + \varepsilon H(\mathbf{b})$ on the transport polytope, because

$$\text{KL}(\mathbf{P} | \mathbf{a} \otimes \mathbf{b}) = -H(\mathbf{P}) + H(\mathbf{a}) + H(\mathbf{b}).$$

Let E_ε be the optimum of the entropy-regularized objective $\langle \mathbf{P}, C \rangle - \varepsilon H(\mathbf{P})$. Since $0 \leq H(\mathbf{P}) \leq \log(nm)$ for any coupling matrix,

$$L_C(\mathbf{a}, \mathbf{b}) - \varepsilon \log(nm) \leq E_\varepsilon \leq L_C(\mathbf{a}, \mathbf{b}).$$

The corrected iterate satisfies $L_{\varepsilon,k} = E_\varepsilon - \Delta_k$, which gives the displayed value bound. The final iteration estimate follows by combining $\Delta_k \leq CR^2/(\varepsilon(k+1))$ with the target $\Delta_k \leq \delta/2$. $\square$

The same identity also yields computable stopping diagnostics. The KL drops are exactly marginal defects: after a row update the row marginal is correct and the remaining drop is measured by the column marginal, and conversely after a column update. Thus marginal violations monitor both feasibility and the remaining dual gap, up to the bounded-radius constant above.

8.4 Sinkhorn Divergences

Sinkhorn divergences remove the entropic self-bias while retaining smoothness. They interpolate between OT-like geometry and kernel-like norms, which explains their statistical behavior.

Entropic bias. A major issue with the value of the Sinkhorn problem (39) is that $\mathcal{L}_c^\varepsilon(\alpha, \beta) > 0$. In particular,

$$\alpha_\varepsilon = \operatorname{argmin}_{\beta} \mathcal{L}_c^\varepsilon(\alpha, \beta)$$

does not satisfy $\alpha_\varepsilon = \alpha$ unless $\varepsilon = 0$. The following proposition shows that the bias induced by this entropic regularization dominates the large ε limit.

Proposition 55 (Large-temperature entropic bias). *Assume that c is bounded and continuous. Then $\mathcal{L}_c^\varepsilon(\alpha, \beta) \rightarrow \iint c(x, y) d\alpha(x) d\beta(y)$ as $\varepsilon \rightarrow +\infty$.*

Proof. Let $(f_\varepsilon, g_\varepsilon)$ be optimal dual potentials, normalized by $\int g_\varepsilon d\beta = 0$. The soft c -transform equation gives

$$f_\varepsilon(x) = -\varepsilon \log \int \exp\left(\frac{g_\varepsilon(y) - c(x, y)}{\varepsilon}\right) d\beta(y).$$

For bounded c , the oscillations of normalized entropic potentials are bounded uniformly in ε by the oscillation of c . Hence the log-sum-exp expansion is uniform:

$$f_\varepsilon(x) = - \int (g_\varepsilon(y) - c(x, y)) d\beta(y) + O(\varepsilon^{-1}) = \int c(x, y) d\beta(y) + O(\varepsilon^{-1}).$$

At optimality the exponential penalty in the dual integrates to zero, so

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \int f_\varepsilon d\alpha + \int g_\varepsilon d\beta = \iint c(x, y) d\alpha(x) d\beta(y) + O(\varepsilon^{-1}).$$

This proves the limit. □

So in the large ε limit, $\mathcal{L}_c^\varepsilon$ behaves like an inner product and not like a norm. For instance, in the case

$$\alpha_\varepsilon \rightarrow \min_{\beta} \langle \int c(x, \cdot) d\alpha(x), \beta \rangle = \delta_{y^*(\alpha)} \quad \text{where} \quad y^*(\alpha) = \operatorname{argmin}_y \int c(x, y) d\alpha(x).$$

For instance, when $c(x, y) = \|x - y\|^2$ then α_ε collapses towards a Dirac located at the mean $\int x d\alpha(x)$ of α .

Sinkhorn divergences. The usual way to go from an inner product to a norm is to use the polarization formula. We thus define the debiased Sinkhorn divergence

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) := \mathcal{L}_c^\varepsilon(\alpha, \beta) - \frac{1}{2} \mathcal{L}_c^\varepsilon(\alpha, \alpha) - \frac{1}{2} \mathcal{L}_c^\varepsilon(\beta, \beta). \quad (86)$$

Although this formula is a debiasing by self-costs, its non-negativity is not automatic from the definition; Proposition 57 proves it below by a kernel Cauchy–Schwarz argument.

We first record a fundamental lemma: at an optimal dual pair, the exponential regularization term integrates to zero, so the entropic cost can be read directly from the potentials. The same cancellation holds along Sinkhorn iterations after each exact block update.

Lemma 3 (Entropic dual cost at optimum). *Let $(f_{\alpha, \beta}, g_{\alpha, \beta})$ be optimal dual potentials, normalized arbitrarily. One has*

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \langle f_{\alpha, \beta}, \alpha \rangle + \langle g_{\alpha, \beta}, \beta \rangle. \quad (87)$$

Proof. We first notice that at optimality, the relation

$$f_{\alpha,\beta} = -\varepsilon \log \int_{\mathcal{Y}} e^{\frac{g_{\alpha,\beta}(y) - c(x,y)}{\varepsilon}} d\beta(y)$$

after taking the exponential, equivalently reads

$$1 = \int_{\mathcal{Y}} e^{\frac{f_{\alpha,\beta}(x) + g_{\alpha,\beta}(y) - c(x,y)}{\varepsilon}} d\beta(y) \implies \int_{\mathcal{X} \times \mathcal{Y}} \left(e^{\frac{f_{\alpha,\beta} \oplus g_{\alpha,\beta} - c}{\varepsilon}} - 1 \right) d(\alpha \otimes \beta) = 0.$$

Plugging this in formula (85), one obtains the result. $\square$

We first record the two limiting regimes of this debiased quantity.

Proposition 56 (Asymptotics of Sinkhorn divergences). *Assume that the two measures are supported on the same space and that c is bounded, continuous, nonnegative and satisfies $c(x, x) = 0$. One has $\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \rightarrow \mathcal{L}_c(\alpha, \beta)$ when $\varepsilon \rightarrow 0$ and*

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \rightarrow \frac{1}{2} \int -cd(\alpha - \beta) \otimes d(\alpha - \beta) \quad \text{when } \varepsilon \rightarrow +\infty.$$

Proof. For discrete measures, the convergence is already proved in (37); we now give a general treatment. **Case $\varepsilon \rightarrow 0$.** The first limit follows from the standard Γ -convergence argument for entropic optimal transport: the entropy term is lower semicontinuous along weakly converging couplings, while any finite-cost coupling can be approximated by couplings with finite entropy. Since $c(x, x) = 0$, the two self-costs in the debiased expression converge to zero, and the cross term converges to $\mathcal{L}_c(\alpha, \beta)$.

Case $\varepsilon \rightarrow +\infty$. We denote by $(f_\varepsilon, g_\varepsilon)$ optimal dual potentials. After normalizing them and using boundedness of c , their oscillations stay uniformly bounded, so the following expansion is uniform. The optimality condition on f_ε (equivalently the Sinkhorn fixed point on f_ε) reads

$$\begin{aligned} f_\varepsilon &= -\varepsilon \log \int \exp \left(\frac{g_\varepsilon(y) - c(\cdot, y)}{\varepsilon} \right) d\beta(y) = -\varepsilon \log \int \left(1 + \frac{g_\varepsilon(y) - c(\cdot, y)}{\varepsilon} + o(1/\varepsilon) \right) d\beta(y) \\ &= -\varepsilon \log \left(1 + \frac{1}{\varepsilon} \int (g_\varepsilon(y) - c(\cdot, y)) d\beta(y) + o(1/\varepsilon) \right) = - \int g_\varepsilon d\beta + \int c(\cdot, y) d\beta(y) + o(1). \end{aligned}$$

Plugging this relation in the dual expression (87)

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \int f_\varepsilon d\alpha + \int g_\varepsilon d\beta = \iint c(x, y) d\alpha(x) d\beta(y) + o(1).$$

Applying this expansion to (α, β) , (α, α) and (β, β) gives

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \rightarrow \int c d\alpha \otimes d\beta - \frac{1}{2} \int c d\alpha \otimes d\alpha - \frac{1}{2} \int c d\beta \otimes d\beta = -\frac{1}{2} \int c d(\alpha - \beta) \otimes d(\alpha - \beta).$$

$\square$

In the case where $-c$ defines a conditionally positive definite kernel, $\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta)$ converges to the square of a Hilbertian kernel norm. A typical example is $c(x, y) = \|x - y\|^p$ for $0 < p < 2$, which corresponds to the energy-distance kernel. This kernel norm is the dual of a homogeneous Sobolev norm.

We now show that this debiased Sinkhorn divergence is positive.

Proposition 57 (Non-negativity of Sinkhorn divergences). *If $k(x, y) = e^{-c(x, y)/\varepsilon}$ is positive definite, then $\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \geq 0$.*

Proof. In the following, we denote by $(f_{\alpha,\beta}, g_{\alpha,\beta})$ optimal dual potentials for the dual Schrödinger problem between α and β . We denote by $f_{\alpha,\alpha} = g_{\alpha,\alpha}$ (one can assume they are equal by symmetry) the solution for the problem between α and itself. Using the suboptimal function $(f_{\alpha,\alpha}, g_{\beta,\beta})$ in the dual maximization problem, and using relation (87) for the simplified expression of the dual cost, one obtains

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) \geq \langle f_{\alpha,\alpha}, \alpha \rangle + \langle g_{\beta,\beta}, \beta \rangle - \varepsilon \langle e^{\frac{f_{\alpha,\alpha} \oplus g_{\beta,\beta} - c}{\varepsilon}} - 1, \alpha \otimes \beta \rangle$$

But one has $\langle f_{\alpha,\alpha}, \alpha \rangle = \frac{1}{2} \mathcal{L}_c^\varepsilon(\alpha, \alpha)$, and similarly for β , so that the previous inequality equivalently reads

$$\frac{1}{\varepsilon} \tilde{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \geq 1 - \langle e^{\frac{f_{\alpha,\alpha} \oplus g_{\beta,\beta} - c}{\varepsilon}}, \alpha \otimes \beta \rangle = 1 - \langle \tilde{\alpha}, \tilde{\beta} \rangle_k$$

where $\tilde{\alpha} = e^{f_{\alpha,\alpha}/\varepsilon} \alpha$, $\tilde{\beta} = e^{g_{\beta,\beta}/\varepsilon} \beta$ and we introduced the inner product (which is valid because k is positive definite) $\langle \tilde{\alpha}, \tilde{\beta} \rangle_k := \int k(x, y) d\tilde{\alpha}(x) d\tilde{\beta}(y)$. One notes that the Sinkhorn fixed point equation, once exponentiated, reads $e^{f_{\alpha,\alpha}/\varepsilon} \odot [k(\tilde{\alpha})] = 1$ and hence

$$\|\tilde{\alpha}\|_k^2 = \langle k(\tilde{\alpha}), \tilde{\alpha} \rangle = \langle e^{f_{\alpha,\alpha}/\varepsilon} \odot k(\tilde{\alpha}), \alpha \rangle = \langle 1, \alpha \rangle = 1$$

and similarly $\|\tilde{\beta}\|_k^2 = 1$. Therefore, by Cauchy–Schwarz, one has $1 - \langle \tilde{\alpha}, \tilde{\beta} \rangle_k \geq 0$. $\square$

Remark 23 (Strict positivity). Under additional assumptions on the kernel, one can furthermore show that $\tilde{\mathcal{L}}_c^\varepsilon(\alpha, \beta) = 0$ implies $\alpha = \beta$, and that this debiased divergence metrizes convergence in law.

8.5 Sample Complexity

This subsection explains why regularization can improve statistical rates. The message is that a biased but smooth distance may be preferable to exact OT in high dimension.

The sample complexity of unregularized OT suffers from the curse of dimensionality. Entropic regularization changes this picture: for a fixed $\varepsilon > 0$, Sinkhorn divergences have parametric $n^{-1/2}$ statistical rates, although the constant deteriorates when $\varepsilon \rightarrow 0$ [95, 28]. Related two-sample-testing viewpoints are developed in [178], and the large- ε kernel limit connects to classical MMD tests [107]. This improvement should be balanced against the regularization bias, which vanishes only when ε is sent to zero.

Proposition 58 (Empirical OT has dimension-dependent rates). *Let α be a probability distribution with a density bounded above and below on $[0, 1]^d$, and let $\hat{\alpha}_n = n^{-1} \sum_{i=1}^n \delta_{X_i}$ be its empirical measure. For $d > 2p$, the typical scale of empirical OT is*

$$\mathbb{E} \mathcal{W}_p(\hat{\alpha}_n, \alpha) \asymp n^{-1/d}.$$

The exponent changes in low dimension, but the important message is that exact OT deteriorates with the ambient dimension.

Proof. We recall the standard multiscale argument, suppressing constants [73, 84, 214]. For the upper bound, partition $[0, 1]^d$ into dyadic cubes. At scale 2^{-j} , the empirical mass fluctuation over the cells is of order $n^{-1/2} 2^{jd/2}$, while moving this excess mass inside cells costs 2^{-j} . Summing the multiscale contributions up to the scale where the expected number of samples per cell is of order one gives 2^{-J} with $2^{Jd} \simeq n$, hence $n^{-1/d}$. For the lower bound, a packing argument shows that a positive fraction of cells at the critical scale are empty or overfull with non-negligible probability, forcing mass to travel at least a distance comparable to $n^{-1/d}$. The matching upper and lower bounds give the displayed rate. $\square$

Proposition 59 (MMD has a parametric empirical rate). *Let k be a bounded positive definite kernel with RKHS $\mathcal{H}_k$, and define*

$$\text{MMD}_k(\alpha, \beta) = \left\| \int k(x, \cdot) d(\alpha - \beta)(x) \right\|_{\mathcal{H}_k}.$$

If $\hat{\alpha}_n$ is the empirical measure of n i.i.d. samples from α , then

$$\mathbb{E} \text{MMD}_k(\hat{\alpha}_n, \alpha)^2 = \frac{1}{n} \left(\mathbb{E} k(X, X) - \mathbb{E} k(X, X') \right),$$

where X, X' are independent with law α . In particular, if $k(x, x) \leq \kappa^2$, then $\mathbb{E} \text{MMD}_k(\hat{\alpha}_n, \alpha) \leq \kappa/\sqrt{n}$.

Proof. Let $\Phi(x) = k(x, \cdot)$ be the feature map and $m_\alpha = \mathbb{E}\Phi(X)$. Then

$$\text{MMD}_k(\hat{\alpha}_n, \alpha) = \left\| \frac{1}{n} \sum_{i=1}^n (\Phi(X_i) - m_\alpha) \right\|_{\mathcal{H}_k}.$$

Independence cancels the cross terms after taking the squared norm and expectation, giving

$$\mathbb{E} \text{MMD}_k(\hat{\alpha}_n, \alpha)^2 = \frac{1}{n} \mathbb{E} \|\Phi(X) - m_\alpha\|_{\mathcal{H}_k}^2 = \frac{1}{n} \left(\mathbb{E} k(X, X) - \mathbb{E} k(X, X') \right).$$

The last bound follows from Jensen's inequality and $k(x, x) \leq \kappa^2$. $\square$

Proposition 60 (Sinkhorn divergences interpolate the rates). *Assume that α and β are supported in a compact subset of $\mathbb{R}^d$ and that the cost is smooth. For fixed $\varepsilon > 0$, debiased Sinkhorn divergences satisfy representative empirical bounds of the form*

$$\mathbb{E} \left| \bar{\mathcal{L}}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_m) - \bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \right| \leq C_{c,d} \varepsilon^{-d/2} \left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}} \right),$$

up to constants and exponents depending on the precise smoothness class and support diameter. Thus regularization removes the $n^{-1/d}$ curse for fixed ε , while the prefactor deteriorates as $\varepsilon \rightarrow 0$.

Proof. The proof follows the empirical-process argument of [95]. By the envelope theorem, the fluctuation of $\mathcal{L}_c^\varepsilon$ with respect to its first marginal is controlled by the class of entropic dual potentials. The soft c -transform smooths these potentials at spatial scale $\sqrt{\varepsilon}$ for a quadratic-type cost. Covering a bounded d -dimensional domain at this scale gives an effective complexity of order $\varepsilon^{-d/2}$. Standard Rademacher or Dudley entropy bounds then give an empirical-process fluctuation of order $\varepsilon^{-d/2}/\sqrt{n}$ for each marginal. Applying the same estimate to the three terms defining the debiased divergence gives the stated bound. $\square$

Remark 24 (No free lunch when approximating exact OT). The parametric rate in Proposition 60 holds for fixed ε . If the goal is to approximate the unregularized OT value, one must also account for the regularization bias. In a typical bounded-cost finite-dimensional regime,

$$\left| \bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) - \mathcal{L}_c(\alpha, \beta) \right| \leq C\varepsilon, \quad \mathbb{E} \left| \bar{\mathcal{L}}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_n) - \bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \right| \leq C_{c,d} \varepsilon^{-d/2} n^{-1/2}.$$

Balancing the two terms gives $\varepsilon \simeq n^{-1/(d+2)}$ and total error of order $n^{-1/(d+2)}$. Equivalently, target accuracy η requires choosing $\varepsilon \simeq \eta$ and $n \simeq \eta^{-(d+2)}$ samples under this bound. Thus entropic smoothing improves the statistical behavior at fixed scale, but approximating exact OT still forces a bias-variance tradeoff whose exponent deteriorates with dimension.

9 Extensions

This section surveys several directions where the basic two-marginal, mass-preserving OT model is too restrictive. The common theme is to keep the transport geometry while changing what is compared: unequal masses, several marginals, projected views, unknown correspondences or matrix-valued objects. Each extension also introduces a new computational compromise.

9.1 Unbalanced OT

Unbalanced OT allows mass creation and destruction by penalizing marginal mismatch. It is essential when histograms are not normalized or when only part of the source should match the target.

Relaxed formulation We consider the following “unbalanced” generalization of OT for $(\alpha, \beta) \in \mathcal{M}_+^1(\mathcal{X}) \times \mathcal{M}_+^1(\mathcal{Y})$

$$\text{UW}_c(\alpha, \beta) = \inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) + \mathcal{D}_{\psi_1}(\pi_1 | \alpha) + \mathcal{D}_{\psi_2}(\pi_2 | \beta) \quad (88)$$

for convex entropy functions ψ_1, ψ_2 . This relaxes the exact conservation of mass $(\pi_1, \pi_2) = (\alpha, \beta)$ by penalizing marginal mismatch with divergence functionals. It is useful to expose the scale of this relaxation by writing $\psi_s = \tau \bar{\psi}_s$ with $\tau > 0$:

$$\text{UW}_{c, \tau}(\alpha, \beta) = \inf_{\pi \geq 0} \int c d\pi + \tau \mathcal{D}_{\bar{\psi}_1}(\pi_1 | \alpha) + \tau \mathcal{D}_{\bar{\psi}_2}(\pi_2 | \beta).$$

Large τ enforces mass conservation and formally recovers balanced OT. Small τ makes creation and destruction cheap; after rescaling by τ , the zero-transport part reveals the pure divergence geometry. The following statement makes this precise on a common compact space, where zero cost forces the two endpoints to coincide.

Proposition 61 (Small-transport-scale limit for marginal penalties). *Assume that α, β are finite measures on a compact metric space $\mathcal{X}$, that c is continuous, $c \geq 0$, and $c(x, y) = 0$ if and only if $x = y$. Assume also that the marginal divergences are non-negative, weak-* lower semicontinuous, and have weak-* compact sublevel sets on $\mathcal{M}_+(\mathcal{X})$. Then*

$$\lim_{\tau \downarrow 0} \frac{1}{\tau} \text{UW}_{c, \tau}(\alpha, \beta) = \inf_{\rho \in \mathcal{M}_+(\mathcal{X})} \mathcal{D}_{\bar{\psi}_1}(\rho | \alpha) + \mathcal{D}_{\bar{\psi}_2}(\rho | \beta).$$

The right-hand side is the infimal gluing divergence obtained by matching the two measures through a common zero-transport marginal ρ . In particular, when $\bar{\psi}_1 = \bar{\psi}_2 = \text{KL}$,

$$\inf_{\rho \in \mathcal{M}_+(\mathcal{X})} \text{KL}(\rho | \alpha) + \text{KL}(\rho | \beta) = \int (\sqrt{\rho_\alpha} - \sqrt{\rho_\beta})^2 d\lambda,$$

when α, β have densities with respect to a common reference measure λ . Thus the KL marginal relaxation contains the squared Hellinger distance as its local mass-variation limit.

Proof. For the upper bound, restrict to diagonal plans $\pi = (\text{Id}, \text{Id})_\# \rho$, whose transport cost is zero and whose two marginals are both ρ . This gives the desired upper bound after optimizing over ρ .

For the lower bound, let $\tau_n \downarrow 0$ and let π_n be almost minimizing plans with bounded scaled values

$$\frac{1}{\tau_n} \text{UW}_{c, \tau_n}(\alpha, \beta).$$

Since the divergences are non-negative, $\int c d\pi_n = O(\tau_n)$, hence $\int c d\pi_n \rightarrow 0$. The bounded scaled values also put the two marginals in compact divergence sublevel sets. Since a coupling has the same total mass as each marginal, the couplings are tight on $\mathcal{X} \times \mathcal{X}$. Thus, up to subsequences, $\pi_n \rightarrow \pi_0$. Lower semicontinuity of the transport cost yields $\int c d\pi_0 = 0$, so π_0 is concentrated on the diagonal. Its two marginals are therefore equal to a common measure ρ . Lower semicontinuity of the marginal divergences gives

$$\liminf_n \frac{1}{\tau_n} \text{UW}_{c, \tau_n}(\alpha, \beta) \geq \mathcal{D}_{\bar{\psi}_1}(\rho | \alpha) + \mathcal{D}_{\bar{\psi}_2}(\rho | \beta),$$

and then the infimum over ρ gives the lower bound. For KL, minimizing pointwise over the density r of ρ gives

$$r \log(r/a) - r + a + r \log(r/b) - r + b.$$

The optimality condition is $\log(r/a) + \log(r/b) = 0$, hence $r = \sqrt{ab}$, and the minimum is $a + b - 2\sqrt{ab} = (\sqrt{a} - \sqrt{b})^2$. $\square$

The following proposition states the dual problem.

Proposition 62 (Dual of unbalanced optimal transport). *One has equality between (88) and*

$$\text{UW}_c(\alpha, \beta) = \sup_{f \oplus g \leq c} -\mathcal{D}_{\psi_1}^*(-f|\alpha) - \mathcal{D}_{\psi_2}^*(-g|\beta).$$

Proof. One gets a dual by using the formula (80) for the dual of a divergence

$$\inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} \sup_{(f, g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y})} \langle c, \pi \rangle + \langle -f, \pi_1 \rangle + \langle -g, \pi_2 \rangle - \mathcal{D}_{\psi_1}^*(-f|\alpha) - \mathcal{D}_{\psi_2}^*(-g|\beta).$$

Exchanging the min and the max gives

$$\sup_{(f, g)} -\mathcal{D}_{\psi_1}^*(-f|\alpha) - \mathcal{D}_{\psi_2}^*(-g|\beta) + \inf_{\pi \geq 0} \langle c - (f \oplus g), \pi \rangle = \sup_{f \oplus g \leq c} -\mathcal{D}_{\psi_1}^*(-f|\alpha) - \mathcal{D}_{\psi_2}^*(-g|\beta).$$

□

Reverse formulation In order to study metric properties of this formulation, we follow Liero, Mielke and Savare [139, 140], who consider a series of equivalent formulations. Assuming for simplicity that (π_1, π_2) have densities with respect to (α, β) and vice-versa, one can write the following factorization formula

$$\begin{aligned} & \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) + \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta) \\ &= \int_{\mathcal{X} \times \mathcal{Y}} \left(c(x, y) + \psi_1 \left(\frac{d\pi_1}{d\alpha}(x) \right) \frac{d\alpha}{d\pi_1}(x) + \psi_2 \left(\frac{d\pi_2}{d\beta}(y) \right) \frac{d\beta}{d\pi_2}(y) \right) d\pi(x, y). \end{aligned}$$

More rigorously, we introduce the following functional

$$\forall (c, r, s) \in \mathbb{R}^3, \quad L_c(r, s) := c + r\psi_1(1/r) + s\psi_2(1/s), \quad (89)$$

where we implicitly assume that $r\psi_1(1/r) = (\psi_1')^\infty$ when $r = 0$, one has

$$\text{UW}_c(\alpha, \beta) = \inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} \int L_{c(x, y)}(F(x), G(y)) d\pi(x, y) + \psi_1(0)\alpha^\perp(\mathcal{X}) + \psi_2(0)\beta^\perp(\mathcal{Y})$$

where the Lebesgue decompositions of the reference measures with respect to the transported marginals are

$$\alpha = F\pi_1 + \alpha^\perp \quad \text{and} \quad \beta = G\pi_2 + \beta^\perp.$$

Thus $F = d\alpha/d\pi_1$ and $G = d\beta/d\pi_2$ on the absolutely continuous parts.

Homogeneous formulation The second formulation is obtained by introducing the perspective transform of L_c

$$H_c(r, s) := \inf_{\theta > 0} \theta L_c(r/\theta, s/\theta). \quad (90)$$

This function H_c is positively 1-homogeneous. The homogeneous formulation is then

$$\text{HW}_c(\alpha, \beta) = \inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} \int H_{c(x, y)}(F(x), G(y)) d\pi(x, y) + \psi_1(0)\alpha^\perp(\mathcal{X}) + \psi_2(0)\beta^\perp(\mathcal{Y}). \quad (91)$$

The easy inequality is $\text{HW} \leq \text{UW}$ since $H_c \leq L_c$ (by taking $\theta = 1$). The proposition shows that the converse inequality also holds:

Proposition 63 (Homogenization does not change the unbalanced cost). *One has $\text{HW}_c(\alpha, \beta) = \text{UW}_c(\alpha, \beta)$.*

Proof. The inequality $\text{HW} \leq \text{UW}$ follows from $H_c \leq L_c$. For the reverse inequality, take a feasible measure π in the homogeneous formulation. By definition of the perspective transform, for every (x, y) and every $\eta > 0$ there exists a scale $\theta(x, y) > 0$ such that

$$H_{c(x,y)}(F(x), G(y)) + \eta \geq \theta(x, y) L_{c(x,y)}(F(x)/\theta(x, y), G(y)/\theta(x, y)).$$

Replacing π by the rescaled measure $\tilde{\pi} = \theta \pi$ and the densities by F/θ and G/θ gives an admissible competitor for the reverse formulation with cost no larger than the homogeneous cost plus $\eta\pi(\mathcal{X} \times \mathcal{Y})$. Letting $\eta \rightarrow 0$ yields $\text{UW} \leq \text{HW}$. The singular terms are unchanged because the same rescaling is performed before taking the Lebesgue decomposition of the marginals. $\square$

Conic lifting We now suppose $\mathcal{X} = \mathcal{Y}$ and $\psi_1 = \psi_2 = \psi$. The last formulation is obtained by lifting the above optimization problem on the cone space $\mathfrak{C}[\mathcal{X}] := (\mathcal{X} \times \mathbb{R}_+)/\sim$ associated to $\mathcal{X}$ (and similarly for $\mathcal{Y}$). Here, the equivalence relation $\sim$ “glues” together all the points $(x, 0)$, which are often called the “apex” of the cone. For some exponent $p \geq 1$, we define the following functional

$$\Delta((x, r), (y, s)) := H_{c(x,y)}(r^p, s^p)^{1/p}.$$

These constructions include the Hellinger–Kantorovich and Wasserstein–Fisher–Rao geometries developed in [59, 61], as well as algorithmic scaling schemes [60]. The interesting case is when c and p are chosen so that Δ defines a distance on $\mathfrak{C}[\mathcal{X}]$. Here are three examples:

- $\mathcal{D}_\psi = \text{KL}$, $p = 2$ and $c(x, y) = -\log \cos^2(d(x, y) \wedge \frac{\pi}{2})$ in which case Δ is the so-called cone metric associated to d and reads

$$\Delta((x, r), (y, s))^2 = r^2 + s^2 - 2rs \cos(d(x, y) \wedge \frac{\pi}{2}).$$

This corresponds to the so-called “Hellinger-Kantorovich”–“Wasserstein-Fisher-Rao” setup.

- $\mathcal{D}_\psi = \text{KL}$, $p = 2$ and $c(x, y) = d(x, y)^2$, so that

$$\Delta((x, r), (y, s))^2 = r^2 + s^2 - 2rs e^{-d(x, y)^2/2}.$$

This corresponds to the so-called “Gaussian Hellinger” setup.

- $\mathcal{D}_\psi = \text{TV}$, $p = 1$ and $c(x, y) = d(x, y)$, so that

$$\Delta((x, r), (y, s)) = r + s - (r \wedge s)(2 - d(x, y))_+.$$

This corresponds to the so-called “partial transport” setup.

The homogeneous formulation (91) is then relaxed as an OT-type problem on the cone

$$\text{CW}(\alpha, \beta) = \inf_{\gamma \in \mathcal{M}_+(\mathfrak{C}[\mathcal{X}]^2)} \left\{ \int \Delta((x, r), (y, s))^p d\gamma((x, r), (y, s)) ; \int r^p d\gamma_1(\cdot, r) = \alpha, \int s^p d\gamma_2(\cdot, s) = \beta \right\}.$$

The following theorem summarizes the key results.

Theorem 6 (Cone formulation of unbalanced OT). *One has $\text{UW} = \text{HW} = \text{CW}$. If Δ is a distance, so is $\text{CW}^{1/p}$.*

Proof. The equality $\text{UW} = \text{HW}$ is the previous proposition. To prove $\text{HW} = \text{CW}$, disintegrate an admissible cone coupling γ with respect to its spatial variables (x, y) and radii (r, s) . The cone marginal constraints say precisely that the spatial marginals are recovered after weighting by r^p and s^p . Because $\Delta((x, r), (y, s))^p = H_{c(x,y)}(r^p, s^p)$, integrating the cone cost gives the homogeneous objective. Conversely, any homogeneous

competitor can be lifted to the cone by placing, over each (x, y) , radii whose p -th powers are the two density factors appearing in H_c . This gives the same cost, hence the same infimum.

If Δ is a distance on the cone, then $CW^{1/p}$ is the usual p -Wasserstein distance between the lifted measures under the linear cone-marginal constraints. Symmetry and the triangle inequality follow from the corresponding Wasserstein properties and the gluing lemma on the cone. If the distance is zero, an optimal cone coupling is concentrated on the diagonal of the cone, so the weighted projections agree and therefore $\alpha = \beta$. $\square$

Entropic regularization A generic regularization of this problem reads

$$\inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) + \mathcal{D}_{\psi_1}(\pi_1 | \alpha) + \mathcal{D}_{\psi_2}(\pi_2 | \beta) + \varepsilon \mathcal{D}_\varphi(\pi | \alpha \otimes \beta)$$

and its dual reads

$$\sup_{f, g} -\mathcal{D}_{\psi_1}^*(-f | \alpha) - \mathcal{D}_{\psi_2}^*(-g | \beta) - \varepsilon \mathcal{D}_\varphi^*\left(\frac{f \oplus g - c}{\varepsilon}\right).$$

When using $\mathcal{D}_\varphi = \text{KL}$, the primal-dual relation reads $\pi = e^{\frac{f \oplus g - c}{\varepsilon}}$ and when also considering $\mathcal{D}_{\psi_1} = \mathcal{D}_{\psi_2} = \tau \text{KL}$, the dual reads

$$\sup_{f, g} -\tau \left(\langle e^{\frac{-f}{\tau}}, \alpha \rangle + \langle e^{\frac{-g}{\tau}}, \beta \rangle - 2 \right) - \varepsilon \iint_{\mathcal{X} \times \mathcal{Y}} e^{\frac{f \oplus g - c}{\varepsilon}} d\alpha \otimes d\beta + \varepsilon.$$

Sinkhorn iterations are obtained by minimizing iteratively with respect to f and g , and read in this special case

$$\begin{aligned} f &= -\frac{\tau \varepsilon}{\tau + \varepsilon} \log \int_{\mathcal{Y}} \exp\left(\frac{g(y) - c(\cdot, y)}{\varepsilon}\right) d\beta(y), \\ g &= -\frac{\tau \varepsilon}{\tau + \varepsilon} \log \int_{\mathcal{X}} \exp\left(\frac{f(x) - c(x, \cdot)}{\varepsilon}\right) d\alpha(x). \end{aligned}$$

9.2 OT Barycenters

Barycenters ask how to average probability measures rather than points. This subsection explains the variational definition, the special closed forms in one dimension and for Gaussians, and the entropic algorithms used in practice.

Fréchet means. For discrete input histograms $\{b_s\}_{s=1}^S$, with $b_s \in \Sigma_{n_s}$, and weights $\lambda \in \Sigma_S$, a Wasserstein barycenter can be computed by minimizing

$$\min_{a \in \Sigma_n} \sum_{s=1}^S \lambda_s L_{C_s}(a, b_s), \quad (92)$$

where the cost matrices $C_s \in \mathbb{R}^{n \times n_s}$ are prescribed.

This barycenter problem was originally introduced by [1] following earlier ideas of [48]. They proved in particular uniqueness of the barycenter for $c(x, y) = \|x - y\|^2$ over $\mathcal{X} = \mathbb{R}^d$, if one of the input measures has a density with respect to the Lebesgue measure (and more generally under the usual hypotheses guaranteeing the existence of a Monge map). Discrete existence, consistency and fixed-point constructions are further studied in [7, 5, 134].

Given a set of input measures $(\beta_s)_s$ defined on some space $\mathcal{X}$, the barycenter problem becomes

$$\min_{\alpha \in \mathcal{M}_+^1(\mathcal{X})} \sum_{s=1}^S \lambda_s \mathcal{L}_c(\alpha, \beta_s). \quad (93)$$

Remark 25 (Two measures give a Wasserstein geodesic). For $S = 2$, $c(x, y) = \|x - y\|^2$ and weights $(1 - t, t)$, the barycenter is the point at time t on the Wasserstein geodesic between β_0 and β_1 . If T is the Brenier map from β_0 to β_1 , this barycenter is $((1 - t)\text{Id} + tT)_\# \beta_0$, the McCann interpolation detailed in Section 10.4. If no Monge map is available, the same construction uses an optimal coupling and the interpolation map $(x, y) \mapsto (1 - t)x + ty$.

Example 9 (Dirac inputs). Problem (93) generalizes the computation of barycenters of points $(x_s)_{s=1}^S \in \mathcal{X}^S$ to arbitrary measures. Indeed, if $\beta_s = \delta_{x_s}$ is a single Dirac mass, then a solution to (93) is δ_{x^*} where x^* is a Fréchet mean of the points $(x_s)_s$.

Remark 26 (Mean of a quadratic barycenter). For $c(x, y) = \|x - y\|^2$, the mean of the barycenter α^* is necessarily the barycenter of the means,

$$\int_{\mathcal{X}} x d\alpha^*(x) = \sum_s \lambda_s \int_{\mathcal{X}} x d\beta_s(x).$$

Indeed, the squared Wasserstein distance decomposes into a squared distance between means plus a centered Wasserstein term. Minimizing the resulting quadratic function of the barycenter mean gives the displayed identity. The support of α^* is also located in the convex hull of the supports of the $(\beta_s)_s$.

In the case where $\mathcal{X} = \mathbb{R}^d$ and $c(x, y) = \|x - y\|^2$, [1] shows that if one of the input measures has a density, then this barycenter is unique.

The following proposition shows that (93) is a convex optimization problem over the infinite dimensional space of measures. However, it is a priori unclear how to solve this infinite-dimensional problem even if α and β are discrete measures. Section 9.3 shows that barycenters α can be computed by solving a linear program, which is a multi-marginal OT problem. This implies in particular that the barycenter of S discrete measures with n Diracs has at most n^S Diracs. The difficulty is that the numerical complexity of solving this problem grows exponentially with S . We will show later that if one fixes a priori the support of α , one can use a fast Sinkhorn algorithm.

Proposition 6 (Convexity of the OT cost). $(\alpha, \beta) \mapsto \mathcal{L}_c(\alpha, \beta)$ is convex.

Proof. Let (α_0, β_0) and (α_1, β_1) be two pairs of probability measures and let $t \in [0, 1]$. For $\varepsilon > 0$, choose couplings $\pi_i \in \mathcal{U}(\alpha_i, \beta_i)$ such that

$$\int c d\pi_i \leq \mathcal{L}_c(\alpha_i, \beta_i) + \varepsilon \quad \text{for } i = 0, 1.$$

Then $\pi_t := (1 - t)\pi_0 + t\pi_1$ is a coupling between $(1 - t)\alpha_0 + t\alpha_1$ and $(1 - t)\beta_0 + t\beta_1$. Hence

$$\mathcal{L}_c((1 - t)\alpha_0 + t\alpha_1, (1 - t)\beta_0 + t\beta_1) \leq (1 - t)\mathcal{L}_c(\alpha_0, \beta_0) + t\mathcal{L}_c(\alpha_1, \beta_1) + \varepsilon.$$

Letting $\varepsilon \rightarrow 0$ gives the claim. $\square$

1-D Case On the line, barycenters become linear after the quantile change of variables. This gives the rare case where the barycenter is explicit rather than the solution of a high-dimensional optimization problem.

Proposition 64 (Quantile barycenters on the line). For $\mathcal{X} = \mathbb{R}$ and $c(x, y) = |x - y|^2$, the quantile function of a Wasserstein barycenter is the weighted average of the input quantile functions:

$$\mathcal{C}_{\alpha^*}^{-1}(r) = \sum_{s=1}^S \lambda_s \mathcal{C}_{\beta_s}^{-1}(r) \quad \text{for } r \in [0, 1].$$

Proof. The one-dimensional formula (11) gives

$$\sum_s \lambda_s \mathcal{W}_2^2(\alpha, \beta_s) = \int_0^1 \sum_s \lambda_s \left| \mathcal{C}_{\alpha}^{-1}(r) - \mathcal{C}_{\beta_s}^{-1}(r) \right|^2 dr.$$

The minimization decouples pointwise in r . For each fixed r , the minimizer of $z \mapsto \sum_s \lambda_s |z - \mathcal{C}_{\beta_s}^{-1}(r)|^2$ is the weighted average $\sum_s \lambda_s \mathcal{C}_{\beta_s}^{-1}(r)$. This function is nondecreasing because it is a positive weighted sum of nondecreasing quantile functions, hence it is itself a valid quantile function. $\square$

Gaussian Case Gaussian barycenters show that the same separation as in Proposition 12 persists: means average linearly, while covariances average according to the Bures–Wasserstein geometry.

Remark 27 (Gaussian barycenters). The barycenter of Gaussian measures is Gaussian. In 1-D, it is obtained by averaging the means and the standard deviations, so the barycenter variance is the square of this averaged standard deviation. In higher dimensions, the covariance Σ minimizes the Bures objective

$$\Sigma \mapsto \sum_s \lambda_s \mathcal{B}(\Sigma, \Sigma_s)^2,$$

and equivalently solves the fixed-point equation

$$\Sigma = \sum_s \lambda_s \left(\Sigma^{1/2} \Sigma_s \Sigma^{1/2} \right)^{1/2}.$$

This is the covariance analogue of the usual Euclidean barycenter equation: the mean part averages linearly, while the covariance part averages through the Bures–Wasserstein geometry [5, 27].

Sinkhorn for barycenters A key difference of problem (93) with respect to the classical Kantorovich formulation is that there is no simple canonical measure that can be used to localize the support of the solution (for classical OT one can use the product measure $\alpha \otimes \beta$). Indeed, from the multimarginal formulation of the barycenter, all we know a priori is that the support of barycenters is constrained to lie among all possible barycenters of sets of S points extracted from all the supports. If they all have n points, this makes n^S possibilities, which is prohibitive.

So in order to reduce this complexity and develop a Sinkhorn solver, one needs to fix a priori some support for the solution, usually with a much smaller size, which introduces some discretization error because this support is likely not to be the support of the target measure.

Once this reference measure is fixed, one can use entropic smoothing and approximate the solution of (92) using

$$\min_{a \in \Sigma_n} \sum_{s=1}^S \lambda_s L_{C_s}^\varepsilon(a, b_s) \quad (94)$$

for some $\varepsilon > 0$. This is a smooth convex minimization problem, which can be tackled using gradient descent [66]. An alternative is to use a descent method, typically quasi-Newton, on the semi-dual [67]; this is useful when adding extra regularization on the barycenter, for instance to impose smoothness. A simple but effective approach, as remarked in [18] is to rewrite (94) as a (weighted) KL projection problem

$$\min_{(P_s)_s} \left\{ \sum_s \lambda_s \text{KL}(P_s | K_s) ; \forall s, P_s^\top \mathbf{1}_n = b_s, P_1 \mathbf{1}_{n_1} = \dots = P_S \mathbf{1}_{n_S} \right\} \quad (95)$$

where we denoted $K_s := e^{-C_s/\varepsilon}$. Here, the barycenter a is implicitly encoded in the row marginals of all the couplings $P_s \in \mathbb{R}^{n \times n_s}$ as $a = P_1 \mathbf{1}_{n_1} = \dots = P_S \mathbf{1}_{n_S}$. As detailed in [18], one can generalize Sinkhorn to this problem, which also corresponds to iterative projection. The optimal couplings $(P_s)_s$ solving (95) are computed in scaling form as

$$P_s = \text{diag}(u_s) K_s \text{diag}(v_s), \quad (96)$$

and the scalings are sequentially updated as

$$\forall s \in \llbracket 1, S \rrbracket, \quad v_s^{(\ell+1)} := \frac{b_s}{K_s^\top u_s^{(\ell)}}, \quad (97)$$

$$\forall s \in \llbracket 1, S \rrbracket, \quad u_s^{(\ell+1)} := \frac{a^{(\ell+1)}}{K_s v_s^{(\ell+1)}}, \quad (98)$$

$$\text{where } a^{(\ell+1)} := \prod_s (K_s v_s^{(\ell+1)})^{\lambda_s}. \quad (99)$$

An alternative way to derive these iterations is to perform alternate minimization on the variables of a dual problem, as detailed in the following proposition.

Proposition 65 (Dual of entropic barycenters). *The optimal (u_s, v_s) appearing in (96) can be written as $(u_s, v_s) = (e^{f_s/\varepsilon}, e^{g_s/\varepsilon})$ where $(f_s, g_s)_s$ are the solutions of the following program (whose value matches the one of (94))*

$$\max_{(f_s, g_s)_s} \left\{ \sum_s \lambda_s \left(\langle g_s, b_s \rangle - \varepsilon \langle K_s e^{g_s/\varepsilon}, e^{f_s/\varepsilon} \rangle \right) ; \sum_s \lambda_s f_s = 0 \right\}. \quad (100)$$

Proof. Introducing Lagrange multipliers in (95) leads to

$$\min_{(P_s)_s, a} \max_{(f_s, g_s)_s} \sum_s \lambda_s \left(\varepsilon \text{KL}(P_s | K_s) + \langle a - P_s \mathbf{1}_{n_s}, f_s \rangle + \langle b_s - P_s^\top \mathbf{1}_n, g_s \rangle \right).$$

Strong duality holds, so one can exchange the minimum and the maximum and obtain

$$\begin{aligned} \max_{(f_s, g_s)_s} \sum_s \lambda_s \left(\langle g_s, b_s \rangle + \min_{P_s} \varepsilon \text{KL}(P_s | K_s) - \langle P_s, f_s \oplus g_s \rangle \right) \\ + \min_a \langle \sum_s \lambda_s f_s, a \rangle. \end{aligned}$$

The explicit minimization on a gives the constraint $\sum_s \lambda_s f_s = 0$ together with

$$\max_{(f_s, g_s)_s} \sum_s \lambda_s \langle g_s, b_s \rangle - \varepsilon \text{KL}^* \left(\frac{f_s \oplus g_s}{\varepsilon} | K_s \right)$$

where $\text{KL}^*(\cdot | K_s)$ is the Legendre transform (80) of the function $\text{KL}(\cdot | K_s)$. This Legendre transform reads

$$\text{KL}^*(U | K) = \sum_{i,j} K_{i,j} (e^{U_{i,j}} - 1), \quad (101)$$

which shows the desired formula. To show (101), since this function is separable, one needs to compute

$$\forall (u, k) \in \mathbb{R} \times \mathbb{R}_+^*, \quad \text{KL}^*(u | k) := \max_{r \geq 0} ur - (r \log(r/k) - r + k)$$

whose optimality condition reads $u = \log(r/k)$, i.e. $r = ke^u$, hence the result. The case $k = 0$ follows by lower semicontinuity and gives zero. $\square$

Minimizing (100) with respect to each g_s , while keeping all the other variable fixed, is obtained in closed form by (97). Minimizing (100) with respect to all the $(f_s)_s$ requires to solve for a using (99) and leads to the expression (98).

Classical applications include 2-D image interpolation, 3-D shape interpolation, and barycenters on surfaces where the ground cost is the square of the geodesic distance. We refer to [200] for more details and other applications to computer graphics and imaging sciences.

9.3 Multimarginal OT

Multi-marginal OT couples more than two measures at once. It is the natural language for barycenters, matching with teams and several-body costs, but its tensor dimension is the main computational obstacle.

Definition, example and properties. The multi-marginal formulation replaces a coupling between two measures by a joint distribution with several prescribed marginals. Given measures $(\alpha_s)_{s=1}^S$ on spaces $(\mathcal{X}_s)_{s=1}^S$ and a cost $c : \mathcal{X}_1 \times \dots \times \mathcal{X}_S \rightarrow \mathbb{R}$, the problem reads

$$\min_{\pi \in \mathcal{U}(\alpha_1, \dots, \alpha_S)} \int_{\mathcal{X}_1 \times \dots \times \mathcal{X}_S} c(x_1, \dots, x_S) d\pi(x_1, \dots, x_S),$$

where $\mathcal{U}(\alpha_1, \dots, \alpha_S)$ is the set of probability measures whose s -th marginal is α_s . This is still a linear program in the discrete setting, but its ambient dimension grows exponentially with the number of marginals.

Multi-marginal formulation of barycenters. Wasserstein barycenters are a central example. For the squared Euclidean cost, one can introduce a latent barycenter point and eliminate it explicitly, leading to a multi-marginal cost of the form

$$c(x_1, \dots, x_S) = \min_{x \in \mathbb{R}^d} \sum_{s=1}^S \lambda_s \|x - x_s\|^2.$$

Proposition 66 (Multi-marginal formula for quadratic barycenters). *Let $\beta_1, \dots, \beta_S \in \mathcal{P}_2(\mathbb{R}^d)$ and $\lambda \in \Sigma_S$. Define*

$$B(x_1, \dots, x_S) = \sum_{s=1}^S \lambda_s x_s, \quad c_{\text{bar}}(x_1, \dots, x_S) = \min_x \sum_s \lambda_s \|x - x_s\|^2.$$

If π^ solves the multi-marginal OT problem with marginals $(\beta_s)_s$ and cost c_{bar} , then $\alpha^* = B_{\#}\pi^*$ is a Wasserstein barycenter. Conversely, every barycenter is obtained this way from an optimal multi-marginal plan.*

Proof. For any candidate barycenter α and couplings $\pi_s \in \mathcal{U}(\alpha, \beta_s)$, glue the couplings along their common α marginal to obtain a joint law of $(X, Y_1, \dots, Y_S)$. Conditioning on $(Y_s)_s$ and minimizing over X gives

$$\sum_s \lambda_s \mathbb{E} \|X - Y_s\|^2 \geq \mathbb{E} \min_x \sum_s \lambda_s \|x - Y_s\|^2 = \mathbb{E} c_{\text{bar}}(Y_1, \dots, Y_S).$$

Taking the infimum over the couplings gives that the barycenter value is at least the multi-marginal value. Conversely, from an optimal multi-marginal plan π^* , set $X = B(Y_1, \dots, Y_S)$. The couplings between X and each Y_s are feasible for the barycenter problem and attain exactly the multi-marginal cost, proving equality and the formula. $\square$

Corollary 5 (Gaussian and discrete barycenters). *Quadratic Wasserstein barycenters of Gaussian measures are Gaussian. If the input measures are discrete, then there exists a barycenter supported on the set of weighted averages $\sum_s \lambda_s x_{s,i_s}$ of one support point from each input; in particular, if the s -th input has n_s atoms, a barycenter exists with at most $\prod_s n_s$ atoms.*

Proof. Let the input Gaussians have means $\mathbf{m}_s$ and covariances Σ_s . For any candidate barycenter α with mean $\mathbf{m}$ and covariance Σ , Gelbrich's inequality [94] gives

$$\mathcal{W}_2^2(\alpha, \beta_s) \geq \|\mathbf{m} - \mathbf{m}_s\|^2 + \mathcal{B}(\Sigma, \Sigma_s)^2,$$

with equality when α is Gaussian. Therefore the barycenter objective is bounded below by a function depending only on $(\mathbf{m}, \Sigma)$, and this lower bound is attained by the Gaussian measure with the minimizing mean and covariance. Hence at least one barycenter is Gaussian, and uniqueness in the usual nondegenerate setting gives the Gaussian barycenter mentioned above. For discrete inputs, any multi-marginal optimizer is supported on the finite product of the input supports, and B maps this product to at most $\prod_s n_s$ points. $\square$

Entropic regularization of multi-marginal. As in the two-marginal case, adding an entropic penalty with respect to the product measure $\alpha_1 \otimes \cdots \otimes \alpha_S$ leads to scaling algorithms:

$$\inf_{\pi \in \mathcal{U}(\alpha_1, \dots, \alpha_S)} \int c d\pi + \varepsilon \text{KL}(\pi | \alpha_1 \otimes \cdots \otimes \alpha_S).$$

The optimizer has the generalized Gibbs form

$$d\pi^*(x_1, \dots, x_S) = \exp\left(\frac{\sum_s f_s(x_s) - c(x_1, \dots, x_S)}{\varepsilon}\right) \prod_s d\alpha_s(x_s),$$

and generalized Sinkhorn iterations alternately update one potential f_s so that the s -th marginal is correct. The bottleneck is that a naive tensor representation has size $\prod_s n_s$ in the discrete case. Practical barycenter solvers therefore exploit separability of the cost, low-rank structure, convolutional kernels, or a fixed barycenter support.

9.4 Sliced Wasserstein

Sliced Wasserstein trades exact high-dimensional geometry for many one-dimensional projections. It is cheap, differentiable after sorting, and often effective in imaging and learning.

For measures on $\mathbb{R}^d$, sliced Wasserstein distances replace one high-dimensional transport problem by many one-dimensional ones. For $\theta \in \mathbb{S}^{d-1}$, let $P_\theta(x) = \langle x, \theta \rangle$ be the projection on direction θ . The sliced distance is

$$\text{SW}_p^p(\alpha, \beta) := \int_{\mathbb{S}^{d-1}} \mathcal{W}_p^p((P_\theta)_\# \alpha, (P_\theta)_\# \beta) d\theta. \quad (102)$$

This is closely related to the Radon transform and is much cheaper to approximate numerically than high-dimensional OT, since each projected problem can be solved by sorting or quantiles. It metrizes weak convergence under moment assumptions, but its geometry differs from that of $\mathcal{W}_p$. The Radon-transform viewpoint and barycentric constructions are developed in [36]; statistical and signal-processing variants are discussed in [128, 127, 159].

Proposition 67 (Metric properties of sliced Wasserstein). *For $p \geq 1$, SW_p is a distance on $\mathcal{P}_p(\mathbb{R}^d)$. Moreover, SW_p metrizes weak convergence together with convergence of the p -th moment. Finally,*

$$\text{SW}_p(\alpha, \beta) \leq \mathcal{W}_p(\alpha, \beta),$$

and, for $p = 2$ with the uniform probability measure on the sphere,

$$\text{SW}_2^2(\alpha, \beta) \leq \frac{1}{d} \mathcal{W}_2^2(\alpha, \beta).$$

Proof. Non-negativity and symmetry follow from the one-dimensional Wasserstein distance. For the triangle inequality, apply the triangle inequality of $\mathcal{W}_p$ for each direction θ and then Minkowski's inequality in $L^p(\mathbb{S}^{d-1})$.

If $\text{SW}_p(\alpha, \beta) = 0$, then $(P_\theta)_\# \alpha = (P_\theta)_\# \beta$ for almost every direction, hence for all directions by continuity of characteristic functions. The Cramer–Wold theorem then implies $\alpha = \beta$. This proves that SW_p is a distance.

The bound $\text{SW}_p \leq \mathcal{W}_p$ follows because P_θ is 1-Lipschitz, so

$$\mathcal{W}_p((P_\theta)_\# \alpha, (P_\theta)_\# \beta) \leq \mathcal{W}_p(\alpha, \beta)$$

for every θ . For $p = 2$, using any coupling π between α and β ,

$$\int_{\mathbb{S}^{d-1}} \int |\langle x - y, \theta \rangle|^2 d\pi(x, y) d\theta = \frac{1}{d} \int \|x - y\|^2 d\pi(x, y).$$

Optimizing over π gives the sharper displayed inequality.

The weak-convergence statement follows from the same Cramer–Wold mechanism plus the moment condition: convergence in SW_p gives convergence of almost all one-dimensional projections and tightness of the p -moments; conversely, weak convergence with p -moment convergence implies convergence of projected $\mathcal{W}_p$ distances and dominated convergence on the sphere. Detailed proofs and statistical refinements can be found in [37, 159]. $\square$

Remark 28 (Hilbert embedding for SW_2). In one dimension, $\mathcal{W}_2$ is the $L^2(0,1)$ distance between quantile functions. Hence

$$\text{SW}_2^2(\alpha, \beta) = \int_{\mathbb{S}^{d-1}} \int_0^1 \left| F_{\theta, \alpha}^{-1}(u) - F_{\theta, \beta}^{-1}(u) \right|^2 du d\theta,$$

where $F_{\theta, \alpha}^{-1}$ is the quantile of $(P_\theta)_\# \alpha$. Thus SW_2 is a Hilbertian distance after embedding each measure into its field of projected quantiles. Consequently, $\exp(-\gamma \text{SW}_2^2)$ is a positive definite kernel on probability measures for every $\gamma > 0$. Conversely, on compact subsets, $\mathcal{W}_p$ can be bounded by a dimension-dependent power of SW_p ; such inequalities are weaker than the direct bound above and explain why sliced distances metrize the same topology without being bi-Lipschitz equivalent to $\mathcal{W}_p$ in high dimension [37].

Definition 17 (Max-sliced Wasserstein). *The max-sliced distance replaces the average over directions by the most discriminating one:*

$$\text{MaxSW}_p(\alpha, \beta) = \sup_{\theta \in \mathbb{S}^{d-1}} \mathcal{W}_p((P_\theta)_\# \alpha, (P_\theta)_\# \beta).$$

It is useful when only a small set of projections carries most of the discrepancy, for instance in generative modeling [71].

Min-sliced lifted transport plans. There is another, conceptually different, use of slicing: one can use a one-dimensional projection to build a feasible high-dimensional transport plan. For equal-weight empirical measures $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$ and $\beta = \frac{1}{n} \sum_i \delta_{y_i}$, sort the projected samples $\langle x_i, \theta \rangle$ and $\langle y_j, \theta \rangle$, and let σ_θ be the monotone matching induced by this sorting. The lifted plan

$$\pi_\theta = \frac{1}{n} \sum_{i=1}^n \delta_{(x_i, y_{\sigma_\theta(i)})}$$

is a genuine coupling between α and β in the original space. Min-SWGG-type methods choose the best projection according to the full-dimensional transport cost,

$$\text{MSWGG}_2^2(\alpha, \beta) := \min_{\theta \in \mathbb{S}^{d-1}} \int \|x - y\|^2 d\pi_\theta(x, y).$$

This is not a projected distance; it is a cheap feasible-plan construction. Consequently it gives an upper bound on $\mathcal{W}_2^2(\alpha, \beta)$, and the interest is algorithmic: the plan is obtained by sorting rather than by solving a high-dimensional linear program [147].

Definition 18 (Subspace-sliced Wasserstein). *Subspace-sliced variants replace one-dimensional lines by k -dimensional orthogonal projections. If $U \in \mathbb{R}^{d \times k}$ satisfies $U^\top U = I_k$, then*

$$\text{SW}_{p,k}^p(\alpha, \beta) = \int \mathcal{W}_p^p((U^\top)_\# \alpha, (U^\top)_\# \beta) dU, \quad \text{MaxSW}_{p,k}(\alpha, \beta) = \sup_{U^\top U = I_k} \mathcal{W}_p((U^\top)_\# \alpha, (U^\top)_\# \beta),$$

where the integral is over the Stiefel manifold. The case $k = 1$ recovers ordinary sliced and max-sliced Wasserstein, while $k = d$ recovers the original Wasserstein distance.

Proposition 68 (Basic bounds for sliced variants). *With normalized surface measure,*

$$\text{SW}_p(\alpha, \beta) \leq \text{MaxSW}_p(\alpha, \beta) \leq \mathcal{W}_p(\alpha, \beta).$$

In the equal-weight empirical setting described above, any min-SWGG-type lifted plan has cost at least $\mathcal{W}_2^2(\alpha, \beta)$, hence gives an upper bound on the optimal quadratic transport cost. For k -dimensional subspace projections, the corresponding projected Wasserstein distance is bounded above by $\mathcal{W}_p$ for every fixed subspace, and so are its average and max over subspaces.

Proof. The first inequality follows because an average is bounded by a supremum. The second and the subspace bound follow because orthogonal projections are 1-Lipschitz, so pushing any admissible coupling forward gives a feasible coupling for the projected problem with no larger cost. A lifted one-dimensional matching is an admissible coupling between the original measures; evaluating the full quadratic cost on any admissible coupling can only be larger than the optimal $\mathcal{W}_2^2$ value. $\square$

9.5 Spectral and Subspace Robust Wasserstein

Spectral OT changes the scalar quadratic cost by measuring the whole displacement covariance through a matrix gauge. The same object admits a robust projected formulation: instead of fixing one projection, one maximizes over the polar set of the gauge. Subspace robust OT is the important non-convex rank-constrained version of this idea [164]; spectral gauges provide its convex minimax counterpart and connect to recent spectral-gradient viewpoints such as Muon dynamics [167].

Definition 19 (Spectral Wasserstein distance). *A monotone spectral gauge on positive semidefinite matrices is a convex, positively 1-homogeneous map γ such that $\gamma(M) = 0$ only for $M = 0$, $\gamma(QMQ^\top) = \gamma(M)$ for every orthogonal matrix Q , and*

$$0 \preceq M \preceq N \implies \gamma(M) \leq \gamma(N).$$

For a coupling $\pi \in \mathcal{U}(\alpha, \beta)$, define its displacement covariance

$$M_\pi = \int (x - y)(x - y)^\top d\pi(x, y).$$

The spectral Wasserstein distance associated with γ is

$$\mathcal{W}_\gamma(\alpha, \beta)^2 := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \gamma(M_\pi). \quad (103)$$

The special case $\gamma(M) = \text{tr}(M)$ gives the usual quadratic Wasserstein distance $\mathcal{W}_2$.

For $A \succeq 0$, define the quadratic projected transport cost

$$\mathcal{W}_{2,A}^2(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int (x - y)^\top A (x - y) d\pi(x, y) = \mathcal{W}_2^2((A^{1/2})_\# \alpha, (A^{1/2})_\# \beta). \quad (104)$$

The polar set of γ is

$$\mathcal{B}_\gamma := \{A \succeq 0 ; \text{tr}(AM) \leq \gamma(M) \text{ for all } M \succeq 0\}. \quad (105)$$

For a closed gauge, $\gamma(M) = \sup_{A \in \mathcal{B}_\gamma} \text{tr}(AM)$.

Proposition 69 (Robust representation and metric equivalence). *Assume, for simplicity, that the measures are compactly supported. Then*

$$\mathcal{W}_\gamma(\alpha, \beta)^2 = \sup_{A \in \mathcal{B}_\gamma} \mathcal{W}_{2,A}^2(\alpha, \beta). \quad (106)$$

If there exist constants $0 < a \leq b < +\infty$ such that

$$aI \in \mathcal{B}_\gamma \quad \text{and} \quad \mathcal{B}_\gamma \subset \{A ; 0 \preceq A \preceq bI\},$$

equivalently $a \operatorname{tr}(M) \leq \gamma(M) \leq b \operatorname{tr}(M)$ on the positive semidefinite cone, then

$$\sqrt{a} \mathcal{W}_2(\alpha, \beta) \leq \mathcal{W}_\gamma(\alpha, \beta) \leq \sqrt{b} \mathcal{W}_2(\alpha, \beta).$$

In particular $\mathcal{W}_\gamma$ is a distance. When γ is the restriction of a norm to the positive semidefinite cone, these bounds hold automatically, so $\mathcal{W}_\gamma$ is equivalent to $\mathcal{W}_2$ on measures with finite second moments.

Proof. Using the polar representation of γ ,

$$\mathcal{W}_\gamma(\alpha, \beta)^2 = \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \sup_{A \in \mathcal{B}_\gamma} \operatorname{tr}(AM_\pi).$$

The coupling set is convex and compact for weak convergence under compact support, $\mathcal{B}_\gamma$ is convex and compact, and the map $(\pi, A) \mapsto \operatorname{tr}(AM_\pi)$ is affine in each variable and continuous. Sion's minimax theorem therefore gives

$$\inf_{\pi} \sup_{A \in \mathcal{B}_\gamma} \operatorname{tr}(AM_\pi) = \sup_{A \in \mathcal{B}_\gamma} \inf_{\pi} \operatorname{tr}(AM_\pi) = \sup_{A \in \mathcal{B}_\gamma} \mathcal{W}_{2,A}^2(\alpha, \beta).$$

For fixed $A \succeq 0$, $\mathcal{W}_{2,A}$ is the Wasserstein pseudodistance associated with the seminorm $x \mapsto \|A^{1/2}x\|$. A supremum of pseudodistances is again symmetric and satisfies the triangle inequality.

If $aI \in \mathcal{B}_\gamma$ and $A \preceq bI$ for all $A \in \mathcal{B}_\gamma$, then

$$a \mathcal{W}_2^2(\alpha, \beta) = \mathcal{W}_{2,aI}^2(\alpha, \beta) \leq \mathcal{W}_\gamma^2(\alpha, \beta) \leq b \mathcal{W}_2^2(\alpha, \beta),$$

which proves both definiteness and the equivalence with $\mathcal{W}_2$. The equivalence between these bounds and $a \operatorname{tr}(M) \leq \gamma(M) \leq b \operatorname{tr}(M)$ follows directly from the polar formula for γ . In finite dimension, any norm restricted to the positive semidefinite cone is equivalent to the trace norm on that cone, giving the final statement. The finite-second-moment case follows by the usual truncation argument when these norm-equivalence bounds hold. $\square$

For the Ky Fan gauge

$$\gamma_k(M) = \sum_{\ell=1}^k \lambda_\ell(M),$$

where the eigenvalues are sorted in decreasing order, the polar set is

$$\mathcal{B}_{\gamma_k} = \{A ; 0 \preceq A \preceq I, \operatorname{tr}(A) \leq k\}.$$

Thus $k = d$ gives $\gamma_d(M) = \operatorname{tr}(M)$ and recovers $\mathcal{W}_2$. The Paty–Cuturi subspace robust Wasserstein distance is obtained by using the non-convex family of rank- k projectors,

$$\operatorname{SRW}_{2,k}(\alpha, \beta) = \sup_{U^\top U = I_k} \mathcal{W}_2((U^\top)_\# \alpha, (U^\top)_\# \beta) = \sup_{\substack{P^2 = P = P^\top \\ \operatorname{tr}(P) = k}} \mathcal{W}_{2,P}(\alpha, \beta).$$

Its convex hull is $\{A ; 0 \preceq A \preceq I, \operatorname{tr}(A) = k\}$, and the associated support function is the Ky Fan gauge γ_k . Thus $\mathcal{W}_{\gamma_k}$ is the convexified spectral counterpart of $\operatorname{SRW}_{2,k}$, while $\operatorname{SRW}_{2,k}$ keeps the original non-convex rank constraint. For $k = 1$, $\gamma_1(M) = \lambda_{\max}(M)$ and

$$\mathcal{B}_{\gamma_1} = \{A \succeq 0 ; \operatorname{tr}(A) \leq 1\}.$$

This top-eigenvalue spectral Wasserstein geometry is the one connected to Muon-type spectral dynamics in [167].

9.6 Linear Optimal Transport

Linear OT replaces a nonlinear transport distance by a Hilbert norm between maps. It is useful when one reference measure is fixed and many nearby distributions must be compared cheaply.

Fix a reference absolutely continuous measure ρ and let T_α be the Brenier map pushing ρ to α . The linear OT embedding is

$$\alpha \mapsto T_\alpha - \text{Id} \in L^2(\rho; \mathbb{R}^d), \quad \text{LOT}_\rho(\alpha, \beta) = \|T_\alpha - T_\beta\|_{L^2(\rho)}. \quad (107)$$

If one of the two targets equals the reference, the linearized distance is exact: for instance, $\text{LOT}_\rho(\rho, \alpha) = \|T_\alpha - \text{Id}\|_{L^2(\rho)} = \mathcal{W}_2(\rho, \alpha)$. For two arbitrary targets it is instead a tangent-space approximation of the Wasserstein geometry [213]. For a family $(\alpha_s)_s$ with weights $(\lambda_s)_s$, the linearized barycenter is obtained by averaging maps,

$$\bar{T} = \sum_s \lambda_s T_{\alpha_s}, \quad \bar{\alpha}_{\text{LOT}} = \bar{T}_\# \rho.$$

This is exact in one dimension, where quantile functions linearize $\mathcal{W}_2$, and it is especially useful when many barycenters with changing weights must be evaluated quickly.

Proposition 70 (Local stability of linear OT). *Assume that the measures are supported on a fixed convex compact set, with densities bounded above and below, and that the Brenier maps from ρ are regular. Then, for α, β in a sufficiently small regular neighborhood of ρ ,*

$$\mathcal{W}_2(\alpha, \beta) \leq \text{LOT}_\rho(\alpha, \beta) \quad \text{and} \quad \text{LOT}_\rho(\alpha, \beta) \leq C \mathcal{W}_2(\alpha, \beta)^\eta$$

for constants $C > 0$ and $\eta \in (0, 1]$ depending on regularity.

Proof. The first inequality is immediate: $(T_\alpha, T_\beta)_\# \rho$ is a feasible coupling between α and β . The reverse local estimate is a standard stability statement for the Monge–Ampère equation under the stated regularity assumptions: changes in the target measure control changes in the Brenier potential in Hölder norms, hence control $T_\alpha - T_\beta$ in $L^2(\rho)$. In simple one-dimensional settings, quantile functions make this exact with $\eta = 1$. In several dimensions one should not read the statement as a global Lipschitz estimate in $\mathcal{W}_2$. Quantitative stability results for semi-discrete and Monge–Ampère maps give Hölder exponents depending on the dimension, density bounds, support geometry and regularity; see for instance the estimates of Mérigot, Delalande and Chazal [154]. Under stronger smooth perturbations of uniformly convex smooth densities, elliptic regularity can give Lipschitz dependence in stronger function norms, but converting those controls to Wasserstein perturbations generally loses powers. $\square$

9.7 Weak Optimal Transport

Weak OT relaxes the cost so that it depends on the conditional distribution of destinations rather than only on pointwise pairs. It is useful when a source point is allowed to choose a randomized response and the model only penalizes some aggregate of that response, such as its conditional mean. The framework was introduced through general transport costs and weak transport inequalities in [104]; existence, duality and optimality conditions on Polish spaces are developed in [14].

Given $\pi \in \mathcal{U}(\alpha, \beta)$, disintegrate it as $\pi(dx, dy) = \pi_x(dy)\alpha(dx)$. For a weak cost $C : \mathcal{X} \times \mathcal{P}(\mathcal{Y}) \rightarrow \mathbb{R} \cup \{+\infty\}$, the weak OT value is

$$\mathcal{W}_C(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int C(x, \pi_x) d\alpha(x). \quad (108)$$

The classical Kantorovich problem is recovered when $C(x, \nu) = \int c(x, y) d\nu(y)$, because the objective then becomes $\int c(x, y) d\pi(x, y)$. The genuinely weak behavior starts when C is nonlinear in ν .

Proposition 71 (Weak Kantorovich duality). *Assume that $\mathcal{X}, \mathcal{Y}$ are compact metric spaces and that $C(x, \nu)$ is lower semicontinuous, bounded from below and convex in ν , with the standard qualification assumptions ensuring Fenchel–Rockafellar duality. For $g \in \mathcal{C}(\mathcal{Y})$ define the weak C -transform*

$$g^C(x) := \inf_{\nu \in \mathcal{P}(\mathcal{Y})} \left\{ C(x, \nu) - \int g(y) d\nu(y) \right\}.$$

Then

$$\mathcal{W}_C(\alpha, \beta) = \sup_{g \in \mathcal{C}(\mathcal{Y})} \left\{ \int g^C(x) d\alpha(x) + \int g(y) d\beta(y) \right\}.$$

When $C(x, \nu) = \int c(x, y) d\nu(y)$, this reduces to the usual Kantorovich dual with $g^C(x) = \inf_y (c(x, y) - g(y))$.

Proof. For any coupling π and any $g \in \mathcal{C}(\mathcal{Y})$, the definition of g^C gives

$$C(x, \pi_x) \geq g^C(x) + \int g(y) d\pi_x(y).$$

After integration with respect to α , the second term becomes $\int g d\beta$ because the second marginal of π is β . This proves weak duality.

For the reverse inequality, consider the convex minimization over probability kernels $x \mapsto \pi_x$ with the affine constraint $\int \pi_x d\alpha(x) = \beta$. Fenchel–Rockafellar duality gives a continuous Lagrange multiplier g for this marginal constraint. Minimizing the Lagrangian over each conditional law gives exactly the pointwise term $g^C(x)$, while the multiplier contributes $\int g d\beta$. The compactness, lower semicontinuity, convexity and qualification assumptions ensure no duality gap. This is the weak-cost analogue of Proposition 35. $\square$

Proposition 72 (Barycentric weak transport is weaker than $\mathcal{W}_2$). *Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ and define*

$$C_{\text{bar}}(x, \nu) = \|x - \int y d\nu(y)\|^2.$$

Then

$$\mathcal{W}_{C_{\text{bar}}}(\alpha, \beta) \leq \mathcal{W}_2^2(\alpha, \beta).$$

Proof. Let π be any coupling and disintegrate it as $\pi_x \alpha$. By Jensen’s inequality,

$$\|x - \int y d\pi_x(y)\|^2 \leq \int \|x - y\|^2 d\pi_x(y).$$

Integrating in x gives $\int C_{\text{bar}}(x, \pi_x) d\alpha(x) \leq \int \|x - y\|^2 d\pi(x, y)$. Taking the infimum over π proves the claim. $\square$

The barycentric cost is the canonical example to keep in mind: it only asks the conditional mean of Y given $X = x$ to be close to x , and ignores the conditional variance. This connects weak OT with martingale transport, Strassen-type convex-order constraints, barycentric projections and learning problems where conditional averages are meaningful objects.

9.8 Gromov Wasserstein

Gromov–Wasserstein compares spaces through their internal distance structures rather than through a fixed ambient ground cost. This is the right extension for graphs, shapes and point clouds whose points are not pre-aligned.

Optimal transport needs a ground cost C to compare histograms (a, b) , and thus cannot be used directly if the histograms are not defined on the same underlying space, or if one cannot pre-register these spaces to define a ground cost. To address this issue, one can instead use a weaker requirement: two matrices $D \in \mathbb{R}^{n \times n}$ and $D' \in \mathbb{R}^{m \times m}$ are available and represent relationships between the points on which the histograms are

defined. A typical scenario is when these matrices are powers of distance matrices. The Gromov–Wasserstein problem reads

$$\text{GW}((a, D), (b, D'))^p := \min_{P \in \mathcal{U}(a, b)} \mathcal{E}_{D, D'}(P) := \sum_{i, j, i', j'} \Delta(D_{i, i'}, D'_{j, j'})^p P_{i, j} P_{i', j'}, \quad (109)$$

where $p \geq 1$ and Δ is a distance on $\mathbb{R}$, typically $\Delta(u, v) = |u - v|$. This is a non-convex problem, which can be recast as a Quadratic Assignment Problem (QAP) [143] and is in full generality NP-hard to solve for arbitrary inputs. It is in fact equivalent to a graph matching problem [144] for a particular cost.

When the matrices D, D' are genuine distance matrices, the general construction below shows that GW satisfies the triangle inequality and defines a distance between metric spaces equipped with a probability distribution, up to measure-preserving isometries. This distance was introduced and studied in detail by Memoli in [151]. An in-depth mathematical exposition (in particular, its geodesic structure and gradient flows) is given in [207]. See also [192] for applications in computer vision. Its relation to Hausdorff and Gromov–Hausdorff distances is discussed at the end of this subsection.

General setting The general setting corresponds to computing couplings between metric measure spaces $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_{\mathcal{Y}}, \beta)$ where $(d_{\mathcal{X}}, d_{\mathcal{Y}})$ are distances and (α, β) are measures on their respective spaces. The natural setting is that of Polish metric spaces; compactness is often assumed in this section to avoid existence and integrability issues. One defines

$$\mathcal{GW}(\mathbb{X}, \mathbb{Y})^p := \min_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X}^2 \times \mathcal{Y}^2} \Delta(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(y, y'))^p d\pi(x, y) d\pi(x', y'). \quad (110)$$

Proposition 73 (Euclidean GW is controlled by Wasserstein). *Let α, β be probability measures on $\mathbb{R}^d$, equipped with the Euclidean distance, and take $\Delta(u, v) = |u - v|$ in (110). Then*

$$\mathcal{GW}((\mathbb{R}^d, \|\cdot\|, \alpha), (\mathbb{R}^d, \|\cdot\|, \beta)) \leq 2\mathcal{W}_p(\alpha, \beta).$$

Proof. Let π be any coupling between α and β . For two independent pairs $(X, Y), (X', Y') \sim \pi$, the reverse triangle inequality gives

$$\|X - X'\| - \|Y - Y'\| \leq \|X - Y\| + \|X' - Y'\|.$$

Taking the L^p norm and using Minkowski gives a bound by $2(\int \|x - y\|^p d\pi)^{1/p}$. Optimizing over π proves the claim. $\square$

In the following, one says that $\mathbb{X}$ and $\mathbb{Y}$ are isometric mm-spaces if there exists $\varphi : \mathcal{X} \rightarrow \mathcal{Y}$ such that $\varphi_{\#}\alpha = \beta$ and which is a bijective isometry between the supports of α and β , i.e. $d_{\mathcal{Y}}(\varphi(x), \varphi(x')) = d_{\mathcal{X}}(x, x')$ on the support of α .

The next theorem explains why the averaged distortion above is not merely a matching score. Once one quotients out measure-preserving isometries, it defines a genuine distance between metric-measure spaces.

Theorem 7 (Gromov–Wasserstein metric modulo isometries). *For compact metric-measure spaces, $p \geq 1$ and $\Delta(u, v) = |u - v|$, $\mathcal{GW}$ defines a distance up to measure-preserving isometries.*

Proof. If $\mathcal{GW}(\mathbb{X}, \mathbb{Y}) = 0$ and π is an optimal plan, then $d_{\mathcal{X}}(x, x') = d_{\mathcal{Y}}(y, y')$ holds $\pi \otimes \pi$ -almost everywhere. By continuity, this equality holds on $\text{supp}(\pi)^2$. We show that $\mathbb{X}$ and $\mathbb{Y}$ are isometric by showing that they are both isometric to the support space $(\text{supp}(\pi), d_{\pi}, \pi)$, where

$$d_{\pi}((x, y), (x', y')) := \frac{1}{2}d_{\mathcal{X}}(x, x') + \frac{1}{2}d_{\mathcal{Y}}(y, y'). \quad (111)$$

We now show that $\psi : (x, y) \in \text{supp}(\pi) \mapsto x \in \mathcal{X}$ is a bijective measure-preserving isometry between $\text{supp}(\pi)$ and $\text{supp}(\alpha)$. One has, for $((x, y), (x', y')) \in \text{supp}(\pi)^2$, $d_{\mathcal{X}}(x, x') = d_{\mathcal{Y}}(y, y')$ and thus

$$d_{\mathcal{X}}(\psi(x, y), \psi(x', y')) = d_{\mathcal{X}}(x, x') = d_{\mathcal{Y}}(y, y') = d_{\pi}((x, y), (x', y')),$$

thus ψ is an isometry, hence injective. We now show surjectivity, i.e. for $x \in \text{supp}(\alpha)$, we need to construct y such that $(x, y) \in \text{supp}(\pi)$. Indeed, since $\psi_{\#}\pi = \alpha$, there exists a sequence $(x_k, y_k) \in \text{supp}(\pi)$ such that $x_k \rightarrow x$. Since $d_{\mathcal{X}}(x_k, x_{k'}) = d_{\mathcal{Y}}(y_k, y_{k'})$ the sequence $(y_k)_k$ is Cauchy. Compactness of the supports gives a convergent subsequence of (x_k, y_k) with limit $(x, y) \in \text{supp}(\pi)$. The same argument for the second projection shows that this support space is also isometric to $\mathbb{Y}$.

Given an optimal π and ξ between three spaces $(\mathcal{X}, \mathcal{Y}, \mathcal{Z})$ with measures (α, β, γ) , we use the gluing Lemma 2 to define σ with marginals $(P_{\mathcal{X}, \mathcal{Y}})_{\#}\sigma = \pi$ and $(P_{\mathcal{Y}, \mathcal{Z}})_{\#}\sigma = \xi$. We denote $\rho := (P_{\mathcal{X}, \mathcal{Z}})_{\#}\sigma$ the composition of the two couplings, so that

$$\begin{aligned} \mathcal{GW}(\mathbb{X}, \mathbb{Z}) &\leq \left(\int_{\mathcal{X} \times \mathcal{Z}} \Delta(d_{\mathcal{X}}(x, x'), d_{\mathcal{Z}}(z, z'))^p d\rho(x, z) d\rho(x', z') \right)^{1/p} \\ &\leq \left(\int_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} (\Delta(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(y, y')) + \Delta(d_{\mathcal{Y}}(y, y'), d_{\mathcal{Z}}(z, z')))^p d\sigma(x, y, z) d\sigma(x', y', z') \right)^{1/p} \\ &\leq \left(\int_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} \Delta(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(y, y'))^p d\sigma(x, y, z) d\sigma(x', y', z') \right)^{1/p} \\ &\quad + \left(\int_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} \Delta(d_{\mathcal{Y}}(y, y'), d_{\mathcal{Z}}(z, z'))^p d\sigma(x, y, z) d\sigma(x', y', z') \right)^{1/p} \\ &\leq \mathcal{GW}(\mathbb{X}, \mathbb{Y}) + \mathcal{GW}(\mathbb{Y}, \mathbb{Z}), \end{aligned}$$

where the first inequality follows from the feasibility of ρ , the second is the triangular inequality of Δ , and the third is the Minkowski inequality. $\square$

The metric structure also gives geodesics. Sturm's construction below is useful conceptually because it allows one to speak about interpolation, barycenters and gradient flows directly on the space of metric-measure spaces, even though the intermediate space lives on a product support and is therefore expensive numerically [207].

Proposition 74 (Gromov–Wasserstein geodesics). *Let $\mathbb{X}_0 = (\mathcal{X}_0, d_{\mathcal{X}_0}, \alpha_0)$ and $\mathbb{X}_1 = (\mathcal{X}_1, d_{\mathcal{X}_1}, \alpha_1)$ be compact metric measure spaces, take $\Delta(u, v) = |u - v|$ in (110), and let π^* be an optimal coupling. Define, on $Z = \mathcal{X}_0 \times \mathcal{X}_1$,*

$$d_t((x_0, x_1), (x'_0, x'_1)) := (1 - t)d_{\mathcal{X}_0}(x_0, x'_0) + td_{\mathcal{X}_1}(x_1, x'_1), \quad \mathbb{X}_t = (Z, d_t, \pi^*).$$

At $t = 0$ and $t = 1$, and possibly in degenerate cases, one quotients Z by the zero-distance relation associated with d_t . Then $t \mapsto \mathbb{X}_t$ is a constant-speed geodesic for $\mathcal{GW}$:

$$\mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) = |t - s| \mathcal{GW}(\mathbb{X}_0, \mathbb{X}_1) \quad \forall s, t \in [0, 1].$$

Proof. Write $D = \mathcal{GW}(\mathbb{X}_0, \mathbb{X}_1)$. For $s < t$, couple $\mathbb{X}_s$ and $\mathbb{X}_t$ by the diagonal coupling induced by the identity map on Z and the measure π^* . For two independent points $z = (x_0, x_1)$ and $z' = (x'_0, x'_1)$ sampled from π^* ,

$$d_t(z, z') - d_s(z, z') = (t - s)(d_{\mathcal{X}_1}(x_1, x'_1) - d_{\mathcal{X}_0}(x_0, x'_0)).$$

Using this feasible coupling in (110) gives

$$\mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) \leq (t - s)D.$$

The same construction with the projections from Z to $\mathcal{X}_0$ and $\mathcal{X}_1$ gives $\mathcal{GW}(\mathbb{X}_0, \mathbb{X}_t) \leq tD$ and $\mathcal{GW}(\mathbb{X}_t, \mathbb{X}_1) \leq (1 - t)D$.

The triangle inequality for $\mathcal{GW}$ now yields the reverse inequalities. Indeed, for $0 \leq s \leq t \leq 1$,

$$D = \mathcal{GW}(\mathbb{X}_0, \mathbb{X}_1) \leq \mathcal{GW}(\mathbb{X}_0, \mathbb{X}_s) + \mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) + \mathcal{GW}(\mathbb{X}_t, \mathbb{X}_1) \leq sD + \mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) + (1 - t)D.$$

Thus $\mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) \geq (t - s)D$, which proves equality. This construction is due to Sturm [207]. Numerically, it is expensive because it lives on the product support; entropic finite-cardinality approximations for geodesics and barycenters are developed in [171]. $\square$

Entropic regularization and iterative solver We describe the common squared distortion, where $\Delta(u, v)^2 = (u - v)^2$. To approximate the computation of GW, and to help convergence of minimization schemes to better minima, one can consider the entropic regularized variant

$$\min_{P \in \mathcal{U}(a, b)} \mathcal{E}_{D, D'}(P) - \varepsilon H(P). \quad (112)$$

As proposed initially in [101, 179], and later revisited in [201] for applications in graphics, one can use iteratively Sinkhorn’s algorithm to progressively compute a stationary point of (112). Indeed, successive linearizations of the objective function lead to the updates

$$P^{(\ell+1)} := \min_{P \in \mathcal{U}(a, b)} \langle P, C^{(\ell)} \rangle - \varepsilon H(P) \quad \text{where} \quad (113)$$

$$C^{(\ell)} := D^{\odot 2} a \mathbf{1}_m^\top + \mathbf{1}_n (D'^{\odot 2} b)^\top - 2D P^{(\ell)} D'^\top,$$

up to an overall factor in the gradient. This can be interpreted as a mirror-descent scheme [201]. Each update can thus be solved using Sinkhorn iterations (33) with cost $C^{(\ell)}$. This entropic Gromov–Wasserstein solver can be used to compute soft maps between domains.

Fused Gromov–Wasserstein. For graphs and structured point clouds, the points often carry features in addition to an internal metric. For featured metric-measure spaces $(\mathcal{X}, d_{\mathcal{X}}, \alpha, f)$ and $(\mathcal{Y}, d_{\mathcal{Y}}, \beta, g)$, with $f : \mathcal{X} \rightarrow \mathcal{F}$, $g : \mathcal{Y} \rightarrow \mathcal{F}$ and feature cost $c_{\mathcal{F}}$, the fused Gromov–Wasserstein objective combines feature matching and structural matching:

$$\begin{aligned} \text{FGW}_{\lambda, p}^p(\mathbb{X}, \mathbb{Y}) = \min_{\pi \in \mathcal{U}(\alpha, \beta)} & \left\{ (1 - \lambda) \int_{\mathcal{X} \times \mathcal{Y}} c_{\mathcal{F}}(f(x), g(y)) d\pi(x, y) \right. \\ & \left. + \lambda \int_{\mathcal{X}^2 \times \mathcal{Y}^2} |d_{\mathcal{X}}(x, x') - d_{\mathcal{Y}}(y, y')|^p d\pi(x, y) d\pi(x', y') \right\} \end{aligned}$$

with $\lambda \in [0, 1]$. The case $\lambda = 0$ is ordinary OT between features, while $\lambda = 1$ is GW. This interpolation is particularly useful for labeled graphs, where node attributes and graph geometry should influence the same soft assignment [210].

Hausdorff and Gromov–Hausdorff viewpoints. If A, B are compact subsets of a common metric space (Z, d_Z) , their Hausdorff distance is

$$d_H^Z(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} d_Z(a, b), \sup_{b \in B} \inf_{a \in A} d_Z(a, b) \right\}.$$

The Gromov–Hausdorff distance removes the common ambient space by minimizing this quantity over all isometric embeddings into a third space:

$$d_{\text{GH}}(X, Y) = \inf_{Z, \varphi, \psi} d_H^Z(\varphi(X), \psi(Y)).$$

Equivalently, it is half the minimal distortion of a correspondence between X and Y [109, 150]. This is a worst-case set distance: every point must be matched with small distortion. Gromov–Wasserstein replaces correspondences by probability couplings and replaces worst-case distortion by an averaged distortion. It is therefore better adapted to noisy sampled shapes and weighted graphs, but it can ignore small sets of mass that would dominate the Hausdorff distance.

9.9 Metric learning and inverse OT

This final subsection points to inverse problems where the ground cost itself is learned. Such problems are typically bilevel and non-convex, but OT provides useful gradients with respect to the cost.

Metric learning and derivatives of OT OT is convex with respect to the measure and concave with respect to the cost. Ground-metric learning was explicitly studied in [65], and it connects to the broader metric-learning literature [131, 16].

Proposition 75 (Derivative with respect to the cost). *In the discrete setting, assume that the optimal coupling for $L_C(a, b)$ is unique and denote it by $P^*(C)$. Then $C \mapsto L_C(a, b)$ is differentiable at C and*

$$\nabla_C L_C(a, b) = P^*(C).$$

Proof. The value is the minimum of affine functions of C ,

$$L_C(a, b) = \min_{P \in \mathcal{U}(a, b)} \langle C, P \rangle.$$

The envelope theorem, or equivalently Danskin’s theorem, states that the subdifferential with respect to C is the convex hull of the optimal couplings. If the optimizer is unique, this subdifferential is the singleton $\{P^*(C)\}$, hence the value is differentiable with the displayed gradient. $\square$

Thus, if the cost is parameterized as C_θ , gradients of losses involving OT values are obtained by back-propagating through the inner product $\langle P^*(C_\theta), \partial_\theta C_\theta \rangle$. The difficulty is not differentiating a solved OT problem, but learning a cost for which the resulting matching has the desired semantic behavior; this is a bilevel and usually non-convex optimization problem.

Inverse Optimal Transport Inverse OT asks for a ground cost that explains observed matchings or flows as optimal transport plans. In its most direct form, one observes a plan $\hat{\pi}$ with marginals (α, β) and seeks a cost c such that $\hat{\pi}$ is optimal for

$$\inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int c(x, y) d\pi(x, y).$$

This is ill-posed without structure: adding potentials $u(x) + v(y)$ to a cost does not change the set of optimal couplings, and many costs can rationalize the same sparse plan.

A useful statistical methodology is to measure the suboptimality of the observed plan through a Fenchel–Young loss. Write the score as $s = -c$ and define the convex regularized prediction value

$$G_\varepsilon(s) = \sup_{\pi \in \mathcal{U}(\alpha, \beta)} \int s d\pi - \varepsilon \text{KL}(\pi | \alpha \otimes \beta).$$

The Fenchel–Young loss

$$\mathcal{L}_\varepsilon(c; \hat{\pi}) = G_\varepsilon(-c) + G_\varepsilon^*(\hat{\pi}) + \int c d\hat{\pi}$$

is nonnegative by Fenchel’s inequality and vanishes exactly when $\hat{\pi} \in \partial G_\varepsilon(-c)$, i.e. when $\hat{\pi}$ satisfies the regularized optimality conditions for c . Sharpened Fenchel–Young losses for inverse problems over measures and inverse entropic/unbalanced OT are developed in [9]; curvature and identifiability of inverse OT with respect to the cost are studied in [173]. Entropic regularization is important here because it makes the forward map smoother and provides gradients with respect to c , at the price of a bias that must be controlled statistically.

In practice, one restricts the cost to a finite-dimensional model class, often affine:

$$C_\theta = \sum_{r=1}^R \theta_r C^{(r)}, \quad \theta \in \Theta,$$

where Θ is convex and the matrices $C^{(r)}$ encode features, graph distances or a Mahalanobis parameterization. This viewpoint appears in low-rank and sparse inverse OT models [74, 8] and in convex formulations for learning OT costs from observed plans [145, 173].

Proposition 76 (Convex dual-gap formulation of inverse OT). *Let $\hat{P} \in \mathcal{U}(a, b)$ be an observed coupling and let C_θ depend affinely on $\theta \in \Theta$, where Θ is convex. The condition that $\hat{P}$ is optimal for the cost C_θ is equivalent to the existence of dual potentials (f, g) such that*

$$f_i + g_j \leq (C_\theta)_{i,j} \quad \text{and} \quad \sum_{i,j} \hat{P}_{i,j} ((C_\theta)_{i,j} - f_i - g_j) = 0.$$

Consequently, for a convex regularizer R , the noisy inverse problem can be relaxed as the convex program

$$\min_{\theta \in \Theta, f, g} R(\theta) + \lambda \sum_{i,j} \hat{P}_{i,j} ((C_\theta)_{i,j} - f_i - g_j) \quad \text{subject to} \quad f_i + g_j \leq (C_\theta)_{i,j} \quad \forall i, j. \quad (114)$$

Proof. For a fixed cost C_θ , Kantorovich duality gives

$$\min_{P \in \mathcal{U}(a,b)} \langle C_\theta, P \rangle = \max_{f_i + g_j \leq (C_\theta)_{i,j}} \langle f, a \rangle + \langle g, b \rangle.$$

Since $\hat{P}$ has marginals (a, b) , every dual feasible pair satisfies

$$\langle C_\theta, \hat{P} \rangle - \langle f, a \rangle - \langle g, b \rangle = \sum_{i,j} \hat{P}_{i,j} ((C_\theta)_{i,j} - f_i - g_j) \geq 0.$$

This nonnegative quantity is exactly the primal-dual gap of $\hat{P}$. It vanishes if and only if $\hat{P}$ reaches the dual value and is therefore optimal. If C_θ is affine and Θ and R are convex, the constraints and objective in (114) are convex, proving the relaxation claim. $\square$

The formulation (114) is useful because it avoids differentiating through a forward OT solver: it learns a cost by making the observed plan satisfy complementary slackness. In statistical settings, $\hat{P}$ is only partially observed or noisy, so one adds sparsity, low-rank, smoothness or metric constraints to select a meaningful cost [74, 8]. For entropic OT, the optimality condition becomes smoother:

$$\hat{P}_{i,j} \approx a_i b_j \exp \left(\frac{f_i + g_j - (C_\theta)_{i,j}}{\varepsilon} \right),$$

which leads to likelihood-based or KL-based convex objectives when C_θ is affine, and connects inverse OT with generalized Sinkhorn iterations and transport-regularized inverse problems [124, 145]. Neural parameterizations of C_θ are more flexible but reintroduce non-convexity; the convex formulation above is the clean mathematical baseline.

9.10 Quantum Optimal Transport

Quantum optimal transport replaces probability vectors by density matrices and scalar couplings by positive operators on a tensor product space. This is the right language when the transported objects are matrix-valued signals, covariance-like descriptors or quantum states, and it also exposes a precise bridge between OT, non-commutative entropy and operator scaling. The finite-dimensional formulation below follows the semidefinite viewpoint developed in matrix-valued and quantum OT [161, 56, 54, 169, 102, 44, 51].

Finite-dimensional states and couplings. Let $\mathbb{H}_n$ be the space of $n \times n$ Hermitian matrices,

$$\mathbb{H}_n^+ = \{A \in \mathbb{H}_n : A \succeq 0\}, \quad \mathbb{H}_n^{+,1} = \{A \in \mathbb{H}_n^+ : \text{tr}(A) = 1\}.$$

Elements of $\mathbb{H}_n^{+,1}$ are density matrices. A joint state between $\mathbb{C}^n$ and $\mathbb{C}^m$ is a matrix $T \in \mathbb{H}_{nm}^+$ acting on $\mathbb{C}^n \otimes \mathbb{C}^m$. Its marginals are the partial traces, defined by duality through

$$\text{tr}(F \text{Tr}_B(T)) = \text{tr}((F \otimes \text{Id}_m)T), \quad \text{tr}(G \text{Tr}_A(T)) = \text{tr}((\text{Id}_n \otimes G)T). \quad (115)$$

Thus $\text{Tr}_B(T) \in \mathbb{H}_n^+$ and $\text{Tr}_A(T) \in \mathbb{H}_m^+$ play exactly the role of the two marginals of a classical coupling. The feasible set is never empty, since $A \otimes B$ has marginals A and B .

Definition 20 (Finite-dimensional quantum OT). *Let $A \in \mathbb{H}_n^{+,1}$, $B \in \mathbb{H}_m^{+,1}$ and let $C \in \mathbb{H}_{nm}$ be a Hermitian cost observable. The quantum OT value is the semidefinite program*

$$\text{QOT}_C(A, B) = \min_{T \in \mathbb{H}_{nm}^+} \{ \text{tr}(CT) : \text{Tr}_B(T) = A, \text{Tr}_A(T) = B \}. \quad (116)$$

Remark 29 (Classical diagonal case). If A, B, C and T are all diagonal in fixed bases, then A and B are probability vectors, T is a nonnegative matrix and the partial-trace constraints reduce to the usual row and column sum constraints. Hence classical Kantorovich OT is the diagonal, commutative subcase of (116); the genuinely quantum feature is that T may contain off-diagonal coherences and entanglement.

Proposition 77 (Quantum Kantorovich duality). *For $A \in \mathbb{H}_n^{+,1}$ and $B \in \mathbb{H}_m^{+,1}$, the dual of (116) is*

$$\text{QOT}_C(A, B) = \max_{F \in \mathbb{H}_n, G \in \mathbb{H}_m} \{ \text{tr}(FA) + \text{tr}(GB) : F \otimes \text{Id}_m + \text{Id}_n \otimes G \preceq C \}. \quad (117)$$

If A and B are positive definite, strong duality follows directly from Slater's condition; the semidefinite case follows by restriction to the supports of A and B or by approximation.

Proof. Introduce Hermitian Lagrange multipliers F and G for the two marginal constraints. The Lagrangian is

$$\text{tr}(CT) + \text{tr}(F(A - \text{Tr}_B T)) + \text{tr}(G(B - \text{Tr}_A T)) = \text{tr}(FA) + \text{tr}(GB) + \text{tr}((C - F \otimes \text{Id}_m - \text{Id}_n \otimes G)T),$$

where (115) was used in the last equality. Minimizing over $T \succeq 0$ gives a finite lower bound if and only if $C - F \otimes \text{Id}_m - \text{Id}_n \otimes G \succeq 0$, in which case the infimum in T is 0. This gives the dual program. When $A, B \succ 0$, the coupling $A \otimes B$ is strictly feasible, so Slater's theorem gives equality of primal and dual values and dual attainment. The general finite-dimensional semidefinite case is obtained by approximation $A_\delta = (1 - \delta)A + \delta \text{Id}_n/n$, $B_\delta = (1 - \delta)B + \delta \text{Id}_m/m$ and by compactness, or equivalently by reducing to the supports of A and B . $\square$

The dual potentials have the usual scalar gauge freedom: replacing (F, G) by $(F + t\text{Id}_n, G - t\text{Id}_m)$ leaves both the constraint and the value unchanged because $\text{tr}(A) = \text{tr}(B) = 1$.

Entropic regularization and Bregman iterations. As in scalar OT, one obtains a smoother problem by adding the von Neumann entropy

$$H(T) = \text{tr}(T(\log T - \text{Id}_{nm})), \quad \nabla H(T) = \log T,$$

with the convention $0 \log 0 = 0$. For $\varepsilon > 0$ define

$$\text{QOT}_C^\varepsilon(A, B) = \min_{T \succeq 0} \{ \text{tr}(CT) + \varepsilon H(T) : \text{Tr}_B(T) = A, \text{Tr}_A(T) = B \}. \quad (118)$$

This is the non-commutative analogue of entropic OT [169, 51]: the Shannon entropy of a coupling is replaced by the trace entropy of a density matrix.

Proposition 78 (Entropic quantum OT duality). *Assume $A \succ 0$, $B \succ 0$ and $\varepsilon > 0$. Then (118) has a unique positive minimizer. Its dual is*

$$\text{QOT}_C^\varepsilon(A, B) = \max_{F \in \mathbb{H}_n, G \in \mathbb{H}_m} \left\{ \text{tr}(FA) + \text{tr}(GB) - \varepsilon \text{tr} \exp \left(\frac{F \otimes \text{Id}_m + \text{Id}_n \otimes G - C}{\varepsilon} \right) \right\}. \quad (119)$$

At optimality, primal and dual variables are linked by the Gibbs formula

$$T_e(F, G) = \exp \left(\frac{F \otimes \text{Id}_m + \text{Id}_n \otimes G - C}{\varepsilon} \right), \quad (120)$$

with $\text{Tr}_B(T_e) = A$ and $\text{Tr}_A(T_e) = B$.

Proof. The trace entropy is strictly convex on positive matrices, hence the regularized primal has at most one positive minimizer. The Fenchel identity

$$\sup_{T \succeq 0} \operatorname{tr}(YT) - \varepsilon H(T) = \varepsilon \operatorname{tr} \exp(Y/\varepsilon)$$

is the matrix analogue of the scalar exponential conjugacy. Applying it to the Lagrangian of (118), with $Y = F \otimes \operatorname{Id}_m + \operatorname{Id}_n \otimes G - C$, gives (119). The stationarity condition of this Fenchel identity gives (120); differentiating the dual objective with respect to F and G yields the two marginal equations. $\square$

Writing $K = e^{-C/\varepsilon}$, the objective in (118) differs by a constant from ε times the quantum KL divergence

$$D_H(T|K) = \operatorname{tr}(T(\log T - \log K) - T + K).$$

The exact quantum analogue of Sinkhorn is an implicit alternating Bregman projection scheme onto the affine marginal sets

$$\mathcal{M}_A = \{T \succeq 0 : \operatorname{Tr}_B(T) = A\}, \quad \mathcal{M}_B = \{T \succeq 0 : \operatorname{Tr}_A(T) = B\}.$$

Proposition 79 (Exact Bregman projections). *Assume $A, B \succ 0$ and let $K = e^{-C/\varepsilon}$. The minimizer of (118) is equivalently the minimizer of $D_H(T|K)$ over $\mathcal{M}_A \cap \mathcal{M}_B$. Moreover, if a current positive definite matrix has Gibbs form $T_e(F, G)$, then its Bregman projection onto $\mathcal{M}_A$ has the form $T_e(F^+, G)$, where F^+ is chosen so that $\operatorname{Tr}_B T_e(F^+, G) = A$. The projection onto $\mathcal{M}_B$ is analogous. Thus, when each one-block marginal equation is solved exactly, alternating Bregman projections are equivalent to alternating block maximization of the dual (119).*

Proof. Since $\log K = -C/\varepsilon$, the identity

$$\operatorname{tr}(CT) + \varepsilon H(T) = \varepsilon D_H(T|K) - \varepsilon \operatorname{tr}(K)$$

holds, so the primal minimizer is the constrained Bregman projection of K up to an additive constant. Since $A \succ 0$, this projection stays in the interior of the positive cone. For the projection of a positive definite matrix S onto $\mathcal{M}_A$, the Lagrangian first variation is

$$\log T - \log S - \Lambda \otimes \operatorname{Id}_m = 0$$

for a Hermitian multiplier Λ . Hence $T = \exp(\log S + \Lambda \otimes \operatorname{Id}_m)$. If $S = T_e(F, G)$, this is again of the form $T_e(F + \varepsilon \Lambda, G)$. The multiplier is fixed by the marginal equation $\operatorname{Tr}_B(T) = A$. The same argument applies to $\mathcal{M}_B$. Finally, the first-order optimality condition for maximizing (119) over one block is exactly the corresponding marginal equation, so the Bregman and block-dual views coincide. $\square$

In the diagonal case this proposition gives the usual multiplicative Sinkhorn updates. In the non-commutative case, however, the exact block equations

$$\operatorname{Tr}_B T_e(F, G) = A, \quad \operatorname{Tr}_A T_e(F, G) = B$$

do not admit scalar division formulas, because the exponential of $F \otimes \operatorname{Id}_m + \operatorname{Id}_n \otimes G - C$ cannot be separated unless the local potential $F \otimes \operatorname{Id}_m + \operatorname{Id}_n \otimes G$ commutes with the cost C . When all matrices are diagonal in the same basis, commutativity restores the scalar form $T_e = \operatorname{diag}(u)K \operatorname{diag}(v)$ and the marginal equations reduce to the usual Sinkhorn divisions.

Gurvits scaling and quantum Sinkhorn. The algorithm often called quantum Sinkhorn comes from the operator-scaling literature of Gurvits and subsequent developments [110, 111, 97, 93]. It replaces the true Gibbs coupling (120) by the symmetric factorization

$$T_s(F, G) = e^{\frac{Z}{2\varepsilon}} e^{-C/\varepsilon} e^{\frac{Z}{2\varepsilon}} = (U \otimes V) K (U \otimes V), \quad Z = F \otimes \text{Id}_m + \text{Id}_n \otimes G, \quad (121)$$

where $U = e^{F/(2\varepsilon)}$, $V = e^{G/(2\varepsilon)}$ and $K = e^{-C/\varepsilon}$. If $[Z, C] = 0$, then $T_s(F, G) = T_e(F, G)$; otherwise this is a Strang-type symmetric surrogate.

Define the completely positive map associated with K by

$$\mathcal{K}(X) = \text{Tr}_B((\text{Id}_n \otimes X)K), \quad \mathcal{K}^*(Y) = \text{Tr}_A((Y \otimes \text{Id}_m)K).$$

For the symmetric coupling, the marginal equations become

$$U \mathcal{K}(V^2) U = A, \quad V \mathcal{K}^*(U^2) V = B.$$

They can be enforced by the explicit congruence normalizations

$$\begin{aligned} R_V &= \mathcal{K}(V^2), \quad U \leftarrow R_V^{-1/2} (R_V^{1/2} A R_V^{1/2})^{1/2} R_V^{-1/2}, \\ S_U &= \mathcal{K}^*(U^2), \quad V \leftarrow S_U^{-1/2} (S_U^{1/2} B S_U^{1/2})^{1/2} S_U^{-1/2}. \end{aligned} \quad (122)$$

These inverse square roots are well-defined when $K \succ 0$ and $U, V, A, B \succ 0$. This is Gurvits/operator scaling with prescribed targets; when all matrices are diagonal it reduces to classical Sinkhorn scaling, and when the targets are proportional to identities it matches the usual bistochastic operator-scaling normalization, up to the conventional trace normalization.

It is important not to identify (122) with the exact Bregman scheme for (118). The exact Bregman step would enforce the marginals of $T_e(F, G) = \exp((Z - C)/\varepsilon)$ and would be a block maximization of the true concave dual (119). Gurvits scaling enforces the marginals of the surrogate $T_s = e^{Z/(2\varepsilon)} e^{-C/\varepsilon} e^{Z/(2\varepsilon)}$. The two coincide in the commuting/diagonal regime, but in general noncommutativity produces commutator errors and the Gurvits iteration should be understood as a tractable symmetric approximation to entropic Q-OT, not as the literal alternating KL projection algorithm.

Remark 30 (Operator-valued couplings). The same definitions extend formally from matrices to separable Hilbert spaces by replacing density matrices with positive trace-class operators of trace one, observables with bounded self-adjoint operators and (115) with partial traces defined by duality against local bounded observables. If $\Pi(A, B)$ denotes positive trace-class operators with partial traces A and B , a bounded cost observable C gives the problem $\inf_{T \in \Pi(A, B)} \text{Tr}(CT)$. For unbounded positive costs one must define the energy through the quadratic form or spectral truncations, and in the entropic case one must ensure that the Gibbs operator $e^{-C/\varepsilon}$ is trace class and that the partial traces of the candidate coupling are well-defined. The matrix formulas above are therefore the clean finite-dimensional core; the operator version adds domain and compactness assumptions rather than a different algebraic structure.

10 Wasserstein (gradient) Flows

This section explains why the Wasserstein distance is not only a way to compare measures, but also a geometry in which measures can move. The stakes are dynamical: continuity equations describe how mass evolves, Benamou–Brenier turns $\mathcal{W}_2$ into a least-action principle, and JKO schemes turn PDEs into gradient descents on probability space. This viewpoint originates in the variational formulation of Fokker–Planck equations [121], Otto’s formal Riemannian calculus [163], and the metric-space theory of gradient flows [6, 191].

The goal is to make precise the connection between optimal transport and evolutions on the space of probability distributions, particularly diffusion PDEs and generative models. We work over $\mathcal{X} = \mathbb{R}^d$ and

emphasize formulas that are rigorous under smoothness and positivity assumptions, while indicating where metric-measure theory is needed for full generality. The section also connects this viewpoint with recent ML applications: training dynamics of wide MLPs, deep transformers viewed as token flows, and flow matching for generative modeling.

10.1 Evolutions over the Space of Measures

We start with the continuity equation because it is the common language for particles, densities and weak measure evolutions. It also makes precise which velocity fields actually move mass.

We consider the evolution $t \mapsto \alpha_t \in \mathcal{P}(\mathbb{R}^d)$. Such evolution can be described in a “Lagrangian” way as the advection of particles along a (time-dependent) vector field $v_t(x)$ in $\mathbb{R}^d$. At the particle level, this advection is governed by

$$\frac{dx(t)}{dt} = v_t(x(t)), \quad (123)$$

such that $x(0)$ is mapped to $x(t)$ by a “transport” mapping $T_t : x(0) \mapsto x(t)$. The fact that α_t is the density of advected particles implies $\alpha_t = (T_t)_\# \alpha_0$. For discrete measures, $\alpha_t = \frac{1}{n} \sum_{i=1}^n \delta_{x_i(t)}$, meaning each $x_i(t)$ solves (123).

In the Eulerian interpretation, over the measure itself, as we show next, the ODE for particles becomes the PDE

$$\frac{\partial \alpha_t}{\partial t} + \operatorname{div}(v_t \alpha_t) = 0. \quad (124)$$

This PDE is often referred to as the advection equation, the continuity equation, or Liouville’s equation when operating over a phase space. This PDE is only valid if α_t has a smooth density, and for general measures, this PDE (124) should be understood in the weak sense, allowing it to be defined even for discrete measures with particles evolving according to (123). Specifically, for any smooth test function $(t, x) \rightarrow \varphi(t, x)$ supported in time (in particular it should vanish at $t = 0, 1$),

$$\int_0^1 \int_{\mathbb{R}^d} (\partial_t \varphi(t, x) + \langle v_t(x), \nabla_x \varphi(t, x) \rangle) d\alpha_t(x) dt = 0. \quad (125)$$

This equation can be obtained from (124) by integration by parts. So if α_t has a smooth density, then these two formulations are equivalent, but (125) makes sense even if $\alpha_{t=0}$ is a discrete measure. We now show rigorously that measures whose particles follow the ODE (123) satisfy the advection PDE.

Proposition 80 (Lagrangian flows solve the continuity equation). *Consider a smooth flow $T_t : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and denote $\alpha_t := (T_t)_\# \alpha_0$. Define the Eulerian velocity field by*

$$v_t(T_t(y)) := \partial_t T_t(y).$$

Then (α_t, v_t) solves the continuity equation in the weak sense (125).

Note that if $\alpha_0 = \frac{1}{n} \sum_i \delta_{x_i(0)}$ is made of particles, then $\alpha_t = \frac{1}{n} \sum_i \delta_{x_i(t)} = (T_t)_\# \alpha_0$ is also made of particles evolving at speed $\dot{x}_i(t) = v_t(x_i(t))$.

Proof. Let $\varphi(t, x)$ be a smooth test function vanishing at $t = 0$ and $t = 1$. Using $\alpha_t = (T_t)_\# \alpha_0$ and the chain rule,

$$\frac{d}{dt} \int \varphi(t, x) d\alpha_t(x) = \frac{d}{dt} \int \varphi(t, T_t(y)) d\alpha_0(y) = \int (\partial_t \varphi(t, T_t(y)) + \langle \nabla_x \varphi(t, T_t(y)), \partial_t T_t(y) \rangle) d\alpha_0(y).$$

By the definition of v_t , this is

$$\int (\partial_t \varphi(t, x) + \langle \nabla_x \varphi(t, x), v_t(x) \rangle) d\alpha_t(x).$$

Integrating in time and using the vanishing boundary values of φ gives (125). $\square$

From measure evolutions to vector fields. It is important to note that for a given evolution α_t , there are infinitely many possible choices of vector fields v_t satisfying

$$\partial_t \alpha_t + \operatorname{div}(v_t \alpha_t) = 0. \quad (126)$$

This is because modifying v_t by a divergence-free field does not affect the density evolution. The linear space of vector fields that leave a measure α invariant is

$$\mathcal{H}_\alpha := \{v : \operatorname{div}(\alpha v) = 0\}.$$

It is non-trivial because it corresponds to the kernel of a “weighted” divergence. For instance, if α is an isotropic Gaussian, $\mathcal{H}_\alpha$ contains vector fields inducing rotations (i.e., associated with anti-symmetric matrices).

Dacorogna and Moser inversion. Consequently, infinitely many particle evolutions result in the same density. Reconstructing particle evolution from observed density evolution (sometimes called “trajectory inference”) is thus an ill-posed inverse problem. A simple choice is to impose that $\alpha_t v_t$ is a gradient field, thus leading to the inversion of a Laplacian (which is possible assuming boundary conditions, for instance, vanishing at infinity)

$$v_t = -\frac{1}{\alpha_t} \nabla \Delta^{-1}(\partial_t \alpha_t). \quad (127)$$

which was initially proposed by Dacorogna and Moser. A difficulty with this choice is that it is not well defined when α_t vanishes, and also it is not a gradient field, which might be desirable in some cases. In the following, we will consider techniques where v_t is a gradient by design, it is well-defined and can be computed more efficiently without explicitly inverting a Laplacian.

Least square inversion and gradient structure. A more fruitful method (corresponding to both flow matching, optimal transport, and Wasserstein flow construction) is to solve the inversion from α_t to v_t in a least-square sense, solving

$$\min_v \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 d\alpha_t(x) dt \quad \text{subject to} \quad \operatorname{div}(\alpha v) + \frac{\partial \alpha}{\partial t} = 0. \quad (128)$$

To understand why the optimal v_t is a gradient field, one can consider an equivalent constrained optimization reformulated as a saddle-point (inf-sup) problem using a Lagrange multiplier $\psi_t(x)$, which plays the role of a dual variable enforcing the continuity equation:

$$\min_v \max_\psi \frac{1}{2} \int_0^1 \left[\int_{\mathbb{R}^d} \|v_t(x)\|^2 d\alpha_t(x) + \int_{\mathbb{R}^d} \psi_t(x) (\operatorname{div}(\alpha_t v_t)(x) + \partial_t \alpha_t(x)) dx \right] dt. \quad (129)$$

Integrating by parts the term involving the divergence moves the derivative onto ψ_t

$$\min_v \max_\psi \int_0^1 \left[\int_{\mathbb{R}^d} \left(\frac{1}{2} \|v_t(x)\|^2 - \nabla \psi_t(x) \cdot v_t(x) \right) d\alpha_t(x) + \int_{\mathbb{R}^d} \psi_t(x) \partial_t \alpha_t(x) dx \right] dt. \quad (130)$$

Minimizing over v_t for fixed ψ_t gives the first-order condition:

$$v_t(x) = \nabla \psi_t(x). \quad (131)$$

Hence, the optimal vector field is necessarily a gradient. Substituting this back, one obtains an equation linking ψ_t and $\partial_t \alpha_t$. This results in the explicit expression:

$$v_t = -\nabla \Delta_\alpha^{-1}(\partial_t \alpha_t), \quad \text{with} \quad \Delta_\alpha(\varphi) := \operatorname{div}(\alpha_t \nabla \varphi). \quad (132)$$

In general, performing this inversion is computationally demanding, but under specific choices for α_t , simpler formulas may be derived.

10.2 Generative Models via Flow Matching

Flow matching constructs a generative map by learning the velocity field of an interpolation. The key computational insight is that a constrained continuity-equation problem can be trained by an unconstrained regression.

Generative models aim to build a transportation map T between a reference distribution α (typically an isotropic Gaussian) and the target data distribution β . Since such reference measures are non-atomic, a measurable map with $T_\# \alpha = \beta$ exists on standard Borel spaces, for instance by identifying both probability spaces with the unit interval and using a quantile-type rearrangement. This abstract existence statement is much weaker than having an explicit and numerically stable construction of T . Optimal transport is one approach to achieving this, but it is computationally expensive and raises questions about how to estimate it from samples. A recent idea, first introduced in diffusion models [117, 202] and later systematically developed in flow matching by Yaron Lipman and collaborators [141], is to obtain T by integrating a time-dependent vector field v_t . This vector field v_t is obtained by constructing an interpolation α_t and then finding v_t using the least square formula (132). As we will explain, for a specific class of interpolation (obtained by a parametric push-forward), this v_t can be obtained by avoiding explicitly inverting a Laplacian and instead computing a simple conditional expectation. This conditional expectation can itself be estimated by solving another least squares, but this time unconstrained, making the estimation feasible from finite samples of α and β .

Stochastic interpolant. We assume that α_t is defined via a “projection” (in a loose sense) of a latent distribution $\pi \in \mathcal{P}(\mathbb{R}^{d'})$, using an operator $P_t : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$ where $d' \gg d$, i.e.

$$\forall t \in [0, 1], \quad \alpha_t := (P_t)_\# \pi. \quad (133)$$

A common case is $d' = 2d$. We write $(x, y) \in \mathbb{R}^{d'} = \mathbb{R}^d \times \mathbb{R}^d$ and assume $P_0(x, y) = x$ and $P_1(x, y) = y$, so that π is a probabilistic coupling between α_0 and α_1 , i.e. π has marginals (α_0, α_1) . For instance, one can use $\pi = \alpha_0 \otimes \alpha_1$, the trivial coupling. An even more special case is the linear interpolation $P_t(x, y) = (1-t)x + ty$. More complex constructions are also possible, provided that sampling from π remains simple; stochastic interpolants provide a general framework unifying deterministic flows and diffusions [2].

If $\pi = \alpha \otimes \beta$ and $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$, $\beta = \frac{1}{m} \sum_j \delta_{y_j}$, then α_t consists of $n \times m$ Dirac masses

$$\alpha_t = \frac{1}{nm} \sum_{i,j} \delta_{P_t(x_i, y_j)}.$$

If $\pi = (\text{Id}, T)_\# \alpha$ is a Brenier-type coupling, then $\alpha_t = ((1-t)\text{Id} + tT)_\# \alpha$ is the so-called McCann OT interpolation.

Flow matching formula. This interpolation is not directly useful for sampling from β , but it can be used to define a flow field v_t so that the Eulerian advection equation (124) holds. This flow field is computed by solving an unconstrained least squares problem, or equivalently, it is a conditional expectation.

Proposition 81 (Flow matching vector field). *For each fixed t , the solution of the following flow-matching problem*

$$\min_{v_t} \int_{\mathbb{R}^{d'}} \|v_t(P_t(u)) - [\partial_t P_t](u)\|^2 d\pi(u). \quad (134)$$

or equivalently the conditional expectation

$$v_t(z) = \mathbb{E}_{u \sim \pi}([\partial_t P_t](u) \mid z = P_t(u)). \quad (135)$$

satisfies the continuity equation (126).

Proof. We now prove the flow matching formula (135), i.e. that it defines a valid flow between α_0 and α_1 . First, let us recall the definition of the interpolated measure α_t and the velocity field v_t . According to (133), the measure α_t can be represented formally as:

$$\alpha_t(z) = \int \delta(z - P_t(u)) \, d\pi(u)$$

and rigorously for any test function $\varphi(z)$ as:

$$\int \varphi(z) \, d\alpha_t(z) = \int \varphi(P_t(u)) \, d\pi(u). \quad (136)$$

Following (135), the velocity field v_t can be written formally as:

$$v_t(z) = \frac{1}{\alpha_t(z)} \int \delta(z - P_t(u)) [\partial_t P_t](u) \, d\pi(u),$$

and rigorously for any vector field $m(z)$ as:

$$\int \langle m(z), v_t(z) \rangle \, d\alpha_t(z) = \int \langle m(P_t(u)), [\partial_t P_t](u) \rangle \, d\pi(u). \quad (137)$$

We aim to prove that the measure α_t satisfies the continuity equation:

$$\frac{\partial \alpha_t}{\partial t} + \operatorname{div}(\alpha_t v_t) = 0.$$

In a rigorous sense, this means showing that for any smooth test function $\varphi(z)$,

$$\frac{d}{dt} \int \varphi(z) \, d\alpha_t(z) - \int \langle v_t(z), \nabla \varphi(z) \rangle \, d\alpha_t(z) = 0. \quad (138)$$

To prove (138), we compute both terms separately and show that they cancel out. First, consider the time derivative of $\int \varphi(z) \, d\alpha_t(z)$. Using (136), we differentiate under the integral sign:

$$\frac{d}{dt} \int \varphi(z) \, d\alpha_t(z) = \int \frac{d}{dt} \varphi(P_t(u)) \, d\pi(u).$$

Applying the chain rule to $t \rightarrow \varphi \circ P_t$:

$$\frac{d}{dt} \varphi(P_t(u)) = \langle \nabla \varphi(P_t(u)), [\partial_t P_t](u) \rangle.$$

Thus:

$$\frac{d}{dt} \int \varphi(z) \, d\alpha_t(z) = \int \langle \nabla \varphi(P_t(u)), [\partial_t P_t](u) \rangle \, d\pi(u). \quad (139)$$

Next, for the term involving $\operatorname{div}(\alpha_t v_t)$, we use the definition of v_t in (137) with $m(z) = \nabla \varphi(z)$. Substituting this into the expression for v_t , we get:

$$\int \langle v_t(z), \nabla \varphi(z) \rangle \, d\alpha_t(z) = \int \langle \nabla \varphi(P_t(u)), [\partial_t P_t](u) \rangle \, d\pi(u).$$

Comparing this result with (139), we see that:

$$\frac{d}{dt} \int \varphi(z) \, d\alpha_t(z) - \int \langle v_t(z), \nabla \varphi(z) \rangle \, d\alpha_t(z) = 0.$$

□

The solution of (134) is the conditional expectation of the velocities $\partial_t P_t$, intuitively, this means that $v_t(z)$ at some point z should be the average velocity of all trajectories passing through z . Numerically, $(x, t) \rightarrow v_t(x)$ can be parameterized by a neural network (e.g., a U-Net for vision tasks) and estimated using stochastic gradient descent on the objective in (134). Once v_t is estimated, integrating the ODE $\dot{x} = v_t(x)$ defines the transport map T_t , ensuring that $\alpha_t = (T_t)_\# \alpha_0$ produces the same interpolation as (133), though with a different particle system. Instead of a coupling, this approach uses a deterministic map. The sampling procedure consists in first drawing $X_0 \sim \alpha$, and then integrating the ODE $\dot{X}_t = v_t(X_t)$ starting with $X_{t=0} = X_0$. The resulting $X_{t=1}$ is distributed according to $\alpha_1 = \beta$.

Connection with diffusion models. In the special case where $P_t(x, y) = (1 - t)x + ty$ is a linear interpolation and $\pi = \alpha \otimes \beta$, then α_t is a convolution of rescaled versions of α_0 and α_1 . Then the flow matching (134) boils down to

$$\min_{(v_t)_t} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|v_t((1 - t)x + ty) - (y - x)\|^2 d\alpha_0(x) d\alpha_1(y).$$

If furthermore, α_0 is an isotropic Gaussian, this is exactly (up to an exponential change of variable in the time variable) the diffusion model method. In this case, the next two propositions give an explicit closed form for v_t and show that it is a gradient field. This means that in this setting, v_t is also the solution of the (constrained) least square problem (132). But note that the least square (134) is much simpler because it is unconstrained. To prove this, we rely on a celebrated formula which gives the expression of the optimal Gaussian denoiser as being the scores (gradient of log-density).

Proposition 7 (Tweedie identity). *Let W be a random vector in $\mathbb{R}^d$ with density β . For $\sigma > 0$, observe*

$$Z = W + \sigma \varepsilon, \quad \text{where } \varepsilon \sim \mathcal{N}(0, I_d) \text{ is independent of } W.$$

Denote by

$$\beta_\sigma = \beta * \mathcal{N}(0, \sigma^2 I_d)$$

the density of Z . Then

$$\mathbb{E}[W \mid Z = z] = z + \sigma^2 \nabla \log \beta_\sigma(z) \quad \text{for all } z \in \mathbb{R}^d.$$

Proof. Bayes' rule gives the conditional density $p_{W|Z}(w \mid z) = \frac{\beta(w) \varphi_\sigma(z - w)}{\beta_\sigma(z)}$ with φ_σ the $\mathcal{N}(0, \sigma^2 I_d)$ density. Hence

$$\mathbb{E}[W \mid Z = z] = \frac{1}{\beta_\sigma(z)} \int_{\mathbb{R}^d} w \beta(w) \varphi_\sigma(z - w) dw.$$

Integrating by parts in w and using $\nabla_z \varphi_\sigma(z - w) = -\sigma^{-2}(z - w) \varphi_\sigma(z - w)$ yields

$$\nabla_z \beta_\sigma(z) = \int \beta(w) \nabla_z \varphi_\sigma(z - w) dw = -\sigma^{-2} \left(z - \mathbb{E}[W \mid Z = z] \right) \beta_\sigma(z).$$

Rearranging finishes the proof. □

Proposition 8 (Optimal vector field). *Let $X \sim \alpha$ and $Y \sim \mathcal{N}(0, I_d)$ be independent. For $t \in (0, 1)$ set*

$$Z_t = (1 - t)X + tY, \quad \alpha_t = \text{Law}(Z_t).$$

The minimizer v^ of*

$$\min_v \int_0^1 \iint_{\mathbb{R}^d \times \mathbb{R}^d} |y - x - v((1 - t)x + ty, t)|^2 d\alpha(x) d\mathcal{N}(y) dt$$

is

$$v^*(x, t) = -\frac{1}{1 - t}x - \frac{t}{1 - t} \nabla \log \alpha_t(x) \quad (x \in \mathbb{R}^d, t \in (0, 1)).$$

Proof. Fix $t \in (0, 1)$ and write $W = (1 - t)X$, $\sigma = t$, so that $Z_t = W + \sigma Y$ matches the setting of Proposition 7. Conditional expectations satisfy $v^*(z, t) = \mathbb{E}[Y - X \mid Z_t = z] = \frac{1}{t} \mathbb{E}[Z_t - W \mid Z_t = z] - \frac{1}{1-t} \mathbb{E}[W \mid Z_t = z]$. Applying Proposition 7 to $\mathbb{E}[W \mid Z_t = z]$ and noting $\mathbb{E}[Y \mid Z_t = z] = -t \nabla \log \alpha_t(z)$ gives the claimed formula. $\square$

When is the induced map optimal? Integrating the learned velocity gives a deterministic map from α_0 to α_1 , but this map is not automatically the Brenier optimal map. It is optimal only in special cases where the accumulated flow remains the gradient of a convex potential. For Gaussian marginals, the flow is affine. If the covariance matrices co-diagonalize, the affine map is symmetric positive definite in the common eigenbasis, hence it coincides with the Gaussian Brenier map. More generally, non-commuting covariance dynamics or mixture-like paths can create rotational or shearing components, and the final map need not be optimal even if every instantaneous velocity looks natural. This distinction is closely related to counterexamples showing that flow maps associated with Fokker–Planck or diffusion-type evolutions do not in general provide optimal transport maps [133]. In particular, starting from an isotropic Gaussian does not by itself guarantee optimality once the target distribution is non-Gaussian; additional structural assumptions on the path or on the coupling are needed.

10.3 Benamou-Brenier dynamic formulation of OT

The dynamic formulation identifies $\mathcal{W}_2$ with the kinetic energy of the cheapest continuity-equation path. It is the point where OT becomes a least-action principle.

Instead of imposing access to the full dynamics $(\alpha_t)_{t=0}^1$, we assume only the knowledge of α_0 and α_1 and seek for an interpolation minimizing the least square energy (128). A fundamental result, proved by Benamou and Brenier [17] is that the value of this “geodesic” energy is equal to the Wasserstein-2 distance. Furthermore, the solution (v_t, α_t) of this geodesic problem can be obtained by evolving particles in straight lines along the Monge map T solving Monge’s problem

Theorem 8 (Benamou–Brenier). *For probability measures α_0, α_1 with finite second moments,*

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{(\alpha_t, v_t)} \int_0^1 \int_{\mathbb{R}^d} |v_t(x)|^2 d\alpha_t(x) dt, \quad (140)$$

where the infimum is over (α_t, v_t) solving $\partial_t \alpha_t + \nabla \cdot (\alpha_t v_t) = 0$ with $\alpha_{t=0} = \alpha_0$ and $\alpha_{t=1} = \alpha_1$. If α_0 has a density and T is the optimal Monge map $T_{\#} \alpha_0 = \alpha_1$, the minimizer is

$$\alpha_t = ((1 - t)\text{Id} + tT)_{\#} \alpha_0, \quad v_t((1 - t)x + tT(x)) = T(x) - x, \quad t \in [0, 1]. \quad (141)$$

Proof. *Dynamic $\leq$ static.* Let T solve the Monge problem and define (α_t, v_t) by (141). As $\dot{T}_t(x) = T(x) - x$ is t -independent,

$$\int_0^1 \int |v_t|^2 d\alpha_t dt = \int |T(x) - x|^2 d\alpha_0(x),$$

so the dynamic cost is no larger.

Static $\leq$ dynamic. Conversely, for a smooth deterministic path take the flow T_t defined by $\dot{T}_t = v_t \circ T_t$, $T_0 = \text{Id}$. Then $\alpha_t = (T_t)_{\#} \alpha_0$ and $(T_1)_{\#} \alpha_0 = \alpha_1$. Jensen’s inequality gives

$$|T_1(x) - x|^2 \leq \int_0^1 |v_t(T_t(x))|^2 dt.$$

Integrating yields

$$\int |T_1(x) - x|^2 d\alpha_0 \leq \int_0^1 \int |v_t|^2 d\alpha_t dt.$$

Thus the Monge cost is no larger than the dynamic one. For general finite-energy solutions of the continuity equation, the same argument is obtained by applying the superposition principle to lift the curve to a probability measure on absolutely continuous paths; this is the path-space formulation stated below. $\square$

While this is not the focus of this chapter, let us note that although the initial problem (140) is non-convex, after the change of variable $m_t := \alpha_t v_t$, it becomes convex in (m_t, α_t) . This convex reformulation enables solving geodesic interpolation using convex optimization techniques once the domain is discretized.

Remark 31 (Path-space formulation). Let $\mathcal{S} = C([0, 1]; \mathbb{R}^d)$ be the space of continuous paths endowed with the uniform topology. For $t \in [0, 1]$ define the evaluation map

$$P_t : \mathcal{S} \rightarrow \mathbb{R}^d, \quad P_t(\gamma) = \gamma(t).$$

The Benamou-Brenier cost admits the equivalent formulation

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{M \in \mathcal{P}(\mathcal{S})} \left\{ \int_{\mathcal{S}} \int_0^1 |\dot{\gamma}(t)|^2 dt dM(\gamma) : (P_0)_\# M = \alpha_0, (P_1)_\# M = \alpha_1 \right\}.$$

If α_0 has a density, the minimizer M^* is unique. Its time marginals reproduce the optimal curve: $\alpha_t = (P_t)_\# M^*$ for all t . Furthermore, denoting $Q_t : \mathcal{S} \rightarrow \mathbb{R}^d$, $Q_t(\gamma) = \dot{\gamma}(t)$, the conditional law of the velocity is deterministic:

$$(P_t, Q_t)_\# M^* = \alpha_t \otimes \delta_{v_t^*(x)}, \quad t \in [0, 1],$$

where v_t^* is the optimal velocity field in the Benamou-Brenier formulation. Hence M^* concentrates on straight-line geodesics, assigning exactly one direction at each spatial point.

Remark 32 (Generalized Benamou–Brenier distances). The dynamic formulation is not specific to $\mathcal{W}_2$. For measures with finite p -th moments and $p > 1$, one has the analogous action formula

$$\mathcal{W}_p^p(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t \alpha_t + \nabla \cdot (\alpha_t v_t) = 0 \\ \alpha_{t=0} = \alpha_0, \alpha_{t=1} = \alpha_1}} \int_0^1 \int_{\mathbb{R}^d} |v_t(x)|^p d\alpha_t(x) dt.$$

When $\alpha_t = \rho_t dx$ and $m_t = \rho_t v_t$, this becomes the convex perspective action

$$\int_0^1 \int_{\mathbb{R}^d} \frac{|m_t(x)|^p}{\rho_t(x)^{p-1}} dx dt, \quad \partial_t \rho_t + \nabla \cdot m_t = 0,$$

with the usual convention that the integrand is 0 if $(\rho, m) = (0, 0)$ and $+\infty$ if $\rho = 0$ but $m \neq 0$.

A second class of variants changes the mobility of the medium: the quadratic action $|m|^2/\rho$ is replaced by $|m|^2/\theta(\rho)$ for a suitable concave mobility θ . Under appropriate structural assumptions, this produces transport metrics adapted to nonlinear diffusions and finite-volume discretizations [72]. On finite graphs and Markov chains, the analogous action uses an edge mobility, often the logarithmic mean of the endpoint densities, and leads to discrete Wasserstein geometries [146, 156]. A third class allows creation and destruction of mass by replacing the continuity equation by a balance equation and penalizing both transport and growth. A representative quadratic action is

$$\partial_t \rho_t + \nabla \cdot m_t = s_t, \quad \int_0^1 \int \left(\frac{|m_t|^2}{\rho_t} + \kappa^2 \frac{s_t^2}{\rho_t} \right) dx dt,$$

with the same perspective convention. This leads, up to normalization conventions, to the Wasserstein–Fisher–Rao or Hellinger–Kantorovich geometry for unbalanced measures [140, 61, 59]. These extensions keep the same variational grammar as Benamou–Brenier: a balance equation, an action density, and geodesics obtained by minimizing an integrated kinetic cost.

10.4 Wasserstein Gradient Flows

Gradient flows turn the metric into dynamics driven by an energy functional. The JKO scheme is the discrete variational construction from which many diffusion PDEs emerge.

We now consider a function $f(\alpha)$ and seek a minimizing evolution $(\alpha_t)_t$. The general strategy of minimizing movement over a metric space is to construct a discrete-time evolution using an implicit Euler scheme:

$$\alpha_{t+\tau} := \arg \min_{\alpha} \frac{1}{2\tau} \mathcal{W}_2(\alpha_t, \alpha)^2 + f(\alpha). \quad (142)$$

Euclidean gradient flows. If we restrict (142) to finite dimensions and assume $\alpha_t = \delta_{x(t)}$ and $\alpha = \delta_x$ (single Dirac measures), this matches the implicit Euler scheme:

$$x(t + \tau) := \arg \min_x \frac{1}{2\tau} \|x - x(t)\|^2 + h(x),$$

where $h(x) = f(\delta_x)$. Its solution is formally given by the implicit Euler formula:

$$x(t + \tau) = (\text{Id} + \tau \nabla h)^{-1}(x(t)).$$

In contrast, the explicit Euler scheme is:

$$x(t + \tau) = (\text{Id} - \tau \nabla h)(x(t)) = x(t) - \tau \nabla h(x(t)).$$

Both schemes converge as $\tau \rightarrow 0$ to:

$$\dot{x}(t) = -\nabla h(x(t)). \quad (143)$$

Wasserstein gradient formula. The implicit Euler scheme has the advantage that it does not require h or f to be smooth. For f , this is crucial to handle evolution over arbitrary measures (with or without densities) seamlessly.

As $\tau \rightarrow 0$, under certain conditions on f , (142) defines a continuous evolution $t \mapsto \alpha_t$. As discussed earlier, this evolution can be described as a Lagrangian evolution (123). A key point is that it provides an explicit vector field v_t (depending on α_t), denoted as $\nabla_{\mathcal{W}} f(\alpha)$, called the Wasserstein gradient. In the weak sense, α_t satisfies:

$$\frac{\partial \alpha_t}{\partial t} + \text{div}(-\nabla_{\mathcal{W}} f(\alpha_t) \alpha_t) = 0. \quad (144)$$

Here, $\nabla_{\mathcal{W}} f(\alpha_t)(x) \in \mathbb{R}^d$ is a vector field and is a gradient (similar to the computation in (128)), computable as:

$$\nabla_{\mathcal{W}} f(\alpha) = \nabla_x \varphi, \quad \text{where } \varphi := \delta f(\alpha).$$

The function $\delta f(\alpha) \in \mathcal{C}(\mathbb{R}^d)$ is known as the first variation or Fréchet (directional) derivative, satisfying for any $\beta \in \mathcal{P}(\mathbb{R}^d)$:

$$f((1 - \tau)\alpha + \tau\beta) = f(\alpha + \tau\rho) = f(\alpha) + \tau \int [\delta f(\alpha)](x) d\rho(x) + o(\tau),$$

where $\rho = \beta - \alpha$ is a zero-mean measure.

Proposition 9 (Formal Wasserstein gradient). *Assume that f admits a smooth first variation $\delta f(\alpha)$ and that α has a smooth positive density. For infinitesimal perturbations generated by a velocity field v through $(\text{Id} + \tau v)_\# \alpha$, the differential of f is*

$$\frac{d}{d\tau} \Big|_{\tau=0} f((\text{Id} + \tau v)_\# \alpha) = \int \langle \nabla \delta f(\alpha)(x), v(x) \rangle d\alpha(x).$$

Hence, for the Riemannian metric $\|v\|_{L^2(\alpha)}^2 = \int \|v\|^2 d\alpha$, the Wasserstein gradient is the vector field

$$\nabla_{\mathcal{W}} f(\alpha) = \nabla \delta f(\alpha).$$

Proof. The push-forward expansion gives, in the sense of distributions,

$$(\text{Id} + \tau v)_\# \alpha = \alpha - \tau \text{div}(\alpha v) + o(\tau).$$

Using the definition of the first variation,

$$f((\text{Id} + \tau v)_\# \alpha) = f(\alpha) - \tau \int \delta f(\alpha) \text{div}(\alpha v) dx + o(\tau).$$

An integration by parts, with either compact support or vanishing boundary flux, gives

$$-\int \delta f(\alpha) \operatorname{div}(\alpha v) dx = \int \langle \nabla \delta f(\alpha), v \rangle d\alpha.$$

By definition of the Riesz representative for the $L^2(\alpha)$ metric, this representative is $\nabla \delta f(\alpha)$. $\square$

The idea of doing gradient flow for the Wasserstein metric was first introduced by John D. Lafferty in his PhD, and published in “The Density Manifold and Configuration Space Quantization”, under the name “density manifold”. It was systematically studied by Felix Otto, who studied the properties of this space [163]. Rigorous metric-space treatments and numerical JKO schemes can be found in [6, 19, 165, 89].

From the JKO step to the velocity field. A first-order expansion of the JKO step explains why (144) uses the vector field $\nabla_{\mathcal{W}} f(\alpha)$. Write (142) as a minimization over displacement fields v such that $\alpha = (\operatorname{Id} + \tau v)_\# \alpha_t$:

$$\min_v \frac{1}{2\tau} \tau^2 \|v\|_{L^2(\alpha_t)}^2 + f((\operatorname{Id} + \tau v)_\# \alpha_t).$$

Then we perform a first-order Taylor expansion of this formulation using

$$\begin{aligned} (\operatorname{Id} + \tau v)_\# \alpha_t &= \alpha_t - \tau \operatorname{div}(v \alpha_t) + o(\tau) \\ f((\operatorname{Id} + \tau v)_\# \alpha_t) &= f(\alpha_t) - \tau \int \delta f(\alpha_t) \operatorname{div}(v \alpha_t) dx + o(\tau) \\ &= f(\alpha_t) + \tau \int \langle \nabla_x \delta f(\alpha_t)(x), v(x) \rangle d\alpha_t(x) + o(\tau) \end{aligned}$$

to obtain the following first-order expansion in τ of the problem minimized in (142)

$$\min_v f(\alpha_t) + \tau \int \left[\frac{1}{2} \|v(x)\|^2 + \langle \nabla_{\mathcal{W}} f(\alpha_t)(x), v(x) \rangle \right] d\alpha_t(x) + o(\tau).$$

The pointwise minimizer is $v = -\nabla_{\mathcal{W}} f(\alpha_t)$, which gives the velocity in the continuity equation. We now detail examples of such Wasserstein gradient flows.

Discrete evolutions. If $f(\alpha)$ can be evaluated on discrete distributions and $\nabla_{\mathcal{W}}$ is continuous in this case, the flow (144) maintains the number of Dirac masses, $\alpha_t = \frac{1}{n} \sum_i \delta_{x_i(t)}$. The particles $X(t) := (x_i(t))_i$ evolve according to a system of coupled ODEs:

$$\dot{x}_i(t) = -n \nabla_{x_i} F(X(t)), \tag{145}$$

where $F(X) := f\left(\frac{1}{n} \sum_i \delta_{x_i}\right)$ and the factor n comes from the empirical Wasserstein metric $\frac{1}{n} \sum_i \|x_i\|^2$.

Linear Functionals. The simplest example of flows is for linear functions

$$f(\alpha) = \int h(x) d\alpha(x). \tag{146}$$

Here, $\delta f(\alpha) = h$ is a fixed function (independent of α). The flow (144) becomes:

$$\frac{\partial \alpha_t}{\partial t} + \operatorname{div}(-\nabla h \alpha_t) = 0.$$

This implies particles move independently according to the usual gradient flow (143).

Shannon Neg-Entropy. A very different behavior is obtained by considering functions which require α_t to have a density, the canonical example being Shannon neg-entropy

$$f(\alpha) = \int \log \left(\frac{d\alpha}{dx}(x) \right) d\alpha(x). \quad (147)$$

Here, $\delta f(\alpha) = \log \left(\frac{d\alpha}{dx} \right)$ up to an additive constant, so $\nabla_{\mathcal{W}} f(\alpha) = \frac{\nabla \alpha}{\alpha}$ (often called the score). The flow (144) becomes the heat equation:

$$\partial_t \alpha_t = \Delta \alpha_t.$$

Other entropy functionals lead to nonlinear diffusion equations; finite-volume and particle discretizations are discussed in [50, 98, 146, 77]. For example, a generalized entropy of the form

$$f(\alpha) = \int g \left(\frac{d\alpha}{dx} \right) dx, \quad (148)$$

for a one-dimensional convex function g , leads to nonlinear diffusions in the smooth-density regime:

$$\frac{\partial \alpha_t}{\partial t} = \Delta(P(\alpha_t)),$$

where the pressure P satisfies $P'(s) = sg''(s)$. For example, $g(s) = s \log(s)$ gives $P(s) = s$ and recovers (147), while $g(s) = s^m/(m-1)$, $m > 1$, gives $P(s) = s^m$ up to an additive constant and yields the porous-medium equation. A celebrated, and non-trivial, theorem by McCann [148] is that a function of the form (148), for $g : \mathbb{R}^+ \rightarrow \mathbb{R}$, is geodesically convex on $\mathcal{P}(\mathbb{R}^d)$ if $g(0) = 0$, g is convex increasing superlinearly and $s \mapsto g(s^{-d})s^d$ is convex decreasing. Examples of such functions are $g(s) = s^q$ for $q > 1$ and Shannon entropy $g(s) = s \log(s)$. Note, however, that $-\log(t)$ (associated with the reverse KL divergence) is not superlinear, so one cannot conclude geodesic convexity from this criterion.

Interaction Energies. In a similar spirit, to obtain nonlinear evolutions without requiring the measure to have density, one can consider

$$f(\alpha) := \iint k(x, y) d\alpha(x) d\alpha(y). \quad (149)$$

For a symmetric kernel k :

$$\delta f(\alpha)(x) = 2 \int k(x, y) d\alpha(y), \quad \nabla_{\mathcal{W}} f(\alpha)(x) = 2 \int \nabla_x k(x, y) d\alpha(y).$$

For $\alpha_0 = \frac{1}{n} \sum_i \delta_{x_i}$, the flow (144) implies particles $(x_i(t))_i$ obey:

$$\dot{x}_i(t) = -\frac{2}{n} \sum_j \nabla k(x_i(t), x_j(t)).$$

If k is positive definite and one minimizes the squared MMD distance to a teacher distribution β , then

$$\|\alpha - \beta\|_k^2 = \iint k d\alpha d\alpha - 2 \int \left(\int k(x, y) d\beta(y) \right) d\alpha(x) + \text{constant}.$$

Thus MMD training energies are exactly an interaction energy plus a linear potential; the teacher distribution appears through the potential $x \mapsto -2 \int k(x, y) d\beta(y)$.

Geodesics and geodesic convexity. Geodesic convexity is the convexity notion adapted to Wasserstein geometry. A constant-speed $\mathcal{W}_2$ geodesic between α_0 and α_1 is obtained from any optimal coupling $\pi^* \in \mathcal{U}(\alpha_0, \alpha_1)$ by the McCann interpolation

$$\alpha_t = ((1-t)P_0 + tP_1)_\# \pi^*, \quad t \in [0, 1],$$

where $P_0(x, y) = x$ and $P_1(x, y) = y$. If the optimal plan is induced by a Brenier map T , this reduces to $((1-t)\text{Id} + tT)_\# \alpha_0$. The coupling formula is important because geodesics exist even when no Monge map exists, for instance when a Dirac mass must split.

Definition 21 (Geodesic convexity). *A functional f on $\mathcal{P}_2(\mathbb{R}^d)$ is geodesically convex if for every $\mathcal{W}_2$ geodesic $(\alpha_t)_t$,*

$$f(\alpha_t) \leq (1-t)f(\alpha_0) + tf(\alpha_1).$$

It is λ -geodesically convex if the right-hand side is improved by $-\frac{\lambda}{2}t(1-t)\mathcal{W}_2^2(\alpha_0, \alpha_1)$.

Proposition 82 (Basic geodesically convex energies). *The following formal statements hold on $\mathcal{P}_2(\mathbb{R}^d)$.*

1. *If h is convex, then $\alpha \mapsto \int h d\alpha$ is geodesically convex; if h is λ -strongly convex, it is λ -geodesically convex.*
2. *If $W(x-y)$ is convex as a function of the displacement, then $\alpha \mapsto \frac{1}{2} \iint W(x-y) d\alpha(x) d\alpha(y)$ is geodesically convex.*
3. *Shannon entropy $\alpha \mapsto \int \rho \log \rho dx$ is geodesically convex.*
4. *The relative entropy $\text{KL}(\alpha|\gamma)$ with $d\gamma = e^{-V} dx/Z$ is λ -geodesically convex when V is λ -strongly convex.*

Proof. Along a Monge geodesic $X_t = (1-t)X_0 + tX_1$, convexity of h gives $h(X_t) \leq (1-t)h(X_0) + th(X_1)$, and strong convexity gives the additional quadratic term; integrating proves the first claim. The interaction claim follows similarly by applying convexity of W to pairwise differences $X_t - X'_t = (1-t)(X_0 - X'_0) + t(X_1 - X'_1)$ and integrating over two independent copies. The entropy claim is McCann's displacement convexity theorem; at the density level it follows from the concavity of the Jacobian determinant under the interpolation of optimal maps. Finally, $\text{KL}(\alpha|\gamma) = \int \rho \log \rho dx + \int V d\alpha + \text{constant}$, so it is the sum of displacement-convex entropy and a λ -geodesically convex linear potential. $\square$

Convergence of the flow. In general, analyzing (144) is delicate. The cleanest case is when f is geodesically convex in the sense above. This condition is the Wasserstein analogue of convexity in Euclidean gradient descent.

Proposition 83 (Energy decay for convex Wasserstein flows). *Assume formally that f is geodesically convex, admits a smooth first variation, and has a minimizer α^* . Let $(\alpha_t)_t$ be a smooth solution of the Wasserstein gradient flow*

$$\partial_t \alpha_t + \text{div}(\alpha_t v_t) = 0, \quad v_t = -\nabla_{\mathcal{W}} f(\alpha_t).$$

Then

$$\frac{d}{dt} f(\alpha_t) = - \int \|\nabla_{\mathcal{W}} f(\alpha_t)(x)\|^2 d\alpha_t(x) \leq 0.$$

If T_t is the optimal map from α_t to α^ , then*

$$f(\alpha_t) - f(\alpha^*) \leq -\frac{d}{dt} \frac{1}{2} \mathcal{W}_2^2(\alpha_t, \alpha^*),$$

and consequently

$$f(\alpha_t) - f(\alpha^*) \leq \frac{\mathcal{W}_2^2(\alpha_0, \alpha^*)}{2t}.$$

If f is λ -geodesically convex with $\lambda > 0$, then

$$f(\alpha_t) - f(\alpha^*) \leq e^{-2\lambda t} (f(\alpha_0) - f(\alpha^*)).$$

Proof. The chain rule and Proposition 9 give

$$\frac{d}{dt}f(\alpha_t) = \int \langle \nabla_{\mathcal{W}}f(\alpha_t)(x), v_t(x) \rangle d\alpha_t(x) = - \int \|\nabla_{\mathcal{W}}f(\alpha_t)(x)\|^2 d\alpha_t(x).$$

Geodesic convexity along the geodesic $((1-s)\text{Id} + sT_t)_{\#}\alpha_t$ gives

$$f(\alpha^*) - f(\alpha_t) \geq \int \langle \nabla_{\mathcal{W}}f(\alpha_t)(x), T_t(x) - x \rangle d\alpha_t(x).$$

Since $v_t = -\nabla_{\mathcal{W}}f(\alpha_t)$, this reads

$$f(\alpha_t) - f(\alpha^*) \leq \int \langle v_t(x), T_t(x) - x \rangle d\alpha_t(x).$$

The standard first-variation formula for the squared Wasserstein distance gives

$$\frac{d}{dt} \frac{1}{2} \mathcal{W}_2^2(\alpha_t, \alpha^*) = \int \langle x - T_t(x), v_t(x) \rangle d\alpha_t(x),$$

which proves the differential inequality. Integrating it from 0 to t and using the monotonicity of $s \mapsto f(\alpha_s)$ gives

$$t(f(\alpha_t) - f(\alpha^*)) \leq \int_0^t (f(\alpha_s) - f(\alpha^*)) ds \leq \frac{1}{2} \mathcal{W}_2^2(\alpha_0, \alpha^*).$$

If f is λ -geodesically convex, the Wasserstein analogue of strong convexity gives the slope inequality

$$\int \|\nabla_{\mathcal{W}}f(\alpha_t)\|^2 d\alpha_t \geq 2\lambda(f(\alpha_t) - f(\alpha^*)).$$

Combining it with the energy dissipation identity yields

$$\frac{d}{dt}(f(\alpha_t) - f(\alpha^*)) \leq -2\lambda(f(\alpha_t) - f(\alpha^*)),$$

and Gronwall's lemma gives the exponential rate. $\square$

Proposition 84 (Convex examples covered by the theory). *The hypotheses of Proposition 83 are satisfied in the following standard cases, at least at the formal smooth level used in this section.*

1. *For the linear energy $f(\alpha) = \int h d\alpha$, geodesic convexity holds when h is convex. If h is λ -strongly convex, then f is λ -geodesically convex and the flow enjoys the exponential rate of Proposition 83.*
2. *For the interaction energy $f(\alpha) = \frac{1}{2} \iint W(x-y) d\alpha(x) d\alpha(y)$, geodesic convexity holds when W is convex and even. This covers repulsive or attractive pairwise models whose displacement cost has no non-convex wells.*
3. *The Shannon entropy $f(\alpha) = \int \rho \log \rho dx$ and, more generally, McCann displacement-convex internal energies generate diffusion-type Wasserstein gradient flows.*
4. *If $\gamma = Z^{-1}e^{-V}dx$ and V is λ -strongly convex, then the relative entropy $\text{KL}(\alpha|\gamma)$ is λ -geodesically convex. Its flow is the Fokker-Planck equation with invariant law γ .*

Proof. Let $(\alpha_t)_t$ be the McCann interpolation between α_0 and α_1 , written with an optimal coupling as $X_t = (1-t)X_0 + tX_1$. For a linear energy, Jensen's inequality gives

$$h(X_t) \leq (1-t)h(X_0) + th(X_1),$$

and the strong convexity version gives the additional term $-\frac{\lambda}{2}t(1-t)\|X_0 - X_1\|^2$. Integrating over the optimal coupling proves geodesic convexity and λ -geodesic convexity.

For interaction energies, use two independent copies of the optimal coupling. The pairwise displacement evolves as

$$X_t - X'_t = (1-t)(X_0 - X'_0) + t(X_1 - X'_1).$$

Convexity of W gives the convexity inequality after integration over the product coupling. Evenness of W ensures that the interaction is symmetric in the two particles and matches the usual factor $1/2$ in (149).

The entropy claim is McCann's displacement-convexity theorem. For smooth positive densities and Brenier maps, it follows from the change-of-variables formula and the concavity of the determinant along positive matrices; the general statement is obtained by approximation. Finally,

$$\text{KL}(\alpha|\gamma) = \int \rho \log \rho \, dx + \int V \, d\alpha + \log Z,$$

so it is the sum of the displacement-convex entropy and the λ -geodesically convex linear potential generated by V . Proposition 83 then applies to all four cases. $\square$

Outside this convex regime, such as in the mean-field neural-network example below, the flow may still be informative but its convergence analysis requires problem-specific arguments.

10.5 Application: Training Two-Layer MLPs as Wasserstein Flows

Mean-field limits recast the training of wide neural networks as transport of a distribution of neurons. This subsection shows how the particle ODE of gradient descent becomes a Wasserstein flow in parameter space.

In this section, we replace the variable x with θ to align with customary notation in machine learning. We consider a two-layer MLP $g_\theta : u \in \mathbb{R}^d \rightarrow \mathbb{R}$ with n neurons:

$$g_\theta(u) := \frac{1}{n} \sum_i \psi(\theta_i, u), \quad \psi(\theta_i, u) := a_i \sigma(\langle u, w_i \rangle),$$

where $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is a pointwise nonlinearity, for instance $\sigma(s) = \max(s, 0)$ for ReLU networks, and $\theta_i := (w_i, a_i) \in \mathbb{R}^{d+1}$ represents the parameters of a neuron. Importantly, these functions are invariant under permutations of the neurons. Thus, we can rewrite it using a probability distribution $\alpha = \frac{1}{n} \sum_i \delta_{\theta_i}$ as:

$$G_\alpha(u) := \int_{\mathbb{R}^{d+1}} \psi(\theta, u) \, d\alpha(\theta).$$

This formulation has the advantage that α can represent a continuous density with an infinite number of neurons.

Now, we consider training the MLP by minimizing an empirical risk to predict labels $y_k \in \mathbb{R}$ from features $u_k \in \mathbb{R}^d$:

$$\min_{\alpha} f(\alpha) := \frac{1}{N} \sum_{k=1}^N \ell(G_\alpha(u_k), y_k).$$

Since $\alpha \mapsto G_\alpha$ is linear, if ℓ is convex, then f is a convex function of α . This convexity is conceptually useful, but it does not by itself yield a finite-dimensional algorithm because α is infinite-dimensional. The typical approach is to perform gradient descent on the neuron parameters, which matches (145) up to the usual time rescaling:

$$\dot{\theta}_i = -n \nabla_{\theta_i} F(\theta), \quad \text{where } F(\theta) := f\left(\frac{1}{n} \sum_i \delta_{\theta_i}\right).$$

This is equivalent to the PDE (144) on the neuron density, where the Wasserstein gradient is used. We now write this PDE in this specific case. For $\ell(s, s') = \frac{1}{2}(s - s')^2$, the first variation of f is:

$$\delta f(\alpha)(\theta) = \frac{1}{N} \sum_{k=1}^N \left(\int \psi(\theta', u_k) d\alpha(\theta') - y_k \right) \psi(\theta, u_k),$$

which can be rewritten as:

$$\delta f(\alpha)(\theta) = \int k(\theta, \theta') d\alpha(\theta') + g(\theta),$$

where the kernel and potential functions are:

$$k(\theta, \theta') := \frac{1}{N} \sum_k \psi(\theta, u_k) \psi(\theta', u_k), \quad (150)$$

$$g(\theta) := -\frac{1}{N} \sum_k y_k \psi(\theta, u_k). \quad (151)$$

Thus, the Wasserstein gradient becomes:

$$\nabla_{\mathcal{W}} f(\alpha)(\theta) = \int \nabla_{\theta} k(\theta, \theta') d\alpha(\theta') + \nabla_{\theta} g(\theta).$$

This corresponds to a Wasserstein flow of the sum of a quadratic and a linear interaction potential. These functions are not geodesically convex because neither k nor g are convex, so the convex-flow guarantees above do not apply directly.

A breakthrough was achieved by Chizat and Bach [57], complemented by mean-field analyses such as [149], who proved that if the initialization has enough Dirac masses, the flow cannot become stuck in local minima or saddle points. This result leverages the classical convexity of the function f and the 1-homogeneity of $\psi((a, w), u)$ with respect to the external weights a .

10.6 Application: One-Step Generative Models

One-step generative models try to keep the geometric training principle of flows while removing the expensive multi-step integration at sampling time. The idea is to evolve the model distribution during training, but to store the final evolution in a single generator evaluation.

Let ζ be a simple latent distribution and let $\alpha_{\theta} = (G_{\theta})_{\#} \zeta$ be the model distribution. Assume that the target data distribution is β . A Wasserstein-flow construction chooses a discrepancy

$$\mathcal{E}_{\beta}(\alpha),$$

for instance a smoothed $\text{KL}(\alpha|\beta)$, an MMD/IPM loss, or the debiased Sinkhorn divergence $\bar{\mathcal{L}}_c^{\varepsilon}(\alpha, \beta)$ introduced in Section 8.4. The associated formal descent is

$$\partial_t \mu_t + \text{div}(\mu_t w_t) = 0, \quad w_t(x) = -\nabla_{\alpha} \mathcal{E}_{\beta}(\mu_t)(x). \quad (152)$$

Instead of integrating (152) at inference time, one fits a parametric residual field U_{η} along the current model distribution:

$$\min_{\eta} \int_0^1 \int \|U_{\eta}(t, x) - w_t(x)\|^2 d\mu_t(x) dt. \quad (153)$$

In a particle or generator implementation, the learned residual is then used to update the current generator by

$$\alpha_{\theta}^{+} = (\text{Id} + \tau U_{\eta})_{\#} \alpha_{\theta}, \quad \text{or equivalently} \quad G_{\theta}^{+}(z) = G_{\theta}(z) + \tau U_{\eta}(G_{\theta}(z)).$$

After many training updates, the accumulated generator is evaluated once at test time. This is the organizing principle behind recent one-step methods based on Wasserstein gradient flows: W-Flow uses such a construction with the Sinkhorn divergence as a tractable global discrepancy [113], while drifting methods evolve the generated distribution during training through a fitted vector field and also admit one-step inference [70]. The gradient-flow interpretation of drifting models, and its relation to KL, MMD, sliced-Wasserstein and Sinkhorn-type discrepancies, is analyzed in [108, 116]. These ideas are also connected to the Sinkhorn-type normalization dynamics used to model attention in Sinkformers [189].

Normalized drifting fields. Drifting methods need not start from an exact Wasserstein gradient. They often prescribe a normalized attraction-minus-repulsion field and then regress this field in $L^2(\mu_t)$. A simple continuous version uses a positive kernel $K_\varepsilon(x, y)$ and defines, for any measure ν ,

$$B_\varepsilon[\nu](x) := \frac{\int (y - x) K_\varepsilon(x, y) d\nu(y)}{\int K_\varepsilon(x, y) d\nu(y)}. \quad (154)$$

For the Gaussian kernel $K_\varepsilon(x, y) = \exp(-\|x - y\|^2/(2\varepsilon))$, this normalized field is a score of a smoothed density:

$$B_\varepsilon[\nu](x) = \varepsilon \nabla \log \left(\int K_\varepsilon(x, y) d\nu(y) \right). \quad (155)$$

The drifting velocity is then

$$u_t(x) = B_\varepsilon[\beta](x) - B_\varepsilon[\mu_t](x) = \varepsilon \nabla \log \frac{\int K_\varepsilon(x, y) d\beta(y)}{\int K_\varepsilon(x, y) d\mu_t(y)}. \quad (156)$$

The first term pulls samples toward data, while the second term corrects self-attraction and prevents all particles from collapsing onto the same high-density region. Sinkhorn drifting replaces these one-sided kernel normalizations by two-sided entropic OT couplings, so that the cross and self terms are normalized by Sinkhorn scaling rather than by a single denominator [116].

Proposition 85 (Drifting as a time-dependent Wasserstein gradient). *Let μ_t be a smooth curve of positive densities and let $u_t = \nabla \varphi_t$ be a smooth time-dependent gradient field. Define the semi-relaxed functional*

$$\mathcal{R}_t(\alpha|\mu_t) := - \int \varphi_t(x) d\alpha(x) + \int \varphi_t(x) d\mu_t(x). \quad (157)$$

Here μ_t and φ_t are frozen when taking the first variation with respect to the first argument α . Then the continuity equation

$$\partial_t \mu_t + \operatorname{div}(\mu_t u_t) = 0$$

is the formal Wasserstein gradient descent of the time-dependent functional $\alpha \mapsto \mathcal{R}_t(\alpha|\mu_t)$.

Proof. Since μ_t and φ_t are fixed in the variation with respect to α , the first variation is

$$\delta_\alpha \mathcal{R}_t(\alpha|\mu_t)(x) = -\varphi_t(x).$$

By Proposition 9,

$$\nabla_{\mathcal{W}} \mathcal{R}_t(\alpha|\mu_t) = \nabla \delta_\alpha \mathcal{R}_t(\alpha|\mu_t) = -\nabla \varphi_t = -u_t.$$

The Wasserstein gradient-descent velocity is the negative of this gradient, namely u_t . Substituting this velocity in the continuity equation gives the claimed flow. $\square$

Example 10 (Kernel drifting as a semi-relaxed divergence). For the Gaussian-kernel drift (156), set

$$\varphi_t(x) = \varepsilon \log \frac{\int K_\varepsilon(x, y) d\beta(y)}{\int K_\varepsilon(x, y) d\mu_t(y)}.$$

Then $u_t = \nabla\varphi_t$, so Proposition 85 shows that kernel drifting is the Wasserstein gradient descent of

$$\mathcal{R}_t^{\text{drift}}(\alpha|\mu_t) = \varepsilon \int \log \frac{\int K_\varepsilon(x, y) d\mu_t(y)}{\int K_\varepsilon(x, y) d\beta(y)} d\alpha(x) + \text{constant}.$$

It is “semi-relaxed” because the current model μ_t is used to build the potential, but it is not varied inside the denominator when computing the first variation in α .

Remark 33 (General fields and projection onto gradients). A general regressed field b_t is not necessarily a Wasserstein gradient, since Wasserstein tangent vectors are represented by gradient fields modulo $L^2(\mu_t)$ -null directions. The gradient component is obtained by the weighted projection

$$\nabla\varphi_t = \operatorname{argmin}_{\nabla\varphi} \int \|\nabla\varphi(x) - b_t(x)\|^2 d\mu_t(x).$$

One may first normalize b_t pointwise, for instance by $b_t/(\|b_t\| + \eta)$, or globally by $\|b_t\|_{L^2(\mu_t)}$, before this projection. Proposition 85 then applies to the projected field. Non-gradient components can still be useful in a parametric model, but they are not descent directions of a scalar functional for the $\mathcal{W}_2$ Riemannian metric.

10.7 Application: Evolution in Depth of Transformers

Deep residual architectures can be read as time discretizations of ODEs or PDEs. For transformers, the transported objects are token measures and the velocity is induced by attention.

We consider very deep transformers, focusing on a single-head attention mechanism for simplicity while ignoring MLP layers, layer normalization, causality, and masking. This framework is best suited to modeling encoders and vision transformers.

Attention as a context-dependent velocity. After tokenization, embedding, and positional encoding, each input (from a set of tokens) is represented as a point cloud $(x_i)_{i=1}^n$ of n points in the space of vectorized tokens. An attention layer with skip connection and rescaling by $1/T$ (where T is the depth) defines a transformation of the tokens:

$$x_i \mapsto x_i + \frac{1}{T} \sum_j \frac{e^{\langle Qx_i, Kx_j \rangle} Vx_j}{\sum_\ell e^{\langle Qx_i, Kx_\ell \rangle}},$$

where $\theta = (K, Q, V)$ are the parameters of the attention layer, represented by three matrices.

Token measure evolution. To handle an arbitrary number of tokens, we define $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$ as the empirical measure of tokens and rewrite the transformer mapping as:

$$x_i \mapsto x_i + \frac{1}{T} \Gamma_\theta[\alpha](x_i),$$

where

$$\Gamma_\theta[\alpha](x) := \int \frac{e^{\langle Qx, Ky \rangle} Vy d\alpha(y)}{\int e^{\langle Qx, Kz \rangle} d\alpha(z)}.$$

In terms of the evolution of the token distribution α , this means α is pushed forward by the “in-context” mapping $\Gamma_\theta[\alpha]$, which depends on the context α , the tokens, and the depth-dependent parameters θ_t . Denoting $t \in [0, 1]$ as the depth and $\tau = 1/T$ as the step size, this gives:

$$\alpha_{t+\tau} = (\text{Id} + \tau \Gamma_{\theta_t}[\alpha_t])_\# \alpha_t.$$

As $\tau \rightarrow 0$, this converges to the following conservation equation:

$$\partial_t \alpha_t + \operatorname{div}(\alpha_t \Gamma[\alpha_t]) = 0.$$

Gradient structure and limitations. When $V = Q^\top K$, the field $\Gamma[\alpha]$ is a gradient vector field, since it is the gradient of the log-partition function associated with the attention weights. It is not, however, the gradient of a first variation, so this PDE is not a Wasserstein gradient flow. This formulation is related to continuous-depth neural networks [53] and was introduced for attention dynamics in the Sinkformers paper [189], building on the transformer architecture of [209]. It models deep transformers as PDEs on token measures. The key challenge is to understand training, which corresponds to optimizing the time-dependent parameters $(\theta_t)_t$.

10.8 Gaussian Closures as Finite-Dimensional Test Cases

Gaussian measures provide a useful testing ground for the preceding dynamics. They are not invariant under a general Wasserstein gradient flow: a nonlinear velocity usually creates non-Gaussian densities immediately. The useful substitute is to either identify exact affine velocities, which really preserve Gaussianity, or to project the dynamics onto the Gaussian manifold. In both cases the measure PDE reduces to matrix ODEs for the mean and covariance. This viewpoint is emphasized in the survey [166] and is useful for comparing diffusion paths, Wasserstein gradient flows, drifting fields and transformer-type dynamics.

For gradient flows, the covariance equation is the finite-dimensional Bures–Wasserstein gradient flow on positive definite matrices. Thus Gaussian closure is not just a computational shortcut: it is the restriction of Wasserstein geometry to the Gaussian submanifold, where affine gradient fields encode tangent vectors.

Proposition 86 (Affine velocities preserve Gaussianity). *Let $\alpha_t = \mathcal{N}(\mathbf{m}_t, \Sigma_t)$, with Σ_t positive definite, solve the continuity equation with an affine velocity*

$$v_t(x) = b_t + A_t(x - \mathbf{m}_t).$$

Then α_t remains Gaussian and its moments solve

$$\dot{\mathbf{m}}_t = b_t, \quad \dot{\Sigma}_t = A_t \Sigma_t + \Sigma_t A_t^\top.$$

Conversely, any smooth Gaussian curve with positive definite covariance can be generated by such an affine velocity. If one wants the velocity to be a Wasserstein tangent gradient, one chooses the unique symmetric solution of the Lyapunov equation

$$A_t \Sigma_t + \Sigma_t A_t = \dot{\Sigma}_t.$$

Proof. Let X_t follow the characteristic ODE $\dot{X}_t = b_t + A_t(X_t - \mathbf{m}_t)$. This linear ODE maps Gaussian random variables to Gaussian random variables. Taking expectation gives $\dot{\mathbf{m}}_t = b_t$. Writing $\tilde{X}_t = X_t - \mathbf{m}_t$, one has $\dot{\tilde{X}}_t = A_t \tilde{X}_t$, hence

$$\dot{\Sigma}_t = \frac{d}{dt} \mathbb{E}(\tilde{X}_t \tilde{X}_t^\top) = A_t \Sigma_t + \Sigma_t A_t^\top.$$

For the converse, set $b_t = \dot{\mathbf{m}}_t$ and choose any matrix A_t satisfying the covariance equation. Since Σ_t is positive definite, the Lyapunov map $A \mapsto A \Sigma_t + \Sigma_t A$ is invertible on symmetric matrices, which gives the unique symmetric choice when a gradient velocity is required. In that case v_t is the gradient of the quadratic potential $x \mapsto \langle b_t, x \rangle + \langle A_t(x - \mathbf{m}_t), x - \mathbf{m}_t \rangle / 2$. $\square$

Example 11 (Flow matching and diffusion paths between Gaussians). Consider a prescribed Gaussian interpolation $\alpha_t = \mathcal{N}(\mathbf{m}_t, \Sigma_t)$. Proposition 86 shows that an exact flow-matching velocity can be taken affine:

$$v_t(x) = \dot{\mathbf{m}}_t + A_t(x - \mathbf{m}_t), \quad A_t \Sigma_t + \Sigma_t A_t = \dot{\Sigma}_t.$$

In the isotropic case $\Sigma_t = s_t^2 \text{Id}$, this reduces to the transparent formula

$$v_t(x) = \dot{\mathbf{m}}_t + \frac{\dot{s}_t}{s_t}(x - \mathbf{m}_t).$$

For instance, the diffusion noising path

$$X_t = a_t X_0 + \sigma_t Z, \quad Z \sim \mathcal{N}(0, \text{Id}),$$

has $\mathbf{m}_t = a_t \mathbf{m}_0$ and $\Sigma_t = a_t^2 \Sigma_0 + \sigma_t^2 \text{Id}$. Thus, in the Gaussian case, diffusion paths and flow-matching paths reduce to the same mean-covariance bookkeeping, although the corresponding training objectives are different.

Proposition 87 (Gaussian-constrained Wasserstein gradients). *Let f be a smooth functional and assume that its restriction to nondegenerate Gaussian measures can be written as*

$$f(\mathcal{N}(\mathbf{m}, \Sigma)) = F(\mathbf{m}, \Sigma).$$

Then the Wasserstein gradient constrained to the Gaussian family is the affine vector field

$$v_F(x) = \nabla_{\mathbf{m}} F(\mathbf{m}, \Sigma) + 2\nabla_{\Sigma} F(\mathbf{m}, \Sigma)(x - \mathbf{m}),$$

where $\nabla_{\Sigma} F$ denotes the symmetric matrix derivative. Equivalently, v_F is the $L^2(\mathcal{N}(\mathbf{m}, \Sigma))$ projection of the ambient Wasserstein gradient onto affine gradient fields, whenever the ambient gradient exists. Hence the gradient descent flow constrained to Gaussian measures satisfies

$$\dot{\mathbf{m}}_t = -\nabla_{\mathbf{m}} F(\mathbf{m}_t, \Sigma_t), \quad \dot{\Sigma}_t = -2(\Sigma_t \nabla_{\Sigma} F(\mathbf{m}_t, \Sigma_t) + \nabla_{\Sigma} F(\mathbf{m}_t, \Sigma_t) \Sigma_t), \quad (158)$$

and the descent velocity is affine.

Proof. Test the functional along a Gaussian tangent vector, represented by an affine gradient field

$$v(x) = b + A(x - \mathbf{m})$$

with A symmetric. The induced first-order variations are $\dot{\mathbf{m}} = b$ and $\dot{\Sigma} = A\Sigma + \Sigma A$. Therefore

$$\text{d}F(\mathbf{m}, \Sigma)[b, A\Sigma + \Sigma A] = \langle \nabla_{\mathbf{m}} F, b \rangle + \text{tr}(\nabla_{\Sigma} F(A\Sigma + \Sigma A)).$$

Since A , Σ and $\nabla_{\Sigma} F$ are symmetric, the second term equals

$$2 \text{tr}(\nabla_{\Sigma} F A \Sigma) = \int \langle 2\nabla_{\Sigma} F(x - \mathbf{m}), A(x - \mathbf{m}) \rangle \text{d}\mathcal{N}(\mathbf{m}, \Sigma)(x).$$

Together with the mean term, this gives

$$\text{d}F(\mathbf{m}, \Sigma)[\dot{\mathbf{m}}, \dot{\Sigma}] = \int \langle v_F(x), v(x) \rangle \text{d}\mathcal{N}(\mathbf{m}, \Sigma)(x)$$

for all affine gradient fields v . This identifies the constrained Wasserstein gradient in the induced $L^2(\alpha)$ metric, or equivalently the projection of the ambient gradient when it exists. Substituting the descent velocity $-v_F$ in Proposition 86 gives (158). $\square$

This proposition should be read as the organizing rule for Gaussian closures: once the scalar energy has been reduced to a function of $(\mathbf{m}, \Sigma)$, its constrained Wasserstein gradient is automatically affine and the covariance follows the Bures-type ODE (158). When the first variation of f is quadratic, this constrained gradient coincides with the full Wasserstein gradient.

Example 12 (Gaussian energies and affine gradients). Proposition 87 turns many standard energies into explicit affine fields:

- *Quadratic potential energy.* If

$$f(\alpha) = \int \left(\frac{1}{2} x^\top H x + \langle \ell, x \rangle \right) d\alpha(x), \quad H = H^\top,$$

then

$$\nabla_{\mathcal{W}} f(\alpha)(x) = Hx + \ell = (H\mathbf{m} + \ell) + H(x - \mathbf{m}).$$

This is the Gaussian form of transport under a quadratic confinement.

- *Quadratic interaction energy.* If

$$f(\alpha) = \frac{1}{4} \iint (x - y)^\top G(x - y) d\alpha(x) d\alpha(y), \quad G = G^\top,$$

then $F(\mathbf{m}, \Sigma) = \frac{1}{2} \text{tr}(G\Sigma)$ and

$$\nabla_{\mathcal{W}} f(\alpha)(x) = G(x - \mathbf{m}).$$

The mean is unchanged and the covariance contracts or expands according to the signs of G .

- *Relative entropy to a Gaussian.* For $\bar{\alpha} = \mathcal{N}(\bar{\mathbf{m}}, \bar{\Sigma})$,

$$f(\alpha) = \text{KL}(\alpha | \bar{\alpha})$$

has

$$\nabla_{\mathcal{W}} f(\alpha)(x) = \bar{\Sigma}^{-1}(\mathbf{m} - \bar{\mathbf{m}}) + (\bar{\Sigma}^{-1} - \Sigma^{-1})(x - \mathbf{m}).$$

The descent equations are the Ornstein–Uhlenbeck moment equations

$$\dot{\mathbf{m}}_t = -\bar{\Sigma}^{-1}(\mathbf{m}_t - \bar{\mathbf{m}}), \quad \dot{\Sigma}_t = 2\text{Id} - \bar{\Sigma}^{-1}\Sigma_t - \Sigma_t\bar{\Sigma}^{-1}.$$

- *Squared Wasserstein distance to a Gaussian.* For

$$f(\alpha) = \frac{1}{2} \mathcal{W}_2^2(\alpha, \bar{\alpha}), \quad \bar{\alpha} = \mathcal{N}(\bar{\mathbf{m}}, \bar{\Sigma}),$$

the Gaussian Brenier map $T_{\alpha \rightarrow \bar{\alpha}}$ is affine,

$$T_{\alpha \rightarrow \bar{\alpha}}(x) = \bar{\mathbf{m}} + M(x - \mathbf{m}), \quad M = \Sigma^{-1/2}(\Sigma^{1/2}\bar{\Sigma}\Sigma^{1/2})^{1/2}\Sigma^{-1/2}.$$

Hence

$$\nabla_{\mathcal{W}} f(\alpha)(x) = x - T_{\alpha \rightarrow \bar{\alpha}}(x) = (\mathbf{m} - \bar{\mathbf{m}}) + (\text{Id} - M)(x - \mathbf{m}),$$

and descent moves each Gaussian infinitesimally along the Bures–Wasserstein geodesic toward $\bar{\alpha}$.

- *Gaussian-only losses.* Sliced SW_2^2 losses to a Gaussian, Gaussian Sinkhorn divergences, and any smooth closed formula depending only on $(\mathbf{m}, \Sigma)$ fit the same constrained-gradient template:

$$v_F(x) = \nabla_{\mathbf{m}} F + 2\nabla_{\Sigma} F(x - \mathbf{m}).$$

For Gaussian Sinkhorn divergences this finite-dimensional flow is studied in [115].

Not every PDE preserves Gaussianity exactly. For instance, Wasserstein flows of relative Fisher information, related to quantum-drift or higher-order diffusion equations, typically require a Gaussian projection to close on $(\mathbf{m}, \Sigma)$. Such projected closures are still useful: they expose the finite-dimensional dynamics predicted by a variational model and make it easy to compare variational flows with non-variational affine dynamics such as drifting fields or the Gaussian transformer closure below.

Proposition 88 (Centered Gaussian covariance catalogue). *Let $\gamma = \mathcal{N}(0, \text{Id})$ and let $\mu_t = \mathcal{N}(0, C_t)$ with $C_t \succ 0$. For the normalizations displayed below, the Wasserstein descent constrained to the centered Gaussian manifold satisfies $\dot{C}_t = h(C_t)$, with*

$$\begin{aligned} \text{KL}(\mu|\gamma) &: h(C) = 2(\text{Id} - C), \\ \frac{1}{2} \mathcal{I}(\mu|\gamma) &: h(C) = 2(C^{-1} - C), \\ \mathcal{W}_2^2(\mu, \gamma) &: h(C) = 4(C^{1/2} - C), \\ \text{MMD}_k^2(\mu, \gamma), \quad k(x, y) = \langle x, y \rangle^2 &: h(C) = 8(C - C^2), \\ S_\varepsilon(\mu, \gamma) &: h(C) = 4 \left(C + \frac{\varepsilon^2}{16} \text{Id} \right)^{1/2} - 2 \left(C^2 + \frac{\varepsilon^2}{16} \text{Id} \right)^{1/2} - 2C - \frac{\varepsilon}{2} \text{Id}, \\ \text{SW}_2^2(\mu, \gamma) &: h(C) = V(C)C + CV(C), \end{aligned}$$

where S_ε is the debiased Sinkhorn divergence for the quadratic cost $\|x - y\|^2$ and KL regularization strength ε , and

$$V(C) = 2 \int_{\mathbb{S}^{d-1}} \left(\frac{1}{\sqrt{\theta^\top C \theta}} - 1 \right) \theta \theta^\top d\sigma(\theta)$$

for the normalized spherical measure σ . Here

$$\mathcal{I}(\mu|\gamma) = \int |\nabla \log \rho(x) + x|^2 \rho(x) dx \quad (\mu = \rho dx).$$

Thus the unhalved Fisher divergence has right-hand side $4(C^{-1} - C)$. Multiplying any of these energies by a constant simply rescales the corresponding right-hand side.

Proof. Each row is obtained by identifying the affine descent velocity $v(x) = M_C x$ generated by the corresponding Gaussian-constrained calculation and then applying Proposition 86, which gives $\dot{C} = M_C C + C M_C^\top$. For $\text{KL}(\cdot|\gamma)$, the Fokker–Planck velocity is $(C^{-1} - \text{Id})x$, hence $\dot{C} = 2(\text{Id} - C)$. For the Fisher row, the restriction of $\frac{1}{2}\mathcal{I}$ to centered Gaussians is

$$\frac{1}{2} (\text{tr}(C) + \text{tr}(C^{-1}) - 2d).$$

Using Proposition 87 gives the descent velocity $(C^{-2} - \text{Id})x$, hence $\dot{C} = 2(C^{-1} - C)$. This row should be read as a Gaussian projected closure of the fourth-order Fisher flow.

For $\mathcal{W}_2^2(\cdot, \gamma)$, the Brenier map from $\mathcal{N}(0, C)$ to γ is $C^{-1/2}x$, so the descent velocity for the unhalved squared distance is $2(C^{-1/2} - \text{Id})x$, giving $4(C^{1/2} - C)$. For the polynomial MMD row, centered Gaussians satisfy $\text{MMD}_k^2(\mu, \gamma) = \|C - \text{Id}\|_{\mathbb{F}}^2$; the first variation is quadratic and its descent velocity is $4(\text{Id} - C)x$, giving $8(C - C^2)$.

Gaussian Sinkhorn dual potentials are quadratic, so the velocity is again linear; differentiating the closed Gaussian formula yields the displayed spectral expression. The square roots are spectral functions of C , hence commute with C , which is why the covariance ODE closes as a matrix function of C alone. For sliced Wasserstein, each one-dimensional projection is a Gaussian transport with velocity $2((\theta^\top C \theta)^{-1/2} - 1)\langle \theta, x \rangle \theta$; averaging these velocities over $\mathbb{S}^{d-1}$ gives $v(x) = V(C)x$ and thus $\dot{C} = V(C)C + CV(C)$. $\square$

The remaining examples leave the static energy list and record affine vector fields that arise directly from models. Some of them still have a time-dependent variational interpretation, but their Gaussian closure is most transparent by applying Proposition 86 to the prescribed field.

Example 13 (Linear mean-field networks). In the two-layer model above with $\psi((a, w), u) = a\langle w, u \rangle$, set

$$S = \frac{1}{N} \sum_{k=1}^N u_k u_k^\top, \quad r = \frac{1}{N} \sum_{k=1}^N y_k u_k, \quad q_t = \int a w d\alpha_t(a, w).$$

For the square loss, set $h_t = Sq_t - r$. The first variation is

$$\delta f(\alpha_t)(a, w) = a \langle w, h_t \rangle.$$

Hence the particle velocity in parameter space is linear:

$$-\nabla_{(a, w)} \delta f(\alpha_t)(a, w) = - \begin{pmatrix} 0 & h_t^\top \\ h_t & 0 \end{pmatrix} \begin{pmatrix} a \\ w \end{pmatrix}.$$

Therefore a Gaussian law of neurons remains Gaussian. Its mean and covariance follow Proposition 86, with a matrix depending only on the current moment q_t . This exact closure is special to the linear activation; for nonlinear activations, Gaussian closures are usually projections rather than invariant families.

Example 14 (Gaussian kernel drifting). Let the target be $\beta = \mathcal{N}(\bar{\mathbf{m}}, \bar{\Sigma})$ and assume $\mu_t = \mathcal{N}(\mathbf{m}_t, \Sigma_t)$. For the Gaussian kernel

$$K_\varepsilon(x, y) = \exp(-\|x - y\|^2 / (2\varepsilon)),$$

the normalized field (154) satisfies

$$B_\varepsilon[\mu_t](x) = -\varepsilon(\Sigma_t + \varepsilon \text{Id})^{-1}(x - \mathbf{m}_t).$$

Thus the drifting velocity (156) is affine and preserves Gaussianity. With

$$A_t = (\Sigma_t + \varepsilon \text{Id})^{-1}, \quad \bar{A} = (\bar{\Sigma} + \varepsilon \text{Id})^{-1},$$

the ODE is

$$\dot{\mathbf{m}}_t = \varepsilon \bar{A}(\bar{\mathbf{m}} - \mathbf{m}_t), \quad \dot{\Sigma}_t = \varepsilon((A_t - \bar{A})\Sigma_t + \Sigma_t(A_t - \bar{A})).$$

This finite-dimensional model explains the stabilizing role of the self-normalized repulsion term in drifting: without it, the covariance equation loses the $A_t \Sigma_t + \Sigma_t A_t$ contribution.

Example 15 (Gaussian closure of attention dynamics). For the transformer PDE, assume $\alpha = \mathcal{N}(\mathbf{m}, \Sigma)$. Since exponential tilting preserves Gaussianity,

$$\frac{\int e^{\langle Qx, Ky \rangle} y \, d\alpha(y)}{\int e^{\langle Qx, Kz \rangle} d\alpha(z)} = \mathbf{m} + \Sigma K^\top Qx.$$

Therefore

$$\Gamma_\theta[\alpha](x) = V\mathbf{m} + V\Sigma K^\top Qx$$

is affine. The Gaussian token law is preserved and satisfies

$$\dot{\mathbf{m}}_t = (V_t + V_t \Sigma_t K_t^\top Q_t) \mathbf{m}_t, \quad \dot{\Sigma}_t = B_t \Sigma_t + \Sigma_t B_t^\top, \quad B_t = V_t \Sigma_t K_t^\top Q_t.$$

When $V_t = Q_t^\top K_t$, the matrix $B_t = Q_t^\top K_t \Sigma_t K_t^\top Q_t$ is symmetric positive semidefinite, matching the gradient-field case mentioned above. This closure is not a convergence theorem for trained transformers. It is instead a tractable model of how attention can shear, amplify or contract a cloud of tokens through its covariance.

Conclusion

Optimal transport is useful because it keeps several viewpoints active at once. It is a linear program over couplings, a duality theory for potentials, a geometry on probability measures, a source of PDEs and gradient flows, and a computational toolbox built around linear programming, Sinkhorn scaling and low-dimensional projections. These viewpoints reinforce each other: Brenier maps explain geodesics, geodesic convexity explains convergence of flows, entropic regularization turns transport into scalable differentiable losses, and dual norms or sliced variants reveal what is gained and lost when OT is simplified.

For machine learning, this interplay is especially stimulating. Modern generative modeling, diffusion and flow matching, inverse problems, robust optimization, and even continuous-depth views of transformers all ask for ways to move, compare or learn distributions in high dimension. These applications do not merely consume existing OT theory; they create difficult mathematical questions because empirical measures are singular, dimensions are large, models are parametrized and non-convex, and computational approximations are inseparable from statistical error. The strength of OT is precisely that it gives a common language for these tensions, while still leaving enough open structure for new mathematics to be needed.

A Notation Table

This appendix collects the main notation used throughout the book. The last column points to the first section, equation, definition, proposition or theorem where the notation is defined or first used in a mathematically meaningful way.

Notation	Meaning	First reference
Ambient spaces, measures and elementary objects		
$\mathbb{R}^d$	Euclidean ambient space.	Section 2.1
$\mathcal{X}, \mathcal{Y}$	Source and target spaces.	Eq. (6)
$\mathcal{M}(\mathcal{X})$	Finite signed Radon measures on $\mathcal{X}$.	Section 2.1
$\mathcal{M}_+(\mathcal{X}), \mathcal{M}_+^1(\mathcal{X})$	Positive finite measures and probability measures.	Section 2.1
$\mathcal{P}(\mathcal{X}), \mathcal{P}_p(\mathcal{X})$	Probability measures, with finite p -moment for $\mathcal{P}_p$.	Section 3.2
Σ_n	Probability simplex of histograms of length n .	Section 2.1
δ_x	Dirac mass at x .	Eq. (2)
α, β, γ	Source, target and auxiliary probability measures.	Eq. (6)
ρ_α	Density of α with respect to a reference measure.	Section 2.1
$d\alpha, dx$	Integration against α and against Lebesgue measure.	Section 2.1
$\mathbb{E}$	Expectation of a random variable.	Eq. (23)
$\text{supp}(\pi)$	Topological support of a measure.	Definition 6
$\text{Supp}(\mathbf{b})$	Index support of a histogram.	Eq. (75)
$\mathcal{C}(\mathcal{X})$	Continuous real-valued functions on $\mathcal{X}$.	Section 2.1
$\ \cdot\ $	Euclidean norm or the norm indicated by a subscript.	Section 1
$\langle \cdot, \cdot \rangle$	Euclidean/Frobenius pairing or measure-function pairing.	Section 2.1
Discrete matching and discrete Kantorovich OT		
$(x_i)_i, (y_j)_j$	Source and target point clouds.	Eq. (1)
$\mathbf{C} = (C_{i,j})$	Cost matrix between source and target points.	Eq. (1)
$\sigma \in \text{Perm}(n)$	Permutation encoding a one-to-one matching.	Eq. (1)
$\mathbf{P}_\sigma, \text{Perm}_n$	Permutation matrix and the set of all such matrices.	Eq. (20)
$\mathcal{B}_n$	Bistochastic matrix polytope.	Theorem 2
$\mathbf{a}, \mathbf{b}$	Discrete probability histograms.	Eq. (18)
$\mathbf{P}$	Discrete transport/coupling matrix.	Eq. (18)
$\mathbf{U}(\mathbf{a}, \mathbf{b})$	Polytope of discrete couplings with marginals $\mathbf{a}, \mathbf{b}$.	Eq. (18)
$\mathbf{1}_n, \mathbf{P}^\top$	All-ones vector and transpose of $\mathbf{P}$.	Eq. (18)
$\mathbf{L}_\mathbf{C}(\mathbf{a}, \mathbf{b})$	Discrete Kantorovich optimal value with cost $\mathbf{C}$.	Eq. (19)
$\mathbf{D}$	Ground distance matrix for discrete Wasserstein distances.	Eq. (25)
$W_p(\mathbf{a}, \mathbf{b})$	Discrete p -Wasserstein distance.	Eq. (25)

Notation	Meaning	First reference
Monge maps, one-dimensional OT and Gaussians		
T, T	Transport map.	Eq. (6)
$T_{\#}\alpha$	Push-forward of α by T .	Definition 2
Id	Identity map.	Definition 2
$\tilde{\mathcal{W}}_p$	Directed Monge transport distance.	Eq. (7)
$\nabla\varphi$	Brenier map for quadratic cost.	Theorem 1
$\mathcal{C}_\alpha$	Cumulative distribution function of a 1-D measure.	Eq. (9)
$\mathcal{C}_\alpha^{-1}$	Quantile function of a 1-D measure.	Eq. (10)
$\mathcal{N}(\mathbf{m}, \Sigma)$	Gaussian law with mean $\mathbf{m}$ and covariance Σ .	Eq. (14)
$\mathbf{m}_\alpha, \Sigma_\alpha$	Mean and covariance of a Gaussian measure α .	Proposition 12
$\mathcal{B}(\Sigma_\alpha, \Sigma_\beta)$	Bures covariance distance.	Eq. (17)
$\text{tr}(\Sigma)$	Trace of a matrix.	Eq. (16)
Continuous Kantorovich OT and Wasserstein distances		
π	Coupling or transport plan.	Definition 5
$\mathcal{U}(\alpha, \beta)$	Set of couplings between α and β .	Eq. (21)
$\mathcal{L}_c(\alpha, \beta)$	Kantorovich optimal value with ground cost c .	Eq. (22)
d	Ground distance on the underlying metric space.	Eq. (26)
$\mathcal{W}_p(\alpha, \beta)$	p -Wasserstein distance.	Eq. (26)
$\mathcal{W}_\infty(\alpha, \beta)$	Worst-displacement Wasserstein distance.	Eq. (28)
Γ	c -cyclically monotone subset of $\mathcal{X} \times \mathcal{Y}$.	Definition 7
ρ	Glued or composed coupling.	Lemma 2
$\rightharpoonup$	Weak* convergence of measures.	Definition 8
$\text{TV}, \ \cdot\ _{\text{TV}}$	Total variation divergence/norm.	Section 2.1
Duality, transforms and weak norms		
f, g	Discrete dual potentials.	Eq. (47)
f, g	Continuous dual potentials.	Eq. (51)
$\mathbf{R}(\mathbf{a}, \mathbf{b})$	Feasible set of discrete dual potentials.	Eq. (48)
$\mathcal{R}(c)$	Feasible set of continuous dual potentials.	Eq. (52)
f^c, g^c	c -transform of a potential.	Definition 12
$\text{Lip}(f)$	Lipschitz constant of f .	Eq. (63)
$\mathcal{W}_1$	Kantorovich–Rubinstein distance/norm.	Eq. (64)
$\mathcal{W}_{1,G}$	Graph Wasserstein-1/transshipment distance.	Proposition 45
$d_G, \nabla_G, \text{div}_G$	Graph geodesic distance, gradient and divergence.	Proposition 45
$\ \cdot\ _B$	Dual norm induced by a discriminator class B .	Eq. (68)
$\mathcal{H}, k$	Reproducing kernel Hilbert space and its kernel.	Definition 14
MMD_k	Maximum mean discrepancy/kernel norm for k .	Definition 14
$\mathcal{D}_\varphi, \text{D}_\varphi$	Continuous and discrete φ -divergences.	Eq. (73)
φ'_∞	Recession slope of an entropy function.	Definition 15
φ^*	Legendre transform of φ .	Eq. (80)
KL, KL	Continuous and discrete Kullback–Leibler divergences.	Eq. (40)
$\mathfrak{h}$	Hellinger divergence/distance.	Section 7.3
JS	Jensen–Shannon divergence.	Section 7.3
Entropic regularization and Sinkhorn algorithms		
ε	Entropic regularization strength.	Eq. (29)
$\text{H}(\mathbf{P})$	Shannon–Boltzmann entropy of a matrix.	Eq. (29)
$\text{L}_G^\varepsilon(\mathbf{a}, \mathbf{b})$	Discrete entropic OT value.	Eq. (29)

Notation	Meaning	First reference
$\mathcal{L}_c^\varepsilon(\alpha, \beta)$	Continuous entropic OT value.	Eq. (39)
K	Gibbs kernel $e^{-C/\varepsilon}$.	Eq. (30)
u, v	Left and right Sinkhorn scalings.	Eq. (30)
$\text{diag}(u)K \text{diag}(v)$	Scaling form of the entropic coupling.	Eq. (4.5)
$\odot$	Entrywise product of vectors.	Eq. (32)
$u^{(\ell)}, u^{(\ell+1)}$	Current and next Sinkhorn iterates.	Eq. (33)
$d_{\mathcal{H}}$	Hilbert projective metric on positive vectors.	Eq. (43)
Proj^{KL}	KL/Bregman projection.	Eq. (41)
$\tilde{\mathcal{L}}_c^\varepsilon(\alpha, \beta)$	Debiased Sinkhorn divergence.	Eq. (86)

Notation	Meaning	First reference
Extensions of OT		
ψ_1, ψ_2	Entropy functions penalizing marginal mismatch.	Eq. (88)
$UW_c, UW_{c,\tau}$	Relaxed unbalanced OT value with marginal penalties.	Eq. (88)
L_c	Reverse-formulation local unbalanced cost.	Eq. (89)
H_c	Homogeneous perspective of the local cost L_c .	Eq. (90)
HW	Homogeneous unbalanced formulation.	Eq. (91)
$\mathfrak{C}[\mathcal{X}]$	Cone over the metric space $\mathcal{X}$.	Section 9.1
CW	Cone formulation of unbalanced OT.	Theorem 6
β_s, λ_s	Input measures and weights in barycenter problems.	Eq. (93)
α^*	Optimal measure, often a barycenter.	Section 9.2
SW_p	Sliced Wasserstein distance.	Eq. (102)
$\mathbb{S}^{d-1}$	Unit sphere of projection directions.	Eq. (102)
P_θ	Projection on direction θ .	Eq. (102)
MaxSW_p	Max-sliced Wasserstein distance.	Definition 17
MSWGG_2	Min-sliced lifted-plan upper bound on $\mathcal{W}_2$.	Section 9.4
$SW_{p,k}, \text{MaxSW}_{p,k}$	Average and max Wasserstein distances over k -dimensional projections.	Section 9.4
$\mathcal{W}_\gamma$	Spectral Wasserstein distance associated with a matrix gauge γ .	Eq. (103)
$\mathcal{B}_\gamma$	Polar set defining the robust projected form of $\mathcal{W}_\gamma$.	Eq. (105)
$\mathcal{W}_{2,A}$	Quadratic Wasserstein pseudodistance after projection by $A^{1/2}$.	Eq. (104)
$\text{SRW}_{2,k}$	Paty–Cuturi subspace robust Wasserstein distance.	Section 9.5
LOT_ρ	Linear OT distance around reference ρ .	Eq. (107)
$\mathcal{W}_C$	Weak OT value with conditional-law cost C .	Eq. (108)
g^C	Weak C -transform in weak OT duality.	Proposition 71
D, D'	Intra-domain distance matrices in discrete GW.	Eq. (109)
Δ	Discrepancy between intra-domain distances.	Eq. (109)
GW	Discrete Gromov–Wasserstein cost.	Eq. (109)
$\mathbb{X}, \mathbb{Y}$	Metric-measure spaces.	Eq. (110)
$\mathcal{GW}$	Continuous Gromov–Wasserstein distance.	Eq. (110)
d_H, d_{GH}	Hausdorff and Gromov–Hausdorff distances.	Section 9.8
$\text{FGW}_{\lambda,p}$	Fused Gromov–Wasserstein distance.	Section 9.8
$\mathbb{H}_n, \mathbb{H}_n^+, \mathbb{H}_n^{+,1}$	Hermitian matrices, positive semidefinite Hermitian matrices and density matrices.	Section 9.10
Tr_A, Tr_B	Partial traces of a bipartite matrix.	Eq. (115)
$\text{QOT}_C(A, B)$	Finite-dimensional quantum OT value with cost observable C .	Eq. (116)
$\text{QOT}_C^\varepsilon(A, B)$	Entropically regularized quantum OT value.	Eq. (118)
$T_e(F, G), T_s(F, G)$	Exact Gibbs coupling and symmetric Gurvits-scaling surrogate.	Eqs. (120), (121)
Dynamic OT and Wasserstein gradient flows		
α_t	Time-dependent curve of probability measures.	Eq. (124)
v_t	Eulerian velocity field transporting α_t .	Eq. (124)
T_t	Lagrangian particle flow map.	Eq. (123)
P_t	Interpolant map in flow matching; later also path evaluation.	Eq. (133)

Notation	Meaning	First reference
$\mathcal{W}_2^2$ via action	Benamou–Brenier dynamic formulation.	Eq. (140)
$\nabla_{\mathcal{W}} f(\alpha)$	Wasserstein gradient of a functional.	Proposition 9
$\delta f(\alpha)$	First variation of f at α .	Proposition 9
$\partial_t \alpha + \operatorname{div}(\alpha v) = 0$	Continuity equation.	Eq. (124)
$\alpha_{t+\tau}$	One JKO/minimizing-movement step.	Eq. (142)
$\mathcal{S} = C([0, 1]; \mathbb{R}^d)$	Path space in the superposition formulation.	Section 10

References

- [1] Martial Agueh and Guillaume Carlier. Barycenters in the Wasserstein space. *SIAM Journal on Mathematical Analysis*, 43(2):904–924, 2011.
- [2] Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions. *Journal of Machine Learning Research*, 26(209):1–80, 2025.
- [3] Syed Mumtaz Ali and Samuel D Silvey. A general class of coefficients of divergence of one distribution from another. *Journal of the Royal Statistical Society. Series B (Methodological)*, 28(1):131–142, 1966.
- [4] Jason Altschuler, Jonathan Weed, and Philippe Rigollet. Near-linear time approximation algorithms for optimal transport via Sinkhorn iteration. In *Advances in Neural Information Processing Systems*, volume 30, pages 1964–1974, 2017.
- [5] Pedro C Álvarez-Esteban, E del Barrio, JA Cuesta-Albertos, and C Matrán. A fixed-point approach to barycenters in Wasserstein space. *Journal of Mathematical Analysis and Applications*, 441(2):744–762, 2016.
- [6] L. Ambrosio, N. Gigli, and G. Savaré. *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. Springer, 2006.
- [7] Ethan Anderes, Steffen Borgwardt, and Jacob Miller. Discrete Wasserstein barycenters: optimal transport for discrete data. *Mathematical Methods of Operations Research*, 84(2):389–409, 2016.
- [8] Francisco Andrade, Gabriel Peyré, and Clarice Poon. Sparsistency for inverse optimal transport. In *International Conference on Learning Representations*, 2024.
- [9] Francisco Andrade, Gabriel Peyré, and Clarice Poon. Learning from samples: Inverse problems over measures via sharpened Fenchel–Young losses. *arXiv preprint arXiv:2505.07124*, 2025.
- [10] Martin Arjovsky, Soumith Chintala, and Léon Bottou. Wasserstein generative adversarial networks. In Doina Precup and Yee Whye Teh, editors, *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 214–223. PMLR, 2017.
- [11] David Arthur and Sergei Vassilvitskii. k -means++: The advantages of careful seeding. In *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 1027–1035, 2007.
- [12] Franz Aurenhammer. Power diagrams: properties, algorithms and applications. *SIAM Journal on Computing*, 16(1):78–96, 1987.
- [13] Franz Aurenhammer, Friedrich Hoffmann, and Boris Aronov. Minkowski-type theorems and least-squares clustering. *Algorithmica*, 20(1):61–76, 1998.
- [14] Julio Backhoff Veraguas, Mathias Beiglböck, and Gudmund Pammer. Existence, duality, and cyclical monotonicity for weak transport costs. *arXiv preprint arXiv:1809.05893*, 2019.

- [15] Martin Beckmann. A continuous model of transportation. *Econometrica*, 20:643–660, 1952.
- [16] Aurélien Bellet, Amaury Habrard, and Marc Sebban. Metric learning. *Synthesis Lectures on Artificial Intelligence and Machine Learning*, 9(1):1–151, 2015.
- [17] Jean-David Benamou and Yann Brenier. A computational fluid mechanics solution to the Monge-Kantorovich mass transfer problem. *Numerische Mathematik*, 84(3):375–393, 2000.
- [18] Jean-David Benamou, Guillaume Carlier, Marco Cuturi, Luca Nenna, and Gabriel Peyré. Iterative Bregman projections for regularized transportation problems. *SIAM Journal on Scientific Computing*, 37(2):A1111–A1138, 2015.
- [19] Jean-David Benamou, Guillaume Carlier, Quentin Mérigot, and Edouard Oudet. Discretization of functionals involving the Monge–Ampère operator. *Numerische Mathematik*, 134(3):611–636, 2016.
- [20] Jean-David Benamou, Francis Collino, and Jean-Marie Mirebeau. Monotone and consistent discretization of the Monge–Ampère operator. *Mathematics of Computation*, 85(302):2743–2775, 2016.
- [21] Jean-David Benamou, Brittany D Froese, and Adam M Oberman. Numerical solution of the optimal transportation problem using the Monge–Ampère equation. *Journal of Computational Physics*, 260:107–126, 2014.
- [22] Alain Berlinet and Christine Thomas-Agnan. *Reproducing Kernel Hilbert Spaces in Probability and Statistics*. Kluwer Academic Publishers, 2003.
- [23] Dimitri P Bertsekas. A new algorithm for the assignment problem. *Mathematical Programming*, 21(1):152–171, 1981.
- [24] Dimitri P Bertsekas. Auction algorithms for network flow problems: a tutorial introduction. *Computational Optimization and Applications*, 1(1):7–66, 1992.
- [25] Dimitri P Bertsekas and Jonathan Eckstein. Dual coordinate step methods for linear network flow problems. *Mathematical Programming*, 42(1):203–243, 1988.
- [26] Dimitris Bertsimas and John N Tsitsiklis. *Introduction to Linear Optimization*. Athena Scientific, 1997.
- [27] Rajendra Bhatia, Tanvi Jain, and Yongdo Lim. On the Bures–Wasserstein distance between positive definite matrices. *Expositiones Mathematicae*, 37(2):165–191, 2019.
- [28] Jérémie Bigot, Elsa Cazelles, and Nicolas Papadakis. Central limit theorems for entropy-regularized optimal transport on finite spaces and statistical applications. *Electronic Journal of Statistics*, 13(2):5120–5150, 2019.
- [29] Garrett Birkhoff. Extensions of jentzsch’s theorem. *Transactions of the American Mathematical Society*, 85(1):219–227, 1957.
- [30] Jose Blanchet and Karthyek Murthy. Quantifying distributional model risk via optimal transport. *Mathematics of Operations Research*, 44(2):565–600, 2019.
- [31] Sergey G. Bobkov. Berry–Esseen bounds and Edgeworth expansions in the central limit theorem for transport distances. *Probability Theory and Related Fields*, 170:229–262, 2018.
- [32] Vladimir I. Bogachev. *Measure Theory*. Springer, Berlin, 2007.
- [33] Emmanuel Boissard, Thibaut Le Gouic, and Jean-Michel Loubes. Distribution’s template estimate with Wasserstein metrics. *Bernoulli*, 21(2):740–759, 2015.

- [34] Francois Bolley, Arnaud Guillin, and Cédric Villani. Quantitative concentration inequalities for empirical measures on non-compact spaces. *Probability Theory and Related Fields*, 137(3):541–593, 2007.
- [35] Nicolas Bonneel, Julie Digne, Jean Feydy, and Gabriel Peyré. A survey of optimal transport for computer graphics and computer vision. *Computer Graphics Forum*, 42(2):439–460, 2023.
- [36] Nicolas Bonneel, Julien Rabin, Gabriel Peyré, and Hanspeter Pfister. Sliced and Radon Wasserstein barycenters of measures. *Journal of Mathematical Imaging and Vision*, 51(1):22–45, 2015.
- [37] Nicolas Bonnotte. *Unidimensional and evolution methods for optimal transportation*. PhD thesis, Université Paris-Sud, 2013.
- [38] Lev M Bregman. The relaxation method of finding the common point of convex sets and its application to the solution of problems in convex programming. *USSR Computational Mathematics and Mathematical Physics*, 7(3):200–217, 1967.
- [39] Yann Brenier. Décomposition polaire et réarrangement monotone des champs de vecteurs. *C. R. Acad. Sci. Paris Sér. I Math.*, 305(19):805–808, 1987.
- [40] Yann Brenier. Polar factorization and monotone rearrangement of vector-valued functions. *Communications on Pure and Applied Mathematics*, 44(4):375–417, 1991.
- [41] Donald Bures. An extension of Kakutani’s theorem on infinite product measures to the tensor product of semifinite w^* -algebras. *Transactions of the American Mathematical Society*, 135:199–212, 1969.
- [42] Luis Caffarelli. The Monge-Ampere equation and optimal transportation, an elementary review. *Lecture Notes in Mathematics*, Springer-Verlag, pages 1–10, 2003.
- [43] Luis Caffarelli, Mikhail Feldman, and Robert McCann. Constructing optimal maps for Monge’s transport problem as a limit of strictly convex costs. *Journal of the American Mathematical Society*, 15(1):1–26, 2002.
- [44] Emanuele Caglioti, Francois Golse, and Thierry Paul. Quantum optimal transport is cheaper. *arXiv preprint arXiv:1908.01829*, 2019.
- [45] Guillermo Canas and Lorenzo Rosasco. Learning probability measures with respect to optimal transport metrics. In F. Pereira, C. J. C. Burges, L. Bottou, and K. Q. Weinberger, editors, *Advances in Neural Information Processing Systems 25*, pages 2492–2500. Curran Associates, Inc., 2012.
- [46] Guillaume Carlier, Victor Chernozhukov, and Alfred Galichon. Vector quantile regression beyond the specified case. *Journal of Multivariate Analysis*, 161:96–102, 2017.
- [47] Guillaume Carlier, Vincent Duval, Gabriel Peyré, and Bernhard Schmitzer. Convergence of entropic schemes for optimal transport and gradient flows. *SIAM Journal on Mathematical Analysis*, 49(2):1385–1418, 2017.
- [48] Guillaume Carlier and Ivar Ekeland. Matching for teams. *Economic Theory*, 42(2):397–418, 2010.
- [49] Guillaume Carlier, Alfred Galichon, and Filippo Santambrogio. From knothe’s transport to Brenier’s map and a continuation method for optimal transport. *SIAM Journal on Mathematical Analysis*, 41(6):2554–2576, 2010.
- [50] José A Carrillo, Alina Chertock, and Yanghong Huang. A finite-volume method for nonlinear nonlocal equations with a gradient flow structure. *Communications in Computational Physics*, 17:233–258, 1 2015.
- [51] Shouvanik Chakrabarti, Yiming Huang, Tongyang Li, Soheil Feizi, and Xiaodi Wu. Quantum Wasserstein generative adversarial networks. *Advances in Neural Information Processing Systems*, 32, 2019.

- [52] Timothy M Chan. Optimal output-sensitive convex hull algorithms in two and three dimensions. *Discrete & Computational Geometry*, 16(4):361–368, 1996.
- [53] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [54] Yongxin Chen, Wilfrid Gangbo, Tryphon T Georgiou, and Allen Tannenbaum. On the matrix Monge-Kantorovich problem. *European Journal of Applied Mathematics*, 31(4):574–600, 2020.
- [55] Yongxin Chen, Tryphon T Georgiou, and Michele Pavon. On the relation between optimal transport and Schrödinger bridges: A stochastic control viewpoint. *Journal of Optimization Theory and Applications*, 169(2):671–691, 2016.
- [56] Yongxin Chen, Tryphon T Georgiou, and Allen Tannenbaum. Matrix optimal mass transport: a quantum mechanical approach. *arXiv preprint arXiv:1610.03041*, 2016.
- [57] Lénaïc Chizat and Francis Bach. On the global convergence of gradient descent for over-parameterized models using optimal transport. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [58] Lénaïc Chizat, Alex Delalande, and Tomas Vaškevičius. Sharper exponential convergence rates for Sinkhorn’s algorithm in continuous settings. *arXiv preprint arXiv:2407.01202*, 2024.
- [59] Lenaïc Chizat, Gabriel Peyré, Bernhard Schmitzer, and Francois-Xavier Vialard. An interpolating distance between optimal transport and Fisher–Rao metrics. *Foundations of Computational Mathematics*, 18(1):1–44, 2018.
- [60] Lenaïc Chizat, Gabriel Peyré, Bernhard Schmitzer, and Francois-Xavier Vialard. Scaling algorithms for unbalanced transport problems. *Mathematics of Computation*, 87:2563–2609, 2018.
- [61] Lenaïc Chizat, Gabriel Peyré, Bernhard Schmitzer, and Francois-Xavier Vialard. Unbalanced optimal transport: geometry and Kantorovich formulation. *Journal of Functional Analysis*, 274(11):3090–3123, 2018.
- [62] Imre Ciszár. Information-type measures of difference of probability distributions and indirect observations. *Studia Scientiarum Mathematicarum Hungarica*, 2:299–318, 1967.
- [63] Michael B Cohen, Aleksander Madry, Dimitris Tsipras, and Adrian Vladu. Matrix scaling and balancing via box constrained Newton’s method and interior point methods. In *58th IEEE Annual Symposium on Foundations of Computer Science*, pages 902–913. IEEE Computer Society, 2017.
- [64] Marco Cuturi. Sinkhorn distances: lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems 26*, pages 2292–2300, 2013.
- [65] Marco Cuturi and David Avis. Ground metric learning. *Journal of Machine Learning Research*, 15:533–564, 2014.
- [66] Marco Cuturi and Arnaud Doucet. Fast computation of Wasserstein barycenters. In *Proceedings of the 31st International Conference on Machine Learning*, volume 32 of *Proceedings of Machine Learning Research*, pages 685–693. PMLR, 2014.
- [67] Marco Cuturi and Gabriel Peyré. A smoothed dual approach for variational Wasserstein problems. *SIAM Journal on Imaging Sciences*, 9(1):320–343, 2016.
- [68] Marco Cuturi and Gabriel Peyré. Semidual regularized optimal transport. *SIAM Review*, 60(4):941–965, 2018.

- [69] George B Dantzig. Application of the simplex method to a transportation problem. *Activity Analysis of Production and Allocation*, 13:359–373, 1951.
- [70] Mingyang Deng, He Li, Tianhong Li, Yilun Du, and Kaiming He. Generative modeling via drifting. *arXiv preprint arXiv:2602.04770*, 2026.
- [71] Ishan Deshpande, Yuan-Ting Hu, Ruoyu Sun, Ayis Pyrros, Nasir Siddiqui, Sanmi Koyejo, Zhizhen Zhao, David A. Forsyth, and Alexander G. Schwing. Max-sliced Wasserstein distance and its use for GANs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10648–10656, 2019.
- [72] Jean Dolbeault, Bruno Nazaret, and Giuseppe Savaré. A new class of transport distances between measures. *Calculus of Variations and Partial Differential Equations*, 34(2):193–231, 2009.
- [73] Richard M. Dudley. The speed of mean Glivenko-Cantelli convergence. *Annals of Mathematical Statistics*, 40(1):40–50, 1969.
- [74] Arnaud Dupuy, Alfred Galichon, and Yifei Sun. Estimating matching affinity matrices under low-rank constraints. *Information and Inference: A Journal of the IMA*, 8(4):677–689, 2019.
- [75] Pavel Dvurechensky, Alexander Gasnikov, and Alexey Kroshnin. Computational optimal transport: Complexity by accelerated gradient descent is better than by sinkhorn’s algorithm. In Jennifer Dy and Andreas Krause, editors, *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1367–1376. PMLR, 10–15 Jul 2018.
- [76] Gintare Karolina Dziugaite, Daniel M Roy, and Zoubin Ghahramani. Training generative neural networks via maximum mean discrepancy optimization. In *Uncertainty in Artificial Intelligence- Proceedings of the 31st Conference, UAI 2015*, pages 258–267, 2015.
- [77] Matthias Erbar. The heat equation on manifolds as a gradient flow in the Wasserstein space. *Annales de l’Institut Henri Poincaré, Probabilités et Statistiques*, 46(1):1–23, 2010.
- [78] Peyman Mohajerin Esfahani and Daniel Kuhn. Data-driven distributionally robust optimization using the wasserstein metric: Performance guarantees and tractable reformulations. *Mathematical Programming*, 171(1-2):115–166, 2018.
- [79] Montacer Essid and Michele Pavon. Traversing the Schrödinger bridge strait: Robert Fortet’s marvelous proof redux. *Journal of Optimization Theory and Applications*, 181:23–60, 2019.
- [80] Mikhail Feldman and Robert McCann. Monge’s transport problem on a Riemannian manifold. *Transactions of the American Mathematical Society*, 354(4):1667–1697, 2002.
- [81] Jean Feydy, Thibault Séjourné, Franccois-Xavier Vialard, Shun-Ichi Amari, Alain Trouvé, and Gabriel Peyré. Interpolating between optimal transport and MMD using Sinkhorn divergences. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 2681–2690. PMLR, 2019.
- [82] Peter J Forrester and Mario Kieburg. Relating the Bures measure to the Cauchy two-matrix model. *Communications in Mathematical Physics*, 342(1):151–187, 2016.
- [83] Robert Fortet. Résolution d’un système d’équations de M. Schrödinger. *Journal de Mathématiques Pures et Appliquées*, 19(1–4):83–105, 1940.
- [84] Nicolas Fournier and Arnaud Guillin. On the rate of convergence in Wasserstein distance of the empirical measure. *Probability Theory and Related Fields*, 162(3-4):707–738, 2015.
- [85] Joel Franklin and Jens Lorenz. On the scaling of multidimensional matrices. *Linear Algebra and its Applications*, 114:717–735, 1989.

- [86] Charlie Frogner, Chiyuan Zhang, Hossein Mobahi, Mauricio Araya, and Tomaso A Poggio. Learning with a Wasserstein loss. In *Advances in Neural Information Processing Systems*, pages 2053–2061, 2015.
- [87] Alfred Galichon. *Optimal Transport Methods in Economics*. Princeton University Press, 2016.
- [88] Alfred Galichon and Bernard Salanié. Matching with trade-offs: revealed preferences over competing characteristics. Technical report, Preprint SSRN-1487307, 2009.
- [89] Thomas O Gallouët and Leonard Monsaingeon. A JKO splitting scheme for Kantorovich–Fisher–Rao gradient flows. *SIAM Journal on Mathematical Analysis*, 49(2):1100–1130, 2017.
- [90] Wilfrid Gangbo and Robert J McCann. The geometry of optimal transportation. *Acta Mathematica*, 177(2):113–161, 1996.
- [91] Wilfrid Gangbo and Andrzej Swiech. Optimal maps for the multidimensional Monge-Kantorovich problem. *Communications on Pure and Applied Mathematics*, 51(1):23–45, 1998.
- [92] Rui Gao and Anton J. Kleywegt. Distributionally robust stochastic optimization with Wasserstein distance. *Mathematics of Operations Research*, 48(2):603–655, 2023.
- [93] Ankit Garg and Rafael Oliveira. Recent progress on scaling algorithms and applications. *arXiv preprint arXiv:1808.09669*, 2018.
- [94] Matthias Gelbrich. On a formula for the l^2 wasserstein metric between measures on euclidean and hilbert spaces. *Mathematische Nachrichten*, 147(1):185–203, 1990.
- [95] Aude Genevay, Lénaïc Chizat, Francis Bach, Marco Cuturi, and Gabriel Peyré. Sample complexity of Sinkhorn divergences. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 1574–1583. PMLR, 2019.
- [96] Aude Genevay, Marco Cuturi, Gabriel Peyré, and Francis Bach. Stochastic optimization for large-scale optimal transport. In *Advances in Neural Information Processing Systems*, pages 3440–3448, 2016.
- [97] Tryphon T Georgiou and Michele Pavon. Positive contraction mappings for classical and quantum Schrödinger systems. *Journal of Mathematical Physics*, 56(3):033301, 2015.
- [98] Ugo Gianazza, Giuseppe Savaré, and Giuseppe Toscani. The Wasserstein gradient flow of the Fisher information and the quantum drift-diffusion equation. *Archive for Rational Mechanics and Analysis*, 194(1):133–220, 2009.
- [99] L Ambrosio-N Gigli. A user’s guide to optimal transport. *Modelling and Optimisation of Flows on Networks, Lecture Notes in Mathematics*, 2013.
- [100] Joan Glaunes, Alain Trounev, and Laurent Younes. Diffeomorphic matching of distributions: a new approach for unlabelled point-sets and sub-manifolds matching. In *Proceedings of the 2004 IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, volume 2, 2004.
- [101] Steven Gold and Anand Rangarajan. A graduated assignment algorithm for graph matching. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 18(4):377–388, April 1996.
- [102] Francoise Golse, Clément Mouhot, and Thierry Paul. On the mean field and classical limits of quantum mechanics. *Communications in Mathematical Physics*, 343(1):165–205, 2016.
- [103] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems*, pages 2672–2680, 2014.

- [104] Nathael Gozlan, Cyril Roberto, Paul-Marie Samson, and Prasad Tetali. Kantorovich duality for general transport costs and applications. *Journal of Functional Analysis*, 273(11):3327–3405, 2017.
- [105] Alexandre Gramfort, Gabriel Peyré, and Marco Cuturi. Fast optimal transport averaging of neuroimaging data. In *Information Processing in Medical Imaging - 24th International Conference, IPMI 2015*, pages 261–272, 2015.
- [106] Arthur Gretton, Karsten M Borgwardt, Malte Rasch, Bernhard Schölkopf, and Alex J Smola. A kernel method for the two-sample-problem. In *Advances in Neural Information Processing Systems*, pages 513–520, 2007.
- [107] Arthur Gretton, Karsten M Borgwardt, Malte J Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(Mar):723–773, 2012.
- [108] Arthur Gretton, Kevin Wenliang Li, Alexandre Galashov, James Thornton, Valentin De Bortoli, and Arnaud Doucet. On the wasserstein gradient flow interpretation of drifting models. *arXiv preprint arXiv:2605.05118*, 2026.
- [109] Mikhail Gromov. *Metric Structures for Riemannian and Non-Riemannian Spaces*. Progress in Mathematics. Birkhäuser, 2001.
- [110] Leonid Gurvits. Classical deterministic complexity of Edmonds’ problem and quantum entanglement. In *Proceedings of the Thirty-Fifth Annual ACM Symposium on Theory of Computing*, pages 10–19. ACM, 2003.
- [111] Leonid Gurvits. Classical complexity and quantum entanglement. *Journal of Computer and System Sciences*, 69(3):448–484, 2004.
- [112] Steven Haker, Lei Zhu, Allen Tannenbaum, and Sigurd Angenent. Optimal mass transport for registration and warping. *International Journal of Computer Vision*, 60(3):225–240, 2004.
- [113] Jiaqi Han, Puheng Li, Qiushan Guo, Renyuan Xu, Stefano Ermon, and Emmanuel J. Candès. One-step generative modeling via wasserstein gradient flows. *arXiv preprint arXiv:2605.11755*, 2026.
- [114] Leonid G Hanin. Kantorovich-Rubinstein norm and its application in the theory of Lipschitz spaces. *Proceedings of the American Mathematical Society*, 115(2):345–352, 1992.
- [115] Mathis Hardion and Théo Lacombe. The wasserstein gradient flow of the sinkhorn divergence between gaussian distributions. *arXiv preprint arXiv:2602.10726*, 2026.
- [116] Ping He, Om Khangaonkar, Hamed Pirsiavash, Yikun Bai, and Soheil Kolouri. Sinkhorn-drifting generative models. *arXiv preprint arXiv:2603.12366*, 2026.
- [117] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.
- [118] Thomas Hofmann, Bernhard Schölkopf, and Alexander J Smola. Kernel methods in machine learning. *Annals of Statistics*, 36(3):1171–1220, 2008.
- [119] Martin Idel. A review of matrix scaling and Sinkhorn’s normal form for matrices and positive maps. *arXiv preprint arXiv:1609.06349*, 2016.
- [120] Hicham Janati, Boris Muzellec, Gabriel Peyré, and Marco Cuturi. Entropic optimal transport between unbalanced Gaussian measures has a closed form. In *Advances in Neural Information Processing Systems*, 2020.
- [121] Richard Jordan, David Kinderlehrer, and Felix Otto. The variational formulation of the Fokker-Planck equation. *SIAM Journal on Mathematical Analysis*, 29(1):1–17, 1998.

- [122] Leonid Kantorovich. On the transfer of masses (in russian). *Doklady Akademii Nauk*, 37(2):227–229, 1942.
- [123] LV Kantorovich and G.S. Rubinstein. On a space of totally additive functions. *Vestn Leningrad Universitet*, 13:52–59, 1958.
- [124] Johan Karlsson and Axel Ringh. Generalized Sinkhorn iterations for regularizing inverse problems using optimal mass transport. *arXiv preprint arXiv:1612.02273*, 2016.
- [125] Abdelwahed Khamis, Russell Tsuchida, Mohamed Tarek, Vivien Rolland, and Lars Petersson. Scalable optimal transport methods in machine learning: A contemporary survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.
- [126] Philip A Knight. The Sinkhorn–Knopp algorithm: convergence and applications. *SIAM Journal on Matrix Analysis and Applications*, 30(1):261–275, 2008.
- [127] Soheil Kolouri, Se Rim Park, Matthew Thorpe, Dejan Slepcev, and Gustavo K Rohde. Optimal mass transport: signal processing and machine-learning applications. *IEEE Signal Processing Magazine*, 34(4):43–59, 2017.
- [128] Soheil Kolouri, Yang Zou, and Gustavo K Rohde. Sliced Wasserstein kernels for probability distributions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 5258–5267, 2016.
- [129] J. Kruithof. Telefoonverkeersrekening. *De Ingenieur*, 52:E15–E25, 1937.
- [130] Harold W. Kuhn. The hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955.
- [131] Brian Kulis. Metric learning: a survey. *Foundations and Trends in Machine Learning*, 5(4):287–364, 2012.
- [132] Naum S. Landkof. *Foundations of Modern Potential Theory*. Springer, 1972.
- [133] Hugo Lavenant and Filippo Santambrogio. The flow map of the Fokker–Planck equation does not provide optimal transport. *Applied Mathematics Letters*, 133:108225, 2022.
- [134] Thibaut Le Gouic and Jean-Michel Loubes. Existence and consistency of Wasserstein barycenters. *Probability Theory and Related Fields*, 168:901–917, 2016.
- [135] Jan Lellmann, Dirk A Lorenz, Carola Schönlieb, and Tuomo Valkonen. Imaging with Kantorovich–Rubinstein discrepancy. *SIAM Journal on Imaging Sciences*, 7(4):2833–2859, 2014.
- [136] Christian Léonard. From the Schrödinger problem to the Monge–Kantorovich problem. *Journal of Functional Analysis*, 262(4):1879–1920, 2012.
- [137] Christian Léonard. A survey of the Schrödinger problem and some of its connections with optimal transport. *Discrete Continuous Dynamical Systems Series A*, 34(4):1533–1574, 2014.
- [138] Christian Léonard. Revisiting Fortet’s proof of existence of a solution to the Schrödinger system. *arXiv preprint arXiv:1904.13211*, 2019.
- [139] Matthias Liero, Alexander Mielke, and Giuseppe Savaré. Optimal transport in competition with reaction: the Hellinger–Kantorovich distance and geodesic curves. *SIAM Journal on Mathematical Analysis*, 48(4):2869–2911, 2016.
- [140] Matthias Liero, Alexander Mielke, and Giuseppe Savaré. Optimal entropy-transport problems and a new hellinger–kantorovich distance between positive measures. *Inventiones Mathematicae*, 211(3):969–1117, 2018.

- [141] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- [142] Stuart Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.
- [143] Eliane Maria Loiola, Nair Maria Maia de Abreu, Paulo Oswaldo Boaventura-Netto, Peter Hahn, and Tania Querido. A survey for the quadratic assignment problem. *European Journal Operational Research*, 176(2):657–690, 2007.
- [144] Vince Lyzinski, Donniell E Fishkind, Marcelo Fiori, Joshua T Vogelstein, Carey E Priebe, and Guillermo Sapiro. Graph matching: relax at your own risk. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 38(1):60–73, 2016.
- [145] Shaojun Ma, Haodong Sun, Xiaojing Ye, Hongyuan Zha, and Haomin Zhou. Learning cost functions for optimal transport. *arXiv preprint arXiv:2002.09650*, 2020.
- [146] Jan Maas. Gradient flows of the entropy for finite Markov chains. *Journal of Functional Analysis*, 261(8):2250–2292, 2011.
- [147] Guillaume Mahey, Laetitia Chapel, Gilles Gasso, Clément Bonet, and Nicolas Courty. Fast optimal transport through sliced Wasserstein generalized geodesics. In *Advances in Neural Information Processing Systems*, 2023.
- [148] Robert J McCann. A convexity principle for interacting gases. *Advances in Mathematics*, 128(1):153–179, 1997.
- [149] Song Mei, Andrea Montanari, and Phan-Minh Nguyen. A mean field view of the landscape of two-layer neural networks. *Proceedings of the National Academy of Sciences*, 115(33):E7665–E7671, 2018.
- [150] Facundo Mémoli. On the use of Gromov–Hausdorff distances for shape comparison. In *Symposium on Point Based Graphics*, pages 81–90. Eurographics Association, 2007.
- [151] Facundo Mémoli. Gromov–Wasserstein distances and the metric approach to object matching. *Foundations of Computational Mathematics*, 11(4):417–487, 2011.
- [152] Quentin Mérigot. A multiscale approach to optimal transport. *Computer Graphics Forum*, 30(5):1583–1592, 2011.
- [153] Quentin Mérigot. A comparison of two dual methods for discrete optimal transport. In *Geometric science of information*, pages 389–396. Springer, 2013.
- [154] Quentin Mérigot, Alex Delalande, and Frédéric Chazal. Quantitative stability of optimal transport maps and linearization of the 2-Wasserstein space. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 3186–3196. PMLR, 2020.
- [155] Quentin Mérigot and Boris Thibert. Optimal transport: discretization and algorithms. *arXiv preprint arXiv:2003.00855*, 2020.
- [156] Alexander Mielke. Geodesic convexity of the relative entropy in reversible Markov chains. *Calculus of Variations and Partial Differential Equations*, 48(1-2):1–31, 2013.
- [157] Gaspard Monge. Mémoire sur la théorie des déblais et des remblais. *Histoire de l’Académie Royale des Sciences*, pages 666–704, 1781.
- [158] Eduardo Fernandes Montesuma, Fred Ngolè Mboula, and Antoine Souloumiac. Recent advances in optimal transport for machine learning. *arXiv preprint arXiv:2306.16156*, 2023.

- [159] Kimia Nadjahi, Alain Durmus, Umut Simsekli, and Roland Badeau. Asymptotic guarantees for learning generative models with the sliced-Wasserstein distance. In *Advances in Neural Information Processing Systems*, 2019.
- [160] Arkadi Nemirovski and Uriel Rothblum. On complexity of matrix scaling. *Linear Algebra and its Applications*, 302:435–460, 1999.
- [161] Lipeng Ning, Tryphon T Georgiou, and Allen Tannenbaum. On matrix-valued Monge–Kantorovich optimal mass transport. *IEEE Transactions on Automatic Control*, 60(2):373–382, 2015.
- [162] James B. Orlin. A polynomial time primal network simplex algorithm for minimum cost flows. *Mathematical Programming*, 78(2):109–129, 1997.
- [163] Felix Otto. The geometry of dissipative evolution equations: the porous medium equation. *Communications in Partial Differential Equations*, 26(1–2):101–174, 2001.
- [164] Francois-Pierre Paty and Marco Cuturi. Subspace robust Wasserstein distances. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 5072–5081, 2019.
- [165] Gabriel Peyré. Entropic approximation of Wasserstein gradient flows. *SIAM Journal on Imaging Sciences*, 8(4):2323–2351, 2015.
- [166] Gabriel Peyré. Optimal and diffusion transports in machine learning. *arXiv preprint arXiv:2512.06797*, 2025. Proc. 2026 International Congress of Mathematicians.
- [167] Gabriel Peyré. Muon dynamics as a spectral Wasserstein flow. *arXiv preprint arXiv:2604.04891*, 2026.
- [168] Gabriel Peyré. Robust sublinear convergence rates for iterative Bregman projections. *arXiv preprint arXiv:2602.01372*, 2026.
- [169] Gabriel Peyré, Lenaïc Chizat, Francois-Xavier Vialard, and Justin Solomon. Quantum entropic regularization of matrix-valued optimal transport. *European Journal of Applied Mathematics*, 30(6):1079–1102, 2019.
- [170] Gabriel Peyré, Marco Cuturi, et al. Computational optimal transport. *Foundations and Trends® in Machine Learning*, 11(5-6):355–607, 2019.
- [171] Gabriel Peyré, Marco Cuturi, and Justin Solomon. Gromov-Wasserstein averaging of kernel and distance matrices. In *International Conference on Machine Learning*, pages 2664–2672, 2016.
- [172] Gabriel Peyré, Jalal Fadili, and Julien Rabin. Wasserstein active contours. In *19th IEEE International Conference on Image Processing*, pages 2541–2544. IEEE, 2012.
- [173] Gabriel Peyré, Clarice Poon, and Oscar Tron. Curvature of optimal transport with respect to the cost and applications to inverse optimal transport. *arXiv preprint arXiv:2604.22670*, 2026.
- [174] Francois Pitié, Anil C. Kokaram, and Rozenn Dahyot. N-dimensional probability density function transfer and its application to color transfer. In *IEEE International Conference on Computer Vision*, pages 1434–1439, 2005.
- [175] Julien Rabin, Gabriel Peyré, Julie Delon, and Marc Bernot. Wasserstein barycenter and its application to texture mixing. In *International Conference on Scale Space and Variational Methods in Computer Vision*, pages 435–446. Springer, 2011.
- [176] Svetlozar T Rachev and Ludger Rüschendorf. *Mass Transportation Problems: Volume I: Theory*. Springer Science & Business Media, 1998.

- [177] Svetlozar T Rachev and Ludger Rüschendorf. *Mass Transportation Problems: Volume II: Applications*. Springer Science & Business Media, 1998.
- [178] Aaditya Ramdas, Nicolás García Trillos, and Marco Cuturi. On Wasserstein two-sample testing and related families of nonparametric tests. *Entropy*, 19(2):47, 2017.
- [179] Anand Rangarajan, Alan L Yuille, Steven Gold, and Eric Mjolsness. Convergence properties of the softassign quadratic assignment algorithm. *Neural Computation*, 11(6):1455–1474, August 1999.
- [180] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.
- [181] Erik Reinhard, Michael Adhikhmin, Bruce Gooch, and Peter Shirley. Color transfer between images. *IEEE Computer Graphics and Applications*, 21(5):34–41, 2001.
- [182] Emmanuel Rio. Asymptotic constants for minimal distance in the central limit theorem. *Electronic Communications in Probability*, 16:96–103, 2011.
- [183] Ralph Tyrell Rockafellar. *Convex Analysis*. Princeton university press, 2015.
- [184] Walter Rudin. *Real and Complex Analysis*. McGraw–Hill, third edition, 1987.
- [185] Ludger Ruschendorf. Convergence of the iterative proportional fitting procedure. *Annals of Statistics*, 23(4):1160–1174, 1995.
- [186] Ludger Rüschendorf and Wolfgang Thomsen. Closedness of sum spaces and the generalized Schrödinger problem. *Theory of Probability and Its Applications*, 42(3):483–494, 1998.
- [187] Hans Samelson et al. On the perron-frobenius theorem. *Michigan Mathematical Journal*, 4(1):57–59, 1957.
- [188] Stefan G. Samko, Anatoly A. Kilbas, and Oleg I. Marichev. *Fractional Integrals and Derivatives: Theory and Applications*. Gordon and Breach, 1993.
- [189] Michael E. Sander, Pierre Ablin, Mathieu Blondel, and Gabriel Peyré. Sinkformers: Transformers with doubly stochastic attention. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, volume 151 of *Proceedings of Machine Learning Research*, pages 3515–3530. PMLR, 2022.
- [190] Filippo Santambrogio. *Optimal transport for applied mathematicians*. Birkhauser, 2015.
- [191] Filippo Santambrogio. {Euclidean, metric, and Wasserstein} gradient flows: an overview. *Bulletin of Mathematical Sciences*, 7(1):87–154, 2017.
- [192] Bernhard Schmitzer and Christoph Schnörr. Modelling convex shape priors and matching based on the Gromov-Wasserstein distance. *Journal of Mathematical Imaging and Vision*, 46(1):143–159, 2013.
- [193] Bernhard Schölkopf and Alexander J Smola. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. MIT Press, 2002.
- [194] Erwin Schrödinger. Über die Umkehrung der Naturgesetze. *Sitzungsberichte Preuss. Akad. Wiss. Berlin. Phys. Math.*, 144:144–153, 1931.
- [195] Soroosh Shafieezadeh Abadeh, Peyman Mohajerin Mohajerin Esfahani, and Daniel Kuhn. Distributionally robust logistic regression. In C. Cortes, N. D. Lawrence, D. D. Lee, M. Sugiyama, and R. Garnett, editors, *Advances in Neural Information Processing Systems 28*, pages 1576–1584, 2015.
- [196] Aman Sinha, Hongseok Namkoong, and John Duchi. Certifying some distributional robustness with principled adversarial training. In *International Conference on Learning Representations*, 2018.

- [197] Richard Sinkhorn. A relationship between arbitrary positive matrices and doubly stochastic matrices. *The Annals of Mathematical Statistics*, 35(2):876–879, 1964.
- [198] Richard Sinkhorn. Diagonal equivalence to matrices with prescribed row and column sums. *American Mathematical Monthly*, 74:402–405, 1967.
- [199] Richard Sinkhorn and Paul Knopp. Concerning nonnegative matrices and doubly stochastic matrices. *Pacific Journal of Mathematics*, 21(2):343–348, 1967.
- [200] Justin Solomon, Fernando De Goes, Gabriel Peyré, Marco Cuturi, Adrian Butscher, Andy Nguyen, Tao Du, and Leonidas Guibas. Convolutional Wasserstein distances: efficient optimal transportation on geometric domains. *ACM Transactions on Graphics*, 34(4):66:1–66:11, 2015.
- [201] Justin Solomon, Gabriel Peyré, Vladimir G Kim, and Suvrit Sra. Entropic metric alignment for correspondence problems. *ACM Transactions on Graphics*, 35(4):72:1–72:13, 2016.
- [202] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.
- [203] Bharath K Sriperumbudur, Kenji Fukumizu, Arthur Gretton, Bernhard Schölkopf, and Gert RG Lanckriet. On integral probability metrics, φ -divergences and binary classification. *arXiv preprint arXiv:0901.2698*, 2009.
- [204] Bharath K Sriperumbudur, Kenji Fukumizu, Arthur Gretton, Bernhard Schölkopf, and Gert RG Lanckriet. On the empirical estimation of integral probability metrics. *Electronic Journal of Statistics*, 6:1550–1599, 2012.
- [205] Bharath K Sriperumbudur, Arthur Gretton, Kenji Fukumizu, Gert Lanckriet, and Bernhard Schölkopf. Injective Hilbert space embeddings of probability measures. In *Proceedings of the 21st Annual Conference on Learning Theory*, pages 111–122. Citeseer, 2008.
- [206] Michael L. Stein. *Interpolation of Spatial Data: Some Theory for Kriging*. Springer, 1999.
- [207] Karl-Theodor Sturm. The space of spaces: curvature bounds and gradient flows on the space of metric measure spaces. Preprint 1208.0434, arXiv, 2012.
- [208] Gábor J Székely and Maria L Rizzo. Testing for equal distributions in high dimension. *InterStat*, 5(16.10), 2004.
- [209] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [210] Titouan Vayer, Nicolas Courty, Romain Tavenard, Laetitia Chapel, and Rémi Flamary. Optimal transport for structured data with application on graphs. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 6275–6284. PMLR, 2019.
- [211] Cedric Villani. *Topics in Optimal Transportation*. Graduate Studies in Mathematics Series. American Mathematical Society, 2003.
- [212] Cedric Villani. *Optimal Transport: Old and New*, volume 338. Springer Verlag, 2009.
- [213] Wei Wang, Dejan Slepčev, Saurav Basu, John A Ozolek, and Gustavo K Rohde. A linear optimal transportation framework for quantifying and visualizing variations in sets of images. *International Journal of Computer Vision*, 101(2):254–269, 2013.

- [214] Jonathan Weed and Francis Bach. Sharp asymptotic and finite-sample rates of convergence of empirical measures in Wasserstein distance. *Bernoulli*, 25(4A):2620–2648, 2019.
- [215] Jonathan Weed, Jonathan Niles-Weed, and Philippe Rigollet. *Statistical Optimal Transport*, volume 2364 of *Lecture Notes in Mathematics*. Springer, Cham, 2025.
- [216] Holger Wendland. *Scattered Data Approximation*. Cambridge University Press, 2005.
- [217] Huan Xu and Shie Mannor. Distributionally robust markov decision processes. *Mathematics of Operations Research*, 37(2):288–300, 2012.
- [218] Insoon Yang. A convex optimization approach to distributionally robust markov decision processes with Wasserstein distance. *IEEE Control Systems Letters*, 1(1):164–169, 2017.
- [219] G Udny Yule. On the methods of measuring association between two attributes. *Journal of the Royal Statistical Society*, 75(6):579–652, 1912.